

Credit Card Services

Using the Simple Order API

August 2019



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Recent Revisions to This Document

Release	Changes
August 2019	All processors: moved reason codes for the Simple Order API to Reason Codes. All processors that support merchant-initiated transactions: updated the note at the beginning of the section by adding a link to a new FAQ. See "Merchant-Initiated Transactions," page 173. CyberSource through VisaNet: updated the TC 33 field names for the following fields. See "Request Fields," page 241. ■ ccAuthService_aggregatorID ■ invoiceHeader_salesOrganizationID ■ invoiceHeader_submerchantID FDI Australia ■ Added support for "Installment Payments," page 145. ■ Added support for "Least-Cost Routing (LCR)," page 164. ■ Added support for "Zero Amount Authorizations," page 229.

Release	Changes
July 2019	All processors that support merchant-initiated transactions: updated the description for subsequentAuthStoredCredential. See "Request Fields," page 241. Barclays: added support for "Merchant-Initiated Transactions," page 173. CyberSource through VisaNet: updated the information about installment payments with Mastercard: ■ In the feature description: • Updated the description for Mastercard in Brazil. See "Installment Payments on CyberSource through VisaNet in Brazil," page 152. • Added "Installment Payments on CyberSource through VisaNet in Croatia," page 155. • Added "Installment Payments on CyberSource through VisaNet in Georgia (Sakartvelo)," page 155. • Added "Installment Payments on CyberSource through VisaNet in Greece," page 156. • Added section for Mastercard in "Installment Payments on CyberSource through VisaNet in Other Countries," page 157. ■ In "Request Fields," page 241: • Added installment_firstInstallmentAmount • Updated descriptions for these fields: - billTo_merchantTaxID - installment_amount - installment_gracePeriodDuration - installment_gracePeriodDurationType - installment_monthlyInterestRate - installment_paymentType - installment_planType - installment_totalCount

(continued)

Release	Changes
July 2019 (continued)	■ In "Reply Fields," page 360: ● Added these new fields: - installment_amount - installment_firstInstallmentDate - installment_totalAmount - installment_totalCount ● Updated descriptions for these fields: - installment_additionalCosts - installment_additionalCostsPercentage - installment_amountFunded - installment_amountRequestedPercentage - installment_annualInterestRate - installment_expenses - installment_expensesPercentage - installment_fees - installment_feesPercentage - installment_firstInstallmentAmount - installment_insurance - installment_insurancePercentage - installment_maximumTotalCount - installment_minimumTotalCount - installment_monthlyInterestRate - installment_taxes - installment_taxesPercentage Elavon Americas: recurring payments with Visa must comply with the Visa merchant-initiated transactions mandate. See "Recurring Payments," page 205. HBoS: added support for "Merchant-Initiated Transactions," page 173. HSBC: added support for "Merchant-Initiated Transactions," page 173. Lloyds-OmniPay: added support for "Merchant-Initiated Transactions," page 173. SIX: added support for "Merchant-Initiated Transactions," page 173. Streamline: added support for "Merchant-Initiated Transactions," page 173.

Release	Changes
June 2019	All processors that support Mastercard SecureCode: changed the name of this feature to <i>Mastercard Identity Check</i>. All processors that support Verified by Visa: changed the name of this feature to <i>Visa Secure</i>. CCS (CAFIS): removed this processor from the documentation. Chase Paymentech Solutions: added support for Discover for "Zero Amount Authorizations," page 229. CyberSource through VisaNet: ■ Added support for "Japanese Payment Options," page 162. ■ Added support for the following payer authentication fields. See "Mastercard Identity Check," page 186. • ccAuthService_directoryServerTransactionID • ccAuthService_paSpecificationVersion ■ Added support for the Sumitomo Mitsui Card Co. acquirer. See Appendix K, "CyberSource through VisaNet Acquirers," on page 473. JCN Gateway: ■ Updated "Japanese Payment Options," page 162. ■ Removed the NICOS and ORICO cards. OmniPay Direct: added support for the following payer authentication fields. See "Mastercard Identity Check," page 186. ■ ccAuthService_directoryServerTransactionID ■ ccAuthService_paSpecificationVersion

Release	Changes
April 2019	Added new processor Getnet. See "Payment Processors," page 32, and Chapter 2, "Payment Card Processing," on page 38. Added new processor Software Express. See "Payment Processors," page 32, and Chapter 2, "Payment Card Processing," on page 38. American Express Direct: added values to Table 57, "Additional Amount Types for Goods and Services," on page 455. Chase Paymentech Solutions: added support for American Express for "Zero Amount Authorizations," page 229. CyberSource through VisaNet: ■ Added enhancements for Mastercard installment payments in Brazil: ● Added two funding methods: co-branded merchant financed and issuer merchant co-financed. See "Overview," page 150. ● Added two values for the installment_planType field for the new funding methods. See "Request Fields," page 241. ● Added two request fields for grace periods. See the installment_gracePeriodDuration and installment_gracePeriodDurationType fields in "Request Fields," page 241. ● Added three reply fields. See the installment_firstInstallmentAmount, installment_minimumTotalCount, and installment_maximumTotalCount fields in "Reply Fields," page 360. ■ Added the following request fields. For details, see "Request Fields," page 241. ● ccAuthService_cryptocurrencyPurchase ● ccAuthService_deferredAuthIndicator ● ccAuthService_lowValueExemptionIndicator ● ccAuthService_riskAnalysisExemptionIndicator ● ccAuthService_secureCorporatePaymentIndicator ● ccAuthService_trustedMerchantExemptionIndicator ● ccAuthService_verificationType Elavon Americas: ■ Added support for "Masterpass," page 169. ■ Updated the maximum length for the merchantReferenceCode field. See "Request Fields," page 241.

Release	Changes
March 2019	Added new processor RuPay. See "Payment Processors," page 32, and Chapter 2, "Payment Card Processing," on page 38. All processors that support merchant descriptors: moved the merchant descriptor information to Merchant Descriptors Using the Simple Order API. Chase Paymentech Solutions: added American Express as a supported card type for full authorization reversals. See "Reversing an Authorization," page 52. Credit Mutuel-CIC: updated the description for cardTypeSelectionIndicator to indicate that the field is required for the authorization service. See "Request Fields," page 241. CyberSource through VisaNet: added American Express as a supported card type for Network International. See Appendix K, "CyberSource through VisaNet Acquirers," on page 473. Ingenico ePayments: added note stating that CyberSource does not support Diners Club or Discover on Ingenico ePayments. See Table 10, "Payment Processors and Card Types," on page 33.
February 2019	Added new processor Rede. See "Payment Processors," page 32, and Chapter 2, "Payment Card Processing," on page 38.

About This Guide

Audience and Purpose

This guide is written for application developers who want to use the CyberSource Simple Order API to integrate payment card processing into their order management system.

Implementing the CyberSource credit card services requires software development skills. You must write code that uses the API request and reply fields to integrate the credit card services into your existing order management system.

Conventions

The following special statements are used in this document:



Note

A *Note* contains helpful suggestions or references to material not contained in this document.



Important

An *Important* statement contains information essential to successfully completing a task or learning a concept.



Warning

A *Warning* contains information or instructions, which, if not heeded, can result in a security risk, irreversible loss of data, or significant cost in time or revenue or both.

The following text conventions are used in this document:

Table 1 Text Conventions

Convention	Meaning
bold	Field and service names in text; for example: Include the ccAuthService_run field.
screen text	■ XML elements ■ Code examples ■ Values for API fields; for example: Set the ccAuthService_run field to <code>true</code>.

Related Documentation

- *Getting Started with CyberSource Advanced for the Simple Order API* describes how to get started using the Simple Order API. ([PDF](#) | [HTML](#))
- The *Classic Reporting Developer Guide* describes how to download reports. ([PDF](#) | [HTML](#))
- The *Secure Acceptance Checkout API Integration Guide* describes how to create a Secure Acceptance Checkout API profile. ([PDF](#) | [HTML](#))
- The *Secure Acceptance Hosted Checkout Integration Guide* describes how to create a Secure Acceptance Hosted Checkout profile. ([PDF](#) | [HTML](#))
- The [CyberSource API Versions](#) page provides information about the CyberSource API versions.

Refer to the Support Center for complete CyberSource technical documentation:

http://www.cybersource.com/support_center/support_documentation

Customer Support

For support information about any CyberSource service, visit the Support Center:

<http://www.cybersource.com/support>

Introduction to the Credit Card Services

Cards and Payment Methods

The credit card services can be used to process the types of cards and payment methods described in the following table.

Table 2 Cards and Payment Methods Processed with Credit Card Services

Card or Payment Method	Description
Credit cards	CyberSource can accept payments made with numerous types of credit cards, including Visa [®] , Mastercard [®] , American Express [®] , Discover [®] , Diners Club [®] , and JCB [®] .
Private label cards	Private label cards are payment cards that are issued by a private company and that can be used only at the issuing company's stores. If you are interested in using CyberSource to process transactions for your company's private label card, contact your CyberSource account representative for information.
Debit cards and prepaid cards	Prepaid cards, Visa-branded debit cards, and Mastercard-branded debit cards can be processed with the credit card services. See Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105.
Quasi-cash	A quasi-cash transaction is a cash-like transaction for the sale of items that are directly convertible to cash. See "Quasi-Cash," page 203.

Cartes Bancaires Cards

Processor:

- Credit Mutuel-CIC

CyberSource supports the following kinds of Cartes Bancaires co-badged cards:

- Visa and Cartes Bancaires
- Mastercard and Cartes Bancaires

**Note**

A co-badged card is not the same as a co-branded card:

- A *co-badged* card supports two or more card types, such as Visa and Cartes Bancaires.
- A *co-branded* card is sponsored by an issuing bank and a retail merchant.

Cartes Bancaires co-badged cards are processed as follows:

- CyberSource can use the card's BIN value to determine whether the card is a co-badged Cartes Bancaires card.
- You must let the customer choose the card type to use for the transaction. You can specify a default card type to use when the customer does not choose a card type or when the customer chooses a card type that is not supported by the card.
- You can include the **cardTypeSelectionIndicator** field in the authorization request.

For more information, contact CyberSource Customer Support.

Debit Cards

The following table provides an overview of the CyberSource services that you can use for processing debit cards.

Table 3 CyberSource Services for Processing Debit Cards

Services	Description
Credit card services	You can use credit card services to process payments with prepaid cards, Visa-branded debit cards, and Mastercard-branded debit cards. See Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105 .
PIN debit services	You can use PIN debit services to process payments if your processor supports PIN debit transactions. See PIN Debit Processing Using the Simple Order API .
PINless debit services	You can use PINless debit services to process payments if your business is in one of the acceptable merchant categories in which a card-not-present debit transaction is low risk. These categories include educational institutions, insurers, and utilities. See PINless Debit Card Services Using the Simple Order API .

Discover Acquisitions and Alliances

Discover has acquired or entered into alliances with the payment card companies shown in the following table.

Table 4 Discover Acquisitions and Alliances

Card Type	Description
China UnionPay Alliance	In 2005, China UnionPay and Discover announced a strategic alliance whereby China UnionPay cards would be routed to the Discover Network. As a result of this alliance: ■ If you have been accepting Discover but not China UnionPay, you are now able to accept and process China UnionPay cards that have been reissued with Discover bank identification numbers (BINs). ■ If you have been accepting China UnionPay but not Discover, you are now able to accept Discover cards.
Diners Club Acquisition	In July 2008, Discover acquired Diners Club International whereby Diners Club cards would be routed to the Discover Network starting October 16, 2009. As a result of this acquisition: ■ If you have been accepting Discover but not Diners Club, you are now able to accept Diners Club cards. ■ If you have been accepting Diners Club but not Discover, you are now able to accept Discover cards.
JCB (US Domestic) Alliance	In December 2006, JCB and Discover announced a strategic alliance whereby JCB cards would be routed to the Discover Network in the U.S. and select U.S. Territories (Puerto Rico, Guam, U.S. Virgin Islands, Northern Mariana Islands) that authorize, process, and fund in USD. As a result of this alliance: ■ If you have been accepting Discover but not JCB, you are now able to accept JCB cards. ■ If you have been accepting JCB but not Discover, you are now able to accept Discover cards.

For some card types on some processors, the information in your CyberSource account must include processor-issued IDs for these transactions to be processed successfully. Call CyberSource Customer Support to update your account information.

As a result of these acquisitions and alliances, the following card types are processed on the Discover Network:

- China UnionPay
- Diners Club
- Discover
- JCB (US Domestic): For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.



Note

Non-U.S. JCB transactions are still routed through JCB.



Note

Your processor takes care of routing your transactions; you do not need to do any additional processing to route these card types to the Discover Network.

mada Debit Cards

Processor:

- CyberSource through VisaNet—National Commercial Bank (NCB) is the only acquirer that supports mada.

CyberSource supports the following kinds of mada co-badged debit cards:

- Visa and mada
- Mastercard and mada



Note

A co-badged card is not the same as a co-branded card:

- A *co-badged* card supports two or more card types, such as Visa and mada.
- A *co-branded* card is sponsored by an issuing bank and a retail merchant.

mada co-badged debit cards are processed as follows:

- Only domestic processing in Saudi Arabia is supported.
- CyberSource sends the transactions directly to the Saudi Arabia Monetary Authority (SAMA) for processing.
- Payer authentication is supported. Visa Secure is supported for co-badged Visa-mada cards. Mastercard Identity Check is supported for co-badged Mastercard-mada cards. For more information, see ["Payer Authentication," page 183](#).

- For acquirers, the card type is identified as MD.
- In reports, the card type is identified as either Visa or Mastercard.
- Dual-message processing is not supported. Only single-message processing is supported. See ["Single-Message Processing," page 76](#).

**Important**

There are important differences between single-message processing and dual-message processing:

- With single-message processing, an authorization+capture request results in a full-financial transaction, which moves funds immediately.
- With single-message processing, the capture amount and authorization amount are the same.
- Several services and features are not supported for single-message processing because they are not applicable or could create reconciliation problems. See ["Unsupported Features," page 78](#).

[Table 5](#) provides additional details about co-badged mada debit cards.

Table 5 Card Types for mada Cards

Card Types Supported by the Card	Card Type Identified in the Transaction Request	Notes
Visa mada	Visa	CyberSource uses the BIN value to determine whether mada restrictions apply to the transaction.
Mastercard mada	Mastercard	CyberSource uses the BIN value to determine whether mada restrictions apply to the transaction.

Mastercard 2-Series Bank Identification Numbers

Mastercard expanded the BIN range by introducing BINs in the 222100-272099 range. Cards containing the 2-series BINs were issued in 2017.

Effective October 2016, Mastercard requires processors, acquirers, issuers, and merchants to support the 2-series BINs. Mastercard transactions on the 2-series primary account numbers (PANs) must be accepted, routed, and processed, and they must operate with the same rules that apply to the existing 5-series BINs.

Types of Transactions

Card-Present Transactions

When a customer uses a card that is physically present to make a purchase, the purchase is known as a *card-present transaction*. This type of transaction typically occurs in a retail environment. To process card-present transactions:

- Use the credit card services described in this guide.
- Provide card-present data as described in [Card-Present Processing Using the Simple Order API](#).

Card-Not-Present Transactions

When a customer provides a card number but you do not have access to the physical card, the purchase is known as a *card-not-present transaction*. This type of transaction typically occurs over the Internet or through a call center. To process card-not-present transactions, use the credit card services described in this guide.

Card-not-present transactions pose an additional level of risk to your business because you cannot directly verify the customer's identification. CyberSource offers features, such as Address Verification System (AVS) and Card Verification Numbers (CVN), in the credit card services that can reduce that risk by checking the validity of the customer's information and notifying you when discrepancies occur. For descriptions of AVS and CVN, see [Chapter 3, "Authorization Features," on page 88](#).

Transactions with Special Data

The credit card services can process these types of special data:

- Airline data: see [Airline Processing Using the Simple Order API](#).
- Level II and Level III data: see [Level II and Level III Processing Using the Simple Order API](#).
- Card-present data: see [Card-Present Processing Using the Simple Order API](#).

International Transactions

Compliance

Accepting payments from a country other than your own requires that you observe the processing rules and practices of the payment systems in that country. The following table describes areas of compliance that have particular focus.

Table 6 Compliance for International Transactions

Area of Compliance	Description
Merchant account descriptor requirements	The merchant account descriptor is a fixed text field that is associated with a payment card account. The purpose of the descriptor is to communicate merchant information to customers to remind them of the circumstances that triggered the payment. Merchant descriptors reduce the possibility of a chargeback. Accordingly, the merchant descriptor displayed on the customer's statement should be a close match to the name on your web site. It is not good practice to consolidate multiple web sites into a single payment card account and use a generic descriptor that more-or-less covers all offerings. For details about merchant descriptors, see "Merchant Descriptors," page 169 .
Excessive chargebacks	You are responsible for maintaining good customer support, rapid problem resolution, a high level of customer satisfaction, and transaction management processes that minimize fraudulent transactions. All of these are required to prevent an excessive number of chargebacks. In the event that credit card chargebacks become excessive, CyberSource can require you to undertake business process changes to reduce chargebacks. If the chargebacks are not reduced to a satisfactory level, CyberSource can terminate the account. If Ingenico ePayments is your processor, see Appendix P, "Ingenico ePayments Credit Card Reversals," on page 485 for more information about chargebacks. Note Ingenico ePayments was previously called <i>Global Collect</i>.

Merchant Remittance Funding

In conjunction with processing international transactions, you can request that CyberSource convert transaction proceeds to a currency other than the currency in which the transaction took place for funding into an operating account. Currency conversion uses a foreign exchange rate to calculate how much the transaction currency is worth in terms of the funding currency. The foreign exchange rate might be explicitly stated as a rate or implicitly stated as a transaction amount, and a funded amount and can vary from day to day. The foreign exchange rate might also include a mark-up for the foreign exchange risk, sales commissions, and handling costs.

Banks and Associations



Note

In this document, the word *processor* can refer to a processor, acquirer, or acquiring processor depending on your location.

Acquiring (Merchant) Banks

An acquiring, or merchant, bank offers accounts to businesses that accept payment cards. Before you can accept payments, you must have a merchant bank account from an acquiring bank. Your merchant bank account must be configured to process card-not-present or mail order/telephone order (MOTO) transactions.



Note

Each acquiring bank has connections to a limited number of payment processors. You must choose a payment processor that your acquiring bank supports. See ["Payment Processors," page 32](#).

Expect to be charged the fees shown in the following table.

Table 7 Fees

Fee	Description
Discount rates	Your acquiring bank charges a fee and collects a percentage of every transaction. The combination of the fee and the percentage is called the discount rate. These charges can be <i>bundled</i> (combined into a single charge) or <i>unbundled</i> (charged separately) depending on your acquiring bank and other factors.
Interchange fees	Visa and Mastercard each have a base fee, called the interchange fee, for each type of transaction. Your acquiring bank and processor can explain how to minimize this fee.
Chargebacks	When customers dispute charges to their accounts, you can incur chargebacks. A chargeback occurs when a charge on a customer's account is reversed. Your merchant bank removes the money from your account and could charge you a fee for the chargeback.

You are responsible for maintaining:

- Good customer support
- Rapid problem resolution
- A high level of customer satisfaction
- Transaction management processes that minimize fraudulent transactions

The items in the preceding list are required to prevent an excessive number of credit card chargebacks. In the event that credit card chargebacks become excessive, CyberSource can require you to undertake business process changes to reduce chargebacks. If the chargebacks are not reduced to a satisfactory level, CyberSource can terminate your account.

If you receive a large number of chargebacks or if a large number of your transactions involve fraud, your acquiring bank might increase your discount rate or revoke your merchant bank account. Contact CyberSource for information about CyberSource products that can help prevent fraud.

Issuing (Consumer) Banks

An issuing, or consumer, bank provides payment cards to and underwrites lines of credit for consumers. The issuing bank provides monthly statements and collects payments. Issuing banks must follow the rules of the payment card companies to which they belong.

Payment Card Companies

Payment card companies manage communications between acquiring banks and issuing banks. They also develop industry standards, support their brands, and establish fees for acquiring banks.

Some payment card companies, such as Visa and Mastercard, are trade associations that do not issue cards. Instead, issuing banks are members of these associations and they issue cards under license from the associations.

Other card companies, such as Discover and American Express, act as the issuing banks for their own cards. Before you use CyberSource to process cards from these companies, you must sign agreements with the companies.

Services

The credit card services are:

- Authorization: see ["Authorizing a Payment," page 38](#).
- Full authorization reversal: see ["Reversing an Authorization," page 52](#).
- Capture: see ["Capturing an Authorization," page 61](#).
- Credit: see ["Crediting a Payment," page 78](#).
- Void: see ["Voiding a Capture or Credit," page 85](#). This service is not restricted to the credit card services; it can also be used for other payment methods.

You can also request an authorization and capture together. See ["Performing a Sale," page 75](#).

Order Tracking

See [Getting Started with CyberSource Advanced for the Simple Order API](#) for information about order tracking. This section provides the names of the API fields that are used for order tracking for the credit card services.

Request IDs

For all CyberSource services, the request ID is returned in the reply messages in **requestID**. The following table lists the fields for the request IDs in request messages.

Table 8 Fields for Request IDs in Request Messages

Service	Request ID Field
Authorization reversal	ccAuthReversalService_authRequestID
Capture	ccCaptureService_authRequestID
Credit	ccCreditService_captureRequestID
Void	voidService_voidRequestID

Reconciliation IDs

The following table lists the fields for the reconciliation IDs, which are returned in the reply messages.

Table 9 Fields for Reconciliation IDs

Service	Reconciliation ID Field	Notes
Authorization	ccAuthReply_reconciliationID	For authorization requests, the reconciliation ID is returned only for these processors: ■ American Express Direct ■ Asia, Middle East, and Africa Gateway ■ Atos ■ BML Direct ■ Chase Paymentech Solutions ■ Cielo ■ CyberSource through VisaNet ■ FDC Compass ■ FDC Nashville Global ■ Moneris ■ Worldpay VAP Worldpay VAP was previously called <i>Little</i>.
Authorization reversal	ccAuthReversalReply_reconciliationID	For authorization reversal requests, the reconciliation ID is returned only for Cielo and Moneris.
Capture	ccCaptureReply_reconciliationID	The reconciliation ID is returned for all capture requests for all processors except JCN Gateway and RBS WorldPay Atlanta. When you perform multiple partial captures for an authorization, each reply includes a different reconciliation ID for each capture request. To learn whether your processor supports multiple partial captures, see "Multiple Partial Captures," page 71.
Credit	ccCreditReply_reconciliationID	The reconciliation ID is returned for all credit requests for all processors except JCN Gateway.

**Note**

On CyberSource through VisaNet, the reconciliation ID is mapped to the purchase identifier field which is sent to your acquirer.

**Note**

JCN Gateway does not support the reconciliation ID for any services.

Payment Processors

**Note**

In this document, the word *processor* can refer to processors, acquirers, or acquiring processors depending on your location.

Payment processors connect CyberSource servers with acquiring banks. Before you can accept payments, you must register with a payment processor. Your acquiring bank might require you to use a payment processor with which the bank has a business relationship.

CyberSource does not necessarily support all the features that are offered by each processor. This guide describes the payment processing features supported by CyberSource. The beginning of each feature description specifies which payment processors support the feature.

Your processor provides you with unique identification numbers for your account. You must provide these identification numbers to CyberSource Customer Support.

The following table lists the processors and corresponding card types that CyberSource supports for the credit card services.

**Note**

Only the card types explicitly listed here are supported.

Table 10 Payment Processors and Card Types

Payment Processor	Supported Card Types & Notes
AIBMS	Visa, Mastercard, Maestro (International), Maestro (UK Domestic)
American Express Brighton	American Express Depending on the country in which your business is located, you might need to get special permission from American Express before you can process transactions with American Express Brighton. For more information, contact American Express.
American Express Direct	American Express
Asia, Middle East, and Africa Gateway	Visa, Mastercard, American Express, Diners Club, JCB
Atos	Visa, Mastercard, Diners Club, JCB, Cartes Bancaires, Maestro (UK Domestic)
Barclays	Visa, Mastercard, JCB, Maestro (International), Maestro (UK Domestic) If you support Maestro (UK Domestic), you must also support Maestro (International), and you must support Mastercard Identity Check for both card types. GBP currency only for JCB and Maestro (UK Domestic).
Chase Paymentech Solutions	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Carte Blanche, Maestro (International)
Cielo	<i>Cielo 1.5</i> Visa, Mastercard, American Express, Discover, Diners Club, JCB, Maestro (International), Elo, Aura, Visa Electron <i>Cielo 3.0</i> Visa, Mastercard, American Express, Discover, Diners Club, JCB, Maestro (International), Elo, Aura, Visa Electron, Hipercard <i>Maestro (International)</i> The Maestro (International) card type is processed the same way that the Mastercard debit card is processed. Use card type value 002 (Mastercard) and use the ccAuthService_overridePaymentMethod field to indicate that it is a debit card. <i>Visa Electron</i> The Visa Electron card type is processed the same way that the Visa debit card is processed. Use card type value 001 (Visa).
Citibank India	For details about the Citibank India processor, contact your CyberSource sales representative.

Table 10 Payment Processors and Card Types (Continued)

Payment Processor	Supported Card Types & Notes
Comercio Latino	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard Note For Hipercard, Comercio Latino supports only 16-digit and 19-digit card numbers. Comercio Latino is the upgraded version of CyberSource Latin American Processing. If Rede is your acquirer, you must inform Comercio Latino of your Rede portal username and password. Important If Banorte is your acquirer, the currency that is stored in the Banorte merchant account database overrides the currency included in the transaction request. The supported currencies are MXN (Mexican peso) and USD (United States dollar).
Credit Mutuel-CIC	Visa, Mastercard, Cartes Bancaires
CyberSource Latin American Processing	Not all card types are supported in all Latin American countries. Contact CyberSource Customer Support for details. For the current processing connection to Latin America, use Comercio Latino. For some countries, you are required to submit the authorization request and the capture request together in the same message. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.
CyberSource through VisaNet	See Appendix K, "CyberSource through VisaNet Acquirers," on page 473 for the list of acquirers that are supported for CyberSource through VisaNet and the card types supported for each acquirer. The Visa Electron card type is processed the same way that the Visa debit card is processed. Use card type value 001 (Visa) for Visa Electron.
Elavon	Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International)
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass	Visa, Mastercard, American Express, Discover, Diners Club, JCB
FDC Germany	Visa, Mastercard, Maestro (UK Domestic), Maestro (International)

Table 10 Payment Processors and Card Types (Continued)

Payment Processor	Supported Card Types & Notes
FDC Nashville Global	Visa, Mastercard, American Express, Discover, Diners Club, JCB, China UnionPay
FDI Australia	Visa, Mastercard, American Express, Diners Club, JCB
FDMS Nashville	Visa, Mastercard, American Express, Discover, Diners Club, Carte Blanche, JCB
FDMS South	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Carte Blanche Important FDMS South does not accept authorization requests. If FDMS South is your processor, you need to either update or migrate your account depending on your settlement currency. If you settle transactions in CAD, you must do the following: ■ Contact CyberSource Customer Support to have your CyberSource account configured to send authorization requests to a third party who will forward the requests to FDMS South on your behalf. ■ Contact First Data to have your First Data account updated. If you settle transactions in USD, CyberSource recommends that you change your processor to FDC Nashville Global, FDMS Nashville, or FDC Compass.
Getnet	Visa, Mastercard, American Express, Elo, Hipercard
GPN GPN is the CyberSource name for Global Payments, Inc.'s East processing platform.	Visa, Mastercard, American Express, Discover, Diners Club, JCB Note USD is the only currency supported with American Express, Discover, Diners Club, and JCB. With Visa and Mastercard, you can use any currency that is supported by both GPN and CyberSource.
HBoS	Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
HSBC HSBC is the CyberSource name for HSBC U.K.	Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
Ingenico ePayments Ingenico ePayments was previously called <i>Global Collect</i> .	Visa, Mastercard, American Express, JCB, Maestro (UK Domestic), Delta, Visa Electron, Dankort, Cartes Bancaires, Carta Si, Eurocard Important CyberSource does not support Diners Club or Discover for payment card processing with Ingenico ePayments. A payment request using one of these card types will not result in a successful transfer of funds to your merchant account. CyberSource does not plan to add support for these card types on Ingenico ePayments.
JCN Gateway	Visa, Mastercard, American Express, Diners Club, JCB

Table 10 Payment Processors and Card Types (Continued)

Payment Processor	Supported Card Types & Notes
Lloyds-OmniPay	Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
LloydsTSB Cardnet	Visa, Mastercard, Maestro (UK Domestic)
Lynk	Visa, Mastercard, American Express, Discover, Diners Club, Carte Blanche, JCB
Moneris	Visa, Mastercard, American Express, Discover
OmniPay Direct	Supported acquirers: - Bank of America Merchant Services: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) - Cardnet International: Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International), Visa Electron - First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) - Global Payments International Acquiring: Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
OmniPay-Ireland OmniPay-Ireland is the CyberSource name for HSBC International.	Visa, Mastercard
PayEase China Processing	Visa, Mastercard, American Express, JCB The information in this guide does not apply to PayEase China Processing. All information required for PayEase China Processing is in the China Processing Implementation Guide .
RBS WorldPay Atlanta	Visa, Mastercard, American Express, Discover, Diners Club, JCB
Rede	Visa, Mastercard, American Express, Diners Club, JCB, Elo, Hipercard
RuPay	RuPay
SIX	Visa, Mastercard, Discover, Diners Club, JCB, Maestro (International), Maestro (UK Domestic), China UnionPay, Visa Electron Use card type value 033 for Visa Electron.
Software Express	Supported acquirer: Getnet Support card types: Visa, Mastercard

Table 10 Payment Processors and Card Types (Continued)

Payment Processor	Supported Card Types & Notes
Streamline	Visa, Mastercard, JCB, Cartes Bancaires, Dankort, Maestro (International), Maestro (UK Domestic) For Maestro (International), Identity Check processing is required.
TSYS Acquiring Solutions	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Carte Blanche
UATP	UATP
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Visa, Mastercard, American Express, Discover, Diners Club, JCB

Payment Card Processing

Authorizing a Payment

CyberSource supports authorizations for all processors.

Online Authorizations

Online authorization means that when you submit an order using a payment card, you receive an immediate confirmation about the availability of the funds. If the funds are available, the issuing bank reduces your customer's open to buy, which is the amount of credit available on the card. Most of the common payment cards are processed online. For online authorizations, you typically start the process of order fulfillment soon after you receive confirmation of the order.

Online authorizations expire with the issuing bank after a specific length of time if they have not been captured and settled. Most authorizations expire within five to seven days. The issuing bank sets the length of time.

**Note**

CyberSource is not informed by the issuing bank when an authorization expires. By default, the authorization remains in the CyberSource system for 60 days after the authorization date, even after it expires with the issuing bank.

When an authorization expires with the issuing bank, your bank or processor might require you to resubmit an authorization request and include a request for capture in the same message.

The following figure shows the steps that occur when you request an online credit card authorization.

Figure 1 Processing an Online Authorization



- 1 The customer places an order and provides the payment card number, the card expiration date, and additional information about the card.
- 2 You send a request for authorization over a secure Internet connection. When the customer buys a digitally delivered product or service, you can request both the authorization and the capture at the same time. When the customer buys a physically fulfilled product, do not request the capture until you ship the product.
- 3 CyberSource validates the order information then contacts your payment processor and requests authorization.
- 4 The processor sends the transaction to the payment card company, which routes it to the issuing bank for the customer’s payment card. Some card companies, including Discover and American Express, act as their own issuing banks.
- 5 The issuing bank approves or declines the request.

Depending on the processor and card type, the issuing bank can use AVS to confirm the billing address and CVN to verify that the customer has possession of the card. See [Chapter 3, "Authorization Features," on page 88](#).

For debit cards and prepaid cards, the issuing bank can approve a partial amount if the balance on the card is less than the requested authorization amount and if the transaction is enabled for partial authorization. For details about partial authorizations and for a list of the processors and card types supported for partial authorizations, see ["Partial Authorizations," page 105](#).

Note	For a limited number of processors and card types, partial authorizations and balance responses are supported for credit cards in addition to debit cards and prepaid cards. See "Partial Authorizations," page 105 , and "Balance Responses," page 111 .
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- 6 CyberSource runs its own tests then tells you whether the authorization succeeded.

Offline Authorizations

Offline authorization means that when you submit an order using a payment card, you do not know whether the funds are available until you capture the order and receive confirmation of payment. You typically do not ship the goods until you receive this payment confirmation. For offline payment cards, it usually takes five days longer to receive payment confirmation than for online cards.

Automatic Captures

Processors:

- Asia, Middle East, and Africa Gateway
- Cielo
- Comercio Latino
- CyberSource Latin American Processing

An *automatic capture* is an authorization that results in an immediate capture if the authorization is approved. A *bundled* request means that an authorization and a capture are requested at the same time.

To enable automatic captures for your account, contact CyberSource Customer Support.

Automatic captures are requested two ways:

- If automatic captures are enabled for your account, submit a bundled request.
- If automatic captures are not enabled for your account, submit a bundled request and set **ccAuthService_authType** to `AUTOCAPTURE`.

If your account is configured to enable automatic captures but you want to process a standard capture for a specific transaction, submit a bundled or standard authorization request and set **ccAuthService_authType** to `STANDARDCAPTURE`. The authorization is processed immediately, and if it is successful, the capture is processed during the next settlement submission cycle.

Authorization reversal and void services are not supported for automatic capture transactions.

Table 11 Automatic Capture Information for Specific Processors

Payment Processor	Automatic Capture Information
Asia, Middle East, and Africa Gateway	Certain acquirers that are connected to the Asia, Middle East, and Africa Gateway require automatic captures. Contact your CyberSource Customer Support representative to learn whether your acquirer uses standard or automatic capture processing.
Cielo	By default, your CyberSource account is configured to support standard capture processing. When you contact Customer Support to set up your account, you can request that the default type of capture be automatic capture instead of standard capture. All Aura Card transactions must be automatic captures.
Comercio Latino	When you contact Customer Support to set up your account, you can request that the default type of capture be automatic capture instead of standard capture.
CyberSource Latin American Processing	With CyberSource Latin American Processing, for some countries you are required to submit an automatic capture. For other countries, you can submit standard authorization and capture requests. Contact CyberSource Customer Support for each country's requirements. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.

Creating an Authorization Request

Step 1 Do not include any of these services in the request:

- Full authorization reversal (**ccAuthReversalService**)
- Credit (**ccCreditService**)
- Services for other payment methods, such as electronic checks or PayPal
- Risk update (**riskUpdateService**)

Step 2 Include the required fields in the request:

- If you are using Apple Pay, see [Apple Pay Using the Simple Order API](#).
- If you are using Google Pay, see [Google Pay Using the Simple Order API](#).
- If you are using Visa Checkout, see [Table 12](#) for the required fields:

Table 12 Required Fields for Authorizations with Visa Checkout

Field	Notes
ccAuthService_run	Set to <code>true</code> .
merchantID	
merchantReferenceCode	
paymentSolution	Set to <code>visacheckout</code> .
purchaseTotals_currency	
purchaseTotals_grandTotalAmount	Either purchaseTotals_grandTotalAmount or item_#_unitPrice must be included in the request.
vc_orderID	
wallet_type	Required only on CyberSource through VisaNet.

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
 - Optional request fields
 - Reply fields
- Otherwise, see [Table 13](#) for the required fields:

Table 13 Required Fields for Authorizations without Visa Checkout

Field	Notes
billTo_city ¹	
billTo_country ¹	
billTo_email ¹	
billTo_firstName ¹	
billTo_lastName ¹	
billTo_postalCode ¹	Required only for transactions in the U.S. and Canada.
billTo_state ¹	Required only for transactions in the U.S. and Canada.
billTo_street1 ¹	

¹ This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

Table 13 Required Fields for Authorizations without Visa Checkout (Continued)

Field	Notes
card_accountNumber	
card_cardType	Required for certain card types. CyberSource strongly recommends that you send the card type even if it is optional for your processor. Omitting the card type can cause the transaction to be processed with the wrong card type.
card_expirationMonth ¹	
card_expirationYear ¹	
ccAuthService_run	Set to <code>true</code> .
merchantID	
merchantReferenceCode	
purchaseTotals_currency	
purchaseTotals_grandTotalAmount	Either purchaseTotals_grandTotalAmount or item_#_unitPrice must be included in the request.
¹ This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting.	

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Optional request fields
- Reply fields

Step 3 If needed, modify the request to accommodate additional information for your processor. See ["Authorization Information for Specific Processors," page 48](#).

Step 4 Include authorization features in the request.

There are several authorization features that can be performed automatically depending on the information included in your request. These features are described in [Chapter 3, "Authorization Features," on page 88](#).

Step 5 Include optional features in the request.

There are several optional features that you can include in your request. These features are described in [Chapter 5, "Optional Features," on page 116](#).

Checking the Status of an Authorization Request

Processor:

- RuPay

Card type:

- RuPay

Typically, when a timeout occurs during an authorization, CyberSource automatically performs an authorization reversal. However, RuPay does not support online authorization reversals. When a timeout occurs during an authorization for a RuPay transaction, CyberSource includes a **ccAuthReply_reasonCode** field set to 151 in the authorization reply message. When you receive this value, check the status of the authorization by requesting the check status service.

The check status service includes a payment status field in the reply message. When the value of the payment status field is AUTHORIZED, proceed with the order by requesting the capture service. When the value of the payment status field is DECLINED, the authorization has been declined. You can request a different form of payment from the customer.

To create a check status request:

- Step 1** Do not include any other services in the request.
- Step 2** Include the required fields in the request:

Table 14 Required Fields for the Check Status Service

Field	Notes
ccCheckStatusService_authRequestID	Set to the request ID in the reply message for the authorization you want to check.
ccCheckStatusService_run	Set to <code>true</code> .
merchantID	
merchantReferenceCode	

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
 - Reply fields
-

Incremental Authorizations

Processor:

- CyberSource through VisaNet

Card types:

- Mastercard
- Visa

Incremental authorizations are useful when a customer adds products and services to an original purchase. After a successful original authorization, the incremental authorization service enables you to:

- Request subsequent authorizations.
- Request one capture for the original and incremental authorizations.

Limitations

- Maximum of 100 incremental authorizations per transaction, in addition to the original authorization.
- Interchange optimization is not supported.
- Split shipments are not supported.

Incremental Authorization Scenario

- 1 The customer reserves a hotel room for two nights at a cost of 200.00 per night. You request an authorization for 400.00. The authorization request is approved.
- 2 The customer orders dinner through room service the first night. You request an incremental authorization of 50.00 for the dinner.
- 3 The customer decides to stay an extra night. You request an incremental authorization of 200.00 for the additional night.
- 4 The customer uses items from the mini-bar. The cost of the mini-bar items is 50.00. You request an incremental authorization of 50.00.
- 5 When the customer ends their stay and checks out, they sign a receipt for 700.00, which is the total of all costs incurred.
- 6 You request a capture for 700.00.

Supported Features

- Authorization reversals with Mastercard. The reversal amount is the amount of the original authorization plus the amounts for all incremental authorizations. In the authorization reversal request, set the request ID to the request ID that was returned in the reply for the original authorization. See ["Reversing an Authorization," page 52](#).
- Authorization reversals with Visa. You can reverse any of the following amounts. See ["Reversing an Authorization," page 52](#).
 - Amount of the original authorization plus the amounts for all incremental authorizations. In the authorization reversal request, set the request ID to the request ID that was returned in the reply for the original authorization.
 - Amount of any one of the incremental authorizations. In the authorization reversal request, set the request ID to the request ID that was returned in the reply for the incremental authorization.
- Final authorization indicator. See ["Final Authorization Indicator," page 138](#).
- Merchant descriptors. When you include merchant descriptors in the original authorization request, CyberSource uses those merchant descriptors for all incremental authorizations that are associated with the original authorization. See ["Merchant Descriptors," page 169](#).
- Merchant-initiated reversals. See ["Merchant-Initiated Reversals and Voids," page 170](#). Merchant-initiated voids are not supported.
- Merchant-initiated transactions. See ["Merchant-Initiated Transactions," page 173](#).
- Partial authorizations for an original authorization. This feature is supported only for an original authorization made with a debit card or prepaid card. You can request an incremental authorization after a partial original authorization, but the incremental authorization cannot result in a partial authorization. See ["Partial Authorizations," page 105](#).

Performing an Incremental Authorization

Step 1 Request an authorization. See ["Creating an Authorization Request," page 41](#).

Step 2 Include the following required fields in each request for an incremental authorization:

Table 15 Required Fields for Incremental Authorizations

Field	Notes
<code>ccIncrementalAuthService_authRequestID</code>	Set to the request ID that was included in the original authorization reply message.
<code>ccIncrementalAuthService_run</code>	Set to <code>true</code> .
<code>merchantID</code>	
<code>merchantReferenceCode</code>	
<code>purchaseTotals_currency</code>	
<code>purchaseTotals_grandTotalAmount</code>	Either <code>purchaseTotals_grandTotalAmount</code> or <code>item_#_unitPrice</code> must be included in the request. CyberSource through VisaNet limits authorization and capture amounts to 12 digits; therefore, the maximum amount is 999999999999.
<code>subsequentAuthReason</code>	

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Optional request fields
- Reply fields

Step 3 When you request the capture, CyberSource recommends that you set the capture amount to the amount of the original authorization plus the amounts for all incremental authorizations.

If the requested capture amount is less than the amount of the original authorization plus the amounts for all incremental authorizations, CyberSource automatically performs a partial authorization reversal. See ["Automatic Partial Authorization Reversals," page 69](#).

If the requested capture amount is more than the amount of the original authorization plus the amounts for all incremental authorizations, CyberSource does not reject the capture request.

See ["Capturing an Authorization," page 61](#).

Authorization Information for Specific Processors

The following table provides additional information about authorizations for specific processors.

Table 16 Authorization Information for Specific Processors

Payment Processor	Authorization Information
American Express Direct	For USD, American Express Direct limits authorization and capture amounts to 9,999,999.00. For other currencies, the maximum amount depends on the currency. Contact American Express for the maximum amounts for the currencies that you are using. Regardless of exponent or currency, the maximum number of digits for the amount value is 12 digits.
Asia, Middle East, and Africa Gateway	The Asia, Middle East, and Africa Gateway limits authorization and capture amounts to four bytes; therefore, the maximum amount is 2147483647. Certain acquirers that are connected to the Asia, Middle East, and Africa Gateway require that an authorization be automatically captured. See "Automatic Captures," page 40.
Atos	Atos limits authorization, capture, and credit amounts to 12 digits; therefore, the maximum amount is 999999999999. Important Authorizations time out after 5 days, 20 hours, and 30 minutes. For Maestro (UK Domestic), when you submit a capture request after 5 days, 20 hours, and 30 minutes, you must reauthorize first. For all other card types, when you submit a capture request after 5 days, 20 hours, and 30 minutes, CyberSource tries to obtain a fresh authorization as described in "Authorization Refresh," page 68.
Barclays	CyberSource rounds the amount to the correct number of decimal places for the currency. Barclays supports enhanced response codes in authorization reply messages. Enhanced response codes provide detailed information about declined transactions. Contact Barclays customer support to have this capability enabled for your Barclays account.

Table 16 Authorization Information for Specific Processors (Continued)

Payment Processor	Authorization Information
Cielo	<i>Automatic Capture and Standard Capture</i> Cielo supports standard captures and automatic captures. ■ By default, your CyberSource account is configured to support standard capture processing. ■ For an Aura Card transaction, you must set the authorization type to <code>AUTOCAPTURE</code>. See "Automatic Captures," page 40. <i>Combo Cards</i> Some card types support two payment methods: they can be processed as credit cards and debit cards. On Cielo: ■ The default payment method is credit card. ■ You can override the default payment method by including the <code>ccAuthService_overridePaymentMethod</code> field, a flag that indicates whether the card is being used as a credit card or debit card, in the authorization request. <i>Debit Cards</i> For debit cards on Cielo: ■ You must request an automatic capture. See "Automatic Captures," page 40. ■ You must include payer authentication data in the request for cards that support it on the Cielo gateway. For a description of payer authentication, see "Payer Authentication," page 183. ■ Some card types must always be processed as debit cards and must be identified with the override payment method field. Cards that must always be processed as debit cards include: • Visa Electron • Maestro (International)

Table 16 Authorization Information for Specific Processors (Continued)

Payment Processor	Authorization Information
Comercio Latino	Regardless of exponent or currency, the maximum number of digits for the amount value is 19 digits. This maximum amount may be subject to further restrictions based on the acquirer requirements. <i>Debit Cards</i> For debit cards on Comercio Latino: ■ You must request an automatic capture. See "Automatic Captures," page 40. ■ You must include payer authentication data in the request for cards that support it on the Comercio Latino gateway. For a description of payer authentication, see "Payer Authentication," page 183. ■ Some card types must always be processed as debit cards and must be identified with the ccAuthService_overridePaymentMethod field. Cards that must always be processed as debit cards include: • Visa Electron • Maestro (International)
CyberSource Latin American Processing	With CyberSource Latin American Processing, for some countries you must submit an automatic capture. See "Automatic Captures," page 40. For other countries, you can submit standard authorization and capture requests. Contact CyberSource Customer Support for each country's requirements. For transactions in Brazil, you must request the follow-on capture within five days of the authorization request. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.
CyberSource through VisaNet	CyberSource through VisaNet limits authorization and capture amounts to 12 digits; therefore, the maximum amount is 999999999999. When you perform a reauthorization or an incremental authorization, your authorization request must include subsequent authorization fields as described in "Merchant-Initiated Transactions," page 173.

Table 16 Authorization Information for Specific Processors (Continued)

Payment Processor	Authorization Information
FDMS South	For the Indonesian rupiah (IDR) and Chilean peso (CLP) currencies only: ■ Rounding occurs, which can cause a minor discrepancy that consists of a maximum of one currency unit between the amount you requested and the amount that is authorized. ■ When a transaction is enabled for partial authorization, you must ensure that the requested amount does not include any digits to the right of the decimal separator. For a description of partial authorizations, see "Partial Authorizations," page 105.
GPN	GPN limits the authorization, capture, and credit amounts to 10 digits.
Ingenico ePayments Ingenico ePayments was previously called <i>Global Collect</i> .	For Cartes Bancaires, the authorization and capture amount must be 0.99 EUR or more.
Moneris	Moneris limits authorization and capture amounts to nine digits; therefore, the maximum amount is 9999999.99.
RBS WorldPay Atlanta	RBS WorldPay Atlanta limits the authorization, capture, and credit amounts to the equivalent of 999,999.99 USD. Depending on the value you send, the decimal is either truncated or appended. For example, if you send 1.123 the decimal is truncated to 1.12. If you send 123 it is converted to 123.00.
RuPay	RuPay requires payer authentication for all authorization requests. See "RuPay PaySecure," page 194. The reply message for the payer authentication check enrollment service indicates whether a card requires single-message processing or dual-message processing. For cards that require single-message processing, see "Single-Message Processing," page 76. In the authorization request, the e-commerce indicator must be rpy. When you do not receive an authorization reply message from CyberSource, request the check status service. See "Checking the Status of an Authorization Request," page 44.
SIX	A request for an authorization must include a capture request.

Table 16 Authorization Information for Specific Processors (Continued)

Payment Processor	Authorization Information
Streamline	Streamline limits authorization and capture amounts to 11 digits; therefore, the maximum amount is 999999999.99. Streamline supports enhanced response codes in authorization reply messages. Enhanced response codes provide detailed information about declined transactions. Contact Streamline customer support to have this capability enabled for your Streamline account.
TSYS Acquiring Solutions	TSYS Acquiring Solutions limits authorization and capture amounts to the equivalent of 99,999.99 USD. To process an amount greater than this, contact TSYS Acquiring Solutions.
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Worldpay VAP limits authorization and capture amounts to eight digits; therefore, the maximum amount is 99999999.

Reversing an Authorization

The full authorization reversal service releases the hold that the authorization placed on the customer's credit card funds. Use this service to reverse an unnecessary or undesired authorization.



Note

Each issuing bank has its own rules for deciding whether a full authorization reversal succeeds or fails. When a reversal fails, contact the issuing bank to learn whether it is possible to reverse the authorization by alternate means.

If your processor supports authorization reversal after void (ARAV), you can reverse an authorization after you void the associated capture. See "[Authorization Reversal after Void \(ARAV\)](#)," [page 59](#). If your processor does not support ARAV, you can use the full authorization reversal service only for an authorization that has not been captured and settled.

Supported Processors and Card Types

The following table lists the processors that are supported for full authorization reversals. For processors that support debit cards and prepaid cards, the full authorization reversal service works for debit cards and prepaid cards in addition to credit cards.

Table 17 Processors That Support Full Authorization Reversals

Processor	Card Types and Notes
AIBMS	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
American Express Direct	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact American Express for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Barclays	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. CyberSource supports enhanced authorization reversals on this processor; therefore, CyberSource sends the processor extra data in the authorization reversal request. You do not need to process or monitor the extra data. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Chase Paymentech Solutions	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Maestro (International), Discover, and Diners Club. Time limit: a full authorization reversal must occur within three days of the authorization. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Cielo	Card types supported for full authorization reversals: Visa, Mastercard, American Express.
Comercio Latino	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard. Time limit: a full authorization reversal must occur by 23:59 BRT (UTC-3) on the same day as the authorization.
Credit Mutuel-CIC	Visa, Mastercard, Cartes Bancaires

Table 17 Processors That Support Full Authorization Reversals (Continued)

Processor	Card Types and Notes
CyberSource through VisaNet	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Diners Club, JCB, Discover, mada as described in "mada Debit Cards," page 24. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59. Important For mada, see "Authorization Reversals for Single-Message Processing," page 59.
Elavon	Card types supported for full authorization reversals: Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International). Time limit: a full authorization reversal must occur within 24 hours of the authorization.
Elavon Americas	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Discover, Diners Club, JCB, and China UnionPay.
FDC Compass	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Discover, Diners Club, and JCB. Time limit: a full authorization reversal must occur within three days of the authorization. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDC Germany	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDC Nashville Global	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic), China UnionPay. For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands. For Discover, Diners Club, and JCB (US Domestic), full authorization reversals are supported for USD transactions only. There are no currency restrictions for full authorization reversals for Visa, Mastercard, and American Express. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDI Australia	Visa, Mastercard, American Express, Diners Club, JCB

Table 17 Processors That Support Full Authorization Reversals (Continued)

Processor	Card Types and Notes
FDMS Nashville	Card types supported for full authorization reversals: Visa, Mastercard, Discover, Diners Club, and JCB (US Domestic). For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDMS South	Card types supported for full authorization reversals: Visa, Mastercard, Discover, and JCB (US Domestic). For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands. Full authorization reversals: ■ Are supported only for transactions that do not go through a currency conversion. ■ Are supported for the following types of merchants and currencies: • Merchants located in the U.S. who authorize, settle, and fund in U.S. dollars. • Merchants located in Canada who authorize, settle, and fund in Canadian dollars. • Merchants located in Latin America or the Caribbean who authorize, settle, and fund in U.S. dollars. • Merchants located in Europe who authorize, settle, and fund in the currency for the country in which the merchant is located. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Getnet	Visa, Mastercard, American Express, Elo, Hipercard
GPN	Card types supported for full authorization reversals: Visa, Mastercard, Discover, Diners Club, and JCB. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
HBoS	Requirement: you are responsible for complying with the processor’s specific requirements for full authorization reversals. Contact the processor for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
HSBC HSBC is the CyberSource name for HSBC U.K.	Card types supported for full authorization reversals: Visa, Mastercard, Maestro (UK Domestic), Maestro (International). Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.

Table 17 Processors That Support Full Authorization Reversals (Continued)

Processor	Card Types and Notes
JCN Gateway	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Diners Club, JCB. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Lloyds-OmniPay	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
LloydsTSB Cardnet	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Moneris	Card types supported for full authorization reversals: Visa, Mastercard, American Express, and Discover.
OmniPay Direct	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. Card types supported for full authorization reversals on each acquirer: ■ Bank of America Merchant Services: Visa, Mastercard, Maestro (UK Domestic), and Maestro (International). ■ Cardnet International: Visa, Mastercard, Maestro (UK Domestic), and Maestro (International). ■ First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), and Maestro (International). ■ Global Payments International Acquiring: Visa, Mastercard, Maestro (UK Domestic), and Maestro (International). On Cardnet International, CyberSource supports enhanced authorization reversals; therefore, CyberSource sends the processor extra data in the authorization reversal request. You do not need to process or monitor the extra data. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
RBS WorldPay Atlanta	Card types supported for full authorization reversals: Visa, Mastercard, American Express, and Discover.
Rede	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Diners Club, JCB, Elo, Hipercard

Table 17 Processors That Support Full Authorization Reversals (Continued)

Processor	Card Types and Notes
SIX	Card types supported for full authorization reversals: Visa, Mastercard, Discover, Diners Club, JCB, Maestro (International), Maestro (UK Domestic), China UnionPay, Visa Electron.
Software Express	Card types supported for full authorization reversals: Visa, Mastercard
Streamline	Requirement: you are responsible for complying with the processor's specific requirements for full authorization reversals. Contact the processor for more information. CyberSource supports enhanced authorization reversals on this processor; therefore, CyberSource sends the processor extra data in the authorization reversal request. You do not need to process or monitor the extra data. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
TSYS Acquiring Solutions	Card types supported for full authorization reversals: Visa, Mastercard, American Express, Discover, Diners Club, and JCB. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Card types supported for full authorization reversals: Visa, Mastercard, Discover, Diners Club, and JCB. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.

Creating a Full Authorization Reversal Request

A full authorization reversal is a follow-on transaction that uses the request ID returned from a previous authorization. The request ID links the full authorization reversal to the authorization. CyberSource uses the request ID to look up the customer's billing and account information from the original authorization, so you are not required to include those fields in your full authorization reversal request.



Note

For American Express aggregator transactions on CtV, CyberSource retrieves the aggregator information for the authorization that is being reversed.

For information about requesting a follow-on service, see [Getting Started with CyberSource Advanced for the Simple Order API.](#)

To create a full authorization reversal request:

Step 1 Do not include any other CyberSource services in the request.

Step 2 Include the required fields in the request:

Table 18 Required Fields for Full Authorization Reversals

Field	Notes
ccAuthReversalService_authRequestID	Set to the request ID that was included in the authorization reply message.
ccAuthReversalService_run	Set to <code>true</code> .
merchantID	
merchantReferenceCode	
paymentSolution	Include this field only when using Visa Checkout.
purchaseTotals_currency	
purchaseTotals_grandTotalAmount	Either purchaseTotals_grandTotalAmount or item_#_unitPrice must be included in the request.
vc_orderID	Include this field only when using Visa Checkout.

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Optional request fields
- Reply fields

Step 3 Make sure that the amount of the reversal is the same as the amount that was authorized:

- You cannot partially reverse an authorization; you can reverse an authorization only for its full amount.
- When you use a debit card or prepaid card and only a partial amount was approved, the amount of the reversal must be the amount that was authorized, not the amount that was requested.

Authorization Reversals for Single-Message Processing

Processors:

- CyberSource through VisaNet—National Commercial Bank (NCB) is the only acquirer that supports single-message processing.

Card Types:

- On CyberSource through VisaNet—all mada card types as described in ["mada Debit Cards," page 24](#).



Note

Authorization reversals are not supported for least-cost routing and single-message processing on Worldpay VAP. To process an aborted transaction or a return, request the credit service. See ["Crediting a Payment," page 78](#).

An authorization reversal returns funds to the customer, either by releasing the hold on the customer's funds or by transferring money from your account to the customer's account. If the authorization reversal request is not successful, perform a credit to refund the money to the customer.

Authorization Reversal after Void (ARAV)

Processors:

- American Express Direct
- Barclays
- Chase Paymentech Solutions
- Comercio Latino
- Credit Mutuel-CIC
- CyberSource through VisaNet
- FDC Compass
- FDC Germany
- FDC Nashville Global
- FDMS Nashville
- FDMS South
- GPN
- HBoS
- HSBC: HSBC is the CyberSource name for HSBC U.K.Lloyds-OmniPay
- LloydsTSB Cardnet

- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- SIX
- Streamline
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.

This feature enables you to reverse an authorization after you void the associated capture.



Important

This functionality enables you to meet the Visa mandate requirements to reverse unused authorizations, and it benefits the customer by releasing the hold on unused credit card funds.

To reverse an authorization after a void:

Step 1 Void a capture. See ["Voiding a Capture or Credit," page 85](#).

Step 2 Reverse the authorization associated with the capture. See ["Reversing an Authorization," page 52](#).



Note

You might need to perform additional steps if you performed multiple partial captures for the authorization. To learn whether your processor supports multiple partial captures, see ["Multiple Partial Captures," page 71](#). For information about multiple captures and ARAV, see ["Multiple Partial Captures and Authorization Reversal after Void," page 73](#).

Automatic ARAV

Processors:

- Credit Mutuel-CIC
- CyberSource through VisaNet
- SIX

Normally, you must send an authorization reversal request after you void the associated capture. With automatic ARAV, CyberSource automatically reverses the authorization after you void the associated capture. To enable automatic ARAV, contact CyberSource Customer Support to have your account configured for this feature.

Capturing an Authorization

CyberSource supports captures for all processors.

When you are ready to fulfill a customer's order and transfer funds from the customer's bank to your bank, capture the authorization for that order.

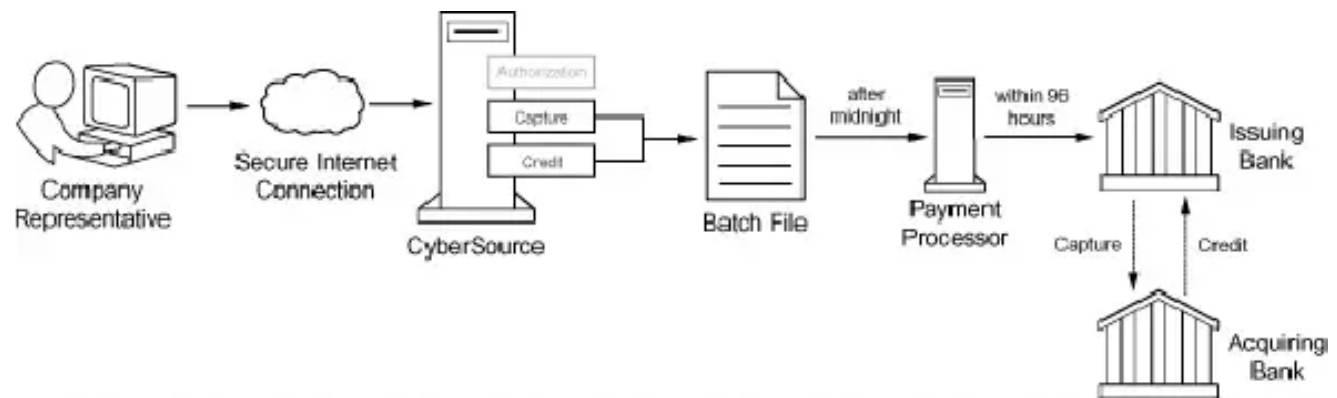
When fulfilling only part of a customer's order, do not capture the full amount of the authorization. Capture only the cost of the items that you ship. When you ship the remaining items, request a new authorization, and then capture the new authorization.

Captures

Unlike authorizations, a capture does not happen in real time. All of the capture requests for a day are placed in a batch file and sent to the processor. In most cases, the batch is settled at night. It usually takes two to four days for your acquiring bank to deposit funds in your merchant bank account.

The following figure shows the steps that occur when you request a capture or credit.

Figure 2 Processing a Capture or Credit



- 1 You send a request for capture or credit over a secure Internet connection.
- 2 CyberSource validates the order information then stores the capture or credit request in a batch file.
- 3 After midnight, CyberSource sends the batch file to your payment processor.

- 4 The processor settles the capture or credit request and transfers funds to the appropriate bank account.

**Note**

The processor does not notify CyberSource when a transaction is declined. To ensure that all captures and credits are processed, reconcile your system's reports with the reports from your processor. See [Getting Started with CyberSource Advanced for the Simple Order API](#) for information about reconciliation.

Due to the potential delay between authorization and capture, the authorization might expire with the issuing bank before you request capture. Most authorizations expire within five to seven days. If an authorization expires with the issuing bank before you request the capture, your bank or processor might require you to resubmit an authorization request and include a request for capture in the same message.

**Note**

CyberSource is not informed by the issuing bank when an authorization expires. By default, the authorization remains in the CyberSource system for 60 days after the authorization date, even after it expires with the issuing bank.

Creating a Capture Request

A capture is a follow-on transaction that uses the request ID returned from a previous authorization. The request ID links the capture to the authorization. CyberSource uses the request ID to look up the customer's billing and account information from the original authorization, so you are not required to include those fields in your capture request.

For information about requesting a follow-on service, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

**Note**

For Atos, your request for a capture must also include the request token returned from a previous authorization in addition to the request ID. Like the request ID, the request token links the capture to the authorization. Send the request token in the **orderRequestToken** field.

To create a capture request:

Step 1 Do not include any of these services in the request:

- Full authorization reversal (**ccAuthReversalService**)
- Credit (**ccCreditService**)
- Services for other payment methods, such as electronic checks or PayPal
- Risk update (**riskUpdateService**)
- Advanced fraud screen (**afsService**)

Step 2 Include the required fields in the request:

Table 19 Required Fields for Captures

Field	Notes
ccCaptureService_run	Set to <code>true</code> .
ccCaptureService_authRequestID	Set to the request ID that was included in the authorization reply message. Optional when ccAuthService and ccCaptureService are in the same request.
merchantID	
merchantReferenceCode	
orderRequestToken	Required only for Atos.
paymentSolution	Include this field only when using Visa Checkout.
purchaseTotals_currency	
purchaseTotals_grandTotalAmount	Either purchaseTotals_grandTotalAmount or item_#_unitPrice must be included in the request.
vc_orderID	Include this field only when using Visa Checkout.

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Optional request fields
- Reply fields

Step 3 If needed, modify the request to accommodate additional information for your processor. See [Table 20, "Capture Information for Specific Processors," on page 64](#).



Note

For Cartes Bancaires cards, your capture request cannot be for less than 0.99 EUR.

Step 4 Include optional features in the request.

There are several optional features that you can include in your request. These features are described in [Chapter 5, "Optional Features,"](#) on page 116.

Capture Information for Specific Processors

The following table provides additional information about captures for some processors.

Table 20 Capture Information for Specific Processors

Payment Processor	Capture Information
AIBMS	Multiple partial captures are supported. See "Multiple Partial Captures," page 71.
American Express Direct	For USD, American Express Direct limits authorization and capture amounts to 9,999,999.00. For other currencies, the maximum amount depends on the currency. Contact American Express for the maximum amounts for the currencies that you are using. Regardless of exponent or currency, the maximum number of digits for the amount value is 12 digits. Multiple partial captures are supported. See "Multiple Partial Captures," page 71. Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73.
Asia, Middle East, and Africa Gateway	The Asia, Middle East, and Africa Gateway limits authorization and capture amounts to four bytes, which is 2147483647. Multiple partial captures are supported. See "Multiple Partial Captures," page 71. Automatic capture requirement: certain acquirers that are connected to the Asia, Middle East, and Africa Gateway require automatic captures. See "Automatic Captures," page 40. Contact your CyberSource Customer Support representative to learn whether your acquirer uses standard or automatic captures.
Atos	Atos limits authorization, capture, and credit amounts to 12 digits; therefore, the maximum amount is 999999999999. Important Authorizations time out after 5 days, 20 hours, and 30 minutes. For Maestro (UK Domestic), when you submit a capture request after 5 days, 20 hours, and 30 minutes, you must reauthorize first. For all other card types, when you submit a capture request after 5 days, 20 hours, and 30 minutes, CyberSource tries to obtain a fresh authorization as described in "Authorization Refresh," page 68.

Table 20 Capture Information for Specific Processors (Continued)

Payment Processor	Capture Information
Barclays	Multiple partial captures are supported. See "Multiple Partial Captures," page 71. Special request fields for multiple partial captures are required. See "Special Request Fields for Multiple Partial Captures," page 72. Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73.
Chase Paymentech Solutions	Multiple partial captures are supported. See "Multiple Partial Captures," page 71. Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73.
Cielo	Cielo supports standard captures and automatic captures. ■ By default, your CyberSource account is configured to support standard capture processing. ■ For an Aura Card transaction, you must set the authorization type to AUTOCAPTURE. See "Automatic Captures," page 40.
Comercio Latino	Comercio Latino supports standard captures and automatic captures. See "Automatic Captures," page 40.
Credit Mutuel-CIC	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
CyberSource Latin American Processing	Automatic capture requirements: payment card company rules generally specify that you must not capture a payment until you have shipped the products to the customer. However, with CyberSource Latin American Processing, for some countries you are required to submit an automatic capture. See "Automatic Captures," page 40. For other countries, you can submit standard authorization and capture requests. Contact CyberSource Customer Support for each country's requirements. For transactions in Brazil: ■ You must request the follow-on capture within five days of the authorization request. ■ The capture amount can be less than the authorization amount. ■ You can request only one capture per authorization. With CyberSource Latin American Processing, it takes 31 days for the funds to be deposited in your merchant bank account. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.

Table 20 Capture Information for Specific Processors (Continued)

Payment Processor	Capture Information
CyberSource through VisaNet	CyberSource through VisaNet limits authorization and capture amounts to 12 digits; therefore, the maximum amount is 999999999999. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
Elavon	Multiple partial captures are supported. See "Multiple Partial Captures," page 71.
Elavon Americas	Multiple partial captures are supported. See "Multiple Partial Captures," page 71.
FDC Compass	Multiple partial captures are supported. See "Multiple Partial Captures," page 71. Special request fields for multiple partial captures are recommended. See "Special Request Fields for Multiple Partial Captures," page 72. Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73.
FDC Germany	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDC Nashville Global	CyberSource always provides merchant descriptor information to the processor for you for all capture and credit transactions. See "Merchant Descriptors," page 169. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDMS Nashville	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
FDMS South	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
GPN	GPN limits the authorization, capture, and credit amounts to 10 digits. Split shipments are supported. See "Split Shipments," page 218. Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
HBoS	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59.
HSBC	Multiple partial captures are supported. See "Multiple Partial Captures," page 71.
HSBC is the CyberSource name for HSBC U.K.	Important This feature has restrictions. Contact CyberSource Customer Support for details.

Table 20 Capture Information for Specific Processors (Continued)

Payment Processor	Capture Information
Ingenico ePayments Ingenico ePayments was previously called <i>Global Collect</i> .	With Cartes Bancaires, the authorization and capture amount must be 0.99 EUR or more. Captures for cards using Ingenico ePayments are not batched. CyberSource submits these captures immediately to Ingenico ePayments when they are received.
JCN Gateway	Multiple partial captures are supported. See "Multiple Partial Captures," page 71 .
Lloyds-OmniPay	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59 .
LloydsTSB Cardnet	Multiple partial captures are supported. See "Multiple Partial Captures," page 71 . Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73 .
Moneris	Moneris limits authorization and capture amounts to nine digits; therefore, the maximum amount is 99999999.99.
OmniPay Direct	Multiple partial captures are supported. See "Multiple Partial Captures," page 71 . Special request fields for multiple partial captures are recommended. See "Special Request Fields for Multiple Partial Captures," page 72 . Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73 .
OmniPay-Ireland	Multiple partial captures are supported. See "Multiple Partial Captures," page 71 . Important This feature has restrictions. Contact CyberSource Customer Support for details.
Streamline	Important ARAV is supported. See "Authorization Reversal after Void (ARAV)," page 59 .
SIX	A request for a capture must include an authorization request.
TSYS Acquiring Solutions	Multiple partial captures are supported. See "Multiple Partial Captures," page 71 . Special request fields for multiple partial captures are required. See "Special Request Fields for Multiple Partial Captures," page 72 . Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73 .

Table 20 Capture Information for Specific Processors (Continued)

Payment Processor	Capture Information
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Worldpay VAP limits authorization and capture amounts to eight digits; therefore, the maximum amount is 99999999. Multiple partial captures are supported. See "Multiple Partial Captures," page 71 . Important ARAV is supported. See "Multiple Partial Captures and Authorization Reversal after Void," page 73 .

Capture Features

Authorization Refresh



Note

On CyberSource through VisaNet and GPN, authorization refresh is performed as part of interchange optimization. See ["Interchange Optimization," page 70](#).

Processor:

- Atos

CyberSource provides authorization refresh functionality to Atos merchants for all card types except Maestro (UK Domestic).

When a capture request occurs more than 5 days, 20 hours, and 30 minutes after the date of the original authorization, CyberSource tries to obtain a fresh authorization for the capture amount by performing a system-generated authorization using the payment data from the original authorization.

Payer authentication data and CVN data are not included in system-generated authorizations. Regardless of whether or not you included payer authentication data in your original authorization request, you will not receive payer authentication protection for a system-generated authorization.

If the system-generated authorization is successful, CyberSource submits the capture request with the information from the new authorization. If the system-generated authorization is not successful, CyberSource submits the capture request with the information from the original authorization.

The system-generated authorization is linked to the original authorization in the Business Center and in reports. The subsequent capture is linked to both authorizations in the Business Center and in reports through the request IDs as with any capture.

Automatic Partial Authorization Reversals

Processors and card types:

See the following table.

Table 21 Processors That Support Automatic Partial Authorization Reversals

Processor	Card Types
Barclays	Visa, Mastercard, JCB, Maestro (International), Maestro (UK Domestic)
Chase Paymentech Solutions ¹	Visa, Mastercard
Credit Mutuel-CIC	Visa, Mastercard, Cartes Bancaires
CyberSource through VisaNet	Visa, Mastercard
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass ¹	Visa, Mastercard
FDC Nashville Global	Visa, Mastercard, Discover, Diners Club, China UnionPay, JCB (US Domestic) ²
FDMS Nashville	Visa, Mastercard, Discover, Diners Club, JCB (US Domestic) ²
FDMS South	Visa, Mastercard, Discover, JCB (US Domestic) ²
GPN	Visa, Mastercard On GPN, automatic partial authorization reversal is performed as part of interchange optimization, which is described in "Interchange Optimization," page 70 .
OmniPay Direct	Cardnet International: Visa
OmniPay-Ireland	Visa
OmniPay-Ireland is the CyberSource name for HSBC International.	
TSYS Acquiring Solutions	Visa, Mastercard, Discover, Diners Club, JCB
¹ The processor performs an automatic partial authorization reversal when there is an interchange benefit. The processor does not allow CyberSource to perform this functionality. ² For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.	

In addition to credit cards, automatic partial authorization reversals are supported for:

- Debit cards and prepaid cards: see [Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105](#).
- Quasi-cash: see ["Quasi-Cash," page 203](#).

If the capture amount is less than the authorization amount, CyberSource automatically performs a partial authorization reversal before it sends the capture request to the processor. The results of a successful partial authorization reversal are:

- The capture amount matches the new authorization amount at the payment card company.
- The hold on the unused credit card funds might be released. The issuing bank decides whether or not to release the hold on unused funds.



Note

Not all issuers act on a request for a partial authorization reversal. Therefore, CyberSource cannot guarantee that the funds will be released.

Interchange Optimization

Processors:

- CyberSource through VisaNet: Visa, Mastercard



Important

Interchange optimization is not available for Mastercard transactions in the IDR currency on CyberSource through VisaNet.

- GPN acquiring merchants: Visa, Mastercard

Interchange optimization helps you reduce your interchange fees. Interchange optimization consists of:

- Automatic authorization refresh: When the capture request occurs more than six days after the date of the original authorization, CyberSource automatically obtains a fresh authorization for the capture amount.

On GPN, the fresh authorization uses the same authorization indicator as the original authorization. For more information, see ["Final Authorization Indicator," page 138](#).

- Automatic partial authorization reversal: If the capture does not need a fresh authorization but the capture amount is less than the authorization amount, CyberSource automatically performs a partial authorization reversal which releases the hold on unused credit card funds and ensures that the settlement amount matches the authorization amount.



Note

Interchange optimization does not work for card-present transactions.

To enable interchange optimization, contact CyberSource Customer Support to have your account configured for this feature.

Multiple Partial Captures

Processors:

- AIBMS
- American Express Direct
- Asia, Middle East, and Africa Gateway
- Barclays
- Chase Paymentech Solutions
- CyberSource through VisaNet: contact CyberSource Customer Support to have your account enabled for this feature. Your account can be enabled for multiple partial captures or split shipments; it cannot be enabled for both features.
- Elavon
- Elavon Americas
- FDC Compass
- FDC Nashville Global: multiple partial captures are supported only for card-not-present transactions; they are not supported for card-present transactions.
- FDMS Nashville: multiple partial captures are supported only for card-not-present transactions; they are not supported for card-present transactions.
- HSBC: HSBC is the CyberSource name for HSBC U.K. To enable multiple partial captures on HSBC, contact CyberSource Customer Support to have your account configured for this feature.
- JCN Gateway
- LloydsTSB Cardnet: to enable multiple partial captures on LloydsTSB Cardnet, contact CyberSource Customer Support to have your account configured for this feature.
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- OmniPay-Ireland: to enable multiple partial captures on OmniPay-Ireland, contact CyberSource Customer Support to have your account configured for this feature.

- Streamline. See ["Multiple Partial Captures on Streamline," page 73.](#)
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.

**Note**

Multiple partial captures and *split shipments* are not the same feature.

- The multiple partial captures feature is provided by the processor. This feature enables you to request multiple partial captures for one authorization.
- The split shipment feature is provided by CyberSource. This feature supports three different scenarios: multiple authorizations, multiple captures, and multiple authorizations with multiple captures. For more information, see ["Split Shipments," page 218.](#)

This feature enables you to request multiple partial captures for one authorization. You must ensure that the total amount of all the captures does not exceed the authorized amount.

[Special Request Fields for Multiple Partial Captures](#)

Processors:

- Barclays. The special request fields are required.
- FDC Compass. To avoid a downgrade for a Visa transaction, the special request fields are required. For other card types, CyberSource strongly recommends that you include the special request fields.
- FDC Nashville Global. The special request fields are required for all card types:
 - For Visa and Mastercard, CyberSource sends the values for the special request fields to the processor.
 - For all card types, CyberSource uses the values for the special request fields to determine whether to initiate an automatic partial authorization reversal as described in ["Automatic Partial Authorization Reversals," page 69.](#)
- FDMS Nashville. The special request fields are required for Visa and Mastercard transactions. They are not supported for other card types.

- OmniPay Direct. CyberSource strongly recommends that you include the special request fields. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- TSYS Acquiring Solutions. The special request fields are required.

Include the following special request fields in each capture request when you are requesting multiple partial captures:

- ccCaptureService_sequence
- ccCaptureService_totalCount

When you do not know the total number of captures that you are going to request, set the capture total count to an estimated value or 99 for all capture requests except the final one. For the final capture request, set the capture total count and the capture sequence to the same value.

Multiple Partial Captures on Streamline

Streamline might consider a partial capture to be a duplicate and reject the transaction when one or more of the following is the same for a merchant ID. You must ensure that you do not submit duplicate transaction information when using multiple partial captures, otherwise Streamline may reject the transaction.

- transaction date
- card_accountNumber
- merchantReferenceCode
- purchaseTotals_grandTotalAmount

Multiple Partial Captures and Authorization Reversal after Void

Processors:

- American Express Direct
- Barclays
- Chase Paymentech Solutions
- CyberSource through VisaNet
- FDC Compass
- FDC Nashville Global
- FDMS Nashville
- LloydsTSB Cardnet

- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.

This feature enables you to reverse an authorization after you void the associated capture.

**Important**

This functionality enables you to meet the Visa mandate requirements to reverse unused authorizations, and it benefits the customer by releasing the hold on unused credit card funds.

For an authorization that has multiple associated captures:

- If you reverse the authorization, CyberSource declines subsequent capture requests.
- If you void only one of the multiple captures, CyberSource declines subsequent authorization reversal requests.
- If you void all of the multiple captures, you can reverse the authorization.

To reverse an authorization after a void for multiple captures:

- Step 1** Void each capture associated with the authorization. See ["Voiding a Capture or Credit," page 85](#).
 - Step 2** Reverse the authorization. See ["Reversing an Authorization," page 52](#).
-

Performing a Sale

A sale is a bundled authorization and capture. Some processors and acquirers require a sale transaction instead of separate authorization and capture requests. For other processors and acquirers, you can request a sale instead of a separate authorization and capture if there is no delay between taking a customer's order and shipping the goods. A sale is typically used for electronic goods and for services that you can turn on immediately.

Dual-Message Processing

To perform a sale, request the authorization and capture services at the same time. Include the request fields that are required for the authorization. No additional fields are required for the capture.

If the authorization is successful, CyberSource processes the capture immediately and the reply message includes results for the authorization and for the capture. If the authorization is declined, CyberSource does not process the capture and the reply message includes results only for the authorization.

For debit cards and prepaid cards, the issuing bank can approve a partial amount if the balance on the card is less than the requested authorization amount and if the transaction is enabled for partial authorization. When this happens, CyberSource does not process the capture. However, you can submit a capture request for the approved amount. For details about partial authorizations and for a list of the processors and card types supported for partial authorizations, see ["Partial Authorizations," page 105](#).

**Note**

For a limited number of processors and card types, partial authorizations are supported for credit cards in addition to debit cards and prepaid cards. See ["Partial Authorizations," page 105](#).

For details about authorizations and captures, see ["Authorizing a Payment," page 38](#), and ["Capturing an Authorization," page 61](#).

Single-Message Processing



Note

See also ["Least-Cost Routing \(LCR\)," page 164](#).

Processors:

- CyberSource through VisaNet—National Commercial Bank (NCB) is the only acquirer that supports single-message processing.
- RuPay

Card Types:

- All mada card types as described in ["mada Debit Cards," page 24](#).
- RuPay



Important

There are important differences between single-message processing and dual-message processing:

- With single-message processing, an authorization+capture request results in a full-financial transaction, which moves funds immediately.
- With single-message processing, the capture amount and authorization amount are the same.
- Several services and features are not supported for single-message processing because they are not applicable or could create reconciliation problems. See ["Unsupported Features," page 78](#).

To perform a sale using SMP, request the authorization and capture services at the same time. Include the request fields that are required for the authorization. No additional fields are required for the capture. On CyberSource through VisaNet, SAMA sends a full-financial transaction request to the issuer.

The reply message that CyberSource sends to you includes authorization and capture results. The authorization amount and capture amount are the same.

For details about authorizations and captures, see ["Authorizing a Payment," page 38](#), and ["Capturing an Authorization," page 61](#).

Partial Authorizations

For SMP, partial authorizations are supported only on CyberSource through VisaNet. If the transaction is enabled for partial authorization, the processor can process a partial amount instead of the entire requested amount. For details about partial authorizations, see ["Partial Authorizations," page 105](#).

Supported Features on CyberSource through VisaNet

The following features are supported for single-message processing:

- AVS. See ["Address Verification System \(AVS\)," page 88](#).
- Balance responses. See ["Balance Responses," page 111](#).
- Credits. See ["Crediting a Payment," page 78](#).
- CVN. See ["Card Verification Numbers \(CVNs\)," page 97](#).
- Debit cards. See [Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105](#).
- Final authorization indicator. See ["Final Authorization Indicator," page 138](#).
- Full authorization reversals. See ["Authorization Reversals for Single-Message Processing," page 59](#).
- Merchant descriptors. See [Merchant Descriptors Using the Simple Order API](#).
- Merchant-initiated reversals and voids. See ["Merchant-Initiated Reversals and Voids," page 170](#).
- Merchant-initiated transactions. See ["Merchant-Initiated Transactions," page 173](#).
- Payer authentication. See ["Payer Authentication," page 183](#).
- Prepaid cards. See [Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105](#).

Supported Features on RuPay

The following features are supported for single-message processing:

- Credits. See ["Crediting a Payment," page 78](#).
- CVN. See ["Card Verification Numbers \(CVNs\)," page 97](#).
- Payer authentication. See ["Payer Authentication," page 183](#).

Unsupported Features

The following features are not supported for single-message processing:

- Authorization reversal after void
- Automatic authorization reversal after void
- Automatic partial authorization reversals
- Interchange optimization
- Multiple partial captures
- Split shipments
- Stand-alone authorizations
- Stand-alone captures
- Verbal authorizations
- Voids

Crediting a Payment

CyberSource supports credits for all processors.

When your request for a credit is successful, the issuing bank for the payment card takes money out of your merchant bank account and returns it to the customer. It usually takes two to four days for your acquiring bank to transfer funds from your merchant bank account.

Warning	Carefully control access to this service to prevent unauthorized credits. Do not request this service directly from your customer interface. Instead, incorporate this service as part of your customer service process.
----------------	--

Credit requests are batched in the same manner as captures. See ["Captures," page 61](#).

Types of Credits

A *follow-on credit* is linked to a capture in the CyberSource system. You can request multiple follow-on credits against a single capture. On CyberSource through VisaNet and SIX, you must request a follow-on credit within 180 days of the authorization. For all other processors, you must request a follow-on credit within 60 days of the authorization.



Note

On Atos, your request for a follow-on credit must also include the request token returned from a previous capture request in addition to the request ID. Like the request ID, the request token links the follow-on credit to the capture. Send the request token in the **orderRequestToken** field.



Important

When you combine a request for a follow-on credit with a request for another service, such as the tax calculation service, you must provide the customer's billing and account information.

A *stand-alone credit* is not linked to a capture. There is no time limit for requesting stand-alone credits. Instead of sending the request ID field in the credit request, the request must include the fields for the customer's billing and account information.



Note

For stand-alone credits, CyberSource does not validate **billTo_postalCode** or **shipTo_postalCode**.

Creating a Credit Request

A follow-on credit uses the request ID returned from a previous capture to link the credit to the capture. CyberSource uses the request ID to look up the customer's billing and account information from the original authorization, so you are not required to include those fields in your credit request. To perform multiple partial follow-on credits, send the same request ID in each follow-on credit request.

For information about requesting a follow-on service, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

To create a credit request:

Step 1 Do not include any of these services in the request:

- Any other credit card services (**ccAuthService**, **ccAuthReversalService**, or **ccCaptureService**)
- Services for other payment methods, such as electronic checks or PayPal
- Risk update (**riskUpdateService**)

Step 2 Include the required fields in the request:

Table 22 Required Fields for Credits

Field	Notes
ccCreditService_run	Set to <code>true</code> .
ccCreditService_captureRequestID	For a follow-on credit, set to the request ID that was included in the capture reply message. Not used for a stand-alone credit.
merchantID	
merchantReferenceCode	
paymentSolution	Include this field only when using Visa Checkout.
purchaseTotals_currency	
purchaseTotals_grandTotalAmount	Either purchaseTotals_grandTotalAmount or item_#_unitPrice must be included in the request.
vc_orderID	Include this field only when using Visa Checkout.

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Optional request fields
- Reply fields

Step 3 For a stand-alone credit, include additional required fields:

Table 23 Additional Required Fields for Stand-Alone Credits

Field	Notes
billTo_city ¹	
billTo_country ¹	
billTo_email ¹	

- ¹ This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91.
Important It is your responsibility to determine whether a field is required for the transaction you are requesting.

Table 23 Additional Required Fields for Stand-Alone Credits (Continued)

Field	Notes
billTo_firstName ¹	
billTo_lastName ¹	
billTo_postalCode ¹	Required only for transactions in the U.S. and Canada.
billTo_state ¹	Required only for transactions in the U.S. and Canada.
billTo_street1 ¹	
card_accountNumber	
card_cardType	Required for certain card types. CyberSource strongly recommends that you send the card type even if it is optional for your processor. Omitting the card type can cause the transaction to be processed with the wrong card type.
card_expirationMonth ¹	
card_expirationYear ¹	
¹ This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting.	

Step 4 If needed, modify the request to accommodate additional information for your processor. See ["Credit Information for Specific Processors," page 81](#).

Step 5 Include optional features in the request. See [Chapter 5, "Optional Features," on page 116](#).

Credit Information for Specific Processors

The following table provides additional information about credits for some processors.

Table 24 Credit Information for Specific Processors

Payment Processor	Credit Information
Atos	Atos supports only follow-on credits. Stand-alone credits are not supported. The credit amount cannot exceed the capture amount. Atos limits authorization, capture, and credit amounts to 12 digits; therefore, the maximum amount is 999999999999. A credit cannot be processed on the same day as the capture that is being credited. You must wait until the day after the capture before requesting a credit.

Table 24 Credit Information for Specific Processors (Continued)

Payment Processor	Credit Information
Cielo	Cielo does not support stand-alone credits. CyberSource recommends that you do not submit a follow-on credit request on the same day as the capture that is being credited.
Comercio Latino	Comercio Latino does not support stand-alone credits. A credit cannot be processed on the same day as the capture that is being credited. You must wait until the day after the capture before requesting a credit. Multiple partial credits cannot exceed the original authorization amount. CyberSource declines credit requests if the associated capture was not successful. Credits must be processed within 180 days of the original authorization. On American Express, multiple partial credits are not supported.
CyberSource Latin American Processing	CyberSource Latin American Processing supports only follow-on credits. Stand-alone credits are not supported. The 60-day limit for follow-on credits does not apply to CyberSource Latin American Processing: you can request a follow-on credit more than 60 days after the original charge. CyberSource Latin American Processing does not support the credit service for Aura Card and Hipercard. You must make manual refunds for these card types. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.
CyberSource through VisaNet	CyberSource recommends that you do not submit a follow-on credit request on the same day as the capture that is being credited.
FDC Nashville Global	CyberSource always provides information to the processor for you for all capture and credit transactions. See "Merchant Descriptors," page 169.
Getnet	Stand-alone credits are not supported. For follow-on credits: ■ A partial credit cannot be processed on the same day as the capture that is being credited. You must wait until the day after the capture before requesting a partial credit. ■ A credit for the full capture amount can be processed on the same day as the capture.

Table 24 Credit Information for Specific Processors (Continued)

Payment Processor	Credit Information
GPN	GPN limits the authorization, capture, and credit amounts to 10 digits.
Ingenico ePayments Ingenico ePayments was previously called <i>Global Collect</i> .	With Ingenico ePayments, you can process only one follow-on credit against a specific captured authorization each day. For example, if you want to process a follow-on credit of 15.00 against an original capture of 50.00, and then later you want to process a follow-on credit of 35.00 against the same capture, you must request the two credits on two separate days. Before performing stand-alone credits with Ingenico ePayments, you must contact CyberSource Customer Support. Credits for cards using Ingenico ePayments are not batched. CyberSource submits these captures immediately to Ingenico ePayments when they are received.
JCN Gateway	JCN Gateway supports stand-alone credits. However, when a request for a stand-alone credit is made, most acquirers make inquiries about the purpose of such a request. CyberSource recommends using follow-on credits instead of stand-alone credits whenever possible.
RBS WorldPay Atlanta	Follow-on credits for verbal authorizations are not supported. You must process these refunds as stand-alone refunds.
Rede	Stand-alone credits are not supported. For follow-on credits: ■ A partial credit cannot be processed on the same day as the capture that is being credited. You must wait until the day after the capture before requesting a partial credit. ■ A credit for the full capture amount can be processed on the same day as the capture.
RuPay	For a single credit, the credit amount cannot exceed the capture amount. For multiple partial credits, the total credit amount cannot exceed the capture amount.
SIX	A credit cannot be processed on the same day as the capture that is being credited. You must wait until the day after the capture before requesting a credit.

Table 24 Credit Information for Specific Processors (Continued)

Payment Processor	Credit Information
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	For a follow-on credit to be successfully processed, the capture that is being credited must have been processed successfully. To ensure that the capture is processed before the follow-on credit request is received, do not batch the follow-on credit on the same day as the capture. If the capture has not been processed yet, CyberSource sends this error message: The follow-on credit cannot be processed because the capture transaction has not been processed yet. If the capture has been processed but was not successful, CyberSource sends this error message: The follow-on credit cannot be processed because the capture transaction failed.

Credit Authorizations

Processor:

- CyberSource through VisaNet

A credit authorization authenticates the card and customer. If you are located in the U.S. or Canada, CyberSource automatically authorizes every credit request. If you are not located in the U.S. or Canada, contact CyberSource Customer Support to enable credit authorizations for your account.



When you use the Simple Order API in XML format, you must use version 1.151 or later of the XML schema to receive the following reply fields.

Note

The following fields are included in the credit reply when CyberSource authorizes the credit request:

- ccCreditReply_authorizationCode
- ccCreditReply_paymentNetworkTransactionID
- ccCreditReply_processorResponse

See [Appendix A, "API Fields," on page 238](#) for descriptions of these fields.

Voiding a Capture or Credit

CyberSource supports voids for all processors except:

- Atos
- Ingenico ePayments—Ingenico ePayments was previously called *Global Collect*.
- Lynk
- SIX



Note

CyberSource Latin American Processing does not support voids for Aura Card and Hipercard because transactions with these cards are captured immediately.

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. The information in this note is for the specific processing connection called *CyberSource Latin American Processing*. It is not for any other Latin American processors that CyberSource supports.



Note

Cielo and Comercio Latino are online gateways. Transactions are batched every four minutes, which provides very little time for you to void a transaction.

A void cancels a capture or credit request that you submitted to CyberSource. A transaction can be voided only when CyberSource has not already submitted the capture or credit request to your processor. CyberSource usually submits capture and credit requests to your processor once a day, so your window for successfully voiding a capture or credit request is small. CyberSource declines your void request when the capture or credit request has already been sent to the processor.

You cannot perform a follow-on credit for a transaction that has been voided.

You cannot undo a void.

When you void a capture, a hold remains on the unused credit card funds. If you are not going to re-capture the authorization as described in ["Capture after Void," page 86](#), and if your processor supports authorization reversal after void as described in ["Authorization Reversal after Void \(ARAV\)," page 59](#), CyberSource recommends that you request an authorization reversal to release the hold on the unused credit card funds.

Capture after Void

If your processor supports multiple captures, you can capture an authorization after you void previous captures associated with the authorization. For example, you can follow these steps:

- 1 Authorize a payment.
- 2 Capture the authorization.
- 3 Void the capture.
- 4 Capture the authorization again.

To learn whether your processor supports multiple captures, see ["Multiple Partial Captures," page 71](#).

On all other processors, when you void a transaction the transaction is at the end of its life and cannot be the source of another follow-on capture or credit. For example, if you authorize and capture a transaction, and then you void the capture, you cannot submit another capture request that uses the authorization code or CyberSource request ID from the original authorization. If you still want to capture that transaction, you must re-authorize the transaction and capture the new authorization.

Creating a Void Request

A void is a follow-on transaction that uses the request ID returned from a capture or credit. The request ID links the void to the service that is being voided. CyberSource uses the request ID to look up the customer's billing and account information from the capture or credit, so you are not required to include those fields in your void request.

For information about requesting a follow-on service, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

To create a void request:

- Step 1
- Do not include any other CyberSource services in the request.
- Step 2
- Include the required fields in the request:

Table 25 Required Fields for Voids

Field	Notes
merchantID	
merchantReferenceCode	
voidService_run	Set to true.
voidService_voidRequestID	Set to the request ID that was included in the capture or credit reply message.

See [Appendix A, "API Fields," on page 238](#) for:

- Detailed descriptions of these required request fields
- Reply fields

Authorization Features

You must support the authorization features that your processor supports.

Address Verification System (AVS)

**Note**

AVS is supported only for cards issued in the U.K., the U.S., and Canada.

Standard AVS

The following table lists the processors and card types for which CyberSource returns standard AVS results.

Table 26 Processors That Support Standard AVS

Processors	Payment Card Types
AIBMS	Visa, Mastercard, Maestro (International), Maestro (UK Domestic)
American Express Brighton	American Express You must contact CyberSource Customer Support to activate standard AVS for American Express Brighton.
American Express Direct	American Express You must contact CyberSource Customer Support to activate standard AVS for American Express Direct.
Atos	Visa and Mastercard: The billing country must be Great Britain.
Barclays	Visa, Mastercard, Maestro (UK Domestic)
Chase Paymentech Solutions	Visa, Mastercard, and American Express: The billing country must be the U.S., Canada, or Great Britain. Discover, Diners Club, and JCB: The billing country must be the U.S.

Table 26 Processors That Support Standard AVS (Continued)

Processors	Payment Card Types
Cielo	Visa, Mastercard, American Express Cielo can charge you additional fees for AVS processing. You must contact Cielo and CyberSource Customer Support to activate standard AVS for Cielo. AVS is supported only for credit card transactions, not debit card transactions. <i>AVS Codes for Cielo 3.0</i> See "AVS Codes for Cielo 3.0 and CyberSource Latin American Processing," page 461. <i>Format for Raw AVS Codes for Cielo 1.5</i> The raw AVS response code is a concatenation of two values: ■ The first value is the raw AVS code for the postal code. ■ The second value is the raw AVS code for the street address. If Cielo returns only one of the values, the missing value is indicated by a question mark (?). Examples: ■ ?N indicates that the raw AVS code for the postal code is missing and that the raw AVS code for the street address is N. ■ T? indicates that the raw AVS code for the postal code is T and that the raw AVS code for the street address is missing.
Comercio Latino	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard Comercio Latino supports AVS input, but does not support AVS response values. Important Because a raw AVS response value is not available, there is a potential impact to the Decision Manager services. You must contact CyberSource Customer Support to activate standard AVS for Comercio Latino.
Credit Mutual-CIC	Visa, Mastercard, Cartes Bancaires
CyberSource Latin American Processing	Visa, Mastercard, American Express, Diners Club In Brazil, AVS is supported only for Redecard. To perform AVS for Redecard in Brazil, you must provide the CPF (Cadastro de Pessoas Fisicas) and the building number. For AVS in Mexico, contact CyberSource Customer Support to have your account enabled for this feature. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.

Table 26 Processors That Support Standard AVS (Continued)

Processors	Payment Card Types
CyberSource through VisaNet	Visa, Mastercard, American Express, Diners Club, JCB, Discover, mada as described in "mada Debit Cards," page 24 Important When you populate billing street address 1 and billing street address 2, CyberSource through VisaNet concatenates the two values. If the concatenated value exceeds 40 characters, CyberSource through VisaNet truncates the value at 40 characters before sending it to Visa and the issuing bank. Truncating this value affects AVS results and therefore might also affect risk decisions and chargebacks.
Elavon	Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) Your country and the billing country must be Great Britain. The currency must be British pounds.
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass	Visa, Mastercard, and American Express: The billing country must be the U.S., Canada, or Great Britain. Discover and Diners Club: The billing country must be the U.S.
FDC Germany	Visa, Mastercard
FDC Nashville Global	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
FDMS Nashville	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
FDMS South	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
GPN	Visa, Mastercard, American Express, Discover, Diners Club, JCB
HBoS	Visa, Mastercard
HSBC	Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
HSBC is the CyberSource name for HSBC U.K.	
Lloyds-OmniPay	Visa, Mastercard

Table 26 Processors That Support Standard AVS (Continued)

Processors	Payment Card Types
LloydsTSB Cardnet	Visa, Mastercard
Lynk	Visa, Mastercard, American Express, Discover, Diners Club
Moneris	Visa, Mastercard, Discover
OmniPay Direct	Bank of America Merchant Services: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) Cardnet International: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) Global Payments International Acquiring: Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
OmniPay-Ireland OmniPay-Ireland is the CyberSource name for HSBC International.	Visa, Mastercard
RBS WorldPay Atlanta	Visa, Mastercard, American Express, Discover, Diners Club
SIX	Visa, Mastercard, Discover, Diners Club
Streamline	Visa, Mastercard, Maestro (UK Domestic), Cartes Bancaires, Dankort You must contact Streamline to activate standard AVS.
TSYS Acquiring Solutions	Visa, Mastercard, American Express, Diners Club: The billing country must be the U.S.
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Visa, Mastercard, American Express, Discover, Diners Club, JCB

Relaxed Requirements for Address Data and Expiration Date

To enable relaxed requirements for address data and expiration date, contact CyberSource Customer Support to have your account configured for this feature. For details about relaxed requirements, see the [Relaxed Requirements for Address Data and Expiration Date page](#).

Processing AVS Codes

When a processor supports AVS for a transaction's card type, the issuing bank uses AVS to confirm that the customer has provided the correct billing address. When a customer provides incorrect information, the transaction might be fraudulent.

AVS occurs automatically with every authorization request. The authorization reply includes the **ccAuthReply_avsCode** field, which contains the AVS code from the issuing bank that indicates whether AVS matched the address and whether the address match was partial or complete. See [Appendix E, "AVS Codes," on page 461](#).

When AVS cannot verify the address, but the authorization is otherwise valid, you might receive an AVS decline. You can capture authorizations that receive an AVS decline. However, you must review these orders to ensure that they are legitimate. Settling authorizations that fail the AVS check might have an impact on the fees charged by your bank. Contact your bank for details about how AVS management might affect your discount rate.

The **ccAuthReply_avsCodeRaw** field is the raw AVS code sent directly from the processor. Do not use this value to handle the AVS response. Use the value only for debugging purposes.

Controlling AVS Results

By default, only the AVS code `N` results in an AVS decline. You can change this behavior by using the **businessRules_declineAVSFlags** field to specify a list of AVS codes that should result in an AVS decline.



Important

When you use **businessRules_declineAVSFlags**, you must include the value `N` in the list if you want to receive declines for AVS code `N`.

When your request includes the **businessRules_ignoreAVSResult** field set to `true`, you receive no AVS declines, even when you use **businessRules_declineAVSFlags**.

Enhanced AVS

Processor:

- American Express Direct

**Note**

You must contact CyberSource Customer Support and American Express to register for Enhanced AVS.

Card type:

- American Express

Enhanced AVS consists of the standard AVS functionality plus verification of some additional fields. The additional fields that are verified for Enhanced AVS are:

- billTo_firstName
- billTo_lastName

Automated Address Verification Plus (AAV+)

Processor:

- American Express Direct

**Note**

You must contact CyberSource Customer Support and American Express to register for AAV+.

Card type:

- American Express

AAV+ consists of the Enhanced AVS functionality plus verification of some additional fields. This service is intended for merchants who deliver physical goods to a different address than the billing address. AAV+ verifies the additional fields only when the standard and Enhanced AVS tests pass first. For information about Enhanced AVS, see ["Enhanced AVS," page 93](#).

The additional fields that are verified for AAV+ are:

- shipTo_firstName
- shipTo_lastName
- shipTo_street1
- shipTo_country
- shipTo_postalCode
- shipTo_phoneNumber
- billTo_phoneNumber: American Express Direct only



Note

For American Express Direct, when your account is enabled for AAV+ and when you include the first name, last name, and phone number in your request message, the reply message includes EV response codes for those fields. See ["Electronic Verification \(EV\)," page 94](#).

Electronic Verification (EV)

Processors:

- American Express Direct
- FDC Nashville Global
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.
For EV, Worldpay VAP verifies only the email address, first name, last name, and phone number.



Note

If Worldpay VAP is your processor, you must contact Worldpay VAP to register for EV.

Card types:

- American Express
- Discover—only on TSYS Acquiring Solutions. Only the first name and last name are checked.

EV confirms the customer's billing information. When a customer provides incorrect information, the transaction might be fraudulent.

**Note**

As part of EV for TSYS Acquiring Solutions and Worldpay VAP, you can provide the IP address in the **billTo_ipAddress** field. When you provide the IP address, American Express does not send a response for it. Instead, American Express uses the IP address to run a check in their internal database to ensure that the IP address does not match previously fraudulent transactions with the same IP address and is not from countries that American Express has determined to be a high risk for fraud. If, based on the IP address, American Express determines that the transaction is fraudulent or is a high risk for fraud, American Express declines the transaction.

Request Fields

To receive an EV response code for a particular value, you must include that value in your authorization request. [Table 27, "Request Fields for Electronic Verification," on page 95](#) lists the request fields for each value that EV can verify. In the table, the R/O column indicates whether the field is required or optional for the authorization service.

**Note**

Some merchants use placeholder data for some required fields, such as addresses and phone numbers, because their customers do not provide them with the required information. The benefit of using certain specific placeholder values is that Decision Manager ignores the values instead of attempting to process them. However, when you use placeholder data in any of the fields that are used for EV, the corresponding EV results are invalid.

Table 27 Request Fields for Electronic Verification

Value That Is Being Verified	R/O for Authorizations	Request Field
Email	R	billTo_email
First name ²	R	billTo_firstName
Last name ²	R	billTo_lastName
Phone number ²	O	billTo_phoneNumber
Postal code	R/O ¹	billTo_postalCode
Street address	R	billTo_street1

¹ Required when the billing country is the U.S. or Canada; otherwise, optional.

² On American Express Direct, to receive EV response codes for the first name, last name, and phone number, your account must be enabled for AAV+. See ["Automated Address Verification Plus \(AAV+\)," page 93](#).

Reply Fields

For each verified value, EV returns a raw response code and a mapped response code:

- The *raw response code* is the value returned by the processor.
- The *mapped response code* is the pre-defined CyberSource value that corresponds to the raw response code. [Appendix M, "Electronic Verification Response Codes," on page 479](#) describes the mapped response codes.

The following table lists the reply fields for each value that EV can verify.

Table 28 API Fields for Electronic Verification Responses

Value That Is Being Verified	API Field for Mapped Response	API Field for Raw Response
Email	ccAuthReply_evEmail	ccAuthReply_evEmailRaw
First name and last name	ccAuthReply_evName	ccAuthReply_evNameRaw
Phone number	ccAuthReply_evPhoneNumber	ccAuthReply_evPhoneNumberRaw
Postal code	ccAuthReply_evPostalCode	ccAuthReply_evPostalCodeRaw
Street address	ccAuthReply_evStreet	ccAuthReply_evStreetRaw

Card Verification Numbers (CVNs)

Table 29 Processors That Support CVNs

Processors	Payment Card Types
AIBMS	Visa, Mastercard, Maestro (International), Maestro (UK Domestic)
American Express Brighton	American Express
American Express Direct	American Express
Asia, Middle East, and Africa Gateway	Visa, Mastercard, American Express, Diners Club
Atos	Visa, Mastercard, Cartes Bancaires
Barclays	Visa, Mastercard, Maestro (UK Domestic)
Chase Paymentech Solutions	Visa, Mastercard, American Express, Discover
Cielo	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Maestro (International), Elo, Aura
Comercio Latino	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard CVN is required for all authorization requests except recurring transactions. CyberSource returns a CVN response value of 3 in the ccAuthReply_cvCode field in the authorization reply, which indicates that the processor did not send a CVN response. When you submit authorizations without CVNs, Comercio Latino or your acquirer declines them, unless you contact Comercio Latino and your acquirer to configure your account to allow transactions without CVNs. When a card fails the CVN check, Comercio Latino declines the authorization.
Credit Mutuel-CIC	Visa, Mastercard, Cartes Bancaires
CyberSource Latin American Processing	Visa, Mastercard, American Express, Elo Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.
CyberSource through VisaNet	Visa, Mastercard, American Express, Diners Club, JCB, Discover, mada as described in "mada Debit Cards," page 24

Table 29 Processors That Support CVNs (Continued)

Processors	Payment Card Types
Elavon	Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) Note Elavon does not return a separate CVN response field in the authorization reply. When the card fails the CVN check, Elavon declines the authorization.
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass	Visa, Mastercard, American Express, Discover
FDC Germany	Visa, Mastercard
FDC Nashville Global	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) Note For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
FDI Australia	Visa, Mastercard, American Express, Diners Club
FDMS Nashville	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) Note For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
FDMS South	Visa, Mastercard, American Express, Discover, Diners Club, JCB (US Domestic) Note For JCB cards, “US Domestic” means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.
GPN	Visa, Mastercard, American Express, Discover, Diners Club
HBoS	Visa, Mastercard
HSBC	Visa, Mastercard, Maestro (International)
HSBC is the CyberSource name for HSBC U.K.	
Ingenico ePayments	Visa, Mastercard
Ingenico ePayments was previously called <i>Global Collect</i> .	Note Do not include the CVN in a request for a recurring payment. See "Recurring Payments," page 205 .
JCN Gateway	Visa, Mastercard, American Express, Diners Club, JCB
Lloyds-Omnipay	Visa, Mastercard
LloydsTSB Cardnet	Visa, Mastercard
Lynk	Visa, Mastercard, American Express, Discover, Diners Club
Moneris	Visa, Mastercard, American Express

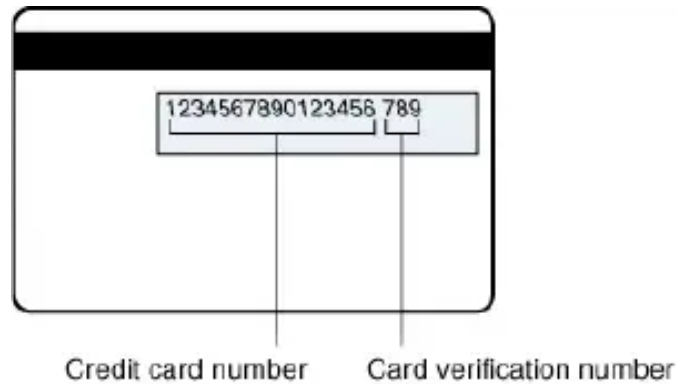
Table 29 Processors That Support CVNs (Continued)

Processors	Payment Card Types
OmniPay Direct	Bank of America Merchant Services: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) Cardnet International: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) Global Payments International Acquiring: Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
OmniPay-Ireland OmniPay-Ireland is the CyberSource name for HSBC International.	Visa, Mastercard
RBS WorldPay Atlanta	Visa, Mastercard, American Express, Discover, Diners Club
RuPay	RuPay
SIX	Visa, Mastercard, Discover, Diners Club, JCB, Maestro (International), Maestro (UK Domestic), China UnionPay, Visa Electron
Streamline	Visa, Mastercard, Maestro (UK Domestic), Cartes Bancaires, Dankort
TSYS Acquiring Solutions	Visa, Mastercard, American Express, Discover, Diners Club
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Visa, Mastercard, American Express, Discover

CVN Locations and Terminology

The CVN, which is printed or embossed on the back of the card, can be sent with the request and verified to help reduce the risk of fraud.

Figure 3 Example of a Visa Card Verification Number



Each payment card company has its own name for this value:

- Visa calls it the *Card Verification Value (CVV2)*.
- American Express and Discover call it the *Card Identification Digits (CID)*.
- Mastercard calls it the *Card Validation Code (CVC2)*.

To use the CVN, include the **card_cvNumber** field in the request. This number is never transferred during card swipes and should be known only by the cardholder.



Important

In Europe, Visa mandates that you must not include a CVN for mail-order transactions and must not record a CVN on any physical format such as a mail-order form.

CVN Codes

The reply message includes a raw response code and a mapped response code:

- The *raw response code* is the value returned by the processor. This value is returned in the **ccAuthReply_cvCodeRaw** field. Use this value only for debugging purposes; do not use it to determine the card verification response.
- The *mapped response code* is the pre-defined CyberSource value that corresponds to the raw response code. This value is returned in the **ccAuthReply_cvCode** field. [Appendix J, "CVN Codes," on page 472](#) describes the mapped response codes.

Even when the CVN does not match the expected value, the issuing bank might still authorize the transaction. You will receive a CVN decline from CyberSource, but you can still capture the transaction because it has been authorized by the bank. However, you must review the order to ensure that it is legitimate.

Settling authorizations that fail the CVN check might have an impact on the fees charged by your bank. Contact your bank for details about how card verification management might affect your discount rate.

When a CVN decline is received for the authorization in a sale request, CyberSource does not process the capture unless you set the **businessRules_ignoreCVResult** field to `true`.

Table 30 CVN Results for Each Card Type

Card Type	CVN Results
American Express	A ccAuthReply_cvCode value of 1 indicates that your account is not configured for CVN. Contact CyberSource Customer Support to have your account enabled for this feature. To use the CVN with American Express, see "Testing American Express Card Verification," page 237.
Discover	For FDC Nashville Global, FDMS Nashville, and FDMS South: ■ CVN results can be returned for any of the card types on the Discover Network as described in "Discover Acquisitions and Alliances," page 23. ■ The CVN results are returned to you and it is your responsibility to decide whether or not to accept the transaction. For all other processors, when the CVN does not match: ■ Discover refuses the card and the request is declined. ■ The reply message does not include the ccAuthReply_cvCode field, which indicates that the CVN failed.
Visa and Mastercard	A CVN code of D or N causes CyberSource to decline the request with reason code 230. You can still capture the transaction, but you must review the order to ensure that it is legitimate. Note CyberSource, not the issuing bank, assigns the CVN decline to the authorization. You can capture any authorization that has a valid authorization code from the issuing bank, even when the request receives a CVN decline. When the issuing bank does not authorize the transaction and the CVN does not match, the request is declined because the card is refused. You cannot capture the transaction.

Verbal Authorizations

CyberSource supports verbal authorizations for these processors:

- AIBMS
- American Express Brighton
- American Express Direct
- Asia, Middle East, and Africa Gateway
- Barclays

- Chase Paymentech Solutions
- CyberSource through VisaNet
- Elavon
- Elavon Americas
- FDC Compass
- FDC Germany
- FDI Australia
- FDC Nashville Global
- FDMS Nashville
- FDMS South
- GPN
- HBoS
- HSBC: HSBC is the CyberSource name for HSBC U.K.
- JCN Gateway
- Lloyds-OmniPay
- LloydsTSB Cardnet
- Lynk
- Moneris
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- OmniPay-Ireland: OmniPay-Ireland is the CyberSource name for HSBC International.
- RBS WorldPay Atlanta
- TSYS Acquiring Solutions
- UATP
- Worldpay VAP—Worldpay VAP was previously called *Little*.

**Note**

Verbal authorizations are not supported for Comercio Latino or CyberSource Latin American Processing.

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. The information in this note is for the specific processing connections called *Comercio Latino* and *CyberSource Latin American Processing*. It is not for any other Latin American processors that CyberSource supports.

**Important**

Do not use Dynamic Currency Conversion with a verbal authorization.

When you request an authorization through CyberSource, the issuing bank might ask you to call the payment processor to answer questions about the transaction. When this happens, the processor gives you a verbal authorization code for the transaction. To capture a verbally authorized transaction, send the verbal authorization code in the capture request. Make sure your customer service and point-of-sale staff can enter verbal authorization codes into your system.

You can use a verbal authorization to capture an authorization that was declined for any of these reasons:

- Verbal authorization required
- Card expired
- Card refused
- Invalid card

**Important**

Do not confuse verbal authorizations with forced captures:

- With a verbal authorization, you obtain the authorization code directly from the processor or issuing bank after requesting an authorization through CyberSource and receiving a CyberSource decline.
- With a forced capture, you get the authorization code by authorizing a payment outside of CyberSource. See ["Forced Captures," page 142](#).

In both cases, you must follow up with a capture that uses the CyberSource system.

A verbal authorization works as follows:

- 1 The authorization reply includes reason code 201, which indicates that the issuing bank is requiring a verbal authorization.

For the American Express card type on FDMS Nashville, the authorization reply also includes a referral response number in **ccAuthReply_referralResponseNumber**. You will be asked for this number, which identifies the failed transaction, when you call American Express for the verbal authorization.

- 2 You call the processor to answer questions about the transaction.
- 3 When the processor verbally authorizes the transaction, the processor gives you a verbal authorization code.

4 You include the verbal authorization code in your capture request:

- Send the verbal authorization code in the **ccCaptureService_verbalAuthCode** field.
- Send the word `verbal` in the **ccCaptureService_authType** field.
When you do not set **ccCaptureService_authType** to `verbal`, CyberSource ignores the **ccCaptureService_verbalAuthCode** field.
- For the American Express card type on American Express Direct or FDMS South, the **ccCaptureService_posData** and **ccCaptureService_transactionID** fields are required to comply with the CAPN requirements.



Note

American Express has indicated that capture requests submitted without a valid transaction ID, including transactions that originated as verbal authorizations, might incur additional transaction charges. Contact your American Express account representative to learn whether your processing is affected by these additional transaction charges.

Features for Debit Cards and Prepaid Cards

Debit cards and prepaid cards are processed using the credit card services described in this document. This chapter describes the special features that are available for debit cards and prepaid cards: partial authorizations and balance responses.

**Note**

To process domestic debit transactions on CyberSource through VisaNet with Mastercard in Canada, you must contact CyberSource Customer Support to have your account configured for this feature.

**Note**

When you use the Simple Order API in XML format, you must use version 1.52 or later of the XML schema to implement partial authorizations or balance responses.

Partial Authorizations

**Note**

The partial authorization functionality does not apply to credit cards.

For debit cards and prepaid cards, the issuing bank can approve a partial amount if the balance on the card is less than the requested authorization amount.

Supported Processors and Card Types

The following table lists the processors and card types for which CyberSource supports partial authorizations. If your processor and card type are not listed in the table, see ["Unsupported Processors and Card Types," page 115](#).

Table 31 Processors Supported for Partial Authorizations

Processor	Card Types for Debit Cards and Prepaid Cards
American Express Direct	American Express
Chase Paymentech Solutions	Visa, Mastercard, American Express, Discover, Diners Club
CyberSource through VisaNet	Visa, Mastercard, American Express, Diners Club, JCB, Discover, mada as described in "mada Debit Cards," page 24 Important Partial authorizations are not available for Mastercard transactions in the IDR currency on CyberSource through VisaNet.
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass ¹	Visa, Mastercard, American Express, Discover
FDC Nashville Global	Visa, Mastercard, American Express, Discover ² , Diners Club ² , China UnionPay, JCB (US Domestic) ^{2,3}
FDMS Nashville	Visa, Mastercard, American Express, Discover ² , Diners Club ² , JCB (US Domestic) ^{2,3}
FDMS South ⁴	Visa, Mastercard, American Express, Discover ² , JCB (US Domestic) ^{2,3}
GPN	Visa, Mastercard, American Express, Discover, Diners Club, JCB
TSYS Acquiring Solutions	Visa, Mastercard, American Express, Discover, Diners Club, JCB
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Visa, Mastercard, American Express, Discover, Diners Club, JCB

1 FDC Compass might support partial authorizations for additional card types in the future so be prepared to handle partial authorizations for all card types if your account is enabled for partial authorizations.

2 For this card type on the specified processor, partial authorizations are supported for credit cards in addition to debit cards and prepaid cards.

3 For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands.

4 FDMS South might support partial authorizations for additional card types in the future so be prepared to handle partial authorizations for all card types if your account is enabled for partial authorizations.

Opting In



Note

If you accept American Express cards and Chase Paymentech Solutions is your processor, see ["Special Processing for American Express Cards on Chase Paymentech Solutions," page 109.](#)

You must opt in to be able to receive and capture partial authorizations. There are two ways to opt in:

- You can call CyberSource Customer Support to have your account enabled for partial authorizations. When you do this, all your authorization requests are enabled for partial authorizations.

or

- You can set **ccAuthService_partialAuthIndicator** to `true` in your authorization or sale request. When you do this, only that specific transaction is enabled for partial authorization.

Opting Out

When your account is enabled for partial authorizations, you can disable partial authorization for a specific transaction by setting **ccAuthService_partialAuthIndicator** to `false` in your authorization or sale request.

How a Partial Authorization Works



Note

Support for your processor and card type does not guarantee a partial authorization. The issuing bank decides whether or not to approve a partial amount.

When the balance on a debit card or prepaid card is less than the requested authorization amount, the issuing bank can approve a partial amount. When this happens, you can accept multiple forms of payment for the order starting with some or all of the approved amount followed by one or more different payment methods:

- 1 If your account is not configured for partial authorizations, you must enable partial authorizations for the transaction by setting **ccAuthService_partialAuthIndicator** to `true` in your request.



Note

If you accept American Express cards and Chase Paymentech Solutions is your processor, see ["Special Processing for American Express Cards on Chase Paymentech Solutions," page 109.](#)

**Note**

If you accept IDR or CLP currencies on FDMS South, see ["Special Processing for IDR and CLP on FDMS South," page 109](#).

- 2 You submit an authorization request or a sale request for a debit card or prepaid card.
- 3 The authorization reply message from CyberSource includes:
 - ccAuthReply_requestAmount: amount you requested
 - ccAuthReply_requestCurrency: currency for the amount you requested
 - ccAuthReply_amount: amount that was authorized
 - purchaseTotals_currency: currency for the amount that was authorized
 - requestID: value you can use to link this authorization request to subsequent transactions

**Note**

If you requested a sale, the authorization was not captured.

- 4 You submit a capture request for the partial authorization.

When you capture only part of the approved amount, CyberSource or your processor might be able to perform an automatic partial authorization reversal for you. See ["Automatic Partial Authorization Reversals," page 69](#).

**Note**

If you do not capture the partial authorization, you must request a full authorization reversal if this service is supported for your processor and card type. See ["Reversing an Authorization," page 52](#).

- 5 You use one or more different payment methods for the rest of the order amount.

When you process these payment methods through CyberSource, you can use the **linkToRequest** field to link the payment requests to the original authorization request. Set **linkToRequest** to the **requestID** value that was returned in the reply message for the original authorization request.

Special Processing for American Express Cards on Chase Paymentech Solutions

If you accept American Express cards and Chase Paymentech Solutions is your processor, perform the following procedure to opt in to partial authorizations.

To opt in to partial authorizations for American Express cards on Chase Paymentech Solutions:

- Step 1** Contact Chase Paymentech Solutions to have your account enabled for partial authorizations for the American Express card type. The transaction division for partial authorizations for American Express should be set to 3.



Important

This step is only for the American Express card type on Chase Paymentech Solutions. For all other card types on Chase Paymentech Solutions, the transaction division for partial authorizations should be set to the default value of 0 (zero).

- Step 2** Contact CyberSource Customer Support to have your account enabled for partial authorizations.
- After your accounts have been enabled for partial authorizations at Chase Paymentech Solutions and at CyberSource, you can disable partial authorizations for a specific transaction by setting **ccAuthService_partialAuthIndicator** to `false` in your authorization or sale request.
-

Special Processing for IDR and CLP on FDMS South

For the Indonesian rupiah (IDR) and Chilean peso (CLP) currencies only:

- Rounding occurs, which can cause a minor discrepancy of up to one currency unit between the amount you requested and the amount that is authorized.
- When a transaction is enabled for partial authorization, you must ensure that the requested amount does not include any digits to the right of the decimal separator.

Real-Time Reversals

There are two kinds of real-time reversals:

- A *full authorization reversal* is a service that you can request.

If you do not capture a partial authorization and if full authorization reversals are supported for your processor and card type, you must request a full authorization reversal to release the hold that the authorization placed on the customer's funds. The amount of the reversal must be the amount that was authorized, not the amount that was requested. For details about this service and to see the processors and card types for which this service is supported, see ["Reversing an Authorization," page 52](#).

- An *automatic partial authorization reversal* is performed automatically by CyberSource or your processor under certain conditions.

When you capture a partial authorization for an amount that is less than the approved amount, CyberSource automatically performs a partial authorization reversal if it is supported for your processor and card type. CyberSource performs the automatic partial authorization reversal before sending the capture request to the processor.



Note

Some processors perform an automatic partial authorization reversal when there is an interchange benefit. These processors do not allow CyberSource to perform this functionality.

For details about automatic partial authorization reversals and for a list of the processors and card types for which it is supported, see ["Automatic Partial Authorization Reversals," page 69](#).

Balance Responses



Note

Balance inquiries and balance responses are two different features:

- Balance responses are supported for prepaid cards and are a partial authorization feature.
 - Balance inquiries are not associated with partial authorizations. See ["Balance Inquiries," page 128](#).
-



Note

Normally, balance responses are not returned for debit cards.



Note

To receive balance responses from Worldpay VAP, your Worldpay VAP account must be enabled for this feature.

When there is a balance remaining on a prepaid card after an authorization, the authorization reply can include the balance amount. Depending on what data your processor sends to CyberSource, the following fields might be included in the reply:

- `ccAuthReply_accountBalance`: balance amount remaining on the prepaid card after the authorization



Note

For Discover, some processors return the balance in the **`ccAuthReply_authorizationCode`** field.

- `ccAuthReply_accountBalanceCurrency`: currency of the balance amount
- `ccAuthReply_accountBalanceSign`: sign for the balance amount

For descriptions of these fields, see [Appendix A, "API Fields," on page 238](#).

The following table lists the processors and card types for which balance responses are supported. Depending on what data your processor sends to CyberSource, the following fields might be included in the reply.

Table 32 Processors Supported for Balance Responses

Processor	Card Type	Balance Field ¹	Currency Field	Sign Field
American Express Direct	American Express	Yes	Yes	No
Chase Paymentech Solutions	Visa	Yes	Yes	No
	Mastercard	Yes	Yes	No
	American Express	Yes	Yes	No
	Discover	Yes	Yes	No
	Diners Club	Yes	Yes	No
	Maestro (International)	Yes	Yes	No
CyberSource through VisaNet	Visa	Yes	Yes	Yes
	Mastercard	Yes	Yes	Yes
	American Express	Yes	Yes	Yes
	Discover	Yes	Yes	Yes
	Diners Club	Yes	Yes	Yes
	JCB	Yes	Yes	Yes
	mada ²	Yes	Yes	Yes
Elavon Americas	Visa	Yes	No	No
	Mastercard	Yes	No	No
	American Express	Yes	No	No
	Discover	Yes	No	No
	Diners Club	Yes	No	No
	China UnionPay	Yes	No	No
	JCB	Yes	No	No
FDC Compass	Visa	Yes	Yes	No
	Mastercard	Yes	Yes	No
	American Express	Yes	Yes	No
	Discover	Yes	Yes	No

¹ For Discover, some processors return the balance in the **ccAuthReply_authorizationCode** field.

² See "mada Debit Cards," page 24.

Table 32 Processors Supported for Balance Responses (Continued)

Processor	Card Type	Balance Field ¹	Currency Field	Sign Field
FDC Nashville Global	Visa	Yes	Yes	Yes
	Mastercard	Yes	Yes	Yes
	American Express	Yes	Yes	Yes
	Discover	Yes	Yes	Yes
	Diners Club	Yes	Yes	Yes
	China UnionPay	Yes	Yes	Yes
	JCB	Yes	Yes	Yes
FDMS Nashville	Visa	Yes	Yes	Yes
	Mastercard	No	No	No
	American Express	Yes	Yes	Yes
	Discover	No	No	No
	Diners Club	No	No	No
	JCB	No	No	No
FDMS South	Visa	Yes	Yes	Yes
	Mastercard	No	No	No
	American Express	Yes	Yes	Yes
	Discover	No	No	No
	Diners Club	No	No	No
	JCB	No	No	No
GPN	Visa	Yes	Yes	Yes
	Mastercard	Yes	Yes	Yes
	American Express	Yes	Yes	Yes
	Discover	Yes	Yes	Yes
	Diners Club	Yes	Yes	Yes
	JCB	Yes	Yes	Yes
TSYS Acquiring Solutions	Visa	Yes	Yes	Yes
	Mastercard	Yes	Yes	Yes
	American Express	Yes	Yes	Yes
	Discover	Yes	Yes	Yes
	Diners Club	Yes	Yes	Yes
	JCB	Yes	Yes	Yes

¹ For Discover, some processors return the balance in the **ccAuthReply_authorizationCode** field.

² See "mada Debit Cards," page 24.

Table 32 Processors Supported for Balance Responses (Continued)

Processor	Card Type	Balance Field ¹	Currency Field	Sign Field
Worldpay VAP	Visa	Yes	Yes	No
Worldpay VAP was previously called <i>Little</i> .	Mastercard	Yes	Yes	No
	American Express	Yes	Yes	No
	Discover	Yes	Yes	No
	Diners Club	Yes	Yes	No
	JCB	Yes	Yes	No
¹ For Discover, some processors return the balance in the ccAuthReply_authorizationCode field. ² See "mada Debit Cards," page 24.				

Features for Maestro (UK Domestic) Cards

To see which processors support Maestro (UK Domestic) cards, see ["Payment Processors," page 32](#).



Note

This section previously covered Solo cards, but Solo cards are being phased out.



Note

Maestro (UK Domestic) cards were previously called Switch cards.

Maestro (UK Domestic) cards are debit cards that originate in the United Kingdom. These cards can have the following features:

- Issue number: A Maestro (UK Domestic) card might have an issue number embossed on it. The issue number can consist of one or two digits; the first digit can be a zero. An issue number of 2 is different from 02.



Note

Effective May 2011, the issue number is no longer required for Maestro (UK Domestic) transactions.

- Start date: A Maestro (UK Domestic) card might have a start date embossed on it. The start date consists of a month and year.



Note

Effective May 2011, the start date is no longer required for Maestro (UK Domestic) transactions.

Unsupported Processors and Card Types

Prepaid cards and debit cards that do not appear in [Table 31, "Processors Supported for Partial Authorizations," on page 106](#) are processed as follows:

- When the card balance is sufficient for the requested transaction, the transaction is successful.
- When the card balance is not sufficient for the requested transaction, the request is declined.

Optional Features

0 Authorizations

See ["Zero Amount Authorizations," page 229](#).

Additional Amounts

Services:

- Capture
- Credit

Processor:

- American Express Direct

This feature enables you to provide detailed information about specific amounts included in a transaction. For example, if a transaction amount includes a gratuity of 5.00, you can include these fields in the capture or credit request:

```
purchaseTotals_additionalAmount0=5.0  
purchaseTotals_additionalAmountType0=058
```

You can include a maximum of five additional amounts in a transaction. For each amount, you must include an amount field and an amount type field:

- purchaseTotals_additionalAmount0 through purchaseTotals_additionalAmount4
- purchaseTotals_additionalAmountType0 through purchaseTotals_additionalAmountType4

The additional amount type values are listed in [Appendix C, "Additional Amount Types," on page 455](#).

Shipping and Handling Fees

Additional amount fields for shipping and handling fees take precedence over item-level fields. See the following example.

Example 1 Shipping and Handling Fees

- 1 You include the following lines in your request:

```
purchaseTotals_additionalAmount0=9.95
purchaseTotals_additionalAmountType0=055
item_0_productCode=shipping_and_handling
item_0_unitPrice=12.95
```

- 2 CyberSource processes the additional amount fields for the shipping and handling amount of 9.95. The item-level fields for the shipping and handling amount are ignored.

Taxes

Additional amount fields for taxes take precedence over item-level fields. See the following example.

Example 2 Taxes

- 1 You include the following lines in your request:

```
purchaseTotals_additionalAmount0=7.95
purchaseTotals_additionalAmountType0=046
item_0_taxAmount=5.95
```

- 2 CyberSource processes the additional amount fields for the tax amount of 7.95. The item-level field for the tax amount is ignored.

Aggregator Support

This feature enables a third-party agent to act as a payment aggregator and process payment card transactions for sub-merchants. Independent sales organizations (ISOs) and member service providers (MSPs) are agents that can also leverage these aggregator features.

Contact CyberSource Customer Support to have your account configured for this feature.

Terminology

Table 33 **Aggregator Terminology**

Term	Definition
aggregator	Also known as <i>payment aggregator</i> . Organization that aggregates sub-merchants under a single account and settles funds directly to the sub-merchants. An aggregator is usually an ISO or MSP.
independent sales organization (ISO)	Organization that does one or more of the following: ■ Works with acquirers to sponsor merchant accounts and usually assumes the risks associated with the merchants' processing. ■ Procures new merchant relationships based on contracts with acquirers. ■ Connects with a gateway to process online payment card transactions for small businesses, usually in exchange for a fee or percentage of sales.
member service provider (MSP)	Same as an ISO although an MSP has no financial responsibility to the merchant.
payment facilitator	Payment aggregator.
service provider	Third-party or outsource provider of payment processing services. A service provider typically provides a single service with no role in settling funds to a merchant.
sub-merchant	Merchant whose transactions are submitted to CyberSource by a payment aggregator.
third-party agent	Umbrella term for independent sales organizations, member service providers, payment aggregators, and payment facilitators.

American Express Direct Aggregators

Services:

- Authorization
- Capture
- Credit

Card type:

- American Express

The following fields are required for aggregator transactions when requesting an authorization, capture, or credit:

- ccAuthService_aggregatorID—required only for the authorization service
- ccAuthService_aggregatorName—required only for the authorization service
- ccCaptureService_aggregatorID—required only for the capture service
- ccCaptureService_aggregatorName—required only for the capture service

- ccCreditService_aggregatorID—required only for the credit service
- ccCreditService_aggregatorName—required only for the credit service
- invoiceHeader_submerchantCity
- invoiceHeader_submerchantCountry
- invoiceHeader_submerchantEmail
- invoiceHeader_submerchantID
- invoiceHeader_submerchantName
- invoiceHeader_submerchantPostalCode
- invoiceHeader_submerchantState
- invoiceHeader_submerchantStreet
- invoiceHeader_submerchantTelephoneNumber
- merchantCategoryCode

The following fields are optional for aggregator transactions:

- invoiceHeader_merchantDescriptor
- invoiceHeader_merchantDescriptorCity
- invoiceHeader_merchantDescriptorContact
- invoiceHeader_merchantDescriptorCountry
- invoiceHeader_merchantDescriptorPostalCode
- invoiceHeader_merchantDescriptorState
- invoiceHeader_merchantDescriptorStreet

All fields except the merchant descriptor fields are described in [Appendix A, "API Fields," on page 238](#). For information about the merchant descriptor fields, see [Merchant Descriptors Using the Simple Order API](#).

Typically, the merchant descriptor field is used to display your business name on the customer's statement. However, when you are a payment aggregator, you can use other values to provide the sub-merchant's business name for capture and credit requests. The following table describes these values. The order of the values in the table is the order that CyberSource uses to determine which values to use.

Table 34 Values for Providing a Sub-Merchant's Business Name on American Express Direct

Option	Values	Description
1	Aggregator Name + Sub-merchant Name	<i>Aggregator Name</i> The aggregator name is an API field you can include in your request. The API fields are ccAuthService_aggregatorName, ccCaptureService_aggregatorName, and ccCreditService_aggregatorName. <i>Sub-merchant Name</i> The sub-merchant name is the value from the invoiceHeader_submerchantName field. <i>Aggregator Name + Sub-merchant Name</i> When you include the aggregator name field in your request and when your CyberSource account information includes a sub-merchant name, CyberSource combines these two values to provide the business name information for the customer's statement. This approach is advantageous because it allows the business name information to be longer than the size of the merchant descriptor field, which has a length of 27 characters. The total length of the value that CyberSource sends to the processor is 36 characters. It is formatted with an asterisk (*) between the aggregator name and the sub-merchant name: <pre>aggregator name*sub-merchant name</pre> Because the asterisk uses one character, 35 characters remain for the combined length of the aggregator name and sub-merchant name. Important If the combined length of the aggregator name and sub-merchant name exceeds 36 characters, CyberSource declines the transaction.
2	Merchant Descriptor	When you do not provide the values for the preceding option, you can provide the business name in the merchant descriptor field invoiceHeader_merchantDescriptor . This field is described in Merchant Descriptors Using the Simple Order API .
3	Merchant Name	When you do not provide the values for the preceding two options, CyberSource uses the merchant name in your CyberSource account. To add this value to your CyberSource account, contact CyberSource Customer Support.

CyberSource through VisaNet Aggregators

Services:

- Authorization
- Capture
- Credit

Card types:

- American Express
- Diners Club
- Discover
- JCB
- Mastercard
- Visa

Aggregator Transactions with American Express

Authorizations

When requesting an authorization, you must include the following fields:

- ccAuthService_aggregatorID
- ccAuthService_aggregatorName
- invoiceHeader_submerchantCity
- invoiceHeader_submerchantName
- invoiceHeader_submerchantStreet

These fields are optional:

- invoiceHeader_submerchantCountry
- invoiceHeader_submerchantEmail
- invoiceHeader_submerchantID
- invoiceHeader_submerchantPostalCode
- invoiceHeader_submerchantState
- invoiceHeader_submerchantTelephoneNumber

The preceding fields are described in [Appendix A, "API Fields," on page 238](#).

Captures and Credits

When requesting a capture or credit, these fields are optional:

- invoiceHeader_merchantDescriptor
- invoiceHeader_merchantDescriptorCity
- invoiceHeader_merchantDescriptorContact
- invoiceHeader_merchantDescriptorCountry
- invoiceHeader_merchantDescriptorPostalCode
- invoiceHeader_merchantDescriptorState
- invoiceHeader_merchantDescriptorStreet
- invoiceHeader_submerchantEmail
- invoiceHeader_submerchantID
- invoiceHeader_submerchantTelephoneNumber
- merchantCategoryCode

All fields except the merchant descriptor fields are described in [Appendix A, "API Fields," on page 238](#). The merchant descriptor fields are described in [Merchant Descriptors Using the Simple Order API](#).

Typically, the merchant descriptor field is used to display your business name on the customer's statement. However, when you are a payment aggregator, you can use other values to provide the sub-merchant's business name for capture and credit requests. When you do not provide a value in the merchant descriptor fields, CyberSource uses the values in your CyberSource account. To add or update the values in your CyberSource account, contact CyberSource Customer Support.

Aggregator Transactions with Mastercard

When requesting an authorization, you must include the following fields:

- ccAuthService_aggregatorID
- invoiceHeader_salesOrganizationID
- invoiceHeader_submerchantID

When requesting an authorization, capture, or credit, these fields are optional:

- invoiceHeader_merchantDescriptor
- invoiceHeader_merchantDescriptorCity
- invoiceHeader_merchantDescriptorContact
- invoiceHeader_merchantDescriptorCountry
- invoiceHeader_merchantDescriptorPostalCode
- invoiceHeader_merchantDescriptorState
- invoiceHeader_merchantDescriptorStreet
- merchantCategoryCode

All fields except the merchant descriptor fields are described in [Appendix A, "API Fields," on page 238](#). The merchant descriptor fields are described in [Merchant Descriptors Using the Simple Order API](#).

Typically, the merchant descriptor field is used to display your business name on the customer's statement. However, when you are a payment aggregator, you can use other values to provide the sub-merchant's business name for capture and credit requests. When you do not provide a value in the merchant descriptor fields, CyberSource uses the values in your CyberSource account. To add or update the values in your CyberSource account, contact CyberSource Customer Support.

Aggregator Transactions with Any Other Card Type

When requesting an authorization, capture, or credit, these fields are optional:

- invoiceHeader_merchantDescriptor
- invoiceHeader_merchantDescriptorCity
- invoiceHeader_merchantDescriptorContact
- invoiceHeader_merchantDescriptorCountry
- invoiceHeader_merchantDescriptorPostalCode
- invoiceHeader_merchantDescriptorState
- invoiceHeader_merchantDescriptorStreet
- merchantCategoryCode

All fields except the merchant descriptor fields are described in [Appendix A, "API Fields," on page 238](#). The merchant descriptor fields are described in [Merchant Descriptors Using the Simple Order API](#).

Typically, the merchant descriptor field is used to display your business name on the customer's statement. However, when you are a payment aggregator, you can use other values to provide the sub-merchant's business name for capture and credit requests. When you do not provide a value in the merchant descriptor fields, CyberSource uses the values in your CyberSource account. To add or update the values in your CyberSource account, contact CyberSource Customer Support.

FDC Compass Aggregators

Services:

- Authorization
- Capture
- Credit

Card types:

- American Express
- Mastercard

The following fields are required for aggregator transactions with American Express or Mastercard when requesting an authorization, capture, or credit:

- `ccAuthService_aggregatorID`—required only for the authorization service
- `ccAuthService_aggregatorName`—required only for the authorization service with Mastercard
- `ccCaptureService_aggregatorID`—required only for the capture service
- `ccCaptureService_aggregatorName`—required only for the capture service with Mastercard
- `ccCreditService_aggregatorID`—required only for the credit service
- `ccCreditService_aggregatorName`—required only for the credit service with Mastercard
- `invoiceHeader_submerchantCity`
- `invoiceHeader_submerchantID`
- `invoiceHeader_submerchantName`
- `invoiceHeader_submerchantTelephoneNumber`

The following fields are optional for aggregator transactions:

- `invoiceHeader_submerchantCountry`
- `invoiceHeader_submerchantEmail`
- `invoiceHeader_submerchantPostalCode`
- `invoiceHeader_submerchantState`
- `invoiceHeader_submerchantStreet`
- `merchantCategoryCode`—supported only for the authorization service

All fields are described in [Appendix A, "API Fields," on page 238](#).

For Mastercard aggregator captures and credits, CyberSource combines the following two values to provide the business name information for the customer's statement:

- Aggregator name in the `ccCaptureService_aggregatorName` or `ccCreditService_aggregatorName` field.
- Sub-merchant name in the `invoiceHeader_submerchantName` field.

The total length of the value that CyberSource sends to the processor is 36 characters. It is formatted with an asterisk (*) between the aggregator name and the sub-merchant name:

```
aggregator name*sub-merchant name
```

Because the asterisk uses one character, 37 characters remain for the combined length of the aggregator name and sub-merchant name.



Important

If the combined length of the aggregator name and sub-merchant name exceeds 37 characters, CyberSource declines the transaction.

FDC Nashville Global Aggregators

Services:

- Authorization
- Capture
- Credit

Card types:

- American Express
- Mastercard

The following fields are required for aggregator transactions with American Express or Mastercard when requesting an authorization, capture, or credit:

- ccAuthService_aggregatorID—required only for the authorization service
- ccAuthService_aggregatorName—required only for the authorization service
- ccCaptureService_aggregatorID—required only for the capture service
- ccCaptureService_aggregatorName—required only for the capture service
- ccCreditService_aggregatorID—required only for the credit service
- ccCreditService_aggregatorName—required only for the credit service
- invoiceHeader_submerchantCity
- invoiceHeader_submerchantCountry
- invoiceHeader_submerchantEmail
- invoiceHeader_submerchantID
- invoiceHeader_submerchantName
- invoiceHeader_submerchantPostalCode
- invoiceHeader_submerchantState
- invoiceHeader_submerchantStreet
- invoiceHeader_submerchantTelephoneNumber
- merchantCategoryCode

The following fields are optional for aggregator transactions:

- `invoiceHeader_submerchantMerchantID`—supported only for American Express
- `invoiceHeader_submerchantRegion`

All fields are described in [Appendix A, "API Fields,"](#) on page 238.

Airline Data

See [Airline Processing Using the Simple Order API](#).

American Express SafeKey

See ["Payer Authentication,"](#) page 183.

Android Pay

The name of Android Pay changed to *Google Pay*. See [Google Pay Using the Simple Order API](#).

Apple Pay

See [Apple Pay Using the Simple Order API](#).

Authorization Only

Service:

- Authorization

Processors:

- American Express Direct
- Elavon Americas

In the authorization reply message, CyberSource provides you with point-of-sale (POS) and transaction ID (TID) values. When you perform authorizations through CyberSource and perform captures and credits through other financial institutions, you can include these values in your capture requests and follow-on credit requests:

- POS data: Get this value from **ccAuthReply_posData**.
- TID: Get this value from **ccAuthReply_transactionID**.

Including these values in your capture requests and follow-on credit requests enables you to comply with the CAPN requirements, thus avoiding noncompliance fees.

**Note**

When you use the Simple Order API in XML format, you must use version 1.63 or later of the XML schema to implement the authorization only feature.

AVS Only

See ["Zero Amount Authorizations," page 229](#).

Balance Inquiries

Service:

- Authorization

Processors:

- CyberSource through VisaNet
- Elavon Americas



Note

Balance inquiries and balance responses are two different features:

- Balance responses are supported for prepaid cards and are a partial authorization feature. See ["Balance Responses," page 111](#).
- Balance inquiries are not associated with partial authorizations.

This feature enables you to request balance information for an account.

To use this feature, include the **balanceInquiry** field in an authorization request. The amount in the request must be zero.

CyberSource returns the following fields:

- ccAuthReply_accountBalance
- ccAuthReply_accountBalanceCurrency
- ccAuthReply_accountBalanceSign
- ccAuthReply_accountType
- ccAuthReply_amountType

These fields are described in ["API Fields," page 238](#).

Bill Payments with Mastercard

See ["Mastercard Bill Payments," page 167](#).

Bill Payments with Visa

See ["Visa Bill Payments," page 227](#).

BNDES

Services:

- Authorization
- Credit

Card types:

- Mastercard
- Visa

Processor:

- CyberSource through VisaNet

BNDES is card product the same way Visa Platinum, Visa Corporate, and Mastercard Corporate are products. BNDES is also a bank named Brazilian Development Bank. The BNDES bank issues the BNDES card products.

To indicate that a transaction on CyberSource through VisaNet is a BNDES Mastercard transaction:

Step 1 You must include the following fields in your authorization or credit request:

- billTo_companyTaxID
- billTo_merchantTaxID
- card_productName

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource through VisaNet is a BNDES Visa transaction:

Step 1 You must include the following fields in your authorization or credit request:

- billTo_personalID or billTo_companyTaxID
- card_productName
- loan_assetType
- loan_type

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Card-Present Data

See [Card-Present Processing Using the Simple Order API](#).

Card Type Indicators (CTIs)

Service:

- Authorization

Processors:

- Chase Paymentech Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.



Note

Contact your processor to have your account configured for this feature.

Contact CyberSource Customer Support to have your account configured for this feature.

This feature enables you to receive CTI information in your authorization reply messages. The processor can provide CTI information for approved or declined transactions, not for rejected transactions.

The CTI information is returned in the following fields:

- ccAuthReply_affluenceIndicator
- ccAuthReply_cardCommercial
- ccAuthReply_cardHealthcare
- ccAuthReply_cardIssuerCountry

- ccAuthReply_cardLevel3Eligible
- ccAuthReply_cardPayroll
- ccAuthReply_cardPINlessDebit
- ccAuthReply_cardPrepaid
- ccAuthReply_cardRegulated
- ccAuthReply_cardSignatureDebit

For Worldpay VAP, CTI information is also returned in the following fields:

- card_prepaidReloadable
- card_prepaidType
- card_virtual

The CTI fields are described in [Appendix A, "API Fields," on page 238](#).

To receive CTI information on Chase Paymentech Solutions:

Your authorization request message must comply with the CTI acceptance criteria as described in the following table.

Table 35 CTI Acceptance Criteria

Card Type	Acceptance Criteria
American Express	CTI is not supported.
Carte Blanche	CTI is not supported.
Diners Club	Currency is USD or CAD.
Discover	Currency is USD or CAD.
JCB	Currency is USD.
Mastercard	Any currency.
Visa	Amount is not 0 (zero). Any currency.

To receive CTI information on Worldpay VAP:

CyberSource returns CTI information in the authorization reply message.

Cash Advances

Services:

- Authorization
- Capture

Processors:

- Barclays
- LloydsTSB Cardnet

A cash advance enables a customer to use a payment card to purchase foreign currency or travelers checks. The currency the customer uses to fund the transactions must be British pounds.

Before processing cash advances, you must:

- Contact the processor to obtain an agreement to process cash advance transactions.
- Contact CyberSource Customer Support to have your account configured for this feature. You must have a separate CyberSource merchant ID that you use only for cash advance transactions.

Process a cash advance transaction the same way you process a regular credit card transaction: with an authorization and a capture.

**Important**

You cannot process a cash advance and airline data in the same transaction.

Chase Pay

See [Chase Pay Using the Simple Order API](#).

Customer Profiles

See "Payment Tokenization," page 202.

Dynamic Currency Conversion with a Third Party Provider

**Note**

This section describes how to include Dynamic Currency Conversion (DCC) data from a third party DCC provider in your requests for CyberSource credit card services. This section covers transaction processing after DCC. For information about DCC, contact your DCC provider.

For information about the DCC service that is provided by CyberSource for card-not-present transactions on FDC Nashville Global and FDMS South, see [Dynamic Currency Conversion for First Data Using the Simple Order API](#).

Services:

- Authorization
- Authorization reversal
- Capture
- Credit

Processors:

- FDC Nashville Global—only for card-present transactions
- SIX—only for card-present transactions

Card types:

- Visa
- Mastercard

DCC converts a foreign customer's purchase from your local pricing currency to the customer's billing currency. This functionality can help you improve or create business relationships with customers who prefer to make purchases in their own currency.

Requirement and Limitations

To include DCC data from a third party provider:

- Contact your acquirer to register for this feature.
- Contact CyberSource Customer Support to have your account configured for this feature.

- You must meet the payment card company rules for DCC. Contact your acquirer for details. For example, you might need to:
 - Provide the customer with a receipt that shows the amount in the local pricing currency, the amount in the billing currency, the rate of exchange used to convert the order amount, and the DCC markup.
 - Have the customer sign an acknowledgment that the customer had a choice to pay in the local pricing currency and that the choice of currency is final.

When you use DCC for an order, you cannot split the order into multiple shipments.

Terminology

Table 36 DCC Terminology

Term	Definition
Billing currency or converted currency	Currency in which the card is denominated and in which transactions are posted to the customer's account.
Converted amount	Amount of the transaction, denominated in the customer's billing currency.
Exchange rate	Factor used to convert an amount in one currency to an amount in another currency.
Local pricing currency	Selling currency that you use for pricing your goods and in which you usually submit transactions for processing.
Original amount	Amount of the transaction, denominated in your local pricing currency.

Examples

See the examples for DCC in [Card-Present Processing Using the Simple Order API](#).

Authorizing a Payment



Important

The value for the **purchaseTotals_originalAmount** field must always be in your local pricing currency.

Except for the original amount, all amounts for the order must be in the converted currency. This requirement includes the total payment amount and any tax amounts or surcharge amounts that you send to CyberSource. Use the exchange rate from your DCC provider to convert these amounts from your local pricing currency to the customer's billing currency.

For information about creating an authorization request, see ["Creating an Authorization Request," page 41](#). Include the following DCC fields in your authorization request:

- **dcc_dccIndicator**: set this field to 1. When you include DCC data in your authorization request and do not set this field to 1, CyberSource rejects the request.
- **dcc_referenceNumber**: unique identifier generated by the DCC provider.
- **item_#_unitPrice** or **purchaseTotals_grandTotalAmount**: converted amount in your customer's billing currency.
- **purchaseTotals_currency**: currency code for your customer's billing currency.
- **purchaseTotals_exchangeRate**: exchange rate.
- **purchaseTotals_exchangeRateTimeStamp**: exchange rate timestamp in GMT in this format: YYYYMMDDhhmmss
- **purchaseTotals_foreignAmount**: converted amount in your customer's billing currency.
- **purchaseTotals_foreignCurrency**: currency code for your customer's billing currency.
- **purchaseTotals_originalAmount**: original amount in your local pricing currency.
- **purchaseTotals_originalCurrency**: currency code for your local pricing currency.

For details about these fields, see [Appendix A, "API Fields," on page 238](#).

Reversing an Authorization



Important

Do not include any DCC fields in a full authorization reversal request. When you include DCC fields in the request, CyberSource ignores them. For full authorization reversals, CyberSource uses the data from the original authorization.



Important

The value for the **purchaseTotals_originalAmount** field must always be in your local pricing currency.

Except for the original amount, all amounts for the order must be in the converted currency. This requirement includes the total payment amount and any tax amounts or surcharge amounts that you send to CyberSource. Use the exchange rate from your DCC provider to convert these amounts from your local pricing currency to the customer's billing currency.

For information about creating an authorization reversal request, see ["Creating a Full Authorization Reversal Request," page 57](#).

Capturing an Authorization



Important

Do not include any DCC fields in a capture request. When you include DCC fields in the request, CyberSource ignores them. For captures, CyberSource uses the data from the original authorization.



Important

The value for the **purchaseTotals_originalAmount** field must always be in your local pricing currency.

Except for the original amount, all amounts for the order must be in the converted currency. This requirement includes the total payment amount and any tax amounts or surcharge amounts that you send to CyberSource. Use the exchange rate from your DCC provider to convert these amounts from your local pricing currency to the customer's billing currency.

For information about creating a capture request, see ["Creating a Capture Request," page 62](#).

Crediting the Payment



Important

Do not include any DCC fields in a credit request. When you include DCC fields in the request, CyberSource ignores them. For credits, CyberSource uses the data from the original authorization.



Important

The value for the **purchaseTotals_originalAmount** field must always be in your local pricing currency.

Except for the original amount, all amounts for the order must be in the converted currency. This requirement includes the total payment amount and any tax amounts or surcharge amounts that you send to CyberSource. Use the exchange rate from your DCC provider to convert these amounts from your local pricing currency to the customer's billing currency.

To credit a payment for a transaction that uses DCC, request a follow-on credit. For information about creating a credit request, see ["Creating a Credit Request," page 79](#). DCC is not supported for stand-alone credits. When you include DCC fields in a request for a stand-alone credit, CyberSource ignores them.

Encoded Account Numbers

Services:

- Authorization
- Credit

Processor:

- Chase Paymentech Solution's Credit Card Encryption program

Depending on your type of business, you might be eligible to acquire from an issuing bank a list of the customers who have payment cards issued by that bank. The list does not include the customers' payment card numbers, but instead includes encoded account numbers. Some processors refer to this type of program as issuer encryption and to the numbers as encrypted account numbers. This type of program is designed to protect customer information according to the provisions of the Gramm-Leach-Bliley Act.

When processing a payment or credit for one of these customers, you use the encoded account number instead of the customer's payment card number. The issuing bank then matches the encoded account number to the customer's payment card number when processing the payment.

You must contact your processor to obtain the information required for the Credit Card Encryption program and you must have a relationship with the bank in order to acquire their list of customers.

Final Authorization Indicator

Services:

- Authorization
- Incremental authorization

Processors:

- Barclays
- Chase Paymentech Solutions—Mastercard and Maestro (International) only. Chase Paymentech Solutions does not support this feature for Maestro (UK Domestic).
- CyberSource through VisaNet
- Elavon
- Elavon Americas—Mastercard only.
- FDC Compass
- FDC Nashville Global
- FDI Australia
- FDMS Nashville
- GPN
- HBoS
- HSBC
- Lloyds-OmniPay
- LloydsTSB Cardnet
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- OmniPay-Ireland—Mastercard only. OmniPay-Ireland does not support Maestro (International) or Maestro (UK Domestic).
- SIX

- Streamline
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*. CyberSource does not take any action to support this feature on Worldpay VAP. The processor sets the indicator.

Card types:

- Mastercard
- Maestro (International)
- Maestro (UK Domestic)
- Co-badged Mastercard and mada as described in ["mada Debit Cards," page 24](#). You must identify the card type as Mastercard. Supported only on CyberSource through VisaNet.

This feature supports a mandate from Mastercard. The purpose of the mandate is to ensure that a customer's funds are available when there is a risk that the order will not be fulfilled.

For an authorization with an amount greater than zero, Mastercard recommends that you indicate whether the authorization is a final authorization, a preauthorization, or an undefined authorization.

Final Authorizations

For a final authorization:

- Authorization amount is greater than zero.
- Authorization amount is the final amount that the customer agrees to pay.
- Authorization should not be cancelled after it is approved except when a system failure occurs.
- Authorization must be submitted for capture within seven calendar days of its request.
- Capture amount and currency must be the same as the authorization amount and currency.
- Chargeback protection is in effect for seven days following the authorization.

Preauthorizations

For a preauthorization:

- Authorization amount is greater than zero.
- Authorization amount can be an estimate when the final amount is unknown, which is typical for hotel, auto rental, e-commerce, and restaurant transactions.
- Authorization must be submitted for capture within 30 calendar days of its request.
- When you do not capture the authorization, you must reverse it.



Note

In the U.S., Canada, Latin America, and Asia Pacific, Mastercard charges an additional fee for a preauthorization that is not captured and not reversed.

In Europe, Russia, Middle East, and Africa, Mastercard charges fees for all preauthorizations.

- Chargeback protection is in effect for 30 days following the authorization.

Undefined Authorizations



Note

Undefined authorizations are supported only in the U.S., Canada, Latin America, and Asia Pacific. They are not supported in Europe, Russia, Middle East, and Africa.



Note

Undefined authorizations are not supported on the following processors:

- Chase Paymentech Solutions
- Elavon Americas
- FDC Compass
- FDC Nashville Global
- FDI Australia
- FDMS Nashville
- SIX

For an undefined authorization:

- Authorization amount is greater than zero.
- Authorization amount can be different from the final transaction amount.

- Authorization should not be cancelled after it is approved except when a system failure occurs.
- Authorization must be submitted for capture within seven calendar days of its request.
- When you do not capture the authorization, you must reverse it; otherwise, Mastercard charges an additional fee for the transaction.
- Chargeback protection is in effect for seven days following the authorization.

**Note**

An authorization is undefined when you set the default authorization type in your CyberSource account to `undefined` and do not include the **authIndicator** field in the authorization request. To set the default authorization type in your CyberSource account, contact CyberSource Customer Support.

Unmarked Authorizations

**Note**

Unmarked authorizations are supported only on the following processors:

- Chase Paymentech Solutions
- CyberSource through VisaNet
- Elavon Americas
- FDC Compass
- FDC Nashville Global
- FDI Australia
- FDMS Nashville
- HBoS
- Lloyds-OmniPay
- LloydsTSB Cardnet
- Streamline
- SIX

For an unmarked authorization:

- CyberSource does not set a mark or indicator for the type of authorization in the request that is sent to the processor.
- Authorization amount is greater than zero.
- Authorization amount can be different from the final transaction amount.

- Your acquirer processes an unmarked authorization as a final authorization, a preauthorization, or an undefined authorization. Contact your acquirer to learn how they process unmarked authorizations.

**Note**

An authorization is unmarked when the default authorization type is not set in your CyberSource account and you do not include the **authIndicator** field in the authorization request. To set the default authorization type in your CyberSource account, contact CyberSource Customer Support.

To indicate whether an authorization is a final authorization or a preauthorization:

- Step 1** Include the **authIndicator** field in your authorization request. See ["Request Fields," page 241](#), for the field description.
- Step 2** For a final authorization on a processor that supports merchant-initiated transactions, your authorization request must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

The **authIndicator** field is included in the reply message for the following processors:

- Chase Paymentech Solutions
 - CyberSource through VisaNet
 - FDC Compass
 - FDC Nashville Global
 - FDI Australia
 - FDMS Nashville
-

Forced Captures

Service:

- Authorization

Processors:

- AIBMS
- American Express Direct
- Asia, Middle East, and Africa Gateway
- Chase Paymentech Solutions

- CyberSource through VisaNet. The supported acquirers are:
 - Banco Safra
 - Bank Sinarmas (Omise Ltd.)
 - BC Card Co., Ltd.
 - Citibank Malaysia
 - CTBC Bank Ltd.
 - Sumitomo Mitsui Card Co.
 - Vietnam Technological and Commercial Joint-stock Bank
- Elavon Americas
- FDC Nashville Global
- FDMS Nashville
- FDMS South
- GPN
- JCN Gateway
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- TSYS Acquiring Solutions

**Note**

Forced captures are not supported for Comercio Latino and CyberSource Latin American Processing.

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. The information in this note is for the specific processing connections called *Comercio Latino* and *CyberSource Latin American Processing*. It is not for any other Latin American processors that CyberSource supports.

A forced capture occurs when you process an authorization outside the CyberSource system but then capture the order through CyberSource.

To perform a forced capture:

After you process the authorization outside the CyberSource system, request the CyberSource authorization and capture services at the same time as described in ["Creating an Authorization Request," page 41](#), and ["Creating a Capture Request," page 62](#):

- Include the request fields that are required for the authorization.
- Include these fields in the request:

```
ccAuthService_authType=verbal
```

```
ccAuthService_verbalAuthCode= the authorization code you received in the  
response for the authorization that was processed outside the CyberSource system
```

- No additional fields are required for the capture.

For the American Express card type on FDMS South, you must include the **ccCaptureService_posData** and **ccCaptureService_transactionID** fields in the capture request to support the CAPN requirements. Obtain the values for these fields from the response for the authorization that was processed outside the CyberSource system.

Full-Financial Transactions

See ["Least-Cost Routing \(LCR\)," page 164](#) and ["Single-Message Processing," page 76](#).

Google Pay

See [Google Pay Using the Simple Order API](#).

Guaranteed Exchange Rates

See ["Multi-Currency Service," page 182](#).

Installment Payments

Services:

- Authorization
- Capture—only on CyberSource through VisaNet and FDC Nashville Global

Processors and card types:

- See the following table.

Table 37 Processors That Support Installment Payments

Processors	Payment Card Types
American Express Direct	American Express See "Installment Payments on American Express Direct," page 148.
Chase Paymentech Solutions	Visa See "Installment Payments on Chase Paymentech Solutions and FDC Compass," page 150.
Cielo	Visa, Mastercard, American Express, Diners Club, JCB, Elo, Aura On Cielo, installment payments are not supported for debit transactions. See "Installment Payments on Processors in Latin America," page 159.
Comercio Latino	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard To enable installment payments, contact CyberSource Customer Support to have your account configured for this feature. On Comercio Latino, the acquirer Banorte requires installment payments be submitted as an automatic capture. See "Automatic Captures," page 40. See "Installment Payments on Processors in Latin America," page 159.
CyberSource Latin American Processing	Visa See "Installment Payments on Processors in Latin America," page 159. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America.

Table 37 Processors That Support Installment Payments (Continued)

Processors	Payment Card Types
CyberSource through VisaNet	Visa, Mastercard, American Express Note Not all card types are supported for all acquirers. The supported acquirers are: • Arab African International Bank (AAIB) • Asia Commercial Bank (ACB) • Auckland Savings Bank (ASB) • Australia and New Zealand Banking Group Limited (ANZ) • Axis Bank Ltd. of India • Banco Nacional de México (Banamex) • Banco Safra • Bangkok Bank Ltd. • Bank of Ayudhya (BAY) • Bank of China (BOC) • Bank Sinarmas (Omise Ltd.) • Banque Pour Le Commerce Exterieur Lao (BCEL) • BC Card Co., Ltd. • Cathay United Bank (CUB) • Citibank Malaysia • Citibank Hongkong and Macau • Commercial Bank of Qatar • CrediMax (Bahrain) • CTBC Bank Ltd. • First Data Merchant Solutions in Brunei • First Data Merchant Solutions in Hong Kong • First Data Merchant Solutions in Malaysia • First Data Merchant Solutions in Singapore • Habib Bank Ltd. (HBL) • HDFC Bank Ltd. of India • Mashreq • National Bank of Abu Dhabi (NBAD) • Overseas Chinese Banking Corp (OCBC) • Promerica in Honduras and Nicaragua • Taishin Bank Ltd. <i>(continued)</i>

Table 37 Processors That Support Installment Payments (Continued)

Processors	Payment Card Types
	• United Overseas Bank (UOB) in Singapore and Vietnam • United Overseas Bank (UOB) in Thailand • Vantiv • Vietcombank • VietinBank • Vietnam Technological and Commercial Joint-stock Bank • Wing Hang Bank • Wing Lung Bank See "Installment Payments on CyberSource through VisaNet," page 150.
FDC Compass	Visa See "Installment Payments on Chase Paymentech Solutions and FDC Compass," page 150.
FDC Nashville Global	Visa, Discover, Diners Club, JCB (US Domestic) For JCB cards, "US Domestic" means that the currency is USD and your location is the U.S., Puerto Rico, Guam, U.S. Virgin Islands, or Northern Mariana Islands. See "Installment Payments on FDC Nashville Global," page 158.
FDI Australia	Visa, Mastercard See "Installment Payments on FDI Australia," page 159.
FDMS Nashville	Visa See "Installment Payments on Other Processors," page 161.
FDMS South	Visa See "Installment Payments on Other Processors," page 161.
Getnet	Visa, Mastercard, American Express, Elo, Hipercard On Getnet, installment payments are not supported for debit transactions. See "Installment Payments on Processors in Latin America," page 159.
OmniPay-Ireland	Visa OmniPay-Ireland is the CyberSource name for HSBC International. See "Installment Payments on Other Processors," page 161.

Table 37 Processors That Support Installment Payments (Continued)

Processors	Payment Card Types
Rede	Visa, Mastercard, American Express, Diners Club, JCB, Elo, Hipercard On Rede, installment payments are not supported for debit transactions. See "Installment Payments on Processors in Latin America," page 159.
Software Express	Supported acquirer: Getnet Support card types: Visa, Mastercard See "Installment Payments on Processors in Latin America," page 159.
TSYS Acquiring Solutions	Visa See "Installment Payments on Other Processors," page 161.
Worldpay VAP	Visa
Worldpay VAP was previously called <i>Little</i> .	See "Installment Payments on Other Processors," page 161.

Installment Payments on American Express Direct

The customer pays for goods or services using an installment plan agreed upon by the customer and you. The following table describes the types of installment payments that American Express Direct supports.

Table 38 Types of Installment Payments on American Express Direct

Type of Installment Payments	Features
Issuer installments American Express calls this arrangement a <i>deferred payment plan</i>.	■ You send one transaction to American Express. ■ American Express charges the amount to the customer in installments. ■ You receive one payment from American Express.
Merchant installments American Express calls this arrangement <i>Plan N</i>.	■ You send one transaction to American Express. ■ American Express charges the amount to the customer in installments. ■ You receive payment from American Express in installments.

The following table lists the countries and regions for which CyberSource supports installment payments on American Express Direct.

Table 39 Country-Specific Information for Installment Payments on American Express Direct

Country or Region	Notes
Argentina	■ Issuer installments and merchant installments are supported. ■ The currency for your installment transactions must be ARS.
Asia Pacific	■ Only issuer installments are supported. Merchant installments are not supported.
Australia	■ Only issuer installments are supported. Merchant installments are not supported.
Mexico	■ Issuer installments and merchant installments are supported. ■ The currency for your installment transactions must be MXN. ■ The purchase amount must be 250 MXN or more.



Important

When you submit an installment transaction that does not meet the American Express Direct requirements for installment payments, American Express Direct processes the transaction as a regular, non-installment transaction.

Before submitting installment transactions:

- Contact American Express Direct to have your account configured for this feature.
- Contact CyberSource Customer Support to have your account configured for this feature.

To indicate that a transaction on American Express Direct is an installment payment:

- Step 1** You must include the **installment_totalCount** field in your authorization request.
- Step 2** You can include the optional **ccAuthService_commerceIndicator** field in your authorization request. Set it to any valid value except `recurring` or `recurring_internet`. For information about the commerce indicator values, see [Appendix I, "Commerce Indicators," on page 470](#).
- Step 3** You must include the **installment_planType** field in your authorization request if the corresponding value is not set in your CyberSource account. If this value is set in your CyberSource account, you can include the field in your authorization request to override the value in your CyberSource account.

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on Chase Paymentech Solutions and FDC Compass

The customer pays for goods or services using an installment plan agreed upon by the customer and you.

To indicate that a transaction on Chase Paymentech Solutions or FDC Compass is an installment payment:

Step 1 Set `ccAuthService_commerceIndicator` to `install`.

Step 2 Include the following required fields in your authorization request:

- `invoiceHeader_merchantDescriptor`
- `invoiceHeader_merchantDescriptorContact`

For information about these fields, see [Merchant Descriptors Using the Simple Order API](#).

Step 3 Include subsequent authorization fields as described in "Merchant-Initiated Transactions," [page 173](#).

Step 4 You can include the following optional fields in your authorization request:

- `installment_sequence`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on CyberSource through VisaNet

Overview

Installment payments, which are called *parcelados* and *parcelas* in Brazil, are a common way to pay for purchases made with payment cards. When a customer pays in installments, the entire payment amount is authorized and captured at the time of the purchase, but the payment is settled in increments over a period of time. Each incremental settlement is an installment payment.

Before you can accept installment payments, you and your acquirer must agree on the maximum number of installments you can accept, which can be different for each card type.

For customers, installment payments provide greater purchasing power and lower impact on their monthly budget. For you, offering installment payments at checkout typically increases the number of successfully completed purchases by 50%.

Types of Funding

There are two standard types of funding for installment payments:

- Issuer-funded installments
- Merchant-funded installments

There are two additional types of funding for installment payments in Brazil:

- Co-branded merchant financed
- Issuer merchant co-financed

CyberSource through VisaNet supports all four types of funding. CyberSource through VisaNet only enables the processing of installment payments. It has no role in setting the terms for the installments.

Issuer-Funded Installment Payments

The customer pays for goods or services using an installment plan agreed upon by the customer and their issuing bank. The issuer controls how the customer's account is debited. Your account is credited for the entire amount in a single transaction. The issuer assumes the risk and establishes credit rates and fees that are charged to the customer. The cardholder pays the funding cost, which is a fee for paying in installments.

In Brazil, a *Crediariorio* is a special type of issuer-funded installment payment plan that enables the customer to request information about the terms of the installment plan before approving the installment payments.

Merchant-Funded Installment Payments

The customer pays for goods or services using an installment plan agreed upon by you and the customer. The issuer controls how the customer's account is debited. Your account is credited periodically for partial amounts as the customer's account is debited. You assume the risk and establish the credit rate and fees that are charged to the customer.

Co-Branded Merchant Financed Installment Payments

You and the issuer determine the terms for this kind of installment plan. The funding varies depending on the agreement between you, the issuer, and the cardholder. This funding method is available only for Mastercard installment payments in Brazil.

Issuer Merchant Co-Financed Installment Payments

The issuer creates the installment plan. You and the issuer determine the service fees that the cardholder pays to you and the issuer. The acquirer is paid in full while the issuer is paid in installments by the cardholder. You or the cardholder pay the funding cost, which is a fee for paying in installments. This funding method is available only for Mastercard installment payments in Brazil.

Installment Payments on CyberSource through VisaNet in Brazil

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Mastercard in Brazil:

- Step 1** You must include the following fields in your authorization or capture request:
- `installment_planType`
 - `installment_totalCount`
- Step 2** When creating an authorization request, you must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource through VisaNet is a Crediario installment payment with Visa in Brazil:

- Step 1** Inquire about Crediario eligibility for the customer by requesting the authorization service. Include the following required fields in the authorization request:
- `eligibilityInquiry`—set to `Crediario`.
 - `installment_amount`
 - `installment_firstInstallmentDate`
 - `installment_frequency`
 - `installment_sequence`
 - `installment_totalAmount`
 - `installment_totalCount`
 - `purchaseTotals_grandTotalAmount`—set to `0`.
- Step 2** The issuer determines whether the customer is eligible for Crediario installment payments. If the customer is eligible, the authorization reply includes the following fields:
- `installment_additionalCosts`
 - `installment_additionalCostsPercentage`

- installment_amountFunded
- installment_amountRequestedPercentage
- installment_annualFinancingCost
- installment_annualInterestRate
- installment_expenses
- installment_expensesPercentage
- installment_fees
- installment_feesPercentage
- installment_insurance
- installment_insurancePercentage
- installment_monthlyInterestRate
- installment_taxes
- installment_taxesPercentage

Step 3 Use the values in the authorization reply to display the installment payment terms to the customer. The customer decides whether to accept the terms.

Step 4 If the customer accepts the terms, request an installment payment:

- a** Request the authorization service. Include the following required fields in the authorization request:
- installment_planType
 - installment_totalAmount
 - installment_totalCount
 - Subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

You can include the following optional fields in the authorization request:

- installment_amount
- installment_firstInstallmentDate
- installment_frequency
- installment_sequence

- b** Request the capture service. Include the following required fields in the capture request:
- installment_additionalCosts
 - installment_additionalCostsPercentage
 - installment_amountFunded
 - installment_amountRequestedPercentage
 - installment_annualFinancingCost
 - installment_annualInterestRate

- installment_expenses
- installment_expensesPercentage
- installment_fees
- installment_feesPercentage
- installment_insurance
- installment_insurancePercentage
- installment_monthlyInterestRate
- installment_planType
- installment_taxes
- installment_taxesPercentage
- installment_totalAmount
- installment_totalCount

You can include the following optional fields in the capture request:

- installment_amount
- installment_firstInstallmentDate
- installment_frequency
- installment_sequence

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource through VisaNet is a non-Crediario installment payment with Visa in Brazil:

Step 1 You must include the following fields in your authorization or capture request:

- installment_planType
- installment_totalCount

Step 2 When creating an authorization request, you must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

Step 3 You can include the following optional field in your authorization or capture request:

- installment_paymentType

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on CyberSource through VisaNet in Croatia

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Mastercard in Croatia:

- Step 1** You must include the following fields in your authorization or capture request:
- billTo_merchantTaxID
 - installment_planType
- Step 2** When creating an authorization request, you must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on CyberSource through VisaNet in Georgia (Sakartvelo)

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Mastercard in Georgia:

- Step 1** You must include the following fields in your authorization or capture request:
- installment_amount
 - installment_firstInstallmentAmount
 - installment_monthlyInterestRate
 - installment_planType
 - installment_totalCount
- Step 2** When creating an authorization request, you must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on CyberSource through VisaNet in Greece

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Mastercard in Greece:

Step 1 You must include the following fields in your authorization or capture request:

- installment_gracePeriodDuration
- installment_gracePeriodDurationType
- installment_planType
- installment_totalCount

Step 2 When creating an authorization request, you must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on CyberSource through VisaNet in Other Countries

To indicate that a transaction on CyberSource through VisaNet is an installment payment with American Express:

- Step 1** Include **installment_planType** or **installment_totalCount** in your authorization or capture request.

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Mastercard:

- Step 1** You must include the following fields in your authorization or capture request:

- **installment_paymentType**
- **installment_planType**

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource through VisaNet is an installment payment with Visa:

- Step 1** Set **ccAuthService_commerceIndicator** to `install` or `install_internet`:

- `install`—U.S. transaction or non-U.S. mail order / telephone order (MOTO) transaction
- `install_internet`—non-U.S. e-commerce (Internet) transaction

- Step 2** Include **installment_planType** in your authorization request.

- Step 3** The authorization request must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

Step 4 You can include the following optional fields in your authorization request:

- `installment_amount`
- `installment_frequency`
- `installment_sequence`
- `installment_totalAmount`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on FDC Nashville Global

The customer pays for goods or services using an installment plan agreed upon by the customer and you.

To indicate that a transaction on FDC Nashville Global is an installment payment:

Step 1 When you request the authorization service, set `ccAuthService_commerceIndicator` to `install`.

Step 2 When you request the capture service, include the following required fields in the request:

- `installment_sequence`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on FDI Australia

The customer pays for goods or services using an installment plan agreed upon by the customer and you.

To indicate that a transaction on FDI Australia is an installment payment:

- Step 1** In the authorization request for the first installment payment, set **ccAuthService_commerceIndicator** to `install`.
- Step 2** In the authorization requests for the subsequent installment payments, include these fields:
- **ccAuthService_commerceIndicator**—set to `install`.
 - **subsequentAuthTransactionID**—set to the value returned in the `ccAuthReply_paymentNetworkTransactionID` field for the previous authorization in the series.

For information about these fields, see [Appendix A, "API Fields,"](#) on page 238.

Installment Payments on Processors in Latin America

The customer pays for goods or services using an installment plan agreed upon by the customer and you.

Before submitting installment transactions, contact CyberSource Customer Support to have your account configured for this feature.

To indicate that a transaction on Cielo or Comercio Latino is an installment payment:

- Step 1** You must include the **installment_totalCount** field in your authorization request.
- Step 2** You can include the optional **ccAuthService_commerceIndicator** field in your authorization request. Set it to one of the following values:
- `install`—U.S. transaction or non-U.S. mail order / telephone order (MOTO) transaction
 - `internet`—e-commerce transaction. This is the default value that CyberSource uses when you do not include the commerce indicator field in the request.

- `spa`—Mastercard Identity Check transaction.
- `vbv`—Visa Secure transaction.

Step 3 You must include the **`installment_planType`** field in your authorization request if the corresponding value is not set in your CyberSource account. If this value is set in your CyberSource account, you can include the field in your authorization request to override the value in your CyberSource account.

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on CyberSource Latin American Processing is an installment payment:



Note

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. The information in this section is for the specific processing connection called *CyberSource Latin American Processing*. It is not for any other Latin American processors that CyberSource supports.

Step 1 Set **`ccAuthService_commerceIndicator`** to `install`.

Step 2 For a transaction in Brazil, you can include the following optional fields in your authorization request:

- `installment_planType`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Step 3 For a transaction in Mexico, installment payments are supported, but conditions vary, so contact CyberSource Customer Support or your CyberSource account manager.

To indicate that a transaction on Getnet or Software Express is an installment payment:

Step 1 You must include the following fields in your authorization request:

- `installment_planType`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

To indicate that a transaction on Rede is an installment payment:

Step 1 You must include the `installment_totalCount` field in your authorization request.

For information about this field, see [Appendix A, "API Fields," on page 238](#).

Installment Payments on Other Processors

The customer pays for goods or services using an installment plan agreed upon by the customer and you.

To indicate that a transaction on any other supported processor is an installment payment:

Step 1 Set `ccAuthService_commerceIndicator` to `install`.

Step 2 If your processor supports merchant-initiated transactions, your authorization request must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

Step 3 Include the following required fields in your authorization request:

- `installment_sequence`
- `installment_totalCount`

For information about these fields, see [Appendix A, "API Fields," on page 238](#).

Japanese Payment Options

Services:

- Authorization
- Capture—supported only on JCN Gateway
- Credit—supported only on JCN Gateway

Processors:

- CyberSource through VisaNet—the only supported acquirer is Sumitomo Mitsui Card Co.
- JCN Gateway

Card types:

- Visa
- Mastercard—supported only on JCN Gateway
- American Express—supported only on JCN Gateway
- Diners Club—supported only on JCN Gateway
- JCB—supported only on JCN Gateway

In addition to standard single payments, Japanese acquirers support the payment options listed in [Table 40, "Japanese Payment Options," on page 163](#).

Before using one of these payment options, you must sign a contract with your acquirer. Additionally, the funding cycle could differ when using these options. Contact your account provider for details about contracts and funding cycles.

Some acquirers might not support all of these payment options. Additionally, a card holder must sign a contract with an issuing bank before using one of these payment options. Therefore, not all card holders take advantage of these payment options. Confirm payment option availability with your account provider and the card holder before implementing one of these payment options.



Important

CyberSource accepts requests with these payment options independently of your agreements with acquirers. When you submit a request with one of these payment options but do not have the necessary contracts and agreements in place, an error might not occur until the acquirer processes the settlement file.

The following table lists the API fields required for each payment option. For details about these fields, see [Appendix A, "API Fields," on page 238](#).

Table 40 Japanese Payment Options

Payment Option	Required Fields
1 (default): Single payment	No Japanese payment option fields are required. When you omit jpo_paymentMethod from a request, CyberSource processes the request as a single payment.
2: Bonus payment	jpo_paymentMethod
4: Installment payment	jpo_installments jpo_paymentMethod
5: Revolving payment	jpo_paymentMethod
6: Combination of bonus payment and installment payment	jpo_installments jpo_paymentMethod

Additional Japanese Payment Option Fields

In addition to the fields listed in the preceding table, CyberSource supports the following fields for transactions in Japan:

- jpo_businessNameAlphanumeric
- jpo_businessNameJapanese
- jpo_businessNameKatakana
- jpo_firstBillingMonth—supported only for JPO payment methods 4 and 6

For details about these fields, see [Appendix A, "API Fields," on page 238](#).

CyberSource also supports the following fields for card-present transactions in Japan:

- jpo_jccaTerminalID
- jpo_jis2TrackData

For details about these fields, see [Card-Present Processing Using the Simple Order API](#).

Verbal Authorizations

When you submit a capture request with a verbal authorization, if the initial authorization included Japanese payment option fields, the capture request must include the same Japanese payment option fields.

Stand-Alone Credits

When you perform a stand-alone credit for a transaction that included Japanese payment option fields, the request for the stand-alone credit must include the same Japanese payment option fields.

On CyberSource through VisaNet, stand-alone credits are supported only for JPO payment method 1.

When a request for a stand-alone credit is made with JCN Gateway, most acquirers make inquiries about the purpose of such a request. CyberSource recommends using follow-on credits instead of stand-alone credits whenever possible.

Additional Information

For more information about the Japanese payment options, contact Customer Support of CyberSource KK (Japan).

JCB J/Secure

See ["Payer Authentication," page 183](#).

Least-Cost Routing (LCR)



Note

See also ["Single-Message Processing," page 76](#).

Processors:

- FDI Australia—supported only for AUD
- Worldpay VAP

Least-cost routing offers the lowest routing costs available. When you request LCR, the processor first determines whether LCR is possible for the transaction, and, if it is possible, the processor uses single-message processing to accomplish LCR.



Important

There are important differences between single-message processing and dual-message processing:

- With single-message processing, an authorization+capture request results in a full-financial transaction, which moves funds immediately.
- With single-message processing, the capture amount and authorization amount are the same.
- Several services and features are not supported for single-message processing because they are not applicable or could create reconciliation problems. See ["Unsupported Features," page 167](#).

To request least-cost routing for all authorization+capture transactions:

- Step 1** Contact CyberSource Customer Support to enable LCR for your account.
- Step 2** For each authorization+capture request:
- a** Request the authorization and capture services at the same time:
 - Include the request fields that are required for an authorization. See ["Authorizing a Payment," page 38](#).
 - If the transaction does not include any optional capture features, such as Level II data, no additional fields are required for the capture. See ["Capturing an Authorization," page 61](#).
 - If the transaction includes an optional capture feature, include the fields that are needed for the feature.
 - b** The reply message from CyberSource includes authorization and capture results. The authorization amount and capture amount are the same.
 - c** To find out whether the processor used LCR for the transaction, see the daily processor report.
- Step 3** To turn off LCR for an authorization+capture transaction, set **ccAuthService_leastCostRouting** to `false` in the authorization+capture request.

For a description of the LCR field, see [Appendix A, "API Fields," on page 238](#).

For an example of an LCR request and reply, see:

- Name-value pair examples: ["Worldpay VAP Examples," page 420](#)
- XML examples: ["Worldpay VAP Examples," page 453](#)

To request least-cost routing for one transaction:

- Step 1** Request the authorization and capture services at the same time:
- Include the request fields that are required for an authorization. See ["Authorizing a Payment," page 38](#).
 - If the transaction does not include any optional capture features, such as Level II data, no additional fields are required for the capture. See ["Capturing an Authorization," page 61](#).

- If the transaction includes an optional capture feature, include the fields that are needed for the feature.
- Set **ccAuthService_leastCostRouting** to `true`.

The reply message from CyberSource includes authorization and capture results. The authorization amount and capture amount are the same.

Step 2 To find out whether the processor used LCR for the transaction, see the daily processor report.

For a description of the LCR field, see [Appendix A, "API Fields," on page 238](#).

For an example of an LCR request and reply, see:

- Name-value pair examples: ["Worldpay VAP Examples," page 420](#)
- XML examples: ["Worldpay VAP Examples," page 453](#)

Partial Authorizations

If the transaction is enabled for partial authorization, the processor can process a partial amount instead of the entire requested amount. For details about partial authorizations, see ["Partial Authorizations," page 105](#).

Supported Features

The following features are supported for least-cost routing and single-message processing on Worldpay VAP:

- AVS. See ["Address Verification System \(AVS\)," page 88](#).
- Balance responses. See ["Balance Responses," page 111](#).
- Card type indicators. See ["Card Type Indicators \(CTIs\)," page 130](#).
- Credits. See ["Crediting a Payment," page 78](#).
- CVN. See ["Card Verification Numbers \(CVNs\)," page 97](#).
- Debit cards. See [Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105](#).
- Merchant descriptors. See [Merchant Descriptors Using the Simple Order API](#).
- Payer authentication. See ["Payer Authentication," page 183](#).
- Prepaid cards. See [Chapter 4, "Features for Debit Cards and Prepaid Cards," on page 105](#).

Unsupported Features

The following features are not supported for least-cost routing and single-message processing:

- Authorization reversal after void
- Full authorization reversals
- Multiple partial captures
- Stand-alone authorizations
- Stand-alone captures
- Verbal authorizations
- Voids

Level II Data

See [Level II and Level III Processing Using the Simple Order API](#).

Level III Data

See [Level II and Level III Processing Using the Simple Order API](#).

Mastercard Bill Payments

Services:

- Authorization

Processor:

- CyberSource through VisaNet



Note

This feature is supported only in Brazil.

Mastercard provides a Bill Payment program that enables customers to use their Mastercard cards to pay their bills. When you participate in this program, Mastercard requests that you flag the bill payments so they can be easily identified. To flag these transactions, include the **billPaymentType** field in your transaction requests. Do not use this indicator if you have not signed up with Mastercard to participate in the program.

Mastercard Expert Monitoring Solutions (EMS)

Service:

- Authorization

Processor:

- CyberSource through VisaNet

Mastercard Expert Monitoring Solutions (EMS) provides a predictive, behavior-based fraud score in real time during authorizations for card-not-present (CNP) transactions on cards issued in the U.S. EMS compares a customer's transaction data to their transaction behavior history and a regional CNP fraud detection model. The resulting score indicates the likelihood that the transaction is fraudulent.

To use EMS, call CyberSource Customer Support to have your account enabled for this feature. After your account is enabled, Mastercard performs EMS on all your CNP authorization requests for U.S.-issued Mastercard cards. For these requests, CyberSource returns the fraud score in the **ccAuthReply_emsTransactionRiskScore** field. For information about this field, see [Appendix A, "API Fields," on page 238](#).

Mastercard Identity Check

See ["Payer Authentication," page 183](#).

Mastercard Send

Mastercard Send is one of the networks supported by CyberSource Payouts.

See [Payouts Using the Simple Order API](#).

Masterpass

Services:

- Authorization
- Credit—Chase Paymentech Solutions and CyberSource through VisaNet only

Processors:

- Chase Paymentech Solutions
- CyberSource through VisaNet
- Elavon Americas
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
- Getnet
- Rede
- SIX

To indicate that a request is for a Masterpass transaction:

Before requesting Masterpass transactions, contact CyberSource Customer Support to have your account configured for this feature.

On Chase Paymentech Solutions or CyberSource through VisaNet, include the **wallet_type** field in your authorization or credit request.

On OmniPay Direct, include the following fields in your authorization request:

- wallet_type
- paymentSolution

For details about these fields, see [Appendix A, "API Fields," on page 238](#).

Merchant Descriptors

See [Merchant Descriptors Using the Simple Order API](#).

Merchant-Initiated Reversals and Voids

Services:

- Authorization
- Capture
- Credit

Processors:

- Chase Paymentech Solutions
- CyberSource through VisaNet
- Elavon Americas
- FDC Nashville Global
- OmniPay Direct—merchant-initiated voids are not supported.
- Rede
- SIX

When you do not receive a reply message after sending a request to CyberSource, this feature enables you to reverse or void the transaction that you requested.

To use merchant-initiated reversals and voids on CyberSource through VisaNet:

- Step 1** Include the **merchantTransactionIdentifier** field in your original request for an authorization, capture, sale, follow-on credit, or stand-alone credit.



Note

The value of the merchant transaction ID must be unique for 60 days.

- Step 2** When you do not receive a reply message for your original transaction request, reverse or void the original transaction as described in the following table.

Transaction to Reverse or Void	Procedure
Authorization	Request the full authorization reversal service as described in " Creating a Full Authorization Reversal Request ," page 57 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your reversal request to your original request.
Capture or sale	Request the void service as described in " Creating a Void Request ," page 86 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your void request to your original request. CyberSource automatically handles authorization reversals on capture and sale requests.

Transaction to Reverse or Void	Procedure
Credit	Request the void service as described in "Creating a Void Request," page 86 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your void request to your original request.

Step 3 If the original transaction failed, the reply message for the reversal or void request includes the following fields:

- originalTransaction_amount
- originalTransaction_reasonCode

To use merchant-initiated reversals on OmniPay Direct:

Step 1 Include the **merchantTransactionIdentifier** field in your original request for an authorization, capture, or sale.



Note

The value of the merchant transaction ID must be unique for 60 days.

Step 2 When you do not receive a reply message for your original transaction request, reverse the original transaction as described in the following table.

Transaction to Reverse or Void	Procedure
Authorization	Request the full authorization reversal service as described in "Creating a Full Authorization Reversal Request," page 57 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your reversal request to your original request.
Capture or sale	1 Request the void service as described in "Creating a Void Request," page 86. Include the request ID in your request message. 2 Request the authorization reversal service as described in "Creating a Full Authorization Reversal Request," page 57. Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your reversal request to your original request.
Credit	Request the void service as described in "Creating a Void Request," page 86 . Include the request ID in your request message.

- Step 3** If the original transaction failed, the reply message for the reversal request includes the following fields:
- originalTransaction_amount
 - originalTransaction_reasonCode

To use merchant-initiated reversals and voids on all other processors:

- Step 1** Include the **merchantTransactionIdentifier** field in your original request for an authorization, capture, sale, follow-on credit, or stand-alone credit.



Note

The value of the merchant transaction ID must be unique for 60 days.

- Step 2** When you do not receive a reply message for your original transaction request, reverse or void the original transaction as described in the following table.

Transaction to Reverse or Void	Procedure
Authorization	Request the full authorization reversal service as described in " Creating a Full Authorization Reversal Request ," page 57 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your reversal request to your original request.
Capture or sale	1 Request the void service as described in "Creating a Void Request," page 86. Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your void request to your original request. 2 Request the authorization reversal service as described in "Creating a Full Authorization Reversal Request," page 57. Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your reversal request to your original request.
Credit	Request the void service as described in " Creating a Void Request ," page 86 . Instead of including the request ID in your request message, include the merchantTransactionIdentifier field. The merchant transaction ID links your void request to your original request.

Step 3 If the original transaction failed, the reply message for the reversal or void request includes the following fields:

- originalTransaction_amount
- originalTransaction_reasonCode

Merchant-Initiated Transactions



Note

This section describes how to perform a merchant-initiated transaction (MIT) yourself. For information about letting Token Management Service (TMS) perform MITs for you, see:

- <https://www.cybersource.com/content/dam/cybersource/MIT-mandate-FAQ.pdf>
 - https://www.cybersource.com/content/dam/cybersource/Token_Management_Service_Stored_Credential_Mandate_FAQ.pdf
-

Service:

- Authorization

Processors:

- AIBMS
- Barclays
- Chase Paymentech Solutions—supports only reauthorizations and unscheduled card-on-file transactions
- CyberSource through VisaNet
- Elavon Americas
- FDC Compass
- FDC Nashville Global
- GPN
- HBoS
- HSBC
- Lloyds-OmniPay
- LloydsTSB Cardnet

- OmniPay Direct—supported for all OmniPay Direct acquirers. See [Table 2, page 21](#), for the list of OmniPay Direct acquirers.
- Rede
- SIX
- Streamline

Card types:

- Mastercard—only on AIBMS and OmniPay Direct
- Visa

Most authorizations are initiated by a cardholder in person, on the phone, or on a web site. A *merchant-initiated transaction* (MIT) is an authorization that you initiate when the cardholder is not present.

Terminology

Table 41 Terminology for Merchant-Initiated Transactions

Term	Description
Cardholder-initiated transaction (CIT)	Transaction that uses payment information provided by the cardholder. A CIT can be any of the following kinds of transactions: ■ Card present: cardholder goes to a brick-and-mortar store in person to make a purchase and provides payment information in the store. ■ Credentials-on-file (COF): cardholder orders an item online and instructs you to use the payment information that is saved in your system. ■ Digital payment: cardholder uses an app to pay for goods or services. ■ E-commerce: cardholder orders an item online and provides payment information during checkout. ■ Mail order or telephone order (MOTO): cardholder orders an item over the telephone and provides payment information to the person who is taking the order.
Credentials on file (COF)	Payment information that you saved in your order management system.
Credentials-on-file transaction	Transaction that uses COF payment information. A COF transaction can be a CIT or an MIT.

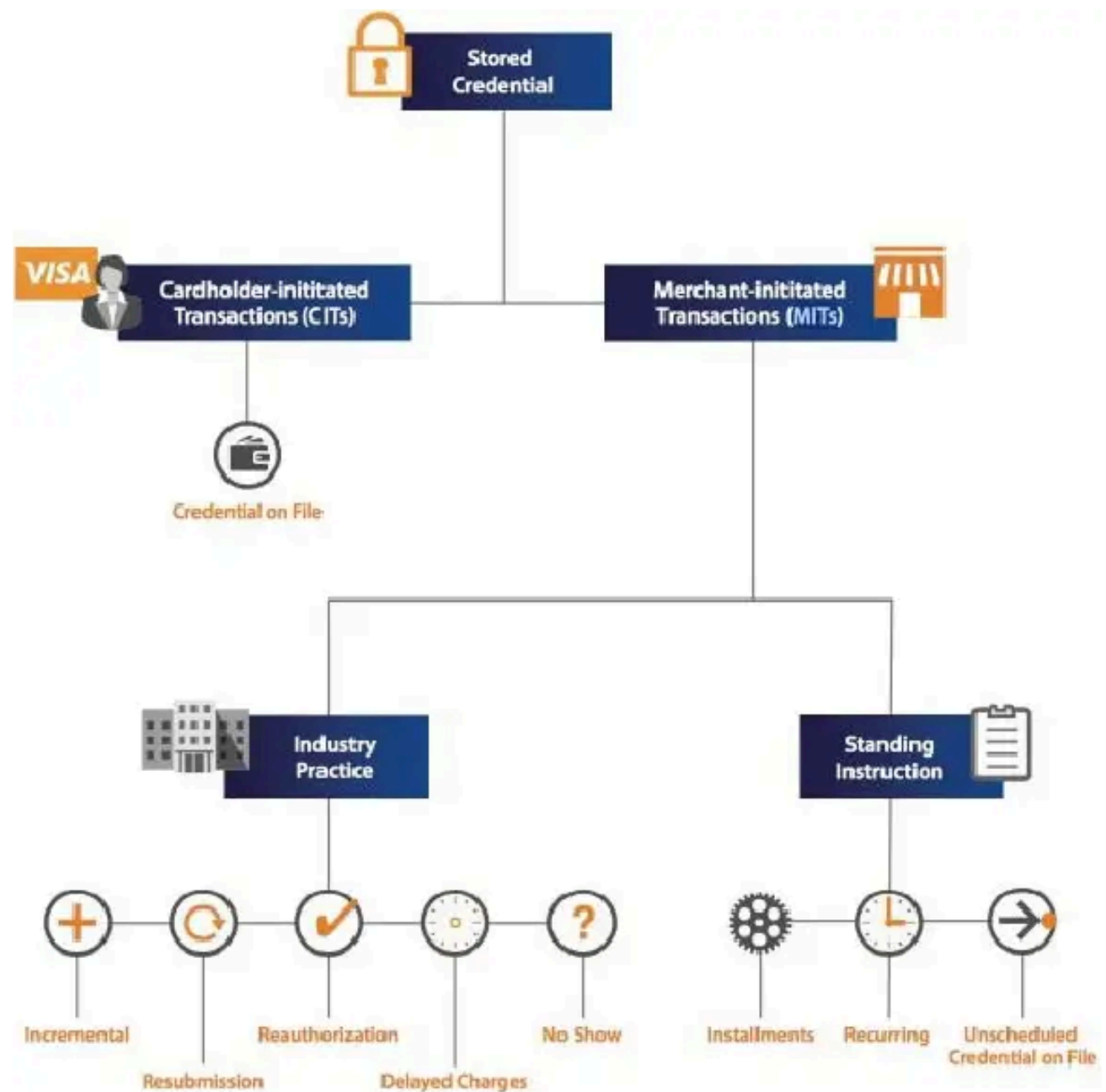
Table 41 Terminology for Merchant-Initiated Transactions (Continued)

Term	Description
Merchant-initiated transaction (MIT)	Transaction that uses payment information from a previous CIT but is conducted without the active participation of the cardholder. There are two main types of MITs: ■ An <i>industry practice</i> transaction: a one-time MIT that derives payment information from a CIT. ■ A <i>standing</i> instruction: one transaction in a series of repeated transactions or a one-time, unscheduled transaction that uses COF payment information.

Overview

Figure 4 illustrates the relationships between stored credentials, CITs, and MITs.

Figure 4 **Stored Credentials and Merchant-Initiated Transactions**



Descriptions

- Account top-up—is the result of instructions between you and the cardholder to charge a specific or variable amount at specified or variable intervals. An account top-up is an unscheduled COF transaction.
- Delayed charge—is associated with an agreement between you and the cardholder for services rendered. Delayed charges are typical for lodging transactions and auto rental transactions.



Note

The CyberSource delayed charge feature is not available on FDC Compass.

- Final authorization—occurs when you need to authorize the final amount after an estimated authorization.
- Incremental authorization—is a continuation of a purchase when the originally approved amount is modified to accommodate additional services. Incremental authorizations are typical for lodging transactions and auto rental transactions.



Note

The CyberSource incremental authorization feature is not available on these processors:

- AIBMS
- FDC Compass
- FDC Nashville Global
- OmniPay Direct

- Installment payment—is the result of instructions governed by a contract between you and a cardholder. The instructions enable you to charge a specific amount at specified intervals. To find out whether your processor is supported for installment payments, see ["Installment Payments," page 145](#).
- No-show transaction—occurs when you and a cardholder have an agreement for a purchase, but the cardholder does not meet the terms of the agreement. No-show transactions are typically used in hotels and motels for a single-night stay.



Note

The CyberSource no-show transaction feature is not available on FDC Compass.

- Reauthorization for split shipment—a split shipment occurs when multiple goods purchased in a single transaction are shipped at separate times. When the goods become available to ship, either you or CyberSource perform a new authorization for the portion of the transaction being delivered. This ensures that the cardholder's funds are still available. A reauthorization is performed in one of the following scenarios:
 - Before requesting a capture, you request an authorization using the saved cardholder credentials.
 - You use the CyberSource split-shipment feature. To find out whether your processor is supported for split shipments, see ["Split Shipments," page 218](#).
- Recurring payment—is the result of instructions governed by a contract between you and a cardholder. The instructions enable you to charge a specific or variable amount at specified intervals. To find out whether your processor is supported for recurring payments, see ["Recurring Payments," page 205](#).
- Resubmission—occurs when a cardholder-initiated purchase occurred, but you could not obtain an authorization at that time. A resubmission is valid only when the original authorization was declined for insufficient funds and only for a limited number of days after the original purchase.

Scenarios

Delayed Charge

A delayed charge is associated with an agreement between you and the cardholder for services rendered. Delayed charges are typical for lodging transactions and auto rental transactions.

To create a delayed charge authorization request:

- Step 1** Include the following required fields in the authorization request:
- subsequentAuth—set the value for this field to `true`.
 - subsequentAuthReason—set the value for this field to `2`.
 - subsequentAuthTransactionID—set the value for this field to the network transaction identifier.
- Step 2** If the payment information is COF information, include the following field in the authorization request:
- subsequentAuthStoredCredential—set the value for this field to `true`.
-

Installment Payment

An installment payment is a COF transaction. A series of installment payments consists of multiple transactions that you bill to a cardholder over a period of time agreed to by you and the cardholder for a single purchase of goods or services. The agreement enables you to charge a specific amount at specified intervals.

To create an installment payment authorization request:

- Step 1** Cardholder consents to terms and establishes service or obtains goods.
- Step 2** You charge the first installment payment as a CIT. Include the following field in the authorization request:
- `subsequentAuthFirst`—set the value for this field to `true`.
- Step 3** You charge subsequent installment payments on a regular basis. Include the following fields in each authorization request:
- `ccAuthService_commerceIndicator`—set the value for this field to `install`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
-

No-Show Transaction

A no-show transaction occurs when you and a cardholder have an agreement for a purchase, but the cardholder does not meet the terms of the agreement. No-show transactions are typically used in hotels and motels for a single-night stay.

To create a no-show transaction authorization request:

- Step 1** Include the following required fields in the authorization request:
- `subsequentAuth`—set the value for this field to `true`.
 - `subsequentAuthReason`—set the value for this field to `4`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
- Step 2** If the payment information is COF information, include the following field in the authorization request:
- `subsequentAuthStoredCredential`—set the value for this field to `true`.
-

Reauthorization

A reauthorization is a purchase made after an original purchase that can reflect a number of specific conditions. Common instances that require reauthorizations include delayed shipments, split shipments, extended stays, and extended rentals.

To create a reauthorization request:

- Step 1** Include the following required fields in the authorization request:
- `subsequentAuth`—set the value for this field to `true`.
 - `subsequentAuthReason`—set the value for this field to `3`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
- Step 2** If the payment information is COF information, include the following field in the authorization request:
- `subsequentAuthStoredCredential`—set the value for this field to `true`.
-

Recurring Payment

A recurring payment is a COF transaction. A series of recurring payments consists of multiple transactions that you bill to a cardholder at fixed, regular intervals not to exceed one year between transactions. The series of recurring payments is the result of an agreement between you and the cardholder.

To create a recurring payment authorization request:

- Step 1** Cardholder consents to terms and establishes service or obtains goods.
- Step 2** You charge the first recurring payment as a CIT. Include the following field in the authorization request:
- `subsequentAuthFirst`—set the value for this field to `true`.
- Step 3** You charge subsequent recurring payments on a regular basis. Include the following fields in each authorization request:
- `ccAuthService_commerceIndicator`—set the value for this field to `recurring`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
-

Resubmission

A resubmission occurs when you cannot obtain an authorization for a cardholder-initiated purchase. A resubmission is valid only when the original authorization was declined for insufficient funds and only for a limited number of days after the original purchase.

To create a resubmission authorization request:

- Step 1** Include the following required fields in the authorization request:
- `subsequentAuth`—set the value for this field to `true`.
 - `subsequentAuthReason`—set the value for this field to `1`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
- Step 2** If the payment information is COF information, include the following field in the authorization request:
- `subsequentAuthStoredCredential`—set the value for this field to `true`.
-

Unscheduled COF Transaction

An unscheduled COF transaction uses stored payment information for a fixed or variable amount that does not occur on a scheduled or regular basis.

To create an unscheduled COF transaction authorization request:

- Step 1** Cardholder consents to terms and establishes service or obtains goods.
- Step 2** You charge the first payment. Include the following field in the authorization request:
- `subsequentAuthFirst`—set the value for this field to `true`.
- Step 3** You charge subsequent payments. Include the following fields in each authorization request:
- `subsequentAuth`—set the value for this field to `true`.
 - `subsequentAuthTransactionID`—set the value for this field to the network transaction identifier.
-

API Field Descriptions

For descriptions of the fields in the preceding scenarios, see [Appendix A, "API Fields,"](#) on page 238.

Micropayments

Services:

- Authorization
- Capture
- Credit

Processors:

- Most of the card types and processors that are supported by CyberSource

Micropayments are payments for less than one unit in the transaction's currency.

Multi-Currency Service

Services:

- Authorization
- Capture
- Credit

Processor:

- Chase Paymentech Solutions

If you sell your products in multiple countries, you might want to list your product prices in your customers' local currencies. The CyberSource multi-currency service provides current, guaranteed exchange rates, which enables customers to pay using their local currencies while enabling you to do business and settle transactions in your desired currency.

For more information about the CyberSource multi-currency service, see the [Multicurrency Service for Chase Paymentech Solutions Using the Simple Order API](#).

Network Tokenization

See ["Payment Network Tokenization,"](#) page 202.

Partial Shipments

See "Split Shipments," page 218.

Payer Authentication



Important

Before you implement payer authentication, you must contact CyberSource Customer Support to have your account configured for this feature.

When you request an authorization using a supported card type and a supported processor, you can include payer authentication data in the request. You can use the CyberSource payer authentication services to add Visa Secure, JCB J/Secure™, ProtectBuy, Mastercard® Identity Check™, RuPay PaySecure, or American Express SafeKey support to your web site without running additional software on your own server. The following table lists the cards supported for each type of payer authentication. For descriptions of the CyberSource payer authentication services, see [Payer Authentication Using the Simple Order API](#).

Table 42 Supported Card Types for Payer Authentication

Type of Payer Authentication	Card Types
American Express SafeKey	American Express
JCB J/Secure	JCB
Mastercard Identity Check	■ Mastercard ■ Co-badged Mastercard and mada as described in "mada Debit Cards," page 24. You must identify the card type as Mastercard. Supported only on CyberSource through VisaNet. ■ Maestro (International) ■ Maestro (UK Domestic)
ProtectBuy	■ Diners Club ■ Discover
RuPay PaySecure	RuPay
Visa Secure	■ Visa ■ Co-badged Visa and mada as described in "mada Debit Cards," page 24. You must identify the card type as Visa. Supported only on CyberSource through VisaNet.

American Express SafeKey

Service:

- Authorization

Processors:

- American Express Direct: mandatory for transactions that originate in Singapore.
- CyberSource through VisaNet
- Elavon Americas
- FDC Nashville Global
- JCN Gateway

American Express SafeKey (AESK) authenticates the Customer during an online purchase and protects payment information as it is transmitted over the Internet.

To request the authorization of an AESK transaction:

Step 1 Add the fields in the following table to your **ccAuthService** request. The values for these fields are in the reply from the validate authentication service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other.

The authorization service returns a raw response code and a mapped response code:

- The *raw response code* is the value returned by the processor. CyberSource returns this value in the **ccAuthReply_cavvResponseCodeRaw** field.
 - The *mapped response code* is the predefined CyberSource value that corresponds to the raw response code. CyberSource returns this value in the **ccAuthReply_cavvResponseCode** field. [Appendix D, "American Express SafeKey Response Codes," on page 460](#) describes the mapped response codes.
-

Table 43 Request Fields for American Express SafeKey

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>CAVV</i> —cardholder authentication verification value. This value is a transaction identifier generated by the issuing bank during American Express SafeKey payer authentication. This value is required.	ccAuthService_cavv	payerAuthValidateReply_cavv
<i>ECI</i> —electronic commerce indicator. This value is required. Possible values: ■ <i>aesk</i>: Successful AESK transaction. ■ <i>aesk_attempted</i>: AESK transaction was attempted but not authenticated.	ccAuthService_commerceIndicator	payerAuthValidateReply_commerceIndicator
<i>XID</i> —transaction identifier. This value is optional.	ccAuthService_xid	payerAuthValidateReply_xid

JCB J/Secure

Service:

- Authorization

Processors:

- CyberSource through VisaNet
- Elavon Americas
- Ingenico ePayments
- JCN Gateway
- SIX
- TSYS Acquiring Solutions

JCB J/Secure authenticates the customer by adding a password identification step to the online shopping process. For details about signing up for and using J/Secure, contact your acquiring bank or go to the JCB web site:

<http://www.jcb-global.com/>

To request the authorization of a JCB J/Secure transaction:

- Step 1** Add the fields listed in [Table 46, "Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy," on page 197](#) to your **ccAuthService** request. The values for these fields are in the reply from the validate authentication service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other.
-

Mastercard Identity Check

Service:

- Authorization

Processors That Support Mastercard Identity Check 1.0:

- AIBMS
- Asia, Middle East, and Africa Gateway
- Atos
- Barclays
- Chase Paymentech Solutions
- Cielo
- Comercio Latino
- Credit Mutuel-CIC
- CyberSource Latin American Processing: Mastercard Identity Check is an emerging feature in the Latin American region. It is not fully supported in all countries. Contact CyberSource Customer Support for details.



Note

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America.

- CyberSource through VisaNet
- Elavon
- Elavon Americas
- FDC Compass
- FDC Germany
- FDI Australia
- FDC Nashville Global
- FDMS Nashville
- FDMS South

- Getnet
- GPN
- HBoS
- HSBC: HSBC is the CyberSource name for HSBC U.K.
- Ingenico ePayments
- JCN Gateway
- LloydsTSB Cardnet
- Moneris
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- OmniPay-Ireland: OmniPay-Ireland is the CyberSource name for HSBC International.

**Note**

On OmniPay-Ireland, Mastercard Identity Check attempts are not considered Identity Check transactions and are downgraded to non-Identity Check transactions. CyberSource recommends that you migrate to the OmniPay Direct processor to use the latest version of the Identity Check feature.

- RBS WorldPay Atlanta
- Rede
- SIX
- Streamline
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.

Processors That Support Mastercard Identity Check 2.0:

- CyberSource through VisaNet
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring

Mastercard Identity Check adds security to online transactions by authenticating Identity Check account holders for specific transactions. Identity Check generates a unique, 32-character transaction token, called the account authentication value (AAV), each time an Identity Check-enabled account holder makes an online purchase. The AAV binds the

account holder to a specific transaction. Identity Check transactions use the universal cardholder authentication field (UCAF) as a standard to collect and pass AAV data. For details about signing up for and using Identity Check or UCAF, contact your acquiring bank or go to the Mastercard web site:

<http://www.mastercard.com/>

To request the authorization of a Mastercard Identity Check transaction:

- Step 1** Add the fields in [Table 44, "Request Fields for Mastercard Identity Check,"](#) to your **ccAuthService** request. The values for these fields are in the reply from the validate authentication service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other.

Table 44 Request Fields for Mastercard Identity Check

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>3D Secure Version</i>—version of 3D Secure that was used to process the Mastercard Identity Check transaction. ■ Used only for these processors: • CyberSource through VisaNet • OmniPay Direct ■ Always required. ■ Examples: 1.0.2 or 2.0.0.	ccAuthService_ paSpecificationVersion	payerAuthEnrollReply_ specificationVersion or payerAuthValidateReply_ specificationVersion

Table 44 Request Fields for Mastercard Identity Check (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>CAVV Algorithm</i>—algorithm for generating the UCAF authentication data. - Used only for these processors: - Atos - Ingenico ePayments when a third-party provider authenticates the transaction - Required when you include the UCAF authentication data in your request. - You must not include the CAVV algorithm value in your request when the UCAF authentication data is not included in your request or when your processor is not Atos or Ingenico ePayments. - Possible values: 0: HMAC (hash-based message authentication code) 1: CVV 2: CVV with ATN 3: Mastercard SPA (secure payment algorithm) Note Ingenico ePayments was previously called <i>Global Collect</i>.	ccAuthService_cavvAlgorithm	payerAuthValidateReply_cavvAlgorithm
<i>Directory Server Transaction ID</i>—Transaction ID generated by the Mastercard directory server during authentication. - Used only for these processors: - CyberSource through VisaNet - OmniPay Direct - Required for Mastercard Identity Check 2.0 transactions. - Example: f38e6948-5388-41a6-bca4-b49723c19437.	ccAuthService_directoryServerTransactionID	payerAuthEnrollReply_directoryServerTransactionID or payerAuthValidateReply_directoryServerTransactionID

Table 44 Request Fields for Mastercard Identity Check (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>ECI</i>—electronic commerce indicator. ■ Used for all processors that support Mastercard Identity Check. ■ Always required. ■ Possible values for a Mastercard Identity Check transaction: • <i>spa</i>: Mastercard Identity Check transaction. • <i>spa_failure</i>: Mastercard Identity Check authentication failed. Available only for Elavon, HSBC, and Streamline. Note The ECI for all Mastercard Identity Check transactions, including authentication attempts, must be set to <i>spa</i>. Otherwise, the transactions will be processed as non-Identity Check transactions.	ccAuthService_commerceIndicator	payerAuthValidateReply_commerceIndicator
<i>ECI Raw</i>—raw electronic commerce indicator. ■ Used for all processors that support Mastercard Identity Check. ■ Required when the payer authentication validation service returns a raw ECI value. ■ Some processors require the raw ECI to guarantee chargeback protection. Contact CyberSource Customer Support for information about your processor's requirements.	ccAuthService_eciRaw	payerAuthValidateReply_eciRaw

Table 44 Request Fields for Mastercard Identity Check (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>PARes Status</i>—payer authentication response status. ■ Used only for these processors: • Asia, Middle East, and Africa Gateway • Atos • Ingenico ePayments when a third-party provider authenticates the transaction ■ For Atos and Ingenico ePayments: required for a successful Mastercard Identity Check transaction, which is indicated when the UCAF collection indicator is 2. ■ For the Asia, Middle East, and Africa Gateway: required unless all of the following are true: • You are requesting the payer authentication and the authorization in separate requests. • This is a successful Mastercard Identity Check transaction, which is indicated when the commerce indicator is <i>spa</i>. • The card is not enrolled, which is indicated when the VERes enrolled status is not <i>Y</i>. When all the preceding conditions are true, do not include the PARes status in the authorization request. If you do, CyberSource sends the value to the processor without modification. CyberSource does not decline the transaction; declines are generated by the processor. ■ Possible values: • <i>Y</i>: Customer was successfully authenticated. • <i>A</i>: Proof of authentication attempt was generated. • <i>N</i>: Customer failed or cancelled authentication. Transaction denied. • <i>U</i>: Authentication not completed regardless of the reason. Note Ingenico ePayments was previously called <i>Global Collect</i>.	ccAuthService_ paresStatus	payerAuthValidateReply_ paresStatus

Table 44 Request Fields for Mastercard Identity Check (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>UCAFAuthenticationData</i>—authentication data for the universal cardholder authentication field. ■ Used for all processors that support Mastercard Identity Check. ■ Required when the UCAF collection indicator is 1, 2, or 5. Do not include UCAF authentication data in the authorization request if the UCAF collection indicator is not 1, 2, or 5. Important Mastercard has indicated that an issuing bank can downgrade an authorization request to a non-secure transaction when the UCAF collection indicator is 1 and UCAF authentication data is not present. An issuing bank can choose not to settle a downgraded Mastercard Identity Check transaction. When UCAF authentication data is not present, set the UCAF collection indicator to 0.	ucaf_authenticationData	payerAuthValidateReply_ucafAuthenticationData
<i>UCAFCollectionIndicator</i>—collection indicator for the universal cardholder authentication field. ■ Used for all processors that support Mastercard Identity Check. ■ Always required. ■ Possible values: ● 0: UCAF collection is not supported at your web site. ● 1: UCAF collection is supported at your web site, and the UCAF was populated. ● 2: UCAF collection is supported at your web site and the UCAF was populated. This value indicates a successful Mastercard Identity Check transaction. ● 5: UCAF collection is supported at your web site, and the UCAF was populated based on the risk assessment that the issuer performed. This value is supported only for Masterpass transactions. ● 6: UCAF collection is supported at your web site, and the UCAF was populated based on the risk assessment that you performed. This value is supported only for Masterpass transactions.	ucaf_collectionIndicator	payerAuthValidateReply_ucafCollectionIndicator

Table 44 Request Fields for Mastercard Identity Check (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>VERes Enrolled</i>—verification response enrollment status. ■ Used only for the Asia, Middle East, and Africa Gateway. ■ Required for all payer authentication transactions. ■ Possible values: • Y: Authentication available. • N: Customer not participating. • U: Unable to authenticate regardless of the reason.	ccAuthService_ veresEnrolled	payerAuthEnrollReply_ veresEnrolled
<i>XID</i>—transaction identifier. Must be 28-character base64 or 40-character hex binary. ■ Used for all processors that support Mastercard Identity Check. ■ For Atos: required for a successful Mastercard Identity Check transaction, which is indicated when the UCAF collection indicator is 2. ■ For all other processors: required when the payer authentication validation service returns an XID value.	ccAuthService_xid	payerAuthValidateReply_xid

ProtectBuy

Service:

- Authorization

Processor:

- SIX

ProtectBuy authenticates the customer by adding a password identification step to the online shopping process. For details about signing up for and using ProtectBuy, contact your acquiring bank or go to the ProtectBuy web site:

<http://protectbuy.nnja.co/>

To request the authorization of a ProtectBuy transaction:

- Step 1** Add the fields listed in [Table 46, "Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy," on page 197](#) to your **ccAuthService** request. The values for these fields are in the reply from the validate authentication service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other.

RuPay PaySecure

Service:

- Authorization

Processor:

- RuPay

RuPay PaySecure authenticates the customer during an online purchase and protects payment information as it is transmitted over the Internet. See the [Payer Authentication RuPay Integration Guide](#).

To request the authorization of a RuPay PaySecure transaction:

- Step 1** Add the fields in the following table to your **ccAuthService** request. The values for these fields are in the reply from the payer authentication validate service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other.

Table 45 Request Fields for RuPay PaySecure

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>CAVV</i> —cardholder authentication verification value. This value is a transaction identifier generated by the issuing bank during RuPay PaySecure payer authentication. This value is required.	ccAuthService_cavv	payerAuthValidateReply_cavv
<i>ECI</i> —electronic commerce indicator. This value is required. Possible value: ■ rpy: Successful RuPay PaySecure transaction.	ccAuthService_commerceIndicator	payerAuthValidateReply_commerceIndicator
<i>XID</i> —transaction identifier. This value is required.	ccAuthService_xid	payerAuthValidateReply_xid

Visa Secure

Service:

- Authorization

Processors:

- AIBMS
- Asia, Middle East, and Africa Gateway
- Atos
- Barclays
- Chase Paymentech Solutions
- Cielo
- Comercio Latino
- Credit Mutuel-CIC
- CyberSource Latin American Processing: Visa Secure is an emerging feature in the Latin American region. It is not fully supported in all countries. Contact CyberSource Customer Support for details.



Note

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America.

- CyberSource through VisaNet
- Elavon
- Elavon Americas
- FDC Compass
- FDC Germany
- FDI Australia
- FDC Nashville Global
- FDMS Nashville
- FDMS South
- Getnet
- GPN
- HBoS
- HSBC: HSBC is the CyberSource name for HSBC U.K.
- Ingenico ePayments
- JCN Gateway

- LloydsTSB Cardnet
- Moneris
- OmniPay Direct. The supported acquirers are:
 - Bank of America Merchant Services
 - Cardnet International
 - First Data Merchant Solutions (Europe)
 - Global Payments International Acquiring
- OmniPay-Ireland: OmniPay-Ireland is the CyberSource name for HSBC International.
- RBS WorldPay Atlanta
- Rede
- SIX
- Streamline
- TSYS Acquiring Solutions
- Worldpay VAP—Worldpay VAP was previously called *Little*.

Visa Secure reduces the risk of unauthorized use of a cardholder account. Visa Secure enables you to verify a customer's identity through the use of a password and provides results to you in real time during the checkout process. For details about signing up for and using Visa Secure, contact your acquiring bank or go to the Visa web site:

<http://visa.com/>



For Visa Secure transactions, use card type 001. Do not use card type 033. For information about card type values, see [Appendix G, "Card Types,"](#) on [page 466](#).



For Visa Checkout transactions, do not map the Visa Secure data from the decrypt Visa Checkout data service reply message to the payer authentication fields in the authorization request. CyberSource maps the data for you. The transaction information that CyberSource sends to the processor includes the Visa Secure data.

To request the authorization of a Visa Secure transaction:

- Step 1** Add the fields listed in the following table to your **ccAuthService** request. The values for these fields are in the reply from the validate authentication service **payerAuthValidateService**. When you request **payerAuthValidateService** and **ccAuthService** together, the data is automatically passed from one service to the other. The authorization service returns a raw response code and a mapped response code:
- The *raw response code* is the value returned by the processor. CyberSource returns this value in the **ccAuthReply_cavvResponseCodeRaw** field.
 - The *mapped response code* is the predefined CyberSource value that corresponds to the raw response code. CyberSource returns this value in the **ccAuthReply_cavvResponseCode** field. [Appendix T, "Visa Secure Response Codes,"](#) on page 498 describes the mapped response codes.

Table 46 Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
CAVV—cardholder authentication verification value. This value is a transaction identifier generated by the issuing bank during Visa Secure, JCB J/Secure, or ProtectBuy payer authentication. Must be 28-character base64 or 40-character hex binary. ■ Used for all processors that support Visa Secure, JCB J/Secure, or ProtectBuy. ■ Required when the commerce indicator is <code>js</code>, <code>pb</code>, <code>pb_attempted</code>, <code>vbv</code>, or <code>vbv_attempted</code>. ■ Optional when the commerce indicator is <code>js_attempted</code>. ■ For Visa Secure on FDC Nashville Global, CyberSource sets this field to the value for the transaction identifier (XID) if the XID is present in the authorization request and the CAVV is not present.	<code>ccAuthService_cavv</code>	<code>payerAuthValidateReply_cavv</code>

Table 46 Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>CAVV Algorithm</i>—algorithm for generating the CAVV. ■ Used only for these processors: • Atos • Ingenico ePayments when a third-party provider authenticates the transaction ■ Required when you include the CAVV in your request. ■ You must not include the CAVV algorithm value in your request when the CAVV is not included in your request or when your processor is not Atos or Ingenico ePayments. ■ Possible values: 0: HMAC (hash-based message authentication code) 1: CVV 2: CVV with ATN Note Ingenico ePayments was previously called <i>Global Collect</i>.	ccAuthService_cavvAlgorithm	payerAuthValidateReply_cavvAlgorithm
<i>ECI</i>—electronic commerce indicator. ■ Used for all processors that support Visa Secure, JCB J/Secure, or ProtectBuy. ■ Always required. ■ Possible values for a Visa Secure, JCB J/Secure, or ProtectBuy transaction: • js: Successful JCB J/Secure transaction. • js_attempted: JCB J/Secure transaction was attempted but not authenticated. • pb: Successful ProtectBuy transaction. • pb_attempted: ProtectBuy transaction was attempted but not authenticated. • vbv: Successful Visa Secure transaction. • vbv_attempted: Visa Secure transaction was attempted but not authenticated. • vbv_failure: Visa Secure authentication failed. Available only for HSBC and Streamline.	ccAuthService_commerceIndicator	payerAuthValidateReply_commerceIndicator

Table 46 Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>ECI Raw</i>—raw electronic commerce indicator. ■ Used for all processors that support Visa Secure, JCB J/Secure, or ProtectBuy. ■ Required when the payer authentication validation service returns a raw ECI value. ■ Some processors require the raw ECI to guarantee chargeback protection. Contact CyberSource Customer Support for information about your processor's requirements.	ccAuthService_eciRaw	payerAuthValidateReply_eciRaw

Table 46 Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>PARes Status</i>—payer authentication response status. ■ Used only for these processors: • Asia, Middle East, and Africa Gateway • Atos • Ingenico ePayments when a third-party provider authenticates the transaction ■ For Atos and Ingenico ePayments: required for a successful Visa Secure transaction, which is indicated when the commerce indicator is <i>vbv</i>. ■ For the Asia, Middle East, and Africa Gateway: required unless all of the following are true: • You are requesting the payer authentication and the authorization in separate requests. • This is a successful or attempted Visa Secure transaction, which is indicated when the commerce indicator is <i>vbv</i> or <i>vbv_attempted</i>. • The card is not enrolled, which is indicated when the <i>VERes enrolled status</i> is not <i>Y</i>. When all the preceding conditions are true, do not include the <i>PARes status</i> in the authorization request. If you do, CyberSource sends the value to the processor without modification. CyberSource does not decline the transaction; declines are generated by the processor. ■ Possible values: • <i>Y</i>: Customer was successfully authenticated. • <i>A</i>: Proof of authentication attempt was generated. • <i>N</i>: Customer failed or cancelled authentication. Transaction denied. • <i>U</i>: Authentication not completed regardless of the reason. Note Ingenico ePayments was previously called <i>Global Collect</i>.	ccAuthService_ paresStatus	payerAuthValidateReply_ paresStatus

Table 46 Request Fields for Visa Secure, JCB J/Secure, and ProtectBuy (Continued)

Value and Requirements	Request Field for the Authorization Service	Get the Value from this Payer Authentication Reply Field
<i>VERes Enrolled</i>—verification response enrollment status. ■ Used only for the Asia, Middle East, and Africa Gateway. ■ Required for all payer authentication transactions. ■ Possible values: • Y: Authentication available. • N: Customer not participating. • U: Unable to authenticate regardless of the reason.	ccAuthService_ veresEnrolled	payerAuthEnrollReply_ veresEnrolled
<i>XID</i>—transaction identifier. Must be 28-character base64 or 40-character hex binary. ■ Used for all processors that support Visa Secure, JCB J/Secure, or ProtectBuy. ■ For Atos: required for a successful Visa Secure transaction, which is indicated when the commerce indicator is <i>vbv</i>. ■ For all other processors: required when the commerce indicator is <i>js</i>, <i>pb</i>, or <i>vbv</i>. ■ Optional when the commerce indicator is <i>js_attempted</i>, <i>pb_attempted</i>, or <i>vbv_attempted</i>. ■ For Visa Secure on FDC Nashville Global, CyberSource sets the cardholder authentication verification value (CAVV) field to the XID value if the XID is present in the authorization request and the CAVV is not present.	ccAuthService_xid	payerAuthValidateReply_xid

Payment Network Tokenization

**Note**

Payment network tokenization and CyberSource payment tokenization are not the same feature.

- With payment network tokenization, the token is created by a token service provider and can be used throughout the financial network.
- With CyberSource payment tokenization, the token is created by CyberSource and can be used only with CyberSource services.

See [Payment Network Tokenization Using the Simple Order API](#).

Payment Tokenization

See ["Token Management Service \(TMS\)," page 226](#).

Payouts

See [Payouts Using the Simple Order API](#).

POS Transactions

See [Card-Present Processing Using the Simple Order API](#).

Quasi-Cash

Services:

- Authorization
- Full authorization reversal
- Capture
- Credit
- Void

Processors:

- Atos: Full authorization reversals and automatic partial authorization reversals are not supported for Atos.
- CyberSource through VisaNet. The supported acquirers are:
 - Auckland Savings Bank (ASB)
 - Australia and New Zealand Banking Group Limited (ANZ)
 - Axis Bank Ltd. of India
 - Banco Safra
 - Bangkok Bank Ltd.
 - Bank Sinarmas (Omise Ltd.)
 - BC Card Co., Ltd.
 - Cathay United Bank (CUB)
 - Citibank Malaysia
 - First Data Merchant Solutions in Brunei
 - First Data Merchant Solutions in Hong Kong
 - First Data Merchant Solutions in Malaysia
 - First Data Merchant Solutions in Singapore
 - Habib Bank Ltd. (HBL)
 - HDFC Bank Ltd. of India
 - Promerica in Honduras and Nicaragua
 - Taishin Bank Ltd.
 - United Overseas Bank (UOB) in Singapore and Vietnam
 - Vantiv
 - Vietnam Technological and Commercial Joint-stock Bank
 - Westpac
- GPN
- TSYS Acquiring Solutions

Before processing quasi-cash transactions, contact CyberSource Customer Support to have your account configured for this feature. If you have questions about the supported card types, contact your processor.

A quasi-cash transaction is a cash-like transaction for the sale of items that are directly convertible to cash, such as:

- Casino gaming chips
- Money orders
- Wire transfers

Automatic partial authorization reversals are supported for quasi-cash transactions. See ["Automatic Partial Authorization Reversals," page 69](#).

Recipients

Service:

- Authorization

Processors:

- Barclays
- Elavon
- HBoS
- LloydsTSB Cardnet
- Streamline

In the United Kingdom there is a regulation that permits customers to use a debit card to pay outstanding debt for another person. This person is referred to as the payment *recipient*. For example, a customer can pay the entire balance or part of the balance on a recipient's payment card or payday loan. To help reduce the high levels of fraud that occur for these kinds of transactions, you must include information about the recipient in the authorization request. The following fields are required in the United Kingdom for Visa debit transactions that are characterized under merchant category code 6012:

- recipient_accountID
- recipient_dateOfBirth
- recipient_lastName
- recipient_postalCode

These fields are described in [Appendix A, "API Fields," on page 238](#).

Recurring Billing

When you use Recurring Billing, you can process an authorization, capture, or credit by using information that is stored in a subscription. CyberSource uses the subscription ID to reference the subscription information in the CyberSource database. Instead of providing all the information that is normally required for a transaction, you only need to provide the following values:

- Merchant ID
- Merchant reference code
- Amount of the payment or credit
- Subscription ID

You can override most of the information stored in the subscription by including the relevant API fields in the payment or credit request. For example, you could provide a different billing or shipping address in the request. You cannot override the payment card account number.

See [Recurring Billing Using the Simple Order API](#).

Recurring Payments

Service:

- Authorization

Processors and card types:

- See the following table.

Table 47 Processors That Support Recurring Payments

Processors	Payment Card Types
AIBMS	Visa, Mastercard, Maestro (International)
American Express Brighton	American Express
American Express Direct	American Express
Asia, Middle East, and Africa Gateway	Visa, Mastercard, American Express, Diners Club, JCB
Atos	Visa, Mastercard
	Before processing recurring payments on Atos, you must:
	■ Contact your acquirer to ensure that you are permitted to accept recurring transactions. ■ Contact Atos to have your account configured to accept recurring transactions.

Table 47 Processors That Support Recurring Payments (Continued)

Processors	Payment Card Types
Barclays	Visa, Mastercard, JCB
Chase Paymentech Solutions	Visa, Mastercard, American Express, Discover
Cielo	Visa, Mastercard, American Express, Diners Club, Discover, JCB, Maestro (International), Elo, Aura On Cielo, recurring payments are not supported for debit transactions.
Comercio Latino	Visa, Mastercard, American Express, Discover, Diners Club, JCB, Elo, Aura, Hipercard When processing transactions in Mexico, you must include the billTo_customerID field in the authorization. Before you request the authorization you must inform the issuer of the customer contract numbers in advance. The supported acquirers are: ■ Banorte—must be submitted as an automatic capture. See "Automatic Captures," page 40. ■ Cielo
Credit Mutuel-CIC	Visa, Mastercard, Cartes Bancaires

Table 47 Processors That Support Recurring Payments (Continued)

Processors	Payment Card Types
CyberSource through VisaNet	Visa, Mastercard, American Express, Diners Club, JCB, Discover Note Not all card types are supported for all acquirers. The supported acquirers are: • Arab African International Bank (AAIB) • Asia Commercial Bank (ACB) • Auckland Savings Bank (ASB) • Australia and New Zealand Banking Group Limited (ANZ) • Axis Bank Ltd. of India • Banco Nacional de México (Banamex) • Banco Safra • Bangkok Bank Ltd. • Bank Muscat of Oman • Bank of Ayudhya (BAY) • Bank of China (BOC) • Bank Sinarmas (Omise Ltd.) • Banque Pour Le Commerce Exterieur Lao (BCEL) • BC Card Co., Ltd. • Cathay United Bank (CUB) • Citibank Hongkong and Macau • Citibank Malaysia • Citibank Singapore Ltd. • Commercial Bank of Qatar • CrediMax (Bahrain) • CTBC Bank Ltd. • First Data Merchant Solutions in Brunei • First Data Merchant Solutions in Hong Kong • First Data Merchant Solutions in Malaysia • First Data Merchant Solutions in Singapore • Global Payments Asia Pacific • Habib Bank Ltd. (HBL) • HDFC Bank Ltd. of India • I&M Bank • ICICI of India • Mashreq

Table 47 Processors That Support Recurring Payments (Continued)

Processors	Payment Card Types
	• National Bank of Abu Dhabi (NBAD) • National Bank of Kuwait (NBK) • Overseas Chinese Banking Corp (OCBC) • Promerica in Honduras and Nicaragua • Qatar National Bank (QNB Group) • Sumitomo Mitsui Card Co. • Taishin Bank Ltd. • United Overseas Bank (UOB) in Singapore and Vietnam • Vantiv • Vietcombank • VietinBank • Vietnam Technological and Commercial Joint-stock Bank • Westpac • Wing Hang Bank
Elavon	Visa, Mastercard, Maestro (UK Domestic), Diners Club
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay When you request a recurring payment transaction with Visa, Elavon Americas requires you to comply with the Visa merchant-initiated transactions mandate by including additional data in the request. You must do one of the following: ■ Include additional data as described in "Merchant-Initiated Transactions," page 173. ■ Make the request using the token management service, which meets the merchant-initiated transactions requirements. See Token Management Service Using the Simple Order API.
FDC Compass	Visa, Mastercard, American Express, Discover, Diners Club, JCB
FDC Germany	Visa, Mastercard
FDC Nashville Global	Visa, Mastercard, American Express, Discover, China UnionPay
FDI Australia	Visa, Mastercard

Table 47 Processors That Support Recurring Payments (Continued)

Processors	Payment Card Types
FDMS South	Visa, Mastercard, Discover On FDMS South, recurring payments are not supported for the CAD currency on the Visa card type.
FDMS Nashville	Visa, Mastercard, American Express, Discover
GPN	Visa, Mastercard, American Express, Discover, Diners Club, JCB
HBoS	Visa, Mastercard
HSBC	HSBC is the CyberSource name for HSBC U.K. To process recurring payments with HSBC, contact the CyberSource European office. For the European office's phone number, go to the CyberSource web site and click the Contact Us link: www.cybersource.com
Ingenico ePayments	Visa, Mastercard, American Express, Cartes Bancaires
Lloyds-OmniPay	Visa, Mastercard
LloydsTSB Cardnet	Visa, Mastercard
Moneris	Visa, Mastercard, American Express, Discover
OmniPay Direct	Bank of America Merchant Services: Visa, Mastercard Cardnet International: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club Global Payments International Acquiring: Visa, Mastercard
OmniPay-Ireland	Visa, Mastercard
OmniPay-Ireland	OmniPay-Ireland is the CyberSource name for HSBC International. To process recurring payments with OmniPay-Ireland, contact the CyberSource European office. For the European office's phone number, go to the CyberSource web site and click the Contact Us link: www.cybersource.com
RBS WorldPay Atlanta	Visa, Mastercard, American Express, Discover, Diners Club, JCB
Rede	Visa, Mastercard, American Express, Diners Club, JCB, Elo, Hipercard On Rede, recurring payments are not supported for debit transactions.

Table 47 Processors That Support Recurring Payments (Continued)

Processors	Payment Card Types
SIX	Visa, Mastercard, Discover, Diners Club, JCB, Maestro (International), Maestro (UK Domestic), China UnionPay, Visa Electron
Streamline	
To process recurring payments with Streamline, contact the CyberSource European office. For the European office's phone number, go to the CyberSource web site and click the Contact Us link: www.cybersource.com	
TSYS Acquiring Solutions	Visa, Mastercard, American Express, Discover
Worldpay VAP	Visa, Mastercard, American Express, Discover, Diners Club, JCB
Worldpay VAP was previously called <i>Little</i> .	

**Note**

American Express and Discover have programs that you must register for if you want to process recurring payments. Contact American Express and Discover for details about their programs.

Depending on the types of products and services you sell, you might want to process recurring payments for a customer. For example, you might want to charge a customer 19.95 USD each month to access a service that you offer.

**Note**

A customer's recurring payment does not have to be the same amount each time.

You must disclose clearly to customers when they make a purchase what the amount will be for the recurring payments. If the amount varies based on usage, make it clear.

To create a recurring payment:

Step 1 For the first payment, the type of request you need to send depends on which processor and card type you are using.

- For all card types on Atos, include the following fields and values in the request for the first payment:

`ccAuthService_commerceIndicator=recurring`

`ccAuthService_firstRecurringPayment=Y`

`card_cvNumber`

- For all card types on Credit Mutuel-CIC, include the following fields and values in the request for the first payment:
 - Set `ccAuthService_commerceIndicator` to `internet, spa, vbv, or vbv_attempted`.
 - Set `ccAuthService_firstRecurringPayment` to `Y`.
- For Mastercard and American Express transactions on FDC Nashville Global, include the following fields and values in the request for the first payment:


```
ccAuthService_commerceIndicator=recurring
ccAuthService_firstRecurringPayment=TRUE
card_cvNumber
```
- For all card types on OmniPay Direct, request a non-recurring transaction and include the following field and value in the request for the first payment:


```
ccAuthService_firstRecurringPayment=Y
```
- For all other processors and card types, request a non-recurring transaction for a credit card authorization.

If the first authorization is successful, you can submit subsequent authorizations for recurring payments using that card. If the first authorization is not successful, do not submit subsequent authorizations using that card.



Important

You must perform Step 1 once per year to verify the account.

- Step 2** For each subsequent recurring payment, send an authorization request using the e-commerce indicator to indicate that the payment is a recurring payment:

```
ccAuthService_commerceIndicator=recurring
```

If your processor supports merchant-initiated transactions, your authorization request must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

CyberSource also offers services that enable you to create a subscription or customer profile for a customer in the CyberSource system and then use that subscription or customer profile later to manually or automatically bill the customer. The CyberSource system eliminates the need for you to handle or store the customer's sensitive payment card information or create your own system for billing the customer on a regular basis. For more information, see ["Payment Tokenization," page 202](#), and ["Recurring Billing," page 205](#).

AVS and Recurring Payments



Note

FDMS Nashville does not support AVS for recurring payments.

If AVS is supported for your processor and card type, AVS is run for every authorization request that you submit. For recurring payments, check the AVS result for the first payment to ensure that the payment information is accurate and to reduce the risk of fraud.

You must decide what to do with the AVS results for subsequent payments. You might want to ignore the AVS results for these payments because you have already confirmed with the first payment that the payment card number is valid and not fraudulent.

When you need to change the payment card number used for a series of recurring payments, follow [Step 1](#) in creating a recurring payment to verify the new account number. Closely evaluate the AVS results. If the first authorization is successful, you can submit subsequent authorizations for recurring payments using that card. If the first authorization is not successful, do not submit subsequent authorizations using that card. For subsequent payments, follow [Step 2](#) in creating a recurring payment. You can choose to ignore the AVS results.

CVN and Recurring Payments



Note

FDMS Nashville does not support CVN for recurring payments.

With Ingenico ePayments, you must not include the CVN in a recurring payment request. If you do, CyberSource rejects the request because of invalid data.



Note

Ingenico ePayments was previously called *Global Collect*.

Replacement Expiration Dates for Recurring Payments

Service:

- Authorization

Processors and card types:

- See the following table.

Table 48 Processors That Support Replacement Expiration Dates for Recurring Payments

Processors	Payment Card Types
AIBMS	Visa, Mastercard, Maestro (International)
American Express Brighton	American Express You must contact American Express Brighton to get approval for using replacement expiration dates before using this feature.
American Express Direct	American Express
Barclays	Visa, Mastercard, JCB
Chase Paymentech Solutions	Visa, Mastercard
CyberSource through VisaNet	Visa, Mastercard, American Express, Diners Club, JCB, Discover Note Not all card types are supported for all acquirers. If an acquirer is supported for recurring payments, the acquirer is also supported for replacement expiration dates for recurring payments. For the list of supported acquirers, see the entry for CyberSource through VisaNet in Table 47, "Processors That Support Recurring Payments," on page 205 .
Elavon Americas	Visa, Mastercard, American Express, JCB, Diners Club, Discover, China UnionPay
FDC Compass	Visa, Mastercard, American Express, Discover, Diners Club
FDC Germany	Visa, Mastercard
FDI Australia	Visa, Mastercard
FDMS South	Visa, Mastercard
HBoS	Visa, Mastercard
HSBC HSBC is the CyberSource name for HSBC U.K.	Visa, Mastercard, Maestro (International)
Lloyds-OmniPay	Visa, Mastercard
LloydsTSB Cardnet	Visa, Mastercard

Table 48 Processors That Support Replacement Expiration Dates for Recurring Payments (Continued)

Processors	Payment Card Types
Streamline	To process recurring payments with Streamline, contact the CyberSource European office. For the European office's phone number, go to the CyberSource web site and click the Contact Us link: www.cybersource.com

Normally when you request a credit card authorization, you must provide a valid expiration date for the payment card. If you are processing a recurring payment, and the payment card that you have on file for the customer has expired, you might still be able to request the authorization depending on which processor you use. Instead of sending the out-of-date expiration date, you can include a replacement expiration date in your request.

**Important**

Do not use a replacement expiration date for cards that have not expired. Use a replacement expiration date only for cards that have expired and only for recurring payments.

Using a replacement expiration date for a recurring payment does not guarantee that the authorization will be successful. The issuing bank determines whether a card is authorized; some issuing banks do not accept an expiration date that does not match the expiration date in the bank's database.

**Important**

Effective October 17, 2014, an issuing bank can decline an authorization request for a recurring transaction with a Visa Europe card if the expiration date is incorrect, invalid, or missing. If you do not provide the correct expiration date for a recurring transaction, the authorization request might be declined.

The replacement expiration date that CyberSource supports is 12/2099. To use this date, include these fields and values in your authorization request:

```
card_expirationMonth=12
card_expirationYear=2099
```

Recurring Profiles

See "Recurring Billing," page 205.

Report Groups

Services:

- Authorization
- Full authorization reversal
- Capture
- Credit

Processor:

- Worldpay VAP—Worldpay VAP was previously called *Little*.

Report group values enable you to define custom groups for your processor reports. You can put your transactions into groups and then request processor reports for each group. This value is case sensitive and space sensitive.

**Note**

When you do not have a specific report group structure in mind, Worldpay VAP recommends that you use your merchant ID as your report group value.

**Important**

To use multiple report groups for your transactions, you must contact Worldpay VAP to have your Worldpay VAP account configured for this feature. When using one report group for all your transactions, you do not need to have your Worldpay VAP account configured for this feature.

The following table describes the logic that CyberSource uses for each kind of request to determine which report group value to use.

Table 49 Determining Which Report Group Value to Use

Kind of Request	Report Group Value
Authorization or Stand-Alone Credit	CyberSource checks the following locations, in the order given, for a report group value and uses the first value it finds: ■ reportGroup field in the authorization or stand-alone credit request ■ Report group value in your CyberSource account: Your CyberSource account can have a different report group value for each currency that you process. CyberSource uses the report group value that corresponds to the currency for the transaction. To create a default report group value in your CyberSource account, contact CyberSource Customer Support. ■ Your Worldpay VAP merchant ID
Capture or Full Authorization Reversal	CyberSource checks the following locations, in the order given, for a report group value and uses the first value it finds: ■ reportGroup field in the capture or full authorization reversal request ■ Report group value that was used for the authorization request
Follow-on Credit	CyberSource checks the following locations, in the order given, for a report group value and uses the first value it finds: ■ reportGroup field in the follow-on credit request ■ Report group value that was used for the capture that is being credited ■ Report group value that was used for the authorization request

Retail POS Data

See [Card-Present Processing Using the Simple Order API](#).

Samsung Pay

See [Samsung Pay Using the Simple Order API](#).

Secure Data

See "Payment Tokenization," page 202.

Service Fees

See [Service Fee Processing Using the Simple Order API](#).

Single-Message Processing

See ["Single-Message Processing," page 76](#).

Soft Descriptors

See ["Merchant Descriptors," page 169](#).

Split Dial/Route

See ["Forced Captures," page 142](#).

Split Shipments

Services:

- Authorization
- Capture

Processors:

- CyberSource through VisaNet



Important

Split shipments are not available for Mastercard transactions in the IDR currency on CyberSource through VisaNet.

- GPN

The split-shipment feature enables you to split an order into multiple shipments with multiple captures.



Note

Multiple partial captures and split shipments are not the same feature.

- The multiple partial captures feature is provided by the processor. This feature enables you to request multiple partial captures for one authorization. For more information, see ["Multiple Partial Captures," page 71](#).
 - The split-shipment feature is provided by CyberSource. This feature supports three different scenarios: multiple authorizations, multiple captures, and multiple authorizations with multiple captures.
-

Benefits of Using Split Shipments

The benefits of using split shipments are:

- All the transactions for a split shipment are linked together in the Business Center and in reports.
- When you split an order into multiple shipments with multiple captures, you do not need to request additional authorizations; CyberSource takes care of the additional authorizations for you.

Requirements

The requirements for using split shipments are:

- You must use CyberSource through VisaNet or GPN.
- You must contact CyberSource Customer Support to have your account configured for this feature. On CyberSource through VisaNet, your account can be enabled for multiple partial captures or split shipments; it cannot be enabled for both features.

How Split Shipments Work

Additional Authorizations

When you need an additional authorization for an order, you can use the link-to-request field to link the additional authorization to the first authorization. For the additional authorization, you must submit an authorization request that includes the link-to-request field in addition to the basic fields required for every authorization request. The additional authorization is linked to the original authorization in the Business Center and in reports. The captures for these authorizations are also linked to the original authorization in the Business Center and in reports.

For an additional authorization on a processor that supports merchant-initiated transactions, your authorization request must include subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).

For scenarios that use an additional authorization, see the following sections:

- ["One Authorization and One Sale," page 220](#)
- ["Two Authorizations and One Capture," page 222](#)

For examples that use an additional authorization, see:

- Name-value pair examples: ["Split-Shipment Examples," page 413](#)
- XML examples: ["Split-Shipment Examples," page 441](#)

Additional Captures

When you need an additional capture for an order, CyberSource performs a system-generated authorization for the additional capture request, using the payment data from the original authorization. The system-generated authorization is linked to the original authorization in the Business Center and in reports. The captures are linked to the authorizations in the Business Center and in reports through the request IDs as with any capture.

On GPN, the system-generated authorization uses the same authorization indicator as the original authorization. For more information, see ["Final Authorization Indicator," page 138](#).

For scenarios that use an additional capture, see the following sections:

- ["One Authorization and Two Captures," page 221](#)
- ["Multiple Captures in a Batch File," page 221](#)

For examples that use an additional capture, see:

- Name-value pair examples: ["Split-Shipment Examples," page 413](#)
- XML examples: ["Split-Shipment Examples," page 441](#)

Split-Shipment Scenarios

One Authorization and One Sale

In this scenario, your customer orders a product that is not available yet.

- 1 You request an authorization to ensure that funds are available.
The product is not available for immediate shipment, so you wait for the product to become available.
- 2 After the product becomes available, you ship the product and request a sale.
For the second authorization, you must submit an authorization request that includes the link-to-request field in addition to the basic fields required for every authorization request. Set the link-to-request field to the request ID from the first authorization's reply:

First Authorization Reply Message: `requestID=SWVdPS5IM`

Second Authorization Request: `linkToRequest=SWVdPS5IM`

Including the link-to-request field in your authorization request triggers the split-shipment functionality. Because you are requesting the second authorization and capture together, you do not need to include the request ID in your capture request.

- 3 CyberSource tries to link the second authorization request to the first authorization:
 - If the link-to-request value is valid, the second authorization is linked to the original authorization in the Business Center and in reports.
 - If the link-to-request value is not valid, the second authorization is not linked to the original authorization in the Business Center and in reports.
- 4 CyberSource links the capture request:
 - If the link-to-request value for the second authorization was valid, all three transactions (first authorization, second authorization, capture) are linked together in the Business Center and in reports.
 - If the link-to-request value for the second authorization was not valid, the second authorization and capture are linked to each other in the Business Center and in reports, but they are not linked to the first authorization.

One Authorization and Two Captures

In this scenario, your customer orders multiple products, one of which is not available yet.

- 1 You request an authorization to ensure that funds are available.
- 2 You ship the available products and request a capture for the amount of the shipped products.
One of the products is not available for immediate shipment, so you ship the available products and wait for the remaining product to become available.
- 3 After the remaining product becomes available, you ship the product and request a capture for the amount of that product.
- 4 CyberSource performs a system-generated authorization for the second capture request.
Because your account is enabled for split shipment, instead of rejecting the capture request as a duplicate capture, CyberSource processes the capture request as a split-shipment request.
The system-generated authorization is linked to the original authorization in the Business Center and in reports.
- 5 CyberSource links the capture request.
The capture is linked to the authorizations in the Business Center and in reports through the request IDs as with any capture. All four transactions (first authorization, system-generated authorization, first capture, second capture) are linked together in the Business Center and in reports.
- 6 You get the status of the second capture request and its associated system-generated authorization.
See ["Obtaining the Status of a System-Generated Authorization," page 223](#).

Multiple Captures in a Batch File



Note

You can also request authorizations in a batch file.

- 1 You create and upload a batch file using one of these methods:
 - Business Center Transaction Batch Functionality: This functionality is described in the [Offline Transaction File Submission Implementation Guide](#) and in the Online Help for the Business Center.
 - Offline Transaction File Submission System: This system is described in the [Offline Transaction File Submission Implementation Guide](#).

- 2 CyberSource processes the batch file.
- 3 You get the status of your batch requests by viewing the Batch Submission Detail Report. Get the report by using one of these methods, both of which are described in the [Offline Transaction File Submission Implementation Guide](#):
 - View the report on the Business Center.
 - Download the report programmatically.
- 4 You get the status of your split-shipment transactions.

Two Authorizations and One Capture

In this scenario, your customer orders a product that is not available yet.

- 1 You request an authorization to ensure that funds are available.
The product is not available for immediate shipment, so you wait for the product to become available.
- 2 After the product becomes available, you request a second authorization to ensure that funds are still available. The authorization request must include:
 - Basic fields required for every authorization request.
 - Link-to-request field. Set the value for this field to the request ID from the first authorization's reply:
First Authorization Reply Message: `requestID=SWVdPS5IM`
Second Authorization Request: `linkToRequest=SWVdPS5IM`
Including the link-to-request field in your authorization request triggers the split-shipment functionality.
 - On a processor that supports merchant-initiated transactions: subsequent authorization fields as described in ["Merchant-Initiated Transactions," page 173](#).
- 3 CyberSource tries to link the second authorization request to the first authorization:
 - If the link-to-request value is valid, the second authorization is linked to the original authorization in the Business Center and in reports.
 - If the link-to-request value is not valid, the second authorization is not linked to the original authorization in the Business Center and in reports.
- 4 You ship the product and request a capture.
Set the request ID in the capture request to the request ID from the second authorization's reply:
Second Authorization Reply Message: `requestID=s139cmdSlkJ`
Capture Request: `ccCaptureService_authRequestID=s139cmdSlkJ`

5 CyberSource links the capture request:

- If the link-to-request value for the second authorization was valid, all three transactions (first authorization, second authorization, capture) are linked together in the Business Center and in reports.
- If the link-to-request value for the second authorization was not valid, the second authorization and capture are linked to each other in the Business Center and in reports, but they are not linked to the first authorization.

Obtaining the Status of a System-Generated Authorization

A system-generated authorization is not performed in real time. The reply message that you receive from CyberSource simply indicates that the request was received; it does not indicate whether the system-generated authorization was approved or declined. A system-generated authorization can be declined for the same reasons that a regular authorization can be declined.

CyberSource recommends that you use one of the methods described in the following table to get the status of the system-generated authorization request before shipping the product.

Table 50 Methods for Obtaining the Status of a System-Generated Authorization

Method	Description
Business Center	Use the capture request ID to search for the second capture. The details for all related transactions are displayed on the Transaction Search Details page. It can take a maximum of six hours for the status of the system-generated authorization request to be available.
On-Demand Single Transaction Report	This report is described in the Classic Reporting Developer Guide . You must use version 1.3 or later and include the parameter includeExtendedDetail in your query. It can take a maximum of six hours for the status of the system-generated authorization request to be available.
Transaction Exception Detail Report	This report is described in the Classic Reporting Developer Guide . CyberSource recommends that you use this report on a daily basis to identify transactions that have been declined.

Additional Information

For descriptions of the required fields for authorization and capture requests, and to see which fields are optional, see [Appendix A, "API Fields," on page 238](#).

For examples of split-shipment requests and replies, see:

- Name-value pair examples: ["Split-Shipment Examples," page 413](#)
- XML examples: ["Split-Shipment Examples," page 441](#)

Staged Digital Wallets

Services:

- Authorization
- Capture

Processor:

- FDC Compass

Terminology

Table 51 Staged Digital Wallet Terminology

Term	Definition
Business application indicator (BAI)	Identifier for the type of transaction.
Digital wallet operator (DWO)	Entity that provides the pass-through digital wallet or staged digital wallet service to the customer.
Merchant verification value (MVV)	Setting in your CyberSource account that is required for processing SDW transactions.
Pass-through wallet or pass-through digital wallet	Digital wallet that replaces the customer's payment information with a token. Examples: Apple Pay, Google Pay, Visa Checkout, Masterpass.
Staged digital wallet (SDW)	Digital wallet that replaces the customer's payment information with credentials provided by a SDWO. Examples: PayPal, AliPay.
Staged digital wallet operator (SDWO)	Entity that provides the staged digital wallet service to the customer and processes the payment between the customer and the merchant.

Requirements

You must:

- Contact CyberSource Customer Support to set the merchant verification value in your CyberSource account.
- Include the **invoiceHeader_businessApplicationID** field in the authorization request and set it to `WT`.

SDW Functionality

The SDW provides functionality that:

- Uses the card or bank account number supplied by the customer to the SDWO.
- Uses a separate account or accounts assigned by the SDWO to the customer.
- Completes a transaction through the SDWO in two stages, in any order:
 - Purchase transaction: customer pays you using the credentials assigned by the SDWO. For example, the typical credentials for PayPal are a username and a password.
 - Funding: uses the account number provided by the customer to fund or reimburse the staged digital wallet.

Subscriptions

See ["Recurring Billing," page 205](#).

Tax Calculation

See [Tax Calculation Service Using the Simple Order API](#).

Tokenization

See ["Payment Network Tokenization," page 202](#), and ["Token Management Service \(TMS\)," page 226](#).

Token Management Service (TMS)

Token Management Service replaces Payment Tokenization. TMS enables you to:

- Tokenize customers' sensitive personal information.
- Eliminate payment data from your order management system to ensure that it is not compromised during a security breach.

When you use TMS, you can process an authorization, capture, or credit by using information that is associated with a customer token. CyberSource uses the customer token to reference customer information in the CyberSource database. Instead of providing all the information that is normally required for a transaction, you only need to provide the following values:

- Merchant ID
- Merchant reference code
- Amount of the payment or credit
- Subscription ID—set this field to the value of the customer token.

You can override most of the information associated with the customer token by including the relevant API fields in the payment or credit request. For example, you could provide a different billing or shipping address in the request. You cannot override the payment card account number.

See [Token Management Service Using the Simple Order API](#).

Type II Cards

See [Level II and Level III Processing Using the Simple Order API](#).

Verbal Authorizations

See "Verbal Authorizations," page 101.

Visa Secure

See "Payer Authentication," page 183.

Visa Bill Payments

Services:

- Authorization
- Credit

Processors:

- Chase Paymentech Solutions
- FDC Compass
- FDC Nashville Global
- FDMS Nashville
- GPN
- OmniPay-Ireland: OmniPay-Ireland is the CyberSource name for HSBC International.
- TSYS Acquiring Solutions

Visa provides a Bill Payment program that enables customers to use their Visa cards to pay their bills. When you participate in this program, Visa requests that you flag the bill payments and credits so they can be easily identified. To flag these transactions, include the **ccAuthService_billPayment** field in your transaction requests.

Although CyberSource accepts the bill payment indicator no matter which processor you are using, do not use this indicator if you have not signed up with Visa to participate in the program.

Visa Checkout

See:

- [Getting Started with Visa Checkout](#)
- ["Creating an Authorization Request," page 41](#)
- ["Visa Checkout Examples," page 419 \(NVP\)](#)
- ["Visa Checkout Examples," page 451 \(XML\)](#)

Visa Debt Repayments

Services:

- Authorization
- Credit

Processors:

- CyberSource through VisaNet—supported only in Australia and New Zealand
- FDC Nashville Global
- FDMS Nashville
- GPN

Visa provides a Debt Repayment program that enables customers to use their Visa debit cards to make a payment towards an existing contractual loan. The types of loans that can qualify for this program are:

- Auto loans
- Payment cards
- Mortgages
- Student loans

To participate in this program, contact your processor for details and requirements.

When you participate in this program, Visa requests that you flag the debt repayments and credits so they can be easily identified. To flag these transactions, include these fields in your transaction requests:

- ccAuthService_billPayment—not required on CyberSource through VisaNet
- debtIndicator



Note

When you use the Simple Order API in XML format, you must use version 1.37 or later of the XML schema to implement Visa debt repayments.

Visa Direct

Visa Direct is one of the networks supported by CyberSource Payouts.

See [Payouts Using the Simple Order API](#).

Zero Amount Authorizations

Service:

- Authorization

Processors and card types:

- See the following table.

Table 52 Processors That Support Zero Amount Authorizations

Processor	AVS	CVN	Card Types and Notes
AIBMS	Yes	Yes	■ Visa ■ Mastercard For zero amount authorizations on AIBMS, the commerce indicator must be <code>internet</code> or <code>moto</code>.
American Express Direct	Yes	No	■ American Express All currencies that are supported for standard authorizations for American Express Direct are also supported for zero amount authorizations.
Barclays	Yes	Yes	■ Visa ■ Mastercard All currencies that are supported for standard authorizations for Barclays are also supported for zero amount authorizations. CyberSource rounds the amount to the correct number of decimal places for the currency. For zero amount authorizations on Barclays, the commerce indicator must be <code>internet</code> or <code>moto</code>. Visa Electron cards are not supported for zero amount authorizations on Barclays.
Chase Paymentech Solutions	Yes	Yes	■ Visa ■ Mastercard ■ American Express ■ Discover ■ Diners Club
Credit Mutuel-CIC	No	Yes	■ Visa ■ Mastercard ■ Cartes Bancaires

Table 52 Processors That Support Zero Amount Authorizations (Continued)

Processor	AVS	CVN	Card Types and Notes
CyberSource through VisaNet	Yes	Yes	■ Visa ■ Mastercard For CyberSource through VisaNet, zero amount authorizations are supported for Internet, MOTO, and card-present transactions. Do not try to perform a zero amount authorization for a recurring, installment, or payer authorization transaction.
Elavon	Yes	Yes	■ Visa ■ Mastercard ■ Maestro (UK Domestic) ■ Maestro (International) All currencies that are supported for standard authorizations for Elavon are also supported for zero amount authorizations.
Elavon Americas	Yes	Yes	■ Visa ■ Mastercard ■ American Express ■ JCB ■ Diners Club ■ Discover ■ China UnionPay
FDC Compass	Yes	Yes	■ Visa ■ Mastercard ■ American Express ■ Diners Club

Table 52 Processors That Support Zero Amount Authorizations (Continued)

Processor	AVS	CVN	Card Types and Notes
FDC Nashville Global	Yes	Yes for all card types except American Express	■ Visa ■ Mastercard ■ American Express ■ Discover ■ Diners Club For a zero amount authorization on FDC Nashville Global: ■ For Visa, Mastercard, and American Express, all currencies that are supported for standard authorizations are also supported for zero amount authorizations. ■ For Discover and Diners Club, only USD is supported for zero amount authorizations.
FDI Australia	Yes	Yes	■ Visa—supported only for AUD. ■ Mastercard—supported only for AUD.
FDMS Nashville	Yes	Yes	■ Visa
FDMS South	Yes	Yes for all card types except American Express	■ Visa ■ Mastercard ■ American Express: ● AVS is required for zero amount authorizations with American Express. ● CVN is not supported for zero amount authorizations with American Express. ■ Diners Club ■ Discover
Getnet	No	No	■ Visa ■ Mastercard ■ Hipercard
GPN	Yes	Yes for all card types except American Express	■ Visa ■ Mastercard ■ American Express: CVN is not supported for zero amount authorizations with American Express. ■ Discover ■ JCB

Table 52 Processors That Support Zero Amount Authorizations (Continued)

Processor	AVS	CVN	Card Types and Notes
HBoS	Yes	Yes	■ Visa ■ Mastercard For zero amount authorizations on HBoS, the commerce indicator must be <code>internet</code> or <code>moto</code>.
HSBC HSBC is the CyberSource name for HSBC U.K.	Yes	Yes	■ Visa ■ Mastercard ■ Maestro (UK Domestic) ■ Maestro (International) For zero amount authorizations on HSBC: ■ The commerce indicator must be <code>internet</code> or <code>moto</code>. ■ The authorization code is not returned.
JCN Gateway	No	Yes	■ Visa ■ Mastercard ■ American Express ■ Diners Club ■ JCB
Lloyds-OmniPay	Yes	Yes	■ Visa ■ Mastercard For zero amount authorizations on Lloyds-OmniPay, the commerce indicator must be <code>internet</code> or <code>moto</code>.
LloydsTSB Cardnet	Yes	Yes	■ Visa ■ Mastercard For zero amount authorizations on LloydsTSB Cardnet, the commerce indicator must be <code>internet</code> or <code>moto</code>.
Moneris	Yes	Yes	■ Visa ■ Mastercard

Table 52 Processors That Support Zero Amount Authorizations (Continued)

Processor	AVS	CVN	Card Types and Notes
OmniPay Direct	Yes	Yes	Bank of America Merchant Services: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) Cardnet International: Visa, Mastercard, Maestro (UK Domestic), Maestro (International) First Data Merchant Solutions (Europe): Visa, Mastercard, Discover, Diners Club, Maestro (UK Domestic), Maestro (International) Global Payments International Acquiring: Visa, Mastercard, Maestro (UK Domestic), Maestro (International)
OmniPay-Ireland	Yes	Yes	■ Visa ■ Mastercard OmniPay-Ireland is the CyberSource name for HSBC International.
RBS WorldPay Atlanta	Yes	Yes	■ Visa ■ Mastercard ■ Diners Club
Rede	No	No	■ Visa ■ Mastercard
SIX	Yes	Yes	■ Visa ■ Mastercard ■ Discover ■ Diners Club ■ JCB ■ Maestro (International) ■ Maestro (UK Domestic) ■ China UnionPay ■ Visa Electron

Table 52 Processors That Support Zero Amount Authorizations (Continued)

Processor	AVS	CVN	Card Types and Notes
Streamline	Yes	Yes	■ Visa ■ Mastercard ■ Maestro (International) ■ Maestro (UK Domestic) ■ Cartes Bancaires ■ Dankort All currencies that are supported for standard authorizations for Streamline are also supported for zero amount authorizations. For a zero amount authorization: ■ The commerce indicator must be <code>internet</code> or <code>moto</code>. ■ Payer authentication is not supported.
TSYS Acquiring Solutions	Yes	Yes for Visa and Mastercard. No for American Express and Discover.	■ Visa ■ Mastercard ■ American Express: CVN is not supported for zero amount authorizations with American Express. ■ Discover: CVN is not supported for zero amount authorizations with Discover.
Worldpay VAP Worldpay VAP was previously called <i>Little</i> .	Yes	Yes	■ Visa ■ Mastercard ■ American Express ■ Discover ■ Diners Club ■ JCB

Authorizing a payment for a zero amount shows whether a payment card account is valid and whether the card is lost or stolen. You cannot capture a zero amount authorization.

Testing the Credit Card Services

To ensure that your requests are processed correctly, you must test the basic success and error conditions for each CyberSource service you plan to use.

Requirements for Testing

**Important**

Before you can test, you must contact CyberSource Customer Support to activate the credit card services and configure your account for testing. You must also contact your processor to set up your processor account.

- Use your regular CyberSource merchant ID when you test your system.
- Unless otherwise specified, use test payment card numbers, not real ones. See [Table 53, "Test Payment Card Numbers," on page 236](#).
- Use a real combination for the city, state, and postal code.
- Use a real combination for the area code and telephone number.
- Use a nonexistent account and domain name for the customer's email address.
- When testing an Ingenico ePayments country-specific payment card, such as Italy's Carta Si, specify the appropriate country code when sending the customer's address and specify the currency used in that country.

**Note**

Ingenico ePayments was previously called *Global Collect*.

- When testing the Simple Order API, use the test URL:
<https://ics2wstesta.ic3.com/commerce/1.x/transactionProcessor>

Note

When you test captures on Ingenico ePayments, you must capture the full amount of the authorization. Although a capture request for a partial amount is not rejected during testing, it will be rejected by the processor in production.

Ingenico ePayments was previously called *Global Collect*.

Testing the Services

Use the payment card numbers in the following table to test the authorization, capture, and credit services. Do not use real payment card numbers. To test card types not listed in the table, use an account number that is within the card's bin range. For best results, try each test with a different CyberSource service request and with different test payment card numbers.

Table 53 Test Payment Card Numbers

Payment Card Type	Test Account Number (Remove spaces when sending to CyberSource.)
American Express	3782 8224 6310 005
Discover	6011 1111 1111 1117
JCB	3566 1111 1111 1113
Maestro (International)	5033 9619 8909 17
Maestro (UK Domestic)	5868 2416 0825 5333 38
	6759 4111 0000 0008
	6759 5600 4500 5727 054
	5641 8211 1116 6669
Note Effective May 2011, the issue number is no longer required for Maestro (UK Domestic) transactions.	
Mastercard	2222 4200 0000 1113
	2222 6300 0000 1125
	5555 5555 5555 4444
UATP	1354 1234 5678 911
Visa	4111 1111 1111 1111

Using Amounts to Simulate Errors

You can simulate the CyberSource error messages by requesting authorization, capture, or credit services with specific amounts that trigger the error messages. These triggers work only on the test server, not on the production server. Each payment processor uses its own error messages.

For trigger amounts and responses, see [Simple Order API and SOAP Toolkit API Testing Information page](#).

Testing American Express Card Verification

Before using CVN with American Express, CyberSource strongly recommends that you perform this procedure.

To test American Express card verification:

- Step 1** Contact CyberSource Customer Support to have your account configured for CVN. Until you do this, you will receive a 1 in the **ccAuthReply_cvCode** reply field.
 - Step 2** Test your system in production using a small currency amount, such as one currency unit. Instead of using the test account numbers, use a real payment card account number, and send an incorrect CVN in the request for authorization. The card should be refused and the request declined.
-

API Fields

Formatting Restrictions

Unless otherwise noted, all field names are case sensitive and all fields accept special characters such as @, #, and %.



Note

The values of the **item_#_** fields must not contain carets (^) or colons (:) because these characters are reserved for use by the CyberSource services.

Values for request-level and item-level fields must not contain new lines or carriage returns. However, they can contain embedded spaces and any other printable characters. CyberSource removes all leading and trailing spaces.

Atos

The **billTo_** fields must not contain colons (:).

Moneris

Values for request-level and item-level fields must not contain these special characters: ampersands (&), single quotes ('), double quotes ("), less than signs (<), and greater than signs (>).

Multibyte Characters

Many international languages – such as Korean, Chinese, and Russian – require more than one ASCII byte per character. Therefore, a string of multibyte characters can exceed the allotted string length of some API fields.

In many international languages, a short name that visually has only 3 or 4 characters may be equivalent to 10 or 15 bytes in UTF-8. Therefore, it is important to take into consideration the overall length of a multibyte string to be sure it fits into the allotted ASCII string length.



Important

For best success when working with international languages, make sure the total size of a multibyte string does not exceed the ASCII string length for a specific API field.

Data Type Definitions

For more information about these data types, see the [World Wide Web Consortium \(W3C\) XML Schema Part 2: Datatypes Second Edition](#).

Table 54 Data Type Definitions

Data Type	Description
Date and time	Format is YYYY-MM-DDThh:mm:ssZ, where: ■ T separates the date and the time ■ Z indicates Coordinated Universal Time (UTC), also known as Greenwich Mean Time (GMT) Example 2019-08-11T22:47:57Z equals August 11, 2019, at 22:47:57 (10:47:57 p.m.).
Integer	Whole number {..., -3, -2, -1, 0, 1, 2, 3, ...}
String	Sequence of letters, numbers, spaces, and special characters

Numbered Elements

The CyberSource XML schema includes several numbered elements. You can include these complex elements more than once in a request. For example, when a customer order includes more than one item, you must include multiple `<item>` elements in your request. Each item is numbered, starting with 0. The XML schema uses an `id` attribute in the item's opening tag to indicate the number. For example:

```
<item id="0">
```

As a name-value pair field name, this tag is represented as **item_0**. In this portion of the field name, the underscore before the number does not indicate hierarchy in the XML schema. The item fields are generically referred to as **item_#_<element name>** in the documentation.

Below is an example of the numbered `<item>` element and the corresponding name-value pair field names. If you are using the Simple Object Access Protocol (SOAP), the client contains a corresponding `Item` class.

Example 3 Numbered XML Schema Element Names and Name-Value Pair Field Names

XML Schema Element Names	Corresponding Name-Value Pair Field Names
<pre><item id="0"> <unitPrice> <quantity> </item></pre>	<pre>item_0_unitPrice item_0_quantity</pre>
<pre><item id="1"> <unitPrice> <quantity> </item></pre>	<pre>item_1_unitPrice item_1_quantity</pre>



Important

When a request is in XML format and includes an `<item>` element, the element must include an `id` attribute. For example: `<item id="0">`.

Request Fields



Note

When you use Payment Tokenization or Recurring Billing and you include a subscription ID in your request, many of the fields in the following table that are normally required for an authorization or credit become optional. See ["Payment Tokenization," page 202](#), and ["Recurring Billing," page 205](#).

Table 55 Request Fields

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
authIndicator	Flag that specifies the purpose of the authorization. Possible values: ■ 0: Preauthorization ■ 1: Final authorization To set the default for this field, contact CyberSource Customer Support. See "Final Authorization Indicator," page 138. <i>Barclays and Elavon</i> The default for Barclays and Elavon is 1 (final authorization). To change the default for this field, contact CyberSource Customer Support. <i>CyberSource through VisaNet</i> When the value for this field is 0, it corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR0 ■ Position: 164 ■ Field: Additional Authorization Indicators When the value for this field is 1, it does not correspond to any data in the TC 33 capture file.	ccAuthService (Optional for Mastercard and Maestro transactions; not used for other card types) ccIncrementalAuthService (O for Mastercard; not used for other card types)	String (1)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
balanceInquiry	Flag indicating whether to return balance information. See "Balance Inquiries," page 128. Possible values: ■ true ■ false	ccAuthService(Required for a balance inquiry; otherwise, not used.)	String (5)
billPaymentType	Reason for the payment. Possible values: ■ 001: Utility payment ■ 002: Government services ■ 003: Mobile phone top-up ■ 004: Coupon payment The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 48-50 ■ Field: Bill Payment Transaction Type Identifier This field is supported only for bill payments in Brazil with Mastercard on CyberSource through VisaNet. See "Mastercard Bill Payments," page 167. Note For information about bill payments with Visa, see "Visa Bill Payments," page 227.	ccAuthService (R for bill payments with Mastercard in Brazil on CyberSource through VisaNet)	String (3)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_buildingNumber	Building number in the street address. For example, if the street address is: Rua da Quitanda 187 then the building number is 187. This field is supported only for: ■ Cielo transactions. ■ Redecard customer validation with CyberSource Latin American Processing.	ccAuthService (O for Cielo. R for Redecard customer validation with CyberSource Latin American Processing. Otherwise, not used.)	String (256)
billTo_city	City of the billing address. <i>Atos</i> This field must not contain colons (:). <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (R) ² ccCaptureService (O) ccCreditService (R) ^{1,2}	Atos: String (32) All other processors: String (50)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_company	Name of the customer's company. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (60)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_companyTaxID	Tax identifier for the customer's company. This field is supported only for these kinds of payments: ■ BNDES transactions on CyberSource through VisaNet. See "BNDES," page 129. ■ Installment payments with Mastercard on CyberSource through VisaNet in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. Set this field to the Cadastro Nacional da Pessoa Jurídica (CNPJ). The request must include this field or billTo_personalID. When a request includes both fields, CyberSource sends the value for the billTo_personalID field to the processor and ignores the billTo_companyTaxID field. For BNDES transactions, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR6 ■ Position: 26-39 ■ Field: Buyer ID For installment payments with Mastercard in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 26-39 ■ Field: Buyer ID	ccAuthService (See description) ccCaptureService (See description) ccCreditService (See the field description; BNDES transactions only.)	String (9)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_country	Country of the billing address. Use the two-character ISO Standard Country Codes. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (R)² ccCaptureService (O) ccCreditService (R)^{1,2}	String (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_customerID	Your identifier for the customer. When a subscription or customer profile is being created, the maximum length for this field for most processors is 30. Otherwise, the maximum length is 100. <i>Comercio Latino</i> For recurring payments in Mexico, the value is the customer's contract number. Note Before you request the authorization, you must inform the issuer of the customer contract numbers that will be used for recurring transactions. <i>Worldpay VAP</i> For a follow-on credit with Worldpay VAP, CyberSource checks the following locations, in the order given, for a customer account ID value and uses the first value it finds: 1 billTo_customerID value in the follow-on credit request 2 Customer account ID value that was used for the capture that is being credited 3 Customer account ID value that was used for the original authorization If a customer account ID value cannot be found in any of these locations, then no value is used.	ccAuthService (Required for recurring transactions in Mexico on Comercio Latino; otherwise, optional.) ccCaptureService (O) ccCreditService (O)	Comercio Latino: String (20) All other processors: String (100)
billTo_district	Customer's neighborhood, community, or region (a <i>barrio</i> in Brazil) within the city or municipality. This field is available only on Cielo.	ccAuthService (O)	String (50)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_email	Customer's email address, including the full domain name. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (R)² ccCaptureService (O) ccCreditService (R)^{1,2}	String (255)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_firstName	Customer's first name. This name must be the same as the name on the card. <i>CyberSource Latin American Processing</i> Important For an authorization request, CyberSource Latin American Processing concatenates billTo_firstName and billTo_lastName. If the concatenated value exceeds 30 characters, CyberSource Latin American Processing declines the authorization request. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America. The information in this field description is for the specific processing connection called <i>CyberSource Latin American Processing</i>. It is not for any other Latin American processors that CyberSource supports. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (R)² ccCaptureService (O) ccCreditService (R)^{1,2}	CyberSource Latin American Processing: see field description Worldpay VAP: String (25) All other processors: String (60)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_hostname	DNS resolved hostname from billTo_ipAddress .	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (60)
billTo_httpBrowserType	Customer's browser as identified from the HTTP header data. For example, <i>Mozilla</i> is the value that identifies the Netscape browser.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (40)
billTo_ipAddress	Customer's IP address.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (15)
billTo_lastName	Customer's last name. This name must be the same as the name on the card. <i>CyberSource Latin American Processing</i> Important For an authorization request, CyberSource Latin American Processing concatenates billTo_firstName and billTo_lastName. If the concatenated value exceeds 30 characters, CyberSource Latin American Processing declines the authorization request. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America. The information in this field description is for the specific processing connection called <i>CyberSource Latin American Processing</i>. It is not for any other Latin American processors that CyberSource supports.	ccAuthService (R) ² ccCaptureService (O) ccCreditService (R) ^{1,2}	CyberSource Latin American Processing: see field description Worldpay VAP: String (25) All other processors: String (60)

(continued on next page)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_lastName (continued)	<i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.		
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_merchantTaxID	This field is supported only on CyberSource through VisaNet. <i>Brazil</i> In Brazil, this field is supported only for BNDES transactions. See "BNDES," page 129. The value for this field is your Cadastro Nacional da Pessoa Jurídica (CNPJ) number. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP07 TCR6 - Position: 40-59 - Field: BNDES Reference Field 1 <i>Croatia</i> In Croatia, this field is supported only for installment payments. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR5 - Position: 79-98 - Field: Mastercard Croatia Domestic Card Acceptor Tax ID	ccAuthService (R for Mastercard BNDES transactions in Brazil or Mastercard installment payments in Croatia; otherwise, not used.) ccCaptureService (R for Mastercard BNDES transactions in Brazil or Mastercard installment payments in Croatia; otherwise, not used.) ccCreditService (R for Mastercard BNDES transactions in Brazil; otherwise, not used.)	String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_personalID	Identifier for the customer. This field is supported only on the processors listed in this description. <i>CyberSource Latin American Processing</i> This field is supported only for Redecard in Brazil. Set this field to the Cadastro de Pessoas Fisicas (CPF), which is required for AVS for Redecard in Brazil. Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America. The information in this field description is for the specific processing connection called <i>CyberSource Latin American Processing</i>. It is not for any other Latin American processors that CyberSource supports. <i>CyberSource through VisaNet</i> This field is supported only for these kinds of payments: ■ BNDES transactions. See "BNDES," page 129. ■ Installment payments with Mastercard in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. Set this field to the Cadastro de Pessoas Fisicas (CPF). The request must include this field or billTo_companyTaxID.	CyberSource Latin American Processing: ccAuthService (See the field description.) CyberSource through VisaNet: ■ ccAuthService (See the field description.) ■ ccCaptureService (See the field description.) ■ ccCreditService (See the field description; BNDES transactions only.)	String (26)

(continued on next page)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_personalID (continued)	<i>CyberSource through VisaNet, continued</i> For BNDES transactions, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR2 ■ Position: 7-21 ■ Field: Cardholder Tax ID For installment payments with Mastercard in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 26-39 ■ Field: Buyer ID		

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_phoneNumber	Customer's phone number. CyberSource recommends that you include the country code when the order is from outside the U.S. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks. For installment payments with Mastercard in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 40-50 ■ Field: Buyer Phone Number	ccAuthService (R for installment payments with Mastercard on CyberSource through VisaNet in Brazil; otherwise, optional) ccCaptureService (R for installment payments with Mastercard on CyberSource through VisaNet in Brazil; otherwise, optional) ccCreditService (O)	Installment payments with Mastercard on CyberSource through VisaNet in Brazil: String (11) All other transactions: String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_postalCode	Postal code for the billing address. The postal code must consist of 5 to 9 digits. When the billing country is the U.S., the 9-digit postal code must follow this format: [5 digits][dash][4 digits] Example 12345-6789 When the billing country is Canada, the 6-digit postal code must follow this format: [alpha][numeric][alpha][space] [numeric][alpha][numeric] Example A1B 2C3 <i>American Express Direct</i> Before sending the postal code to the processor, CyberSource removes all non-alphanumeric characters and, if the remaining value is longer than nine characters, truncates the value starting from the right side. <i>Atos</i> This field must not contain colons (:). <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (Required when the billing country is the U.S. or Canada; otherwise, optional.)² ccCaptureService (O) ccCreditService (Required when the billing country is the U.S. or Canada; otherwise, optional.)^{1,2}	Comercio Latino and CyberSource through VisaNet: String (9) All other processors: String (10)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_state	State or province of the billing address. Use the State, Province, and Territory Codes for the United States and Canada. <i>CyberSource through VisaNet</i> Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (Required when the billing country is the U.S. or Canada; otherwise, optional.)² ccCaptureService (O) ccCreditService (Required when the billing country is the U.S. or Canada; otherwise, optional.)^{1,2}	String (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_street1	First line of the billing street address as it appears on the payment card issuer's records. <i>Atos</i> This field must not contain colons (:). <i>CyberSource through VisaNet</i> Important When you populate billing street address 1 and billing street address 2, CyberSource through VisaNet concatenates the two values. If the concatenated value exceeds 40 characters, CyberSource through VisaNet truncates the value at 40 characters before sending it to Visa and the issuing bank. Truncating this value affects AVS results and therefore might also affect risk decisions and chargebacks. Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (R)² ccCaptureService (O) ccCreditService (R)^{1,2} <i>JCN Gateway</i> Required for ccAuthService and ccCreditService when Decision Manager is included in the request. Otherwise, optional.	Atos: String (29) CyberSource through VisaNet: String (40) Moneris: String (50) Worldpay VAP: String (35) All other processors: String (60)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
billTo_street2	Additional address information. Example Attention: Accounts Payable <i>Atos</i> This field must not contain colons (:). <i>Chase Paymentech Solutions, FDC</i> <i>Compas, and TSYS Acquiring Solutions</i> This value is used for AVS. <i>CyberSource through VisaNet</i> Important When you populate billing street address 1 and billing street address 2, CyberSource through VisaNet concatenates the two values. If the concatenated value exceeds 40 characters, CyberSource through VisaNet truncates the value at 40 characters before sending it to Visa and the issuing bank. Truncating this value affects AVS results and therefore might also affect risk decisions and chargebacks. Credit card networks cannot process transactions that contain non-ASCII characters. CyberSource through VisaNet accepts and stores non-ASCII characters correctly and displays them correctly in reports. However, the limitations of the credit card networks prevent CyberSource through VisaNet from transmitting non-ASCII characters to the credit card networks. Therefore, CyberSource through VisaNet replaces non-ASCII characters with meaningless ASCII characters for transmission to the credit card networks.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	Atos: String (29) CyberSource through VisaNet: String (40) Moneris: String (50) Worldpay VAP: String (35) All other processors: String (60)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
businessRules_ declineAVSFlags	List of AVS codes that cause the request to be declined for AVS reasons. Use a space to separate the values in the list. Important To receive declines for the AVS code N, include the value N in the list.	ccAuthService (O)	String (255)
businessRules_ ignoreAVSResult	Flag for a sale request that indicates whether to allow the capture service to run even when the authorization receives an AVS decline. Possible values: ■ <code>true</code>: Ignore the results of AVS checking and run the capture service. ■ <code>false</code> (default): If the authorization receives an AVS decline, do not run the capture service. When the value of this field is <code>true</code> , the list in the businessRules_declineAVSFlags field is ignored.	ccAuthService (O)	String (5)
businessRules_ ignoreCVResult	Flag for a sale request that indicates whether to allow the capture service to run even when the authorization receives a CVN decline, as indicated by a ccAuthReply_cvCode value of D or N. Possible values: ■ <code>true</code>: Ignore the results of CVN checking and run the capture service. ■ <code>false</code> (default): If the authorization receives a CVN decline, do not run the capture service.	ccAuthService (O)	String (5)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_accountEncoderID	Identifier for the issuing bank that provided the customer's encoded account number. Contact your processor for the bank's ID. See "Encoded Account Numbers," page 137 .	ccAuthService(Required when processing encoded account numbers; otherwise, not used.) ccCreditService (Required when processing encoded account numbers; otherwise, not used.) ¹	String (3)
card_accountNumber	Customer's payment card number. <i>Encoded Account Numbers</i> When processing encoded account numbers, use this field for the encoded account number.	ccAuthService (R) ccCreditService (R) ¹	String with numbers only (20)
card_cardType	Type of card to authorize. See Appendix G, "Card Types," on page 466 for a list of valid values. To see which cards are supported for each processor, see "Payment Processors," page 32 . <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file ⁵ : ■ Record: CP01 TCR1 ■ Position: 70-71 ■ Field: Card ID	ccAuthService ccCreditService ¹ Important CyberSource strongly recommends that you send the card type even if it is optional for your processor and card type. Omitting the card type can cause the transaction to be processed with the wrong card type.	String (3)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_cvIndicator	Flag indicating whether a CVN code was sent. Possible values: 0 (default): CVN service not requested. CyberSource uses this default value when you do not include card_cvNumber in the request. 1 (default): CVN service requested and supported. CyberSource uses this default value when you include card_cvNumber in the request. 2: CVN on payment card is illegible. 9: CVN was not imprinted on payment card.	ccAuthService (O)	String with numbers only (1)
card_cvNumber	CVN. See "Card Verification Numbers (CVNs)," page 97, for a list of processors that support CVN. <i>Ingenico ePayments</i> Do not include this field when ccAuthService_commerceIndicator=recurring. Note Ingenico ePayments was previously called <i>Global Collect</i>.	ccAuthService (O)	String with numbers only (4)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_expirationMonth	Two-digit month in which the payment card expires. Format: MM. Possible values: 01 through 12. <i>Barclays and Streamline</i> For Maestro (UK Domestic) and Maestro (International) cards on Barclays and Streamline, this must be a valid value (01 through 12) but is not required to be a valid expiration date. In other words, an expiration date that is in the past does not cause CyberSource to reject the request. However, an invalid expiration date might cause the issuer to reject your request. <i>Encoded Account Numbers</i> For encoded account numbers (card_cardType=039), use 12 if there is no expiration date available.	ccAuthService (R)² ccCreditService (R)^{1,2}	String (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_expirationYear	Four-digit year in which the payment card expires. Format: YYYY. <i>Barclays and Streamline</i> For Maestro (UK Domestic) and Maestro (International) cards on Barclays and Streamline, this must be a valid value (1900 through 3000) but is not required to be a valid expiration date. In other words, an expiration date that is in the past does not cause CyberSource to reject the request. However, an invalid expiration date might cause the issuer to reject your request. <i>Encoded Account Numbers</i> For encoded account numbers (card_cardType=039), if there is no expiration date available, use 2021. <i>FDC Nashville Global and FDMS South</i> You can send in 2 digits or 4 digits. When you send in 2 digits, they must be the last 2 digits of the year.	ccAuthService (R)² ccCreditService (R)^{1,2}	FDC Nashville Global and FDMS South: String (See description) All other processors: String (4)
card_issueNumber	Number of times a Maestro (UK Domestic) card has been issued to the account holder. The card might or might not have an issue number. The number can consist of one or two digits, and the first digit might be a zero. When you include this value in your request, include exactly what is printed on the card. A value of 2 is different than a value of 02. Do not include the field, even with a blank value, if the card is not a Maestro (UK Domestic) card. Note The issue number is not required for Maestro (UK Domestic) transactions.	ccAuthService (O) ccCreditService (O)	String (5)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_productName	Name of the card product. Possible value: ■ BNDES This field is supported only for BNDES transactions on CyberSource through VisaNet. See "BNDES," page 129. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 115-120 ■ Field: Brazil Country Data	ccAuthService (O)	String (15)
card_startMonth	Month of the start of the Maestro (UK Domestic) card validity period. Do not include the field, even with a blank value, if the card is not a Maestro (UK Domestic) card. Format: MM. Possible values: 01 through 12. Note The start date is not required for Maestro (UK Domestic) transactions.	ccAuthService (O) ccCreditService (O)	String (2)
card_startYear	Year of the start of the Maestro (UK Domestic) card validity period. Do not include the field, even with a blank value, if the card is not a Maestro (UK Domestic) card. Format: YYYY. Note The start date is not required for Maestro (UK Domestic) transactions.	ccAuthService (O) ccCreditService (O)	String (4)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
card_usage	Indicates how to use the card for the requested transaction. Possible values: ■ C: Credit transaction ■ D: Debit transaction The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 51 ■ Field: Combination Card Transaction Identifier This field is supported only for Mastercard transactions in Brazil on CyberSource through VisaNet.	ccAuthService (O)	String (1)
cardTypeSelectionIndicator	Identifies how the card type was selected. Possible values: ■ 0: Card type was selected based on default acquirer settings. ■ 1: Customer selected the card type. See "Cartes Bancaires Cards," page 21. This field is supported only on Credit Mutuel-CIC.	ccAuthService (R)	String (1)
category_affiliate	Indicates an affiliate organization. You can use this information for selective analytics and reporting. Contact your processor to have your account enabled for this feature. This field is supported only for Worldpay VAP.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (25)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
category_campaign	Indicates a marketing campaign. You can use this information for selective analytics and reporting. Contact your processor to have your account enabled for this feature. This field is supported only for Worldpay VAP.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (25)
category_group	Indicates a merchant-defined category or group. You can use this information for selective analytics and reporting. Contact your processor to have your account enabled for this feature. This field is supported only for Worldpay VAP.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (25)
ccAuthReversalService_authRequestID	Request ID for the authorization that you want to reverse.	ccAuthReversal Service (R)	String (26)
ccAuthReversalService_authRequestToken	Value of the request token returned from a previous request for ccAuthService . The field is an encoded string that contains no confidential information, such as an account number or card verification number. The string can contain a maximum of 256 characters.	ccAuthReversal Service (O)	String (256)
ccAuthReversalService_reversalReason	Reason for the authorization reversal. Possible value: ■ 34: Suspected fraud CyberSource ignores this field for processors that do not support this value.	ccAuthReversal Service (O)	String (3)
ccAuthReversalService_run	Whether to include ccAuthReversalService in your request. Possible values: ■ <code>true</code> : Include the service in your request. ■ <code>false</code> (default): Do not include the service in your request.	ccAuthReversal Service (R)	String (5)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ aggregatorID	Value that identifies you as a payment aggregator. Get this value from the processor. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR6 - Position: 95-105 - Field: Market Identifier / Payment Facilitator ID <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: R for Mastercard aggregator transactions and for American Express aggregator authorizations; otherwise, not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (20) CyberSource through VisaNet with American Express: String (20) CyberSource through VisaNet with Mastercard: String (11) FDC Compass: String (20) FDC Nashville Global: String (15)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ aggregatorName	Your payment aggregator business name. See "Aggregator Support," page 117. <i>American Express Direct</i> The maximum length of the aggregator name depends on the length of the sub-merchant name. The combined length for both values must not exceed 36 characters. <i>CyberSource through VisaNet</i> With American Express, the maximum length of the aggregator name depends on the length of the sub-merchant name. The combined length for both values must not exceed 36 characters. The value for this field does not map to the TC 33 capture file⁵. <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: R for American Express aggregator authorizations; otherwise, not used. FDC Compass: R for Mastercard aggregator transactions; otherwise, not used. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (see description) CyberSource through VisaNet: String (see description) FDC Compass: String (37) FDC Nashville Global: String (12)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_authType	Authorization type. Possible values: ■ AUTOCAPTURE: automatic capture; see "Automatic Captures," page 40. ■ STANDARDCAPTURE: standard capture; see "Automatic Captures," page 40. ■ verbal: forced capture; see "Forced Captures," page 142. <i>Asia, Middle East, and Africa Gateway; Cielo; Comercio Latino; and CyberSource Latin American Processing</i> Set this field to AUTOCAPTURE and include it in a bundled request to indicate that you are requesting an automatic capture. If your account is configured to enable automatic captures, set this field to STANDARDCAPTURE and include it in a standard authorization or bundled request to indicate that you are overriding an automatic capture. For more information, see "Automatic Captures," page 40. <i>Forced Capture</i> Set this field to verbal and include it in the authorization request to indicate that you are performing a forced capture; therefore, you receive the authorization code outside the CyberSource system. For more information, see "Forced Captures," page 142.	ccAuthService (See description)	Comercio Latino: String (15) All other processors: String (11)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_billPayment	Flag indicating that this is a payment for a bill or for an existing contractual loan. See "Visa Bill Payments," page 227, and "Visa Debt Repayments," page 228, for lists of processors that support these features. This value is case sensitive. Possible values: ■ <code>true</code>: Bill payment or loan payment. ■ <code>false</code> (default): Not a bill payment or loan payment. Note For information about bill payments with Mastercard, see "Mastercard Bill Payments," page 167.	ccAuthService (O)	String (5)
ccAuthService_captureDate	Date on which you want the capture to occur. This field is supported only for CyberSource through VisaNet. Format: MMDD	ccAuthService (O)	String (4)
ccAuthService_cavv	Cardholder authentication verification value (CAVV). For the description and requirements, see "Payer Authentication," page 183.	ccAuthService	String (40)
ccAuthService_cavvAlgorithm	Algorithm used to generate the CAVV for Visa Secure or the UCAF authentication data for Mastercard Identity Check. For the description and requirements, see "Payer Authentication," page 183.	ccAuthService	String (1)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ commerceIndicator	Type of transaction. Some payment card companies use this information when determining discount rates. When you omit this field for Ingenico ePayments, the processor uses the default transaction type they have on file for you instead of the default value listed here. <i>Ingenico ePayments</i> Ingenico ePayments was previously called <i>Global Collect</i>. <i>Payer Authentication Transactions</i> For the possible values and requirements, see "Payer Authentication," page 183. <i>Other Types of Transactions</i> See Appendix I, "Commerce Indicators," on page 470.	ccAuthService (Required for payer authentication transactions; otherwise, optional.)	String (20)
ccAuthService_ cryptocurrencyPurchase	Indicates whether the payment is for the purchase of cryptocurrency. Possible values: ■ <code>true</code>: The payment is for the purchase of cryptocurrency. ■ <code>false</code> (default): The payment is not for the purchase of cryptocurrency. This field is supported only for Visa transactions on CyberSource through VisaNet.	ccAuthService (O)	String (5)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ deferredAuthIndicator	Indicates whether the authorization request was delayed because connectivity was interrupted. Possible values: ■ <code>true</code>: Deferred authorization. ■ <code>false</code> (default): Not a deferred authorization. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR0 ■ Position: 160-163 ■ Field: Message Reason Code This field is supported only on CyberSource through VisaNet.	ccAuthService (O)	String (5)
ccAuthService_ directoryServerTransaction ID	Transaction ID generated by the Mastercard directory server during authentication. For more information, see "Payer Authentication," page 183. This field is supported only on CyberSource through VisaNet and OmniPay Direct. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR7 ■ Position: 114-149 ■ Field: MC AVV Verification—Directory Server Transaction ID	ccAuthService (R for Mastercard Identity Check 2.0 transactions)	String (36)
ccAuthService_eciRaw	Raw electronic commerce indicator (ECI). For the description and requirements, see "Payer Authentication," page 183.	ccAuthService	String (2)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ firstRecurringPayment	Flag indicating whether this transaction is the first in a series of recurring payments. See "Recurring Payments," page 205. This field is supported only for Atos, FDC Nashville Global, and OmniPay Direct. <i>Atos and OmniPay Direct</i> Possible values: ■ Y: Yes, this is the first payment in a series of recurring payments. ■ N (default): No, this is not the first payment in a series of recurring payments. <i>FDC Nashville Global</i> Possible values: ■ TRUE: Yes, this is the first payment in a series of recurring payments. ■ FALSE (default): No, this is not the first payment in a series of recurring payments.	ccAuthService (O)	String (5)
ccAuthService_ leastCostRouting	Indicates whether to request that the processor use least-cost routing for the transaction. When the request includes this field, this value overrides the information in your CyberSource account. See "Least-Cost Routing (LCR)," page 164. Possible values: ■ true: Request that the processor use least-cost routing for the transaction. ■ false (default): Do not request that the processor use least-cost routing for the transaction. This field is supported only on Worldpay VAP.	ccAuthService (O)	String (5)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ lowValueExemption Indicator	Indicates whether the transaction is exempt from strong customer authentication (SCA) requirements in Europe because the payment amount is low. Possible values: ■ 0 (default): Not exempt. ■ 1: Exempt from SCA requirements because the payment amount is low. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR8 ■ Position: 126 ■ Field: Low Value Exemption Indicator This field is supported only in Europe on CyberSource through VisaNet.	ccAuthService (O)	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ overridePayment Details	Type of account that is being used when the value for the ccAuthService_overridePaymentMethod field is line of credit (LI) or prepaid card (PP). Possible values for line of credit: ■ AGRC: Visa Agro Custeio ■ AGRE: Visa Agro Electron ■ AGRI: Visa Agro Investimento ■ AGRO: Visa Agro Possible values for prepaid card: ■ VVA: Visa Vale Alimentacao ■ VVF: Visa Vale Flex ■ VVR: Visa Vale Refeicao This field is supported only for combo card transactions in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 44-47 ■ Field: Account Accessed	ccAuthService (R for combo card transactions in Brazil on CyberSource through VisaNet; otherwise, not used.)	String (4)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ overridePayment Method	Flag that specifies the type of account associated with the card. The cardholder provides this information during the payment process. Note Combo cards in Brazil contain credit and debit functionality in a single card. Visa systems use a credit bank identification number (BIN) for this type of card. Using the BIN to determine whether a card is debit or credit can cause transactions with these cards to be processed incorrectly. <i>Cielo and Comercio Latino</i> Possible values: ■ CR: Credit card ■ DB: Debit card <i>CyberSource through VisaNet</i> Possible values: ■ CH: Checking account ■ CR: Credit card account ■ LI: Line of credit or credit portion of combo card ■ PP: Prepaid card or prepaid portion of combo card ■ SA: Savings account (continued on next page)	ccAuthService (R for combo card transactions in Brazil on Cielo, Comercio Latino, and CyberSource through VisaNet; otherwise, not used.)	String (2)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ overridePaymentMethod (continued)	For combo card transactions with Mastercard in Brazil on CyberSource through VisaNet, the card_usage field is also supported. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP07 TCR0 - Positions: 42-43 - Field: Account Selection This field is supported only for transactions with Brazilian-issued combo cards on Cielo, Comercio Latino, and CyberSource through VisaNet.		
ccAuthService_paresStatus	Payer authentication response status. For the description and requirements, see "Payer Authentication," page 183 .	ccAuthService	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ partialAuthIndicator	Flag indicating whether the transaction is enabled for partial authorization. When the request includes this field, this value overrides the information in your CyberSource account. Possible values: ■ <code>true</code>: Enable the transaction for partial authorization. ■ <code>false</code>: Do not enable the transaction for partial authorization. See "Partial Authorizations," page 105. <i>CyberSource through VisaNet</i> To set the default for this field, contact CyberSource Customer Support. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR0 ■ Position: 164 ■ Field: Additional Authorization Indicators	ccAuthService (O)	String (5)
ccAuthService_ paSpecificationVersion	3D Secure version that was used to process the Mastercard Identity Check transaction. Examples: 1.0.2 or 2.0.0. For more information, see "Payer Authentication," page 183. This field is supported only on CyberSource through VisaNet and OmniPay Direct. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR7 ■ Position: 113 ■ Field: MC AVV Verification—Program Protocol	ccAuthService (R for Mastercard Identity Check transactions)	String (20)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ riskAnalysisExemption Indicator	Indicates whether the transaction is exempt from strong customer authentication (SCA) requirements in Europe because it is a low-risk transaction according to the Payments Service Directive 2/Regulatory Technical Standards (PSD2/RTS) regulations. Possible values: 0 (default): Not exempt. 1: Exempt from SCA requirements because the transaction is low risk. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR8 - Position: 127 - Field: Transaction Risk Analysis Exemption Indicator This field is supported only in Europe on CyberSource through VisaNet.	ccAuthService (O)	String (1)
ccAuthService_run	Whether to include ccAuthService in your request. Possible values: - true: Include the service in your request. - false (default): Do not include the service in your request.	ccAuthService (R)	String (5)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ secureCorporatePayment Indicator	Indicates whether the transaction is exempt from strong customer authentication (SCA) requirements in Europe because the payment is a secure corporate payment. Possible values: ■ 0 (default): Not exempt. ■ 1: Exempt from SCA requirements because the payment is a secure corporate payment. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR8 ■ Position: 129 ■ Field: Secure Corporate Payment Indicator This field is supported only in Europe on CyberSource through VisaNet.	ccAuthService (O)	String (1)
ccAuthService_ trustedMerchantExemption Indicator	Indicates whether the transaction is exempt from strong customer authentication (SCA) requirements in Europe because the merchant is trusted by the customer. Possible values: ■ 0 (default): Not exempt. ■ 1: Exempt from SCA requirements because the merchant is trusted by the customer. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR8 ■ Position: 128 ■ Field: Trusted Merchant Exemption Indicator This field is supported only in Europe on CyberSource through VisaNet.	ccAuthService (O)	String (1)
ccAuthService_ verbalAuthCode	Authorization code you received from an authorization that you performed outside the CyberSource system. See "Forced Captures," page 142.	ccAuthService (Required for a forced capture; otherwise, not used.)	String (6)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccAuthService_ veresEnrolled	Verification response enrollment status. For the description and requirements, see "Payer Authentication," page 183 .	ccAuthService	String (1)
ccAuthService_ verificationType	Type of customer verification data you are requesting. After receiving the authorization request, the issuer sends you the type of data you requested by means of a communication method or network that does not include CyberSource through VisaNet. Use the data to verify the customer's identity. See "Installment Payments," page 145 . Possible values: 01: Customer's personal data 02: Customer's age This field is supported only for Mastercard installment payments in Poland on CyberSource through VisaNet.	ccAuthService (O)	String (2)
ccAuthService_ xid	Transaction identifier. For the description and requirements, see "Payer Authentication," page 183 .	ccAuthService	String (40)
ccCaptureService_ aggregatorID	Value that identifies you as a payment aggregator. Get this value from the processor. See "Aggregator Support," page 117 . <i>FDC Compass</i> This value must consist of uppercase characters.	ccCaptureService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (20) FDC Compass: String (20) FDC Nashville Global: String (15)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCaptureService_ aggregatorName	Your payment aggregator business name. See "Aggregator Support," page 117. <i>American Express Direct</i> The maximum length of the aggregator name depends on the length of the sub-merchant name. The combined length for both values must not exceed 36 characters. <i>FDC Compass</i> This value must consist of uppercase characters.	ccCaptureService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: not used. FDC Compass: R for Mastercard aggregator transactions; otherwise, not used. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (see description) FDC Compass: String (37) FDC Nashville Global: String (12)
ccCaptureService_ authRequestID	Value of the request ID returned from a previous ccAuthReply .	ccCaptureService Required unless ccAuthService and ccCaptureService are both called in the same request.	String (26)
ccCaptureService_ authRequestToken	Value of the request token returned from a previous request for ccAuthService. The field is an encoded string that contains no confidential information, such as an account number or card verification number. The string can contain a maximum of 256 characters.	ccCaptureService (Required for Atos; otherwise, optional.) <i>Atos</i> When you request the authorization and capture services together, the capture request does not require a request token.	String (256)
ccCaptureService_authType	Authorization type. When the transaction contains a verbally authorized transaction, this field must contain the value <code>verbal</code>.	ccCaptureService (O)	String (6)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCaptureService_ dpdeBillingMonth	Dynamic payment descriptor extension (DPDE) that specifies the month for which you are billing the customer. Depending on your business model, you might bill for a service that has already been provided, such as a telephone service, or you might bill for a service that is going to be provided, such as a subscription to investment information. This value lets the customer know which month the payment is for. Format: YYMM This field is supported only for JCN Gateway and is not supported for all Japanese acquirers.	ccCaptureService (O)	String (4)
ccCaptureService_posData	Point-of-sale data. On FDMS South, this field is required for verbal authorizations and forced captures with the American Express card type to comply with the CAPN requirements: - Forced capture: Obtain the value for this field from the authorization response. - Verbal authorization: You cannot obtain a value for this field so CyberSource uses the default value. The default value is generated by CyberSource based on various factors of the transaction such as e-commerce or not, card present or not, and swiped or keyed. See "Verbal Authorizations," page 101.	ccCaptureService (See the field description.)	String (12)
ccCaptureService_run	Whether to include ccCaptureService in your request. Possible values: <code>true</code>: Include the service in your request. <code>false</code> (default): Do not include the service in your request.	ccCaptureService (R)	String (5)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCaptureService_ sequence	Capture number when requesting multiple partial captures for one authorization. Used along with ccCaptureService_totalCount to track which capture is being processed. For example, the second of five captures would be passed to CyberSource as ccCaptureService_sequence = 2 and ccCaptureService_totalCount = 5. For the list of processors that support this field, see "Special Request Fields for Multiple Partial Captures," page 72.	ccCaptureService (See "Special Request Fields for Multiple Partial Captures," page 72.)	Integer (2)
ccCaptureService_ totalCount	Total number of captures when requesting multiple partial captures for one authorization. Used along with ccCaptureService_sequence to track which capture is being processed. For example, the second of five captures would be passed to CyberSource as ccCaptureService_sequence = 2 and ccCaptureService_totalCount = 5. For the list of processors that support this field, see "Special Request Fields for Multiple Partial Captures," page 72.	ccCaptureService (See "Special Request Fields for Multiple Partial Captures," page 72.)	Integer (2)
ccCaptureService_ transactionID	Transaction ID (TID). On FDMS South, this field is required for verbal authorizations and forced captures with the American Express card type to comply with the CAPN requirements: - Forced capture: Obtain the value for this field from the authorization response. - Verbal authorization: You cannot obtain a value for this field so CyberSource uses the default value of 0000000000000000 (15 zeros). See "Verbal Authorizations," page 101, for important information about using this default value.	ccCaptureService (See the field description.)	String (15)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCaptureService_ verbalAuthCode	Verbally received authorization code.	ccCaptureService (O)	JCN Gateway: String (7) All other processors: String (6)
ccCheckStatusService_ authRequestID	Request ID for the authorization that you want to check.	ccCheckStatusService (R)	String (26)
ccCheckStatusService_run	Whether to include ccCheckStatusService in your request. Possible values: ■ <code>true</code>: Include the service in your request. ■ <code>false</code> (default): Do not include the service in your request.	ccCheckStatusService (R)	String (5)
ccCreditService_ aggregatorID	Value that identifies you as a payment aggregator. Get this value from the processor. See "Aggregator Support," page 117 . <i>FDC Compass</i> This value must consist of uppercase characters.	ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (20) FDC Compass: String (20) FDC Nashville Global: String (15)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCreditService_ aggregatorName	Your payment aggregator business name. See "Aggregator Support," page 117. <i>American Express Direct</i> The maximum length of the aggregator name depends on the length of the sub-merchant name. The combined length for both values must not exceed 36 characters. <i>FDC Compass</i> This value must consist of uppercase characters.	ccCaptureService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: not used. FDC Compass: R for Mastercard aggregator transactions; otherwise, not used. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (see description) FDC Compass: String (37) FDC Nashville Global: String (12)
ccCreditService_billPayment	Flag indicating whether this is a credit for a bill that the customer paid with a Visa card. See "Visa Bill Payments," page 227, for a list of processors that support bill payments with Visa. This value is case sensitive. Possible values: ■ <code>true</code>: Credit for a bill payment. ■ <code>false</code> (default): Not a credit for a bill payment	ccCreditService (O)	String (5)
ccCreditService_ captureRequestID	Value of the request ID returned from a previous request for ccCaptureService. Creates a follow-on credit by linking the credit to the previous capture. When you send this field, you do not need to send several other credit request fields. See "Crediting a Payment," page 78, for a description of follow-on credits.	ccCreditService (O)	String (26)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCreditService_ captureRequestToken	Value of the request token returned from a previous request for ccCaptureService . The field is an encoded string that contains no confidential information, such as an account number or card verification number. The string can contain a maximum of 256 characters.	ccCreditService (Required for Atos; otherwise, optional)	String (256)
ccCreditService_ commerceIndicator	Type of transaction. Use with stand-alone credits. Some payment card companies use this information when determining discount rates. Possible values: ■ <code>internet</code> (default) ■ <code>moto</code> ■ <code>recurring</code> ■ <code>recurring_internet</code> For details about these values, see Appendix I, "Commerce Indicators," on page 470 .	ccCreditService (O)	String (13)
ccCreditService_ dpdeBillingMonth	Dynamic payment descriptor extension (DPDE) that specifies the month for which you are billing the customer. Depending on your business model, you might bill for a service that has already been provided, such as a telephone service, or you might bill for a service that is going to be provided, such as a subscription to investment information. This value lets the customer know which month the payment is for. Format: YYMM This field is supported only for JCN Gateway and is not supported for all Japanese acquirers.	ccCreditService (O)	String (4)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCreditService_ overridePayment Details	Type of account that is being used when the value for the ccCreditService_overridePaymentMethod field is line of credit (LI) or prepaid card (PP). Possible values for line of credit: ■ AGRC: Visa Agro Custeio ■ AGRE: Visa Agro Electron ■ AGRI: Visa Agro Investimento ■ AGRO: Visa Agro Possible values for prepaid card: ■ VVA: Visa Vale Alimentacao ■ VVF: Visa Vale Flex ■ VVR: Visa Vale Refeicao This field is supported only for combo card transactions in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 44-47 ■ Field: Account Accessed	ccCreditService (R for combo card transactions in Brazil on CyberSource through VisaNet; otherwise, not used.)	String (4)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCreditService_ overridePayment Method	Flag that specifies the type of account associated with the card. The cardholder provides this information during the payment process. Note Combo cards in Brazil contain credit and debit functionality in a single card. Visa systems use a credit bank identification number (BIN) for this type of card. Using the BIN to determine whether a card is debit or credit can cause transactions with these cards to be processed incorrectly. Possible values: ■ CH: Checking account ■ CR: Credit card account ■ LI: Line of credit or credit portion of combo card ■ PP: Prepaid card or prepaid portion of combo card ■ SA: Savings account For combo card transactions with Mastercard in Brazil on CyberSource through VisaNet, the card_usage field is also supported. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Positions: 42-43 ■ Field: Account Selection This field is supported only for transactions with Brazilian-issued combo cards on CyberSource through VisaNet.	ccCreditService (R for combo card transactions in Brazil on CyberSource through VisaNet. Otherwise, not used.)	String (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ccCreditService_run	Whether to include ccCreditService in your request. Possible values: ■ <code>true</code>: Include the service in your request. ■ <code>false</code> (default): Do not include the service in your request.	ccCreditService (R)	String (5)
cclIncrementalAuthService_authRequestID	Request ID for the original authorization that you want to increment. See "Incremental Authorizations," page 45 .	cclIncrementalAuthService (R)	String (26)
cclIncrementalAuthService_duration	Number of days that the customer plans to stay at the lodging. For a lodging deposit, the value for this field specifies the number of days covered by the advance payment. See "Incremental Authorizations," page 45 . Possible values: 01 through 99.	cclIncrementalAuthService (O)	String (2)
cclIncrementalAuthService_run	Whether to include cclIncrementalAuthService in your request. Possible values: ■ <code>true</code>: Include the service in your request. ■ <code>false</code> (default): Do not include the service in your request.	cclIncrementalAuthService (R)	String (5)
cclIncrementalAuthService_transactionLocalDateTime	Local date and time at your physical location. Include both the date and time in this field or leave it blank. Format: YYYYMMDDhhmmss where: ■ YYYY = year ■ MM = month ■ DD = day ■ hh = hour ■ mm = minutes ■ ss = seconds	cclIncrementalAuthService (O)	String (14)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
dcc_dccIndicator	Flag indicating that DCC is being used for the transaction. Set this field to 1. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133.	ccAuthService (R for DCC transactions)	String (1)
dcc_referenceNumber	Unique identifier generated by the DCC provider. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133.	ccAuthService (O)	String (60)
debtIndicator	Flag indicating whether this is a payment towards an existing contractual loan. See "Visa Debt Repayments," page 228 , for a list of processors that support this feature. Possible values: ■ <code>true</code>: Loan payment ■ <code>false</code> (default): Not a loan payment	ccAuthService (O) ccCreditService (O)	String (5)
eligibilityInquiry	Indicates whether the authorization request is a Crediaro eligibility inquiry. See "Installment Payments on CyberSource through VisaNet," page 150. Set the value for this field to <code>Crediaro</code> . This field is supported only for Crediaro installment payments in Brazil on CyberSource through VisaNet.	ccAuthService (R for Crediaro eligibility inquiry; otherwise, not used.)	String (9)
extendedCreditTotalCount	Number of months over which the cardholder can pay for the purchase. You can use this field when offering extended credit to a cardholder at a retail location. The cardholder provides this value. The issuer pays you for the purchase in one payment, and then the cardholder pays the issuer in the number of monthly payments specified by this value. Note This field is supported only for acquirers in South Africa and only for CyberSource through VisaNet.	ccAuthService (O)	String (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_additionalCosts	Additional costs charged by the issuer to fund the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 128-139 ■ Field: Total Other Costs	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ additionalCostsPercentage	Additional costs divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 140-143 ■ Field: Percent of Total Other Costs	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_amount	Amount for the current installment payment. This field is supported only on CyberSource through VisaNet. <i>Brazil</i> In Brazil, this field is supported for Crediario installment payments. See "Installment Payments on CyberSource through VisaNet," page 150. When you do not include this field in a request for a Crediario installment payment, CyberSource sends a value of 0 to the processor. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 26-37 ■ Field: Amount of Each Installment <i>Georgia (Sakartvelo)</i> In Georgia, this field is supported for installment payments. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR5 ■ Position: 63-74 ■ Field: Mastercard Subsequent Installment Amount	ccAuthService (R for Crediario eligibility inquiries in Brazil and installment payments in Georgia; otherwise, O) ccCaptureService (O)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_amountFunded	Amount funded. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 48-59 - Field: Total Amount Funded	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)
installment_amountRequested Percentage	Amount requested divided by the amount funded. For example: - A value of 90.0 specifies 90%. - A value of 93.7 specifies 93.7%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 60-63 - Field: Percent of Amount Requested	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ annualFinancingCost	Annual cost of financing the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 158-164 - Field: Annual Total Cost of Financing	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (7)
installment_ annualInterestRate	Annual interest rate. For example: - A value of 1.0 specifies 1%. - A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 151-157 - Field: Annual Interest Rate	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (7)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_downPayment	Down payment. This field is supported only for airline transactions on Getnet.	ccAuthService (O)	String (12)
installment_expenses	Expenses charged by the issuer to fund the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file ⁵ : ■ Record: CP01 TCR9 ■ Position: 64-75 ■ Field: Total Expenses	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ expensesPercentage	Expenses divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 76-79 ■ Field: Percent of Total Expenses	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_fees	Fees charged by the issuer to fund the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 80-91 ■ Field: Total Fees	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_feesPercentage	Fees divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 92-95 ■ Field: Percent of Total Fees	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)
installment_firstInstallmentAmount	Amount of the first installment payment. This field is supported only for Mastercard installment payments in Georgia (Sakartvelo) on CyberSource through VisaNet. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR5 ■ Position: 23-34 ■ Field: Amount of Each Installment	ccAuthService (R for installment payments in Georgia; otherwise, not used)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ firstInstallmentDate	Date of the first installment payment. Format: YYMMDD. When you do not include this field, CyberSource sends a string of six zeros (000000) to the processor. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 42-47 - Field: Date of First Installment	ccAuthService (R for Crediario eligibility inquiry; otherwise, O.) ccCaptureService (O)	String (6)
installment_frequency	Frequency of the installment payments. When you do not include this field in a request for a Crediario installment payment, CyberSource sends a space character to the processor. This field is supported only for CyberSource through VisaNet. Possible values: - B : Biweekly - M : Monthly - W : Weekly See "Installment Payments on CyberSource through VisaNet," page 150. For Crediario installment payments, the value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 41 - Field: Installment Frequency	ccAuthService (R for Crediario eligibility inquiries; otherwise, O) ccCaptureService (O)	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ gracePeriodDuration	Grace period requested by the customer before the first installment payment is due. When you include this field in a request, you must also include the grace period duration type field. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR5 - Position: 100-101 - Field: Mastercard Grace Period Details This field is supported only for Mastercard installment payments in Brazil and Greece on CyberSource through VisaNet.	ccAuthService (R for Mastercard installment payments in Greece and O for Mastercard installment payments in Brazil; otherwise, not used)	String (2)
installment_ gracePeriodDurationType	Unit for the requested grace period duration. See "Installment Payments on CyberSource through VisaNet," page 150. Possible values: - D: Days - W: Weeks - M: Months The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR5 - Position: 99 - Field: Mastercard Grace Period Details This field is supported only for Mastercard installment payments in Brazil and Greece on CyberSource through VisaNet.	ccAuthService (R when the grace period duration field is included; otherwise, not used.)	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_insurance	Insurance charged by the issuer to fund the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 112-123 ■ Field: Total Insurance	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ insurancePercentage	Insurance costs divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 124-127 ■ Field: Percent Of Total Insurance	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)
installment_invoiceData	Invoice information that you want to provide to the issuer. This value is similar to a tracking number and is the same for all installment payments for one purchase. This field is supported only for installment payments with Mastercard on CyberSource through VisaNet in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 51-70 ■ Field: Purchase Identification	ccAuthService (O) ccCaptureService (O)	String (20)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ monthlyInterestRate	Monthly interest rate. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported only for installment payments on CyberSource through VisaNet. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 144-150 ■ Field: Monthly Interest Rate <i>Georgia (Sakartvelo)</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR5 ■ Position: 41-45 ■ Field: Mastercard Interest Rate	ccCaptureService (R for Crediario installment payments in Brazil and for installment payments in Georgia)	String (7)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_paymentType	Payment plan for the installments. This field is supported only for installment payments on CyberSource through VisaNet. See "Installment Payments on CyberSource through VisaNet," page 150. Possible values: ■ 0 (default): Regular installment. This value is not allowed for airline transactions. ■ 1: Installment payment with down payment. ■ 2: Installment payment without down payment. This value is supported only for airline transactions. ■ 3: Installment payment; down payment and boarding fee will follow. This value is supported only for airline transactions. ■ 4: Down payment only; regular installment payment will follow. ■ 5: Boarding fee only. This value is supported only for airline transactions. <i>Brazil</i> In Brazil, this field is supported only for installment payments with Visa. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR1 ■ Position: 9 ■ Field: Merchant Installment Supporting Information	ccAuthService (R for installment payments in all countries except Brazil, Croatia, Georgia, and Greece; O for installment payments in Brazil; otherwise, not used.) ccCaptureService (R for installment payments in all countries except Brazil, Croatia, Georgia, and Greece; O for installment payments in Brazil; otherwise, not used.)	String (1)

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- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_paymentType (continued)	<i>Rest of World</i> In all countries except Brazil, Croatia, Georgia, and Greece, this field is supported only for installment payments with Mastercard. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR5 ■ Position: 35-37 ■ Field: Installment Payment Number		
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_planType	<i>American Express Direct, Cielo, and CyberSource Latin American Processing</i> Flag indicating the type of funding for the installment plan associated with the payment. Possible values: ■ 1 : Merchant-funded installment plan ■ 2 : Issuer-funded installment plan When you do not include this field in the request, CyberSource uses the value in your CyberSource account. To change the value in your CyberSource account, contact CyberSource Customer Service. See "Installment Payments," page 145. <i>CyberSource through VisaNet with American Express</i> American Express-defined code that indicates the type of installment plan for this transaction. Contact American Express for: ■ Information about the kinds of installment plans that American Express provides ■ Values for this field See "Installment Payments on CyberSource through VisaNet," page 150. For installment payments with American Express in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR3 ■ Position: 5-6 ■ Field: Plan Type	ccAuthService (R for installment payments with Visa or Mastercard on CyberSource through VisaNet in Brazil; otherwise, optional) ccCaptureService (R for installment payments with Visa or Mastercard on CyberSource through VisaNet in Brazil; otherwise, optional)	CyberSource through VisaNet: String (2) All other processors: String (1)

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- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_planType (continued)	<i>CyberSource through VisaNet with Visa or Mastercard</i> Flag indicating the type of funding for the installment plan associated with the payment. Possible values: ■ 1 or 01: Merchant-funded installment plan ■ 2 or 02: Issuer-funded installment plan ■ 3 or 03: Co-branded merchant financed—only with Mastercard in Brazil ■ 4 or 04: Issuer merchant co-financed—only with Mastercard in Brazil ■ 43: Crediario installment plan—only with Visa in Brazil See "Installment Payments on CyberSource through VisaNet," page 150. For installment payments with Visa in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR1 ■ Position: 5-6 ■ Field: Installment Type For all other kinds of installment payments, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR5 ■ Position: 39-40 ■ Field: Installment Plan Type		

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_sequence	Installment number when making payments in installments. Used along with installment_totalCount to track which payment is being processed. For example, the second of 5 payments would be passed to CyberSource as installment_sequence = 2 and installment_totalCount = 5. See "Installment Payments," page 145. <i>Chase Paymentech Solutions and FDC Compass</i> This field is optional because this value is required in the merchant descriptors. See Merchant Descriptors Using the Simple Order API. <i>CyberSource through VisaNet</i> When you do not include this field in a request for a Crediario installment payment, CyberSource sends a value of 0 to the processor. For Crediario installment payments, the value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR9 - Position: 38-40 - Field: Installment Payment Number	ccAuthService: - Chase Paymentech Solutions and FDC Compass: O. - CyberSource Latin American Processing in Brazil: Not used. - CyberSource through VisaNet: R for Crediario eligibility inquiries; otherwise, O. - All other processors: R for installment payments. ccCaptureService (R for Crediario eligibility inquiries on CyberSource through VisaNet; otherwise, not used.)	Integer (2)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_taxes	Taxes collected by the issuer to fund the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 96-107 ■ Field: Total Taxes	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_ taxesPercentage	Taxes divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 108-111 ■ Field: Percent of Total Taxes	ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (4)
installment_totalAmount	Total amount of the loan that is being paid in installments. This field is supported only for CyberSource through VisaNet. See "Installment Payments," page 145.	ccAuthService (R for Crediario eligibility inquiries and installment payments in Brazil on CyberSource through VisaNet) ccCaptureService (R for Crediario installment payments in Brazil on CyberSource through VisaNet)	String (12)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_totalCount	Total number of installments when making payments in installments. See "Installment Payments," page 145. <i>Chase Paymentech Solutions and FDC Compass</i> This field is optional because this value is required in the merchant descriptors. See Merchant Descriptors Using the Simple Order API. <i>American Express Direct, Cielo, and Comercio Latino</i> This value is the total number of installments that you approved. <i>CyberSource Latin American Processing in Brazil</i> This value is the total number of installments that you approved. The default is 1. <i>All Other Processors</i> This value is used along with installment_sequence to track which payment is being processed. For example, the second of 5 payments would be passed to CyberSource as installment_sequence = 2 and installment_totalCount = 5. <i>CyberSource through VisaNet in Brazil</i> For installment payments with Mastercard and for Crediario installment payments with Visa, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR9 ■ Position: 23-25 ■ Field: Number of Installments	ccAuthService ccCaptureService Chase Paymentech Solutions, CyberSource Latin American Processing, and FDC Compass: O. CyberSource through VisaNet in Brazil: R for Visa and for Crediario eligibility inquiries; otherwise, O. CyberSource through VisaNet in all countries except Brazil: R in Georgia and Greece; otherwise, O. All other processors: R for installment payments.	Integer (2)

(continued on next page)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
installment_totalCount (continued)	For installment payments with American Express or with Visa, the value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP07 TCR3 - Position: 7-8 - Field: Number of Installments <i>CyberSource through VisaNet in All Countries Except Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR5 - Position: 20-22 - Field: Number of Installments		
invoiceHeader_amexDataTAA1	Four Transaction Advice Addendum (TAA) fields. These fields are used to display descriptive information about a transaction on the customer's American Express card statement. When you send TAA fields, start with invoiceHeader_amexDataTAA1, then ...TAA2, and so on. Skipping a TAA field causes subsequent TAA fields to be ignored. To use these fields, contact CyberSource Customer Support to have your account enabled for this feature. For information about merchant descriptors, including which processors support this field, see "Merchant Descriptors," page 169. These fields are frequently used for Level II transactions. See Level II and Level III Processing Using the Simple Order API.	ccCaptureService (O)	String (40)
invoiceHeader_amexDataTAA2		ccCreditService (O)	
invoiceHeader_amexDataTAA3			
invoiceHeader_amexDataTAA4			
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ businessApplicationID	Type of transaction. For a list of possible values, see Appendix F, "Business Application Identifiers," on page 465. This field is a <i>pass-through</i>, which means that CyberSource does not verify the value or modify it in any way before sending it to the processor. When the request includes this field, this value overrides the information in your CyberSource account. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR8 ■ Position: 108-109 ■ Field: Business Application Identifier (BAI)	ccAuthService (O)	String (2)
invoiceHeader_ customData_1	Free form data about the transaction. CyberSource forwards this value to your processor, which then forwards the value to your acquirer during settlement. When you send this field for both the authorization and capture services, CyberSource forwards the value sent in the capture request. This field is supported only on FDC Nashville Global.	ccAuthService (O) ccCaptureService (O)	String (32)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ salesOrganizationID	Company ID assigned to an independent sales organization. Get this value from Mastercard. See "Aggregator Support," page 117 . <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file ⁵ : ■ Record: CP01 TCR6 ■ Position: 106-116 ■ Field: Independent Sales Organization ID	ccAuthService (Required for Mastercard aggregator transactions on CyberSource through VisaNet; otherwise, not used.)	Nonnegative integer (11)
invoiceHeader_ submerchantCity	Sub-merchant's city. See "Aggregator Support," page 117 . <i>CyberSource through VisaNet</i> The value for this field does not map to the TC 33 capture file ⁵ . <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: R for American Express aggregator authorizations; otherwise, not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (14) CyberSource through VisaNet: String (14) FDC Compass: String (21) FDC Nashville Global: String (11)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantCountry	Sub-merchant's country. Use the two-character ISO Standard Country Codes. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> The value for this field does not map to the TC 33 capture file⁵. <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for American Express aggregator authorizations; otherwise, not used. FDC Compass: O for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	String (3)
invoiceHeader_ submerchantEmail	Sub-merchant's email address. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> With American Express, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCRB ■ Position: 25-64 ■ Field: American Express Seller E-mail Address	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for all aggregator transactions with American Express; otherwise, not used. FDC Compass: O for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (40) CyberSource through VisaNet: String (40) FDC Compass: String (40) FDC Nashville Global: String (19)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantID	The ID you assigned to your sub-merchant. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> With American Express, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCRB ■ Position: 65-84 ■ Field: American Express Seller ID With Mastercard, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR6 ■ Position: 117-131 ■ Field: Sub-Merchant ID <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: ■ O for all American Express aggregator transactions; ■ R for all Mastercard aggregator authorizations; ■ otherwise, not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.W	American Express Direct: String (20) CyberSource through VisaNet with American Express: String (20) CyberSource through VisaNet with Mastercard: String (15) FDC Compass: String (20) FDC Nashville Global: String (14)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_submerchant MerchantID	Unique identifier assigned by the payment card company to the sub-merchant. See " Aggregator Support ," page 117.	ccAuthService American Express Direct: not used. CyberSource through VisaNet: not used. FDC Compass: not used. FDC Nashville Global: O for American Express aggregator authorizations; otherwise, not used.	String (15)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantName	Sub-merchant's business name. See "Aggregator Support," page 117. <i>American Express Direct</i> The maximum length of the sub-merchant name depends on the length of the aggregator name. The combined length for both values must not exceed 37 characters. <i>CyberSource through VisaNet</i> With American Express, the maximum length of the sub-merchant name depends on the length of the aggregator name. The combined length for both values must not exceed 37 characters. The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCRB - Position: 109-146 - Field: American Express Seller Name <i>FDC Compass</i> This value must consist of uppercase characters. <i>FDC Nashville Global</i> With Mastercard, the maximum length of the sub-merchant name depends on the length of the aggregator name: - If aggregator name length is 1 through 3, maximum sub-merchant name length is 21. - If aggregator name length is 4 through 7, maximum sub-merchant name length is 17. - If aggregator name length is 8 through 12, maximum sub-merchant name length is 12.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: R for American Express aggregator authorizations; otherwise, not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (see description) CyberSource through VisaNet: String (see description) FDC Compass with American Express: String (19) FDC Compass with Mastercard: String (37) FDC Nashville Global with American Express: String (12) FDC Nashville Global with Mastercard: String (see description)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantPostalCode	Partial postal code for the sub-merchant's address. See "Aggregator Support," page 117 . <i>CyberSource through VisaNet</i> The value for this field does not map to the TC 33 capture file ⁵ . <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for American Express aggregator authorizations; otherwise, not used. FDC Compass: O for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (10) CyberSource through VisaNet: String (10) FDC Compass: String (15) FDC Nashville Global: String (9)
invoiceHeader_ submerchantRegion	Sub-merchant's region. See "Aggregator Support," page 117 . Example NE indicates that the sub-merchant is in the northeast region.	ccAuthService American Express Direct: not used. CyberSource through VisaNet: not used. FDC Compass: not used. FDC Nashville Global: O for all aggregator authorizations; otherwise, not used.	String (3)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantState	Sub-merchant's state or province. Use the State, Province, and Territory Codes for the United States and Canada. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR4 - Position: 164-166 - Field: Region Code <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for American Express aggregator authorizations; otherwise, not used. FDC Compass: O for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	String (3)
invoiceHeader_ submerchantStreet	First line of the sub-merchant's street address. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> The value for this field does not map to the TC 33 capture file⁵. <i>FDC Compass</i> This value must consist of uppercase characters.	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: R for American Express aggregator authorizations; otherwise, not used. FDC Compass: O for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (29) CyberSource through VisaNet: String (29) FDC Compass: String (38) FDC Nashville Global: String (25)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
invoiceHeader_ submerchantTelephone Number	Sub-merchant's telephone number. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> With American Express, the value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCRB - Position: 5-24 - Field: American Express Seller Telephone Number <i>FDC Compass</i> This value must consist of uppercase characters. Use one of these recommended formats: NNN-NNN-NNNN NNN-AAAAAAA	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for all aggregator transactions with American Express; otherwise, not used. FDC Compass: R for all aggregator transactions. FDC Nashville Global: R for all aggregator transactions.	American Express Direct: String (20) CyberSource through VisaNet: String (20) FDC Compass: String (13) FDC Nashville Global: String (10)
issuer_additionalData	Data defined by the issuer. For more information, see Appendix N, "Formats for Discretionary Data," on page 480. This field is supported only for Visa and Mastercard on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP08 TCR1 - Position: 9-168 - Field: Free Form Text	ccAuthService (O) ccAuthReversalService (O) ccCaptureService (O)	String (255) For an authorization, the maximum length for this value is 255 characters. In the capture file, the value is truncated at 161 characters.

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
item_#_productCode	Type of product. This value is used to determine the category that the product is in: electronic, handling, physical, service, or shipping. The default value is <code>default</code> . See Table 74, "Product Codes," on page 493 for a list of valid values. For ccAuthService , when you set this field to a value other than <code>default</code> or any of the values related to shipping and/or handling, the item_#_quantity , item_#_productName , and item_#_productSKU fields are required. See "Numbered Elements," page 240 .	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (255)
item_#_productName	For ccAuthService and ccCaptureService , this field is required when item_#_productCode is not <code>default</code> or one of the values related to shipping and/or handling. See "Numbered Elements," page 240 .	ccAuthService (See the field description.) ccCaptureService (See the field description.)	String (255)
item_#_productSKU	Identification code for the product. For ccAuthService and ccCaptureService , this field is required when item_#_productCode is not <code>default</code> or one of the values related to shipping and/or handling. See "Numbered Elements," page 240 .	ccAuthService (See the field description.) ccCaptureService (See the field description.)	String (255)
item_#_quantity	The default is 1. For ccAuthService and ccCaptureService , this field is required when item_#_productCode is not <code>default</code> or one of the values related to shipping and/or handling. See "Numbered Elements," page 240 .	ccAuthService (See the field description.) ccAuthReversal Service (O) ccCaptureService (See the field description.) ccCreditService (O)	Integer (10)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
item_#_taxAmount	Total tax to apply to the product. This value cannot be negative. The tax amount and the unit price must be in the same currency. The tax amount field is additive. The following example uses a two-exponent currency such as USD: - You include the following items in your request: <pre> item_0_unitPrice=10.00 item_0_quantity=1 item_0_taxAmount=0.80 item_1_unitPrice=20.00 item_1_quantity=1 item_1_taxAmount=1.60 </pre> - The total amount authorized will be 32.40, not 30.00 with 2.40 of tax included. If you want to include the tax amount and also request the taxService service, see Tax Calculation Service Using the Simple Order API. This field is frequently used for Level II and Level III transactions. See Level II and Level III Processing Using the Simple Order API. See "Numbered Elements," page 240.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (15)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
item_#_unitPrice	Per-item price of the product. This value cannot be negative. You can include a decimal point (.), but you cannot include any other special characters. CyberSource truncates the amount to the correct number of decimal places. See "Numbered Elements," page 240. Important Some processors have specific requirements and limitations, such as maximum amounts and maximum field lengths. This information is covered in: ■ Merchant Descriptors Using the Simple Order API ■ Table 20, "Capture Information for Specific Processors," on page 64 ■ Table 24, "Credit Information for Specific Processors," on page 81 <i>DCC with a Third-Party Provider</i> Set this field to the converted amount that was returned by the DCC provider. You must include either this field or purchaseTotals_grandTotalAmount in your request. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133. <i>FDMS South</i> If you accept IDR or CLP currencies, see the entry for FDMS South in Merchant Descriptors Using the Simple Order API. <i>Zero Amount Authorizations</i> If your processor supports zero amount authorizations, you can set this field to 0 for the authorization to check if the card is lost or stolen. See "Zero Amount Authorizations," page 229.	ccAuthService ³ ccAuthReversalService ³ ccCaptureService ³ ccCreditService ³	String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
jpo_ businessNameAlphanumeric	Business name in alphanumeric characters. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (O) ccCaptureService (O on JCN Gateway; not used on CyberSource through VisaNet) ccCreditService (O on JCN Gateway; not used on CyberSource through VisaNet)	String (25)
jpo_ businessNameJapanese	Business name in Japanese characters. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (O) ccCaptureService (O on JCN Gateway; not used on CyberSource through VisaNet) ccCreditService (O on JCN Gateway; not used on CyberSource through VisaNet)	String (25)
jpo_businessNameKatakana	Business name in Katakana characters. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (O) ccCaptureService (O on JCN Gateway; not used on CyberSource through VisaNet) ccCreditService (O on JCN Gateway; not used on CyberSource through VisaNet)	String (25)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
jpo_firstBillingMonth	First billing month for a transaction that uses installment payments. Possible values: 01 through 12. When you do not include this field in a request that uses installment payments for one of the Japanese payment options, CyberSource sets the first billing month to the calendar month that follows the transaction date. See "Japanese Payment Options," page 162. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (O when jpo_paymentMethod is 4 or 6; otherwise, not used.)	Integer (2)
jpo_installments	Japanese payment option installments: Number of installment payments. See "Japanese Payment Options," page 162. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (Required when jpo_paymentMethod is 4 or 6; otherwise, not used.) ccCaptureService (Required on JCN Gateway when jpo_paymentMethod is 4 or 6; otherwise, not used.) ccCreditService (Required on JCN Gateway when jpo_paymentMethod is 4 or 6; otherwise, not used.)	Integer (2)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
jpo_paymentMethod	Japanese payment option payment method: type of payment option. Possible values: ■ 1 (default): Single payment ■ 2: Bonus payment ■ 4: Installment payment ■ 5: Revolving payment ■ 6: Combination of bonus payment and installment payment See "Japanese Payment Options," page 162. This field is supported only on JCN Gateway and for the Sumitomo Mitsui Card Co. acquirer on CyberSource through VisaNet.	ccAuthService (O) ccCaptureService (O on JCN Gateway; not used on CyberSource through VisaNet) ccCreditService (O on JCN Gateway; not used on CyberSource through VisaNet)	Integer (1)
linkToRequest	Value that links the current authorization request to the original authorization request. Set this value to the request ID that was returned in the reply message from the original authorization request. This value is used for: ■ Partial authorizations: see "Partial Authorizations," page 105. ■ Split shipments: see "Split Shipments," page 218.	ccAuthService (O)	String (26)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
loan_assetType	Indicates whether a loan is for a recoverable item or a non-recoverable item. Possible values: ■ N: non-recoverable item ■ R: recoverable item This field is supported only for BNDES transactions on CyberSource through VisaNet. See "BNDES," page 129. The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR2 ■ Position: 26 ■ Field: Asset Indicator	ccAuthService (R when using a Visa card issued by BNDES) ccCaptureService (R when using a Visa card issued by BNDES) ccCreditService (R when using a Visa card issued by BNDES)	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
loan_type	Type of loan based on an agreement between you and the issuer. Examples: AGRO-CUSTEIO, AGRO-INVEST, BNDES-Type1, CBN, FINAME. This field is supported only for these kinds of payments: ■ BNDES transactions on CyberSource through VisaNet. See "BNDES," page 129. ■ Installment payments with Mastercard on CyberSource through VisaNet in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. For BNDES transactions, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR2 ■ Position: 27-46 ■ Field: Loan Type For installment payments with Mastercard in Brazil, the value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR4 ■ Position: 5-24 ■ Field: Financing Type	ccAuthService (R for BNDES transactions on CyberSource through VisaNet and installment payments with Mastercard on CyberSource through VisaNet in Brazil) ccCaptureService (R for BNDES transactions on CyberSource through VisaNet and installment payments with Mastercard on CyberSource through VisaNet in Brazil)	String (20)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
merchantCategoryCode	Four-digit number that the payment card industry uses to classify merchants into market segments. A payment card company assigned one or more of these values to your business when you started accepting the payment card company's cards. When you do not include this field in your request, CyberSource uses the value in your CyberSource account. See "Aggregator Support," page 117. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR4 - Position: 150-153 - Field: Merchant Category Code	ccAuthService ccCaptureService ccCreditService American Express Direct: R for all aggregator transactions. CyberSource through VisaNet: O for all aggregator transactions. FDC Compass: O for all aggregator authorizations; otherwise, not used. FDC Nashville Global: R for all aggregator transactions.	Integer (4)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
merchantCategoryCode Domestic	Merchant category code for domestic transactions. The value for this field is a four-digit number that the payment card industry uses to classify merchants into market segments. A payment card company assigned one or more of these values to your business when you started accepting the payment card company's cards. Including this field in a request for a domestic transaction might reduce interchange fees. When you include this field in a request: - Do not include the merchantCategoryCode field. - The value for this field overrides the value in your CyberSource account. This field is supported only for: - Domestic transactions with Mastercard in Spain. <i>Domestic</i> means that you and the customer are in the same country. - Merchants enrolled in the OmniPay Direct interchange program. - First Data Merchant Solutions (Europe) on OmniPay Direct.	ccAuthService (O)	Integer (4)
merchantDefinedData_field1 to merchantDefinedData_ field20	Fields that you can use to store information. Important These fields have been replaced by merchantDefinedData_mddField_1 to 100. CyberSource recommends that you update your order management system to use the new fields. Warning Merchant-defined fields <i>must not</i> be used to capture personally identifying information as stated in the warning under the following field description for merchantDefinedData_mddField_1 to 100.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	String (255)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
merchantDefinedData_ mddField_1 to merchantDefinedData_ mddField_100	Fields that you can use to store information. Important These fields override the old merchant-defined data fields. For example, if you use the obsolete field merchantDefinedData_field15 and the new field merchantDefinedData_mddField_15 in the same request, the value for the new field overwrites the value for the obsolete field. Warning Merchant-defined data fields are not intended to and <i>must not</i> be used to capture personally identifying information. Accordingly, merchants are prohibited from capturing, obtaining, and/or transmitting any personally identifying information in or via the merchant-defined data fields. Personally identifying information includes, but is not limited to, address, payment card number, social security number, driver's license number, state-issued identification number, passport number, and card verification numbers (CVV, CVC2, CVV2, CID, CVN). In the event CyberSource discovers that a merchant is capturing and/or transmitting personally identifying information via the merchant-defined data fields, whether or not intentionally, CyberSource will immediately suspend the merchant's account, which will result in a rejection of any and all transaction requests submitted by the merchant after the point of suspension.	ccAuthService (O) ccCaptureService (O) ccCreditService (O)	Installment payments with Mastercard on CyberSource through VisaNet in Brazil: String (20) All other transactions: String (255)

(continued on next page)

- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
merchantDefinedData_ mddField_1 to merchantDefinedData_ mddField_100 (continued)	<i>CyberSource through VisaNet</i> For installment payments with Mastercard in Brazil, use merchantDefinedData_mddField_1 and merchantDefinedData_mddField_2 for data that you want to provide to the issuer to identify the transaction. See "Installment Payments on CyberSource through VisaNet," page 150. For installment payments with Mastercard in Brazil: ■ The value for merchantDefinedData_mddField_1 corresponds to the following data in the TC 33 capture file⁵: • Record: CP07 TCR5 • Position: 25-44 • Field: Reference Field 2 ■ The value for merchantDefinedData_mddField_2 corresponds to the following data in the TC 33 capture file⁵: • Record: CP07 TCR5 • Position: 45-64 • Field: Reference Field 3		
merchantID	Your CyberSource merchant ID. Use the same merchant ID for evaluation, testing, and production.	Required for all CyberSource services.	String (30)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
merchantReferenceCode	Merchant-generated order reference or tracking number. CyberSource recommends that you send a unique value for each transaction so that you can perform meaningful searches for the transaction. For information about tracking orders, see Getting Started with CyberSource Advanced for the Simple Order API. <i>FDC Nashville Global</i> Certain circumstances can cause the processor to truncate this value to 15 or 17 characters for Level II and Level III processing, which can cause a discrepancy between the value you submit and the value included in some processor reports.	Required for all CyberSource services.	Asia, Middle East, and Africa Gateway: String (40) Atos: String (32) Elavon Americas: String (39) RuPay: ■ Credits: String (24) ■ Other services: String (50) All other processors: String (50)
merchantTransaction Identifier	Identifier that you assign to the transaction. See "Merchant-Initiated Reversals and Voids," page 170 .	ccAuthService (O) ccAuthReversal Service (O) ccCaptureService (O) ccCreditService (O) voidService (O)	String (30)
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see Getting Started with CyberSource Advanced for the Simple Order API. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
nationalNetDomesticData	Supplementary domestic transaction information provided by the acquirer for National Net Settlement Service (NNSS) transactions. NNSS is a settlement service that Visa provides. For transactions on CyberSource through VisaNet in countries that subscribe to NNSS: ■ VisaNet clears transactions; VisaNet transfers funds to the acquirer after deducting processing fees and interchange fees. ■ VisaNet settles transactions in the local pricing currency through a local financial institution. This field is supported only on CyberSource through VisaNet for domestic data in Colombia.	ccAuthService (O) ccAuthReversal Service (O) ccCaptureService (O) ccCreditService (O)	String (123)
orderRequestToken	The request token value returned from a previous request. This value links the previous request to the current follow-on request. This field is an encoded string that does not contain any confidential information, such as account numbers or card verification numbers. The string can contain a maximum of 256 characters.	ccAuthReversal Service (O) ccCaptureService (R for Atos; otherwise, O. When you request the authorization and capture services together, the capture request does not require a request token.) ccCreditService (R for Atos; otherwise, O.) voidService (Optional.)	String (256)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
paymentSolution	Type of payment solution that is being used for the transaction. Possible Values: 005: Masterpass. This value is required for Masterpass transactions on OmniPay Direct. See "Masterpass," page 169. - visacheckout: Visa Checkout. This value is required for Visa Checkout transactions. See Visa Checkout Using the Simple Order API.	ccAuthService (See description.) ccAuthReversal Service ⁴ ccCaptureService ⁴ ccCreditService ⁴	Masterpass: String (3) Visa Checkout: String (12)
personalID_number	Identifier for the customer. This field is supported only on the processors listed in this description. <i>Cielo 3.0</i> Set this field to the Cadastro Nacional da Pessoa Jurídica (CNPJ) or Cadastro de Pessoas Físicas (CPF). <i>Comercio Latino</i> Set this field to the Cadastro de Pessoas Físicas (CPF).	ccAuthService (O)	String (18)
personalID_type	Type of value in the personalID_number field. Possible values: - BR_CNPJ: Cadastro Nacional da Pessoa Jurídica (CNPJ) - BR_CPF: Cadastro de Pessoas Físicas (CPF) This field is supported only on Cielo 3.0.	ccAuthService (R if personalID_number is included; otherwise, not used.)	String (7)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
pos_environment	Operating environment. Possible values for all card types except Mastercard: ■ 0: No terminal used or unknown environment. ■ 1: On merchant premises, attended. ■ 2: On merchant premises, unattended. Examples: oil, kiosks, self-checkout, mobile telephone, personal digital assistant (PDA). ■ 3: Off merchant premises, attended. Examples: portable POS devices at trade shows, at service calls, or in taxis. ■ 4: Off merchant premises, unattended. Examples: vending machines, home computer, mobile telephone, PDA. ■ 5: On premises of cardholder, unattended. ■ 9: Unknown delivery mode. ■ S: Electronic delivery of product. Examples: music, software, or eTickets that are downloaded over the internet. ■ T: Physical delivery of product. Examples: music or software that is delivered by mail or by a courier.	ccAuthService (O)	String (1)

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- 1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- 3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- 4 Required for Visa Checkout transactions. Otherwise, not used.
- 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
pos_environment (continued)	Possible values for Mastercard: ■ 2: On merchant premises, unattended, or cardholder terminal. Examples: oil, kiosks, self-checkout, home computer, mobile telephone, personal digital assistant (PDA). Cardholder terminal is supported only for Mastercard transactions on CyberSource through VisaNet. ■ 4: Off merchant premises, unattended, or cardholder terminal. Examples: vending machines, home computer, mobile telephone, PDA. Cardholder terminal is supported only for Mastercard transactions on CyberSource through VisaNet. This field is supported only on American Express Direct and CyberSource through VisaNet.		
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
postdatedTransaction_ guaranteeAmount	Amount that is guaranteed for the post-dated transaction. The amount is specified in an agreement between you and the issuer. Format: 8 digits including 2 minor units, BRL currency only. A post-dated transaction enables a customer to make a purchase using a debit Mastercard and to have the debit post to the customer's account at a later date. This field is supported only for post-dated transactions with Mastercard in Brazil on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 5-12 ■ Field: Mastercard Post-Dated Transaction Details	ccAuthService (R for post-dated transactions with Mastercard)	String (8)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
postdatedTransaction_ guaranteeIndicator	Indicates whether the post-dated transaction has a guarantee defined by an agreement between you and the issuer. The customer tells you whether they would like the transaction to be post-dated. The issuer provides the guarantee. Possible values: ■ <code>true</code>: The post-dated transaction has a guarantee. ■ <code>false</code>: The post-dated transaction does not have a guarantee. A post-dated transaction enables a customer to make a purchase using a debit Mastercard and to have the debit post to the customer's account at a later date. This field is supported only for post-dated transactions with Mastercard in Brazil on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 55 ■ Field: Mastercard Post-Dated Transaction Details	ccAuthService (R for post-dated transactions with Mastercard)	String (5)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
postdatedTransaction_ settlementDate	Date you plan to send a capture request to CyberSource for the post-dated transaction. Format: YYYYMMDD A post-dated transaction enables a customer to make a purchase using a debit Mastercard and to have the debit post to the customer's account at a later date. This field is supported only for post-dated transactions with Mastercard in Brazil on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP07 TCR0 ■ Position: 13-18 ■ Field: Mastercard Post-Dated Transaction Details	ccAuthService (R for post-dated transactions with Mastercard)	String (8)
processorID	Value that identifies the acquirer to use for the transaction. This value is supported only for CyberSource through VisaNet. Contact CyberSource Customer Support to get the value for this field.	ccAuthService (O) ccCreditService (O for stand-alone credits; otherwise, not used.)	String (3)
purchaseTotals_ additionalAmount0	Additional amount. This field is supported only for American Express Direct. See "Additional Amounts," page 116.	ccCaptureService (O)	String (12)
purchaseTotals_ additionalAmount1		ccCreditService (O)	
purchaseTotals_ additionalAmount2			
purchaseTotals_ additionalAmount3			
purchaseTotals_ additionalAmount4			
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
purchaseTotals_ additionalAmountType0 purchaseTotals_ additionalAmountType1 purchaseTotals_ additionalAmountType2 purchaseTotals_ additionalAmountType3 purchaseTotals_ additionalAmountType4	Additional amount type. This field is supported only for American Express Direct. See "Additional Amounts," page 116 , for a description of this feature. For the possible values for this field, see Appendix C, "Additional Amount Types," on page 455 .	ccCaptureService (O) ccCreditService (O)	String (3)
purchaseTotals_currency	Currency used for the order. Use the three-character ISO Standard Currency Codes . For ccAuthReversalService and ccCaptureService , you must use the same currency that you used in your request for ccAuthService . <i>DCC with a Third-Party Provider</i> Your customer's billing currency. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (R) ccAuthReversal Service (R) ccCaptureService (R) ccCreditService (R) ccIncrementalAuth Service (R)	String (5)
purchaseTotals_ exchangeRate	Exchange rate returned by the DCC provider. Includes a decimal point and a maximum of 4 decimal places. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (R for DCC transactions)	String (16)
purchaseTotals_ exchangeRateTimeStamp	Time stamp for the exchange rate. This value is returned by the DCC provider. This value must be in GMT. Format: YYYYMMDDhhmmss For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (O)	String (14)
purchaseTotals_ foreignAmount	Set this field to the converted amount that was returned by the DCC provider. See "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (O)	String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
purchaseTotals_ foreignCurrency	Your customer's billing currency. See "Dynamic Currency Conversion with a Third Party Provider," page 133.	ccAuthService (O)	String (5)
purchaseTotals_ grandTotalAmount	Grand total for the order. This value cannot be negative. You can include a decimal point (.), but you cannot include any other special characters. CyberSource truncates the amount to the correct number of decimal places. Important Some processors have specific requirements and limitations, such as maximum amounts and maximum field lengths. This information is covered in: ■ Merchant Descriptors Using the Simple Order API ■ Table 20, "Capture Information for Specific Processors," on page 64 ■ Table 24, "Credit Information for Specific Processors," on page 81 If your processor supports zero amount authorizations, you can set this field to 0 for the authorization to check if the card is lost or stolen. See "Zero Amount Authorizations," page 229. <i>DCC with a Third-Party Provider</i> Set this field to the converted amount that was returned by the DCC provider. You must include either this field or item_#_unitPrice in your request. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133. <i>FDMS South</i> If you accept IDR or CLP currencies, see the entry for FDMS South in Merchant Descriptors Using the Simple Order API.	ccAuthService ³ ccAuthReversal Service ³ ccCaptureService ³ ccCreditService ³ ccIncrementalAuth Service (R)	Comercio Latino: String (19) All other processors: String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date,"](#) page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
purchaseTotals_ originalAmount	Amount in your original local pricing currency. This value cannot be negative. You can include a decimal point (.) in this field to denote the currency exponent, but you cannot include any other special characters. If needed, CyberSource truncates the amount to the correct number of decimal places. For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (R when DCC is used for the purchase.) ccCaptureService (R when DCC is used for the purchase.) ccCreditService (R when DCC is used for the purchase.)	String (15)
purchaseTotals_ originalCurrency	Your local pricing currency code. For the possible values, see the ISO Standard Currency Codes . For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133 .	ccAuthService (R for DCC transactions)	String (5)
recipient_accountID	Identifier for the recipient's account. Use the first six digits and last four digits of the recipient's account number. This field is a <i>pass-through</i> , which means that CyberSource does not verify the value or modify it in any way before sending it to the processor. If the field is not required for the transaction, CyberSource does not forward it to the processor. See "Recipients," page 204 .	ccAuthService (Required in recipient transactions; otherwise, not used)	String with numbers only (10)
recipient_dateOfBirth	Recipient's date of birth. Format: YYYYMMDD. This field is a <i>pass-through</i> , which means that CyberSource ensures that the value is eight numeric characters but otherwise does not verify the value or modify it in any way before sending it to the processor. If the field is not required for the transaction, CyberSource does not forward it to the processor. See "Recipients," page 204 .	ccAuthService (Required in recipient transactions; otherwise, not used)	String with numbers only (8)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See ["Relaxed Requirements for Address Data and Expiration Date," page 91](#). **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
recipient_lastName	Recipient's last name. This field is a <i>pass-through</i> , which means that CyberSource does not verify the value or modify it in any way before sending it to the processor. If the field is not required for the transaction, CyberSource does not forward it to the processor. See "Recipients," page 204 .	ccAuthService (Required in recipient transactions; otherwise, not used)	String with letters and numbers only (6)
recipient_postalCode	Partial postal code for the recipient's address. For example, if the postal code is NN5 7SG, the value for this field should be the first part of the postal code: NN5. This field is a <i>pass-through</i> , which means that CyberSource does not verify the value or modify it in any way before sending it to the processor. If the field is not required for the transaction, CyberSource does not forward it to the processor. See "Recipients," page 204 .	ccAuthService (Required in recipient transactions; otherwise, not used)	String with letters and numbers only (6)
recurringSubscriptionInfo_subscriptionID	When you use Payment Tokenization or Recurring Billing and you include this value in your request, many of the fields that are normally required for an authorization or credit become optional. See "Payment Tokenization," page 202 , and "Recurring Billing," page 205 .	ccAuthService (O) ccCreditService (O)	String (26)
reportGroup	Attribute that lets you define custom grouping for your processor reports. This field is supported only for Worldpay VAP. See "Report Groups," page 215 .	ccAuthService (O) ccAuthReversal Service (O) ccCaptureService (O) ccCreditService (O)	String (25)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
shipFrom_postalCode	Postal code for the address from which the goods are shipped, which is used to establish nexus. The default is the postal code associated with your CyberSource account. The postal code must consist of 5 to 9 digits. When the billing country is the U.S., the 9-digit postal code must follow this format: [5 digits][dash][4 digits] Example 12345-6789 When the billing country is Canada, the 6-digit postal code must follow this format: [alpha][numeric][alpha][space] [numeric][alpha][numeric] Example A1B 2C3 This field is frequently used for Level II and Level III transactions. See Level II and Level III Processing Using the Simple Order API. <i>American Express Direct</i> Before sending the postal code to the processor, CyberSource removes all non-alphanumeric characters and, if the remaining value is longer than nine characters, truncates the value starting from the right side.	ccCaptureService (O) ccCreditService (O)	String (10)
shipTo_buildingNumber	Building number in the street address. For example, the building number is 187 in the following address: Rua da Quitanda 187	ccAuthService (O)	String (15)
shipTo_city	City of the shipping address.	ccAuthService Required if any shipping address information is included in the request and shipping to the U.S. or Canada; otherwise, optional.	String (50)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
shipTo_country	Country of the shipping address. Use the two-character <i>ISO Standard Country Codes</i> .	ccAuthService ccCaptureService ccCreditService Required if any shipping address information is included in the request; otherwise, optional.	String (2)
shipTo_district	Neighborhood, community, or region within a city or municipality.	ccAuthService (O)	String (50)
shipTo_firstName	First name of the recipient.	ccAuthService (O)	Worldpay VAP: String (25) All other processors: String (60)
shipTo_lastName	Last name of the recipient.	ccAuthService (O)	Worldpay VAP: String (25) All other processors: String (60)
shipTo_phoneNumber	Phone number for the shipping address.	ccAuthService (O)	String (15)
- Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. - This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. - You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. - Required for Visa Checkout transactions. Otherwise, not used. - The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
shipTo_postalCode	Postal code for the shipping address. The postal code must consist of 5 to 9 digits. When the shipping country is the U.S., the 9-digit postal code must follow this format: [5 digits][dash][4 digits] Example 12345-6789 When the shipping country is Canada, the 6-digit postal code must follow this format: [alpha][numeric][alpha][space] [numeric][alpha][numeric] Example A1B 2C3 <i>American Express Direct</i> Before sending the postal code to the processor, CyberSource removes all non-alphanumeric characters and, if the remaining value is longer than nine characters, truncates the value starting from the right side.	ccAuthService ccCaptureService ccCreditService Required if any shipping address information is included in the request and shipping to the U.S. or Canada; otherwise, optional.	String (10)
shipTo_shippingMethod	Shipping method for the product. Possible values: ■ lowcost: Lowest-cost service ■ sameday: Courier or same-day service ■ oneday: Next-day or overnight service ■ twoday: Two-day service ■ threeday: Three-day service ■ pickup: Store pick-up ■ other: Other shipping method ■ none: No shipping method because product is a service or subscription	ccAuthService (O)	String (10)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
shipTo_state	State or province of the shipping address. Use the State, Province, and Territory Codes for the United States and Canada .	ccAuthService Required if any shipping address information is included in the request and shipping to the U.S. or Canada; otherwise, optional.	String (2)
shipTo_street1	First line of the shipping address.	ccAuthService Required if any shipping address information is included in the request; otherwise, optional.	Worldpay VAP: String (35) All other processors: String (60)
shipTo_street2	Second line of the shipping address.	ccAuthService (O)	Worldpay VAP: String (35) All other processors: String (60)
subsequentAuth	Indicates whether the transaction is a merchant-initiated transaction. Possible values: ■ <code>true</code>: Merchant-initiated transaction ■ <code>false</code>: Not a merchant-initiated transaction This field is supported for: ■ All merchant-initiated transactions. ■ Subsequent authorizations on FDC Nashville Global and Streamline only. <i>CyberSource through VisaNet</i> The value for this field does not correspond to any data in the TC 33 capture file.⁵ <i>All Processors</i> See "Merchant-Initiated Transactions," page 173.	ccAuthService (R for merchant-initiated transactions; otherwise, not used)	String (5)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
subsequentAuthFirst	Indicates whether the transaction is the first merchant-initiated transaction in a series, which means that the customer initiated the previous transaction. Possible values: ■ <code>true</code>: First merchant-initiated transaction ■ <code>false</code>: Not the first merchant-initiated transaction This field is supported only for merchant-initiated transactions. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR1 ■ Position: 136 ■ Field: POS Environment <i>All Processors</i> See "Merchant-Initiated Transactions," page 173.	ccAuthService (R for merchant-initiated transactions; otherwise, not used)	String (5)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
subsequentAuthReason	Reason for the merchant-initiated transaction or incremental authorization. Possible values: ■ 1: Resubmission ■ 2: Delayed charge ■ 3: Reauthorization for split shipment ■ 4: No show ■ 5: Account top up This field is required only for the five kinds of transactions in the preceding list. This field is supported only for merchant-initiated transactions and incremental authorizations. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: ■ Record: CP01 TCR0 ■ Position: 160-163 ■ Field: Message Reason Code <i>All Processors</i> See "Merchant-Initiated Transactions," page 173. For CyberSource through VisaNet, see also "Incremental Authorizations," page 45.	ccAuthService (See description) ccIncrementalAuthService (R)	String (1)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
subsequentAuthStored Credential	Indicates whether you obtained the payment information from the credentials on file (COF) instead of from the customer. Possible values: ■ <code>true</code>: Transaction uses COF ■ <code>false</code>: Transaction does not use COF When you use the Token Management Service, CyberSource sets this field to <code>true</code> for you. For information about the Token Management Service, see Token Management Service Using the Simple Order API. For information about merchant-initiated transactions, see "Merchant-Initiated Transactions," page 173.	ccAuthService (R for transactions that use COF information; otherwise, not used)	String (5)
subsequentAuthTransaction ID	Network transaction identifier that was returned in the ccAuthReply_paymentNetworkTransactionID field in the reply message for either the original authorization in the series or the previous authorization in the series. <i>CyberSource through VisaNet</i> The value for this field does not correspond to any data in the TC 33 capture file.⁵ <i>FDI Australia</i> See "Installment Payments on FDI Australia," page 159. <i>All Processors Other Than FDI Australia</i> See "Merchant-Initiated Transactions," page 173.	ccAuthService (R for installment payments on FDI Australia and for merchant-initiated transactions; otherwise, not used)	String (15)
surchargeAmount	The surcharge amount is included in the total transaction amount but is passed in a separate field to the issuer and acquirer for tracking. The issuer can provide information about the surcharge amount to the customer. This field is supported only for CyberSource through VisaNet.	ccAuthService (O)	String (15)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see [Getting Started with CyberSource Advanced for the Simple Order API](#).

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
surchargeSign	Sign for the surcharge amount. Possible values: ■ C: The surcharge amount will be credited to the customer's account. ■ D: The surcharge amount will be debited from the customer's account. This field is supported only for CyberSource through VisaNet.	ccAuthService (O)	String (1)
thirdPartyCertification Number	Value that identifies the application vendor and application version for a third party gateway. CyberSource provides you with this value during testing and validation. This field is supported only on CyberSource through VisaNet.	ccAuthService (O) ccAuthReversalService (O) ccCaptureService (O) ccCreditService (O) ccIncrementalAuthService (O) voidService (O)	String (12)
transactionLocalDateTime	Local date and time at your physical location. Include both the date and time in this field or leave it blank. This field is supported only for CyberSource through VisaNet. Format: YYYYMMDDhhmmss where: ■ YYYY = year ■ MM = month ■ DD = day ■ hh = hour ■ mm = minutes ■ ss = seconds	ccAuthService (O)	String (14)
ucaf_authenticationData	Universal cardholder authentication field (UCAF) data. For the description and requirements, see "Payer Authentication," page 183 .	ccAuthService	String (32)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
ucaf_collectionIndicator	Universal cardholder authentication field (UCAF) collection indicator. For the description and requirements, see "Payer Authentication," page 183. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR7 - Position: 5 - Field: Mastercard Electronic Commerce Indicators—UCAF Collection Indicator	ccAuthService	String with numbers only (1)
vc_orderID	Identifier for the Visa Checkout order. Visa Checkout provides a unique order ID for every transaction in the Visa Checkout callID field. See Visa Checkout Using the Simple Order API.	ccAuthService ⁴ ccAuthReversalService ⁴ ccCaptureService ⁴ ccCreditService ⁴	String (48)
voidService_run	Whether to include voidService in your request. Possible values: <code>true</code>: Include the service in your request. <code>false</code> (default): Do not include the service in your request.	voidService (R)	String (5)
voidService_voidRequestID	Request ID of the capture or credit you want to void.	voidService (R)	String (26)
voidService_voidRequestToken	Value of the request token returned from a previous request for a service that you want to void. The field is an encoded string that contains no confidential information, such as an account number or card verification number. The string can contain a maximum of 256 characters.	voidService (Required for Atos; otherwise, optional.)	String (256)

1 Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.

2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.

3 You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.

4 Required for Visa Checkout transactions. Otherwise, not used.

5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
wallet_type	Type of wallet. For possible values, see Appendix U, "Values for the Wallet Type Field," on page 499. For Visa Checkout transactions, the way CyberSource processes the value for this field depends on the processor. See the Visa Checkout section below. For all other values, this field is a <i>passthrough</i>; therefore, CyberSource does not verify the value or modify it in any way before sending it to the processor. For incremental authorizations, this field is supported only for Mastercard and the supported values are 101 and 102. Payment card companies can introduce new values without notice. Your order management system should be able to process new values without problems. <i>CyberSource through VisaNet</i> When the value for this field is 101, 102, 103, 216, or 217, it corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR6 - Position: 88-90 - Field: Mastercard Wallet Identifier When the value for this field is VCIND, it corresponds to the following data in the TC 33 capture file⁵: - Record: CP01 TCR8 - Position: 72-76 - Field: Agent Unique ID	<i>Masterpass</i> (101, 102, 103, 216, and 217) ccAuthService (O) ccCreditService (O on Chase Paymentech Solutions and CyberSource through VisaNet. Not used for credits on OmniPay Direct.) ccIncrementalAuthService (O for Mastercard on CyberSource through VisaNet; otherwise, not used.) <i>Staged Digital Wallet (SDW)</i> ccAuthService (O) ccCreditService (O) <i>Visa Checkout (VCIND)</i> ccAuthService (See description) ccCreditService (O for stand-alone credits. Not used for follow-on credits.)	String (5)

(continued on next page)

- Optional for a follow-on credit request, which must include **ccCreditService_captureRequestID**.
- This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. **Important** It is your responsibility to determine whether a field is required for the transaction you are requesting.
- You must include either **item_#_unitPrice** or **purchaseTotals_grandTotalAmount** in your request. For information about items and grand totals, see *Getting Started with CyberSource Advanced for the Simple Order API*.
- Required for Visa Checkout transactions. Otherwise, not used.
- The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.

Table 55 Request Fields (Continued)

Field	Description	Used By: Required (R) or Optional (O)	Data Type & Length
wallet_type (continued)	<i>Masterpass (101, 102, 103, 216, and 217)</i> The Masterpass platform generates the wallet type value and passes it to you along with the customer's checkout information. <i>Visa Checkout</i> This field is optional for Visa Checkout authorizations on FDI Australia. For all other processors, this field is required for Visa Checkout authorizations. For Visa Checkout transactions on the following processors, CyberSource sends the value that the processor expects for this field: ■ FDC Compass ■ FDC Nashville Global ■ FDI Australia ■ TSYS Acquiring Solutions For all other processors, this field is a <i>passthrough</i>; therefore, CyberSource does not verify the value or modify it in any way before sending it to the processor.		
1 Optional for a follow-on credit request, which must include ccCreditService_captureRequestID. 2 This field is optional if your CyberSource account is configured for relaxed requirements for address data and expiration date. See "Relaxed Requirements for Address Data and Expiration Date," page 91. Important It is your responsibility to determine whether a field is required for the transaction you are requesting. 3 You must include either item_#_unitPrice or purchaseTotals_grandTotalAmount in your request. For information about items and grand totals, see <i>Getting Started with CyberSource Advanced for the Simple Order API</i>. 4 Required for Visa Checkout transactions. Otherwise, not used. 5 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment card companies.			

Reply Fields

Table 56 Reply Fields

Field	Description	Returned By	Data Type & Length
additionalData	This field might contain information about a decline. This field is supported only for CyberSource through VisaNet.	ccAuthReply ccIncrementalAuthReply	String (255)
additionalProcessor Response	Processor-defined response category code. The associated detail error code is in the ccAuthReply_processorResponse field or the ccAuthReversalReply_processorResponse field depending on which service you requested. This field is supported only for: - Japanese issuers - Domestic transactions in Japan - Comercio Latino—processor transaction ID required for troubleshooting	ccAuthReply ccAuthReversalReply	Comercio Latino: Integer (32) All other processors: Integer (3)
authIndicator	Flag indicating the type of authorization that was performed. See "Final Authorization Indicator," page 138. This field is not returned for unmarked authorizations. Possible values for all processors except CyberSource through VisaNet: 0: Preauthorization 1: Final authorization Some processors that support the final authorization indicator do not return this field. For a list of the processors that support this field, see the procedure at the end of "Final Authorization Indicator," page 138. <i>CyberSource through VisaNet</i> Possible value for Visa transactions: 0: Authorization for an estimated amount Possible values for Mastercard transactions: 0: Preauthorization 1: Final authorization 2: Undefined authorization	ccAuthReply	String (1)

1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
card_prepaidReloadable	Indicates whether the prepaid card is reloadable. This field is supported only for Worldpay VAP. See "Card Type Indicators (CTIs)," page 130 . Possible values: ■ Yes ■ No ■ Unknown	ccAuthReply	String (7)
card_prepaidType	Type of prepaid card, such as: ■ GIFT ■ PAYROLL ■ GENERAL_PREPAID This field is supported only for Worldpay VAP. See "Card Type Indicators (CTIs)," page 130 .	ccAuthReply	String (50)
card_virtual	Indicates whether the card number corresponds to a virtual card number. This field is supported only for Worldpay VAP. See "Card Type Indicators (CTIs)," page 130 . Possible values: ■ true ■ false	ccAuthReply	String (5)
ccAuthReply_accountBalance	Remaining balance on the account. See "Balance Responses," page 111 , and "Balance Inquiries," page 128 .	ccAuthReply	String (12)
ccAuthReply_accountBalanceCurrency	Currency of the remaining balance on the account. For the possible values, see the ISO Standard Currency Codes . Also see "Balance Responses," page 111 , and "Balance Inquiries," page 128 .	ccAuthReply	String (5)
ccAuthReply_accountBalanceSign	Sign for the remaining balance on the account. Returned only when the processor returns this value. See "Balance Inquiries," page 128 . Possible values: ■ positive ■ negative	ccAuthReply	String (8)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ accountType	Type of account. This value is returned only when you request a balance inquiry. See "Balance Inquiries," page 128. Possible values: ■ 00: Not applicable or not specified ■ 10: Savings account ■ 20: Checking account ■ 30: Credit card account ■ 40: Universal account	ccAuthReply	String (2)
ccAuthReply_ affluenceIndicator	<i>Chase Paymentech Solutions</i> Indicates whether a customer has high credit limits. This information enables you to market high cost items to these customers and to understand the kinds of cards that high income customers are using. This field is supported for Visa, Mastercard, Discover, and Diners Club. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130. <i>Worldpay VAP</i> Flag indicating that a Visa cardholder or Mastercard cardholder is in one of the affluent categories. Possible values: ■ AFFLUENT: High income customer with high spending pattern (>100k USD annual income and >40k USD annual card usage). ■ MASS AFFLUENT: High income customer (>100k USD annual income).	ccAuthReply	Chase Paymentech Solution: String (1) Worldpay VAP: String (13)
ccAuthReply_ amount	Amount that was authorized. <i>FDMS South</i> If you accept IDR or CLP currencies on FDMS South, see the entry for FDMS South in Merchant Descriptors Using the Simple Order API.	ccAuthReply	String (15)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ amountType	Type of amount. This value is returned only when you request a balance inquiry. The issuer determines the value that is returned. See "Balance Inquiries," page 128. Possible values for deposit accounts: 01: Current ledger (posted) balance. 02: Current available balance, which is typically the ledger balance less outstanding authorizations. Some depository institutions also include pending deposits and the credit or overdraft line associated with the account. Possible values for credit card accounts: 01: Credit amount remaining for customer (open to buy). 02: Credit limit.	ccAuthReply	String (2)
ccAuthReply_ authorizationCode	Authorization code. Returned only when the processor returns this value. <i>Elavon Encrypted Account Number Program</i> The returned value is OFFLINE. See "Encoded Account Numbers," page 137. <i>TSYS Acquiring Solutions</i> The returned value for a successful zero amount authorization is 000000. See "Zero Amount Authorizations," page 229.	ccAuthReply	String The length of this value depends on your processor.
ccAuthReply_ authorizedDateTime	Time of authorization. Format: YYYY-MM-DDThh:mm:ssZ Example 2018-08-11T22:47:57Z equals August 11, 2018, at 22:47:57 (10:47:57 p.m.). The T separates the date and the time. The Z indicates UTC.	ccAuthReply	String (20)
ccAuthReply_ avsCode	AVS results. See "Address Verification System (AVS)," page 88, for a description of AVS. See Appendix E, "AVS Codes," on page 461, for the list of AVS codes.	ccAuthReply	String (1)
ccAuthReply_ avsCodeRaw	AVS result code sent directly from the processor. Returned only when the processor returns this value. Important Do not use this field to evaluate the result of AVS. Use for debugging purposes only.	ccAuthReply	String (10)

1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ cardCategory	<i>CyberSource through VisaNet</i> Visa product ID. For descriptions of the Visa product IDs, see the Product ID table on the Visa Request & Response Codes web page at https://developer.visa.com/guides/request_response_codes. <i>GPN</i> Visa or Mastercard product ID. For descriptions of the Visa product IDs, see the Product ID table on the Visa Request & Response Codes web page at https://developer.visa.com/guides/request_response_codes. For descriptions of the Mastercard product IDs, see Appendix S, "Product IDs," on page 494. <i>Worldpay VAP</i> Important Before using this field on Worldpay VAP, you must contact CyberSource Customer Support to have your account configured for this feature. Type of card used in the transaction. The only possible value is: ■ PREPAID: Prepaid Card <i>RBS WorldPay Atlanta</i> Type of card used in the transaction. Possible values: ■ B: Business Card ■ O: Noncommercial Card ■ R: Corporate Card ■ S: Purchase Card ■ Blank: Purchase card not supported	ccAuthReply	CyberSource through VisaNet: String (3) GPN: String (3) RBS WorldPay Atlanta: String (1) Worldpay VAP: String (7)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ cardCommercial	Indicates whether the card is a commercial card, which enables you to include Level II data in your transaction requests. This field is supported for Visa and Mastercard on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cardGroup	Type of commercial card. This field is supported only for CyberSource through VisaNet. Possible values: ■ B: Business card ■ R: Corporate card ■ S: Purchasing card ■ 0: Noncommercial card	ccAuthReply	String (1)
ccAuthReply_ cardHealthcare	Indicates whether the card is a healthcare card. This field is supported for Visa and Mastercard on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cardIssuerCountry	Country in which the card was issued. This information enables you to determine whether the card was issued domestically or internationally. Use the two-character ISO Standard Country Codes. This field is supported for Visa, Mastercard, Discover, Diners Club, JCB, and Maestro (International) on Chase Paymentech Solutions. See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (3)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ cardLevel3Eligible	Indicates whether the card is eligible for Level III interchange fees, which enables you to include Level III data in your transaction requests. This field is supported for Visa and Mastercard on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cardPayroll	Indicates whether the card is a payroll card. This field is supported for Visa, Discover, Diners Club, and JCB on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cardPINlessDebit	Indicates whether the card is a PINless debit card. This field is supported for Visa and Mastercard on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)

1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ cardPrepaid	Indicates whether the card is a prepaid card. This information enables you to determine when a gift card or prepaid card is presented for use when establishing a new recurring, installment, or deferred billing relationship. This field is supported for Visa, Mastercard, Discover, Diners Club, and JCB on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cardRegulated	Indicates whether the card is regulated according to the Durbin Amendment. If the card is regulated, the card issuer is subject to price caps and interchange rules. This field is supported for Visa, Mastercard, Discover, Diners Club, and JCB on Chase Paymentech Solutions. Possible values: ■ Y: Yes (assets greater than 10B USD) ■ N: No (assets less than 10B USD) ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ cardSignatureDebit	Indicates whether the card is a signature debit card. This information enables you to alter the way an order is processed. For example, you might not want to reauthorize a transaction for a signature debit card, or you might want to perform reversals promptly for a signature debit card. This field is supported for Visa, Mastercard, and Maestro (International) on Chase Paymentech Solutions. Possible values: ■ Y: Yes ■ N: No ■ X: Not applicable / Unknown See "Card Type Indicators (CTIs)," page 130.	ccAuthReply	String (1)
ccAuthReply_ cavvResponseCode	Mapped response code for Visa Secure and American Express SafeKey: ■ See "Visa Secure," page 195, and Appendix T, "Visa Secure Response Codes," on page 498. ■ See "American Express SafeKey," page 184, and Appendix D, "American Express SafeKey Response Codes," on page 460.	ccAuthReply	String (3)
ccAuthReply_ cavvResponseCode Raw	Raw response code sent directly from the processor for Visa Secure and American Express SafeKey: ■ See "Visa Secure," page 195. ■ See "American Express SafeKey," page 184.	ccAuthReply	String (3)
ccAuthReply_cvCode	CVN result code. See "Card Verification Numbers (CVNs)," page 97, for a description of the card verification check. See Appendix J, "CVN Codes," on page 472 for the list of CVN codes.	ccAuthReply	String (1)
ccAuthReply_ cvCodeRaw	CVN result code sent directly from the processor. Returned only when the processor returns this value. Important Do not use this field to evaluate the result of card verification. Use for debugging purposes only.	ccAuthReply	String (11)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_emsTransactionRiskScore	Fraud score for a Mastercard transaction. For a description of this feature, see "Mastercard Expert Monitoring Solutions (EMS)," page 168. - Positions 1-3: Fraud score. This value ranges from 001 to 998, where 001 indicates the least likely fraudulent transaction and 998 indicates the most likely fraudulent transaction. - Positions 4-5: Reason code that specifies the reason for the fraud score. See Appendix L, "Expert Monitoring Solutions (EMS) Reason Codes," on page 477. - Positions 6-32: Reserved for future use. This field is supported only on CyberSource through VisaNet.	ccAuthReply	String (32)
ccAuthReply_evEmail	Mapped Electronic Verification response code for the customer's email address. See Appendix M, "Electronic Verification Response Codes," on page 479 .	ccAuthReply	String (1)
ccAuthReply_evEmailRaw	Raw Electronic Verification response code from the processor for the customer's email address.	ccAuthReply	String (1)
ccAuthReply_evName	Mapped Electronic Verification response code for the customer's name. See Appendix M, "Electronic Verification Response Codes," on page 479 .	ccAuthReply	String (1)
ccAuthReply_evNameRaw	Raw Electronic Verification response code from the processor for the customer's last name.	ccAuthReply	String (1)
ccAuthReply_evPhoneNumber	Mapped Electronic Verification response code for the customer's phone number. See Appendix M, "Electronic Verification Response Codes," on page 479 .	ccAuthReply	String (1)
ccAuthReply_evPhoneNumberRaw	Raw Electronic Verification response code from the processor for the customer's phone number.	ccAuthReply	String (1)
ccAuthReply_evPostalCode	Mapped Electronic Verification response code for the customer's postal code. See Appendix M, "Electronic Verification Response Codes," on page 479 .	ccAuthReply	String (1)
ccAuthReply_evPostalCodeRaw	Raw Electronic Verification response code from the processor for the customer's postal code.	ccAuthReply	String (1)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_evStreet	Mapped Electronic Verification response code for the customer's street address. See Appendix M, "Electronic Verification Response Codes," on page 479 .	ccAuthReply	String (1)
ccAuthReply_evStreetRaw	Raw Electronic Verification response code from the processor for the customer's street address.	ccAuthReply	String (1)
ccAuthReply_forwardCode	Name of the Japanese acquirer that processed the transaction. Returned only for JCN Gateway. Contact the CyberSource Japan Support Group for more information.	ccAuthReply	String (32)
ccAuthReply_merchantAdviceCode	Reason the recurring payment transaction was declined. For some processors, this field is used only for Mastercard. For other processors, this field is used for Visa and Mastercard. And for other processors, this field is not implemented. Possible values: 00: Response not provided. 01: New account information is available. Obtain the new information. 02: Try again later. 03: Do not try again. Obtain another type of payment from the customer. 04: Problem with a token or a partial shipment indicator. 21: Recurring payment cancellation service. 99: An unknown value was returned from the processor.	ccAuthReply	String (2)
ccAuthReply_merchantAdviceCode Raw	Raw merchant advice code sent directly from the processor. This field is used only for Mastercard. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file ¹ : - Record: CP01 TCR7 - Position: 96-99 - Field: Response Data—Merchant Advice Code	ccAuthReply	String (2)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ownerMerchantID	Merchant ID that was used to create the subscription or customer profile for which the service was requested. <i>Payment Tokenization</i> When your account is enabled for Payment Tokenization, this field is returned only when you use profile sharing and when your merchant ID is in the same merchant ID pool as the owner merchant ID. See the profile sharing information in Payment Tokenization Using the Simple Order API. <i>Recurring Billing</i> When your account is enabled for Recurring Billing, this field is returned only when you use subscription sharing and when your merchant ID is in the same merchant ID pool as the owner merchant ID. See the subscription sharing information in Recurring Billing Using the Simple Order API.	ccAuthReply	String (30)
ccAuthReply_paymentNetworkTransactionID	Network transaction identifier (TID). You can use this value to identify a specific transaction when you are discussing the transaction with your processor. Not all processors provide this value. <i>Cielo</i> For Cielo, this value is the non-sequential unit (NSU) and is supported for all transactions. The value is generated by Cielo or the issuing bank. <i>Comercio Latino</i> For Comercio Latino, this value is the proof of sale or non-sequential unit (NSU) number generated by the acquirers Cielo and Rede, or the issuing bank. <i>CyberSource through VisaNet and GPN</i> For details about this value for CyberSource through VisaNet and GPN, see Appendix Q, "Network Transaction Identifiers," on page 491.	ccAuthReply	Comercio Latino:String (20) All other processors: String (15)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ personalIDCode	Personal identifier result. This field is supported only for Redecard in Brazil for CyberSource Latin American Processing. If you included billTo_personalID in the request, this value indicates whether billTo_personalID matched a value in a record on file. Returned only when the personal ID result is returned by the processor. Possible values: ■ Y: Match ■ N: No match ■ K: Not supported ■ U: Unknown ■ Z: No response returned Note <i>CyberSource Latin American Processing</i> is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, <i>CyberSource Latin American Processing</i> does not refer to the general topic of processing in Latin America. The information in this field description is for the specific processing connection called <i>CyberSource Latin American Processing</i>. It is not for any other Latin American processors that CyberSource supports.	ccAuthReply	String (1)
ccAuthReply_posData	Point-of-sale details for the transaction. This value is returned only for American Express Direct. CyberSource generates this value, which consists of a series of codes that identify terminal capability, security data, and specific conditions present at the time the transaction occurred. To comply with the CAPN requirements, this value must be included in all subsequent follow-on requests, such as captures and follow-on credits. When you perform authorizations, captures, and credits through CyberSource, CyberSource passes this value from the authorization service to the subsequent services for you. However, when you perform authorizations through CyberSource and perform subsequent services through other financial institutions, you must ensure that your requests for captures and credits include this value. See "Authorization Only," page 127.	ccAuthReply	String (12)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_processorResponse	For most processors, this is the error message sent directly from the bank. Returned only when the processor returns this value. Important Do not use this field to evaluate the result of the authorization. <i>A/B MS</i> If this value is 08, you can accept the transaction if the customer provides you with identification. <i>Atos</i> This value is the response code sent from Atos and it might also include the response code from the bank. Format: aa,bb with the two values separated by a comma and where: ■ aa is the two-digit error message from Atos. ■ bb is the optional two-digit error message from the bank. <i>Comercio Latino</i> This value is the status code and the error or response code received from the processor separated by a colon. Format: [status code]:E[error code] or [status code]:R[response code] Example 2:R06 <i>JCN Gateway</i> Processor-defined detail error code. The associated response category code is in the additionalProcessorResponse field.	ccAuthReply	JCN Gateway: String (3) All other processors: String (10)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_ processorTransactionID	Processor transaction ID. <i>Cielo and CyberSource Latin American Processing</i> This value is a unique identifier for the transaction. <i>Moneris</i> This value identifies the transaction on a host system. It contains the following information: ■ Terminal used to process the transaction ■ Shift during which the transaction took place ■ Batch number ■ Transaction number within the batch You must store this value. If you give the customer a receipt, display this value on the receipt. Example For the value 66012345001069003: ■ Terminal ID = 66012345 ■ Shift number = 001 ■ Batch number = 069 ■ Transaction number = 003	ccAuthReply	Cielo and CyberSource Latin American Processing: String (50) Moneris: Positive Integer (18)
ccAuthReply_ reasonCode	Numeric value corresponding to the result of the authorization request. See Reason Codes .	ccAuthReply	Integer (5)
ccAuthReply_ reconciliationID	Reference number for the transaction. This value is not returned for all processors. See Table 9, "Fields for Reconciliation IDs," on page 31 for the list of processors for which this value is returned. See Getting Started with CyberSource Advanced for the Simple Order API for information about order tracking and reconciliation.	ccAuthReply	Atos: Integer (6) All other processors: String (60)
ccAuthReply_ reconciliationReference Number	<i>Ingenico ePayments</i> Unique number generated by CyberSource that identifies the transaction. You can use this value to identify transactions in the Ingenico ePayments Collections Report, which provides settlement information. <i>CyberSource through VisaNet</i> Retrieval request number.	ccAuthReply	String (20)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_referralResponseNumber	Referral response number for a verbal authorization with FDMS Nashville when using an American Express card. Give this number to American Express when you call them for the verbal authorization.	ccAuthReply	String (6)
ccAuthReply_requestAmount	Amount you requested to be authorized. This value is returned for partial authorizations as described in "Partial Authorizations," page 105 .	ccAuthReply	String (15)
ccAuthReply_requestCurrency	Currency for the amount you requested to be authorized. This value is returned for partial authorizations as described in "Partial Authorizations," page 105 . For the possible values, see the ISO Standard Currency Codes .	ccAuthReply	String (5)
ccAuthReply_transactionID	Transaction identification (TID) that is used to identify and track a transaction throughout its life cycle. This value is returned only for American Express Direct. American Express generates this value. To comply with the CAPN requirements, this value must be included in all subsequent follow-on requests, such as captures and follow-on credits. When you perform authorizations, captures, and credits through CyberSource, CyberSource passes this value from the authorization service to the subsequent services for you. However, when you perform authorizations through CyberSource and perform subsequent services through other financial institutions, you must ensure that your requests for captures and credits include this value. See "Authorization Only," page 127.	ccAuthReply	String (15)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReply_transactionIntegrity	Transaction integrity classification provided by Mastercard. This value specifies Mastercard's evaluation of the transaction's safety and security. This field is returned only for CyberSource through VisaNet. For card-present transactions, possible values: ■ A1: EMV or token in a secure, trusted environment ■ B1: EMV or chip equivalent ■ C1: Magnetic stripe ■ E1: Key entered ■ U0: Unclassified For card-not-present transactions, possible values: ■ A2: Digital transactions ■ B2: Authenticated checkout ■ C2: Transaction validation ■ D2: Enhanced data ■ E2: Generic messaging ■ U0: Unclassified For information about these values, contact Mastercard or your acquirer. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR6 ■ Position: 136-137 ■ Field: Mastercard Transaction Integrity Classification	ccAuthReply	String (2)
ccAuthReversalReply_amount	Amount that was reversed.	ccAuthReversal Reply	String (15)
ccAuthReversalReply_authorizationCode	Authorization code. Returned only when the authorization code is returned by the processor.	ccAuthReversal Reply	String (6)
ccAuthReversalReply_forwardCode	Name of the Japanese acquirer that processed the transaction. Returned only for JCN Gateway. Contact the CyberSource Japan Support Group for more information.	ccAuthReversal Reply	String (32)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccAuthReversalReply_processorResponse	Processor response code. <i>JCN Gateway</i> Processor-defined detail error code. The associated response category code is in the additionalProcessorResponse field.	ccAuthReversal Reply	JCN Gateway: String (3) All other processors: String (10)
ccAuthReversalReply_processorTransactionID	Processor transaction ID. This field is supported only for Moneris. This value identifies the transaction on a host system. It contains the following information: - Terminal used to process the transaction - Shift during which the transaction took place - Batch number - Transaction number within the batch You must store this value. If you give the customer a receipt, display this value on the receipt. Example For the value 66012345001069003: - Terminal ID = 66012345 - Shift number = 001 - Batch number = 069 - Transaction number = 003	ccAuthReversal Reply	Positive Integer (18)
ccAuthReversalReply_reasonCode	Numeric value corresponding to the result of the full authorization reversal request. See Reason Codes .	ccAuthReversal Reply	Integer (5)
ccAuthReversalReply_reconciliationID	Reference number for the transaction. This value is not returned for all processors. See Table 9, "Fields for Reconciliation IDs," on page 31 for the list of processors for which this value is returned. See Getting Started with CyberSource Advanced for the Simple Order API for information about order tracking and reconciliation.	ccAuthReversal Reply	String (60)
ccAuthReversalReply_requestDateTime	Date and time when the service was requested.	ccAuthReversal Reply	String (20)
ccCaptureReply_amount	Amount that was captured.	ccCaptureReply	String (15)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccCaptureReply_ processorTransactionID	Processor transaction ID. This value identifies the transaction on a host system. This value is supported only for Moneris. It contains this information: - Terminal used to process the transaction - Shift during which the transaction took place - Batch number - Transaction number within the batch You must store this value. If you give the customer a receipt, display this value on the receipt. Example For the value 66012345001069003: - Terminal ID = 66012345 - Shift number = 001 - Batch number = 069 - Transaction number = 003	ccCaptureReply	Positive Integer (18)
ccCaptureReply_ reasonCode	Numeric value corresponding to the result of the capture request. See Reason Codes .	ccCaptureReply	Integer (5)
ccCaptureReply_ reconciliationID	Reference number that you use to reconcile your CyberSource reports with your processor reports. See Getting Started with CyberSource Advanced for the Simple Order API for information about order tracking and reconciliation.	ccCaptureReply	Atos: Integer (6) FDC Nashville Global: String (8) All other processors: String (60)
ccCaptureReply_ reconciliationReference Number	Unique number generated by CyberSource that identifies the transaction. You can use this value to identify transactions in the Ingenico ePayments Collections Report, which provides settlement information. This field is supported only on Ingenico ePayments.	ccCaptureReply	String (20)
ccCaptureReply_ requestDateTime	Date and time when the service was requested.	ccCaptureReply	String (20)
ccCheckStatusReply_ authorizationCode	Authorization code. Returned only when the payment status is AUTHORIZED.	ccCheckStatus Reply	String (6)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccCheckStatusReply_ paymentStatus	Authorization status. Possible values: ■ AUTHORIZED ■ DECLINED	ccCheckStatus Reply	String (20)
ccCheckStatusReply_ reasonCode	Numeric value corresponding to the result of the check status request. See Reason Codes .	ccCheckStatus Reply	Integer (5)
ccCreditReply_ amount	Amount that was credited.	ccCreditReply	String (15)
ccCreditReply_ authorizationCode	Credit authorization code. Returned only when the issuer returns this value and CyberSource authorizes the credit as described in " Credit Authorizations ," page 84. This field is supported only on CyberSource through VisaNet.	ccCreditReply	String (6)
ccCreditReply_ forwardCode	Name of the Japanese acquirer that processed the transaction. Returned only for JCN Gateway. Contact the CyberSource Japan Support Group for more information.	ccCreditReply	String (32)
ccCreditReply_ ownerMerchantID	Merchant ID that was used to create the subscription or customer profile for which the service was requested. <i>Payment Tokenization</i> When your account is enabled for Payment Tokenization, this field is returned only when you use profile sharing and when your merchant ID is in the same merchant ID pool as the owner merchant ID. See the profile sharing information in Payment Tokenization Using the Simple Order API . <i>Recurring Billing</i> When your account is enabled for Recurring Billing, this field is returned only when you use subscription sharing and when your merchant ID is in the same merchant ID pool as the owner merchant ID. See the subscription sharing information in Recurring Billing Using the Simple Order API .	ccCreditReply	String (30)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccCreditReply_ paymentNetwork TransactionID	Network transaction identifier (TID). You can use this value to identify a specific transaction when you are discussing the transaction with your processor. Returned only when CyberSource authorizes the credit as described in " Credit Authorizations ," page 84 . This field is supported only on CyberSource through VisaNet.	ccCreditReply	String (15)
ccCreditReply_ processorResponse	Error message. Returned only when CyberSource authorizes the credit as described in " Credit Authorizations ," page 84 . This field is supported only on CyberSource through VisaNet.	ccCreditReply	String (10)
ccCreditReply_ processorTransactionID	Processor transaction ID. This value identifies the transaction on a host system. This value is supported only for Moneris. It contains this information: ■ Terminal used to process the transaction ■ Shift during which the transaction took place ■ Batch number ■ Transaction number within the batch You must store this value. If you give the customer a receipt, display this value on the receipt. Example For the value 66012345001069003: ■ Terminal ID = 66012345 ■ Shift number = 001 ■ Batch number = 069 ■ Transaction number = 003	ccCreditReply	Positive Integer (18)
ccCreditReply_ reasonCode	Numeric value corresponding to the result of the credit request. See Reason Codes .	ccCreditReply	Integer (5)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ccCreditReply_ reconciliationID	Reference number that you use to reconcile your CyberSource reports with your processor reports. See Getting Started with CyberSource Advanced for the Simple Order API for information about order tracking and reconciliation.	ccCreditReply	Atos: Integer (6) FDC Nashville Global: String (8) All other processors: String (60)
ccCreditReply_ reconciliationReference Number	Unique number generated by CyberSource that identifies the transaction. You can use this value to identify transactions in the Ingenico ePayments Collections Report, which provides settlement information. This field is supported only on Ingenico ePayments.	ccCreditReply	String (20)
ccCreditReply_ requestDateTime	Date and time when the service was requested.	ccCreditReply	String (20)
ccIncrementalAuthReply_ amount	Amount that was authorized.	ccIncrementalAuth Reply	String (15)
ccIncrementalAuthReply_ authorizationCode	Authorization code. Returned only when the processor returns this value.	ccIncrementalAuth Reply	String (7)
ccIncrementalAuthReply_ authorizedDateTime	Time of authorization. Format: YYYY-MM-DDThh:mm:ssZ Example 2018-08-11T22:47:57Z equals August 11, 2018, at 22:47:57 (10:47:57 p.m.). The T separates the date and the time. The Z indicates UTC.	ccIncrementalAuth Reply	String (20)
ccIncrementalAuthReply_ cardCategory	Visa product ID. For descriptions of the Visa product IDs, see the Product ID table on the Visa Request & Response Codes web page at https://developer.visa.com/guides/request_response_codes .	ccIncrementalAuth Reply	String (3)
ccIncrementalAuthReply_ paymentNetwork TransactionID	Network transaction identifier (TID). You can use this value to identify a specific transaction when you are discussing the transaction with your processor.	ccIncrementalAuth Reply	String (15)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
cclIncrementalAuthReply_processorResponse	Error message sent directly from the issuing bank. Returned only when the processor returns this value. Important Do not use this field to evaluate the result of the incremental authorization.	cclIncrementalAuth Reply	String (10)
cclIncrementalAuthReply_reasonCode	Numeric value corresponding to the result of the incremental authorization request. See Reason Codes .	cclIncrementalAuth Reply	Integer (5)
cclIncrementalAuthReply_reconciliationID	Reference number for the transaction. See Getting Started with CyberSource Advanced for the Simple Order API for information about order tracking and reconciliation.	cclIncrementalAuth Reply	String (60)
decision	Summarizes the result of the overall request. Possible values: ■ ACCEPT ■ ERROR ■ REJECT ■ REVIEW: Returned only when you use CyberSource Decision Manager. For details about these values, see the information about handling replies in Getting Started with CyberSource Advanced for the Simple Order API .	All CyberSource services	String (6)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ additionalCosts	Additional costs charged by the issuer to fund the installment payments. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 128-139 ■ Field: Total Other Costs	ccAuthReply	String (12)
installment_ additionalCosts Percentage	Additional costs divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 140-143 ■ Field: Percent of Total Other Costs	ccAuthReply	String (4)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_amount	Amount of the current installment payment. This field is supported for Mastercard installment payments on CyberSource through VisaNet in all countries except Croatia and Georgia. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 26-37 ■ Field: Amount of Each Installment <i>Greece</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 23-34 ■ Field: Amount of Each Installment <i>Other Countries</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 63-74 ■ Field: Mastercard Subsequent Installment Amount	ccAuthReply	String (12)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ amountFunded	Amount funded. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 48-59 ■ Field: Total Amount Funded	ccAuthReply	String (12)
installment_ amountRequested Percentage	Amount requested divided by the amount funded. For example: ■ A value of 90.0 specifies 90%. ■ A value of 93.7 specifies 93.7%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 60-63 ■ Field: Percent of Amount Requested	ccAuthReply	String (4)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ annualFinancingCost	Annual cost of financing the installment payments. This field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments in Brazil. See "Installment Payments on CyberSource through VisaNet," page 150. This field is supported only for Crediario installment payments in Brazil on CyberSource through VisaNet. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 158-164 ■ Field: Annual Total Cost of Financing	ccAuthReply	String (7)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ annualInterestRate	Annual interest rate. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments on CyberSource through VisaNet: ■ Crediario with Visa in Brazil: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments. ■ Mastercard in all countries except Brazil, Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 151-157 ■ Field: Annual Interest Rate <i>Other Countries</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 58-62 ■ Field: Mastercard Annual Percentage Rate	ccAuthReply	String (7)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_expenses	Expenses charged by the issuer to fund the installment payments. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 64-75 ■ Field: Total Expenses	ccAuthReply	String (12)
installment_expensesPercentage	Expenses divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 76-79 ■ Field: Percent of Total Expenses	ccAuthReply	String (4)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_fees	Fees charged by the issuer to fund the installment payments. This field is supported for two kinds of installment payments on CyberSource through VisaNet: ■ Crediario with Visa in Brazil: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments. ■ Mastercard in all countries except Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 80-91 ■ Field: Total Fees <i>Other Countries</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 46-57 ■ Field: Mastercard Installment Fee	ccAuthReply	String (12)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_feesPercentage	Fees divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 92-95 ■ Field: Percent of Total Fees	ccAuthReply	String (4)
installment_firstInstallmentAmount	Amount of the first installment payment. The issuer provides this value when the first installment payment is successful. This field is supported for Mastercard installment payments on CyberSource through VisaNet in all countries except Brazil, Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 23-34 ■ Field: Amount of Each Installment	ccAuthReply	String (12)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ firstInstallmentDate	Date of the first installment payment. Format: YYMMDD. This field is supported for Mastercard installment payments on CyberSource through VisaNet in Greece. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR5 - Position: 102-107 - Field: Mastercard Greece First Installment Due Date	ccAuthReply	String (6)
installment_insurance	Insurance charged by the issuer to fund the installment payments. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: - Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments - Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR9 - Position: 112-123 - Field: Total Insurance	ccAuthReply	String (12)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ insurancePercentage	Insurance costs divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 124-127 ■ Field: Percent Of Total Insurance	ccAuthReply	String (4)
installment_ maximumTotalCount	Maximum number of installments offered by the issuer for this purchase. The issuer provides this value when the first installment payment is successful. This field is supported for installment payments with Mastercard on CyberSource through VisaNet in all countries except Brazil, Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 77-78 ■ Field: Mastercard Maximum Number Of Installments	ccAuthReply	String (2)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ minimumTotalCount	Minimum number of installments offered by the issuer for this purchase. The issuer provides this value when the first installment payment is successful. This field is supported for installment payments with Mastercard on CyberSource through VisaNet in all countries except Brazil, Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 75-76 ■ Field: Mastercard Minimum Number Of Installments	ccAuthReply	String (2)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_ monthlyInterestRate	Monthly interest rate. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments on CyberSource through VisaNet: ■ Crediario with Visa in Brazil: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments. ■ Mastercard in all countries except Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 144-150 ■ Field: Monthly Interest Rate <i>Other Countries</i> The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR5 ■ Position: 41-45 ■ Field: Mastercard Interest Rate	ccAuthReply	String (7)

¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_taxes	Taxes collected by the issuer to fund the installment payments. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 96-107 ■ Field: Total Taxes	ccAuthReply	String (12)
installment_taxesPercentage	Taxes divided by the amount funded. For example: ■ A value of 1.0 specifies 1%. ■ A value of 4.0 specifies 4%. This field is supported for two kinds of installment payments in Brazil on CyberSource through VisaNet: ■ Crediario with Visa: this field is included in the authorization reply for the Crediario eligibility request when the issuer approves the customer's request for Crediario installment payments ■ Mastercard See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: ■ Record: CP01 TCR9 ■ Position: 108-111 ■ Field: Percentage of Total Taxes	ccAuthReply	String (4)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
installment_totalAmount	Total amount of the loan that is being paid in installments. This field is supported for installment payments with Mastercard on CyberSource through VisaNet in all countries except Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. <i>Brazil</i> The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR9 - Position: 8-19 - Field: Installment Total Amount <i>Other Countries</i> The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR5 - Position: 5-16 - Field: Installment Payment Total Amount	ccAuthReply	String (12)
installment_totalCount	Total number of installments. This field is supported for installment payments with Mastercard on CyberSource through VisaNet in all countries except Brazil, Croatia, Georgia, and Greece. See "Installment Payments on CyberSource through VisaNet," page 150. The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR5 - Position: 20-22 - Field: Number Of Installments	ccAuthReply	Integer (2)
invalidField_0 through invalidField_N	Fields in the request that have invalid data. For information about missing or invalid fields, see Getting Started with CyberSource Advanced for the Simple Order API. Note These fields are included as an aid to software developers only. Do not use these fields to interact with your customers.	All CyberSource services	String (100)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
issuer_additionalData	Data defined by the issuer. The value for this reply field will probably be the same as the value that you submitted in the authorization request, but it is possible for the processor, issuer, or acquirer to modify the value. For more information, see Appendix N, "Formats for Discretionary Data," on page 480. This field is supported only for Visa transactions on CyberSource through VisaNet.	ccAuthReply ccAuthReversalReply ccCaptureReply	String (255)
issuer_message	Text message from the issuer. If you give the customer a receipt, display this value on the receipt.	ccIncrementalAuthReply	String (250)
merchantReferenceCode	Order reference or tracking number that you provided in the request. If you included multi-byte characters in this field in the request, the returned value might include corrupted characters. <i>FDC Nashville Global</i> Order reference or tracking number that you provided in the request. If the request did not include a merchant reference number, this value is provided by the client software that is installed on the POS terminal. There are some special circumstances in which the processor truncates this value to 15 or 17 characters for Level II and Level III processing. This can cause a discrepancy between the value you submit and the value included in some processor reports. <i>SIX</i> Order reference or tracking number that you provided in the request. If the request did not include a merchant reference number, this value is provided by the client software that is installed on the POS terminal.	All CyberSource services	String (50)
missingField_0 through missingField_N	Required fields that were missing from the request. For information about missing or invalid fields, see Getting Started with CyberSource Advanced for the Simple Order API. Note These fields are included as an aid to software developers only. Do not use these fields to interact with your customers.	All CyberSource services	String (100)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
originalTransaction_amount	Amount of the original transaction. See "Merchant-Initiated Reversals and Voids," page 170.	ccAuthReversal Reply voidReply	String (15)
originalTransaction_reasonCode	Reason code for the original transaction. See "Merchant-Initiated Reversals and Voids," page 170, and Reason Codes .	ccAuthReversal Reply voidReply	Integer (5)
purchaseTotals_currency	Currency used for the order. For the possible values, see the ISO Standard Currency Codes . <i>DCC with a Third-Party Provider</i> Your customer's billing currency For details, see "Dynamic Currency Conversion with a Third Party Provider," page 133.	ccAuthReply ccAuthReversal Reply ccCaptureReply ccCreditReply ccIncrementalAuth Reply	String (5)
reasonCode	Numeric value corresponding to the result of the overall request. See Reason Codes .	All CyberSource services	Integer (5)
receiptNumber	This field is returned only for American Express Direct and CyberSource through VisaNet. <i>American Express Direct</i> System trace audit number (STAN). This value identifies the transaction and is useful when investigating a chargeback dispute. <i>CyberSource through VisaNet</i> System trace number that must be printed on the customer's receipt.	ccAuthReply ccIncrementalAuth Reply	String (6)
requestID	Identifier for the request.	All CyberSource services	String (26)
requestToken	Request token data created by CyberSource for each reply. The field is an encoded string that contains no confidential information such as an account or card verification number. The string can contain a maximum of 256 characters. When you request the authorization and capture services together, the request token is for the capture reply only. <i>Atos</i> You must store the contents of this field so that you can retrieve and send it in follow-on requests.	All CyberSource services	String (256)

1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
ucaf_collectionIndicator	Universal cardholder authentication field (UCAF) collection indicator to which the transaction was downgraded. For the description and requirements, see "Payer Authentication," page 183. This field is returned only for downgraded Mastercard Identity Check transactions on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR7 - Position: 5 - Field: Mastercard Electronic Commerce Indicators—UCAF Collection Indicator	ccAuthService	String with numbers only (1)
ucaf_downgradeReasonCode	Reason the transaction was downgraded. When you set the e-commerce indicator to a value that indicates that Mastercard Identity Check data is included in the request, Mastercard provides this response value when the transaction is downgraded. See "Payer Authentication," page 183. Possible values: 0: The ucaf_authenticationData field is missing. 1: The value for the ucaf_authenticationData field is invalid. This field is returned only for downgraded Mastercard Identity Check transactions on CyberSource through VisaNet. <i>CyberSource through VisaNet</i> The value for this field corresponds to the following data in the TC 33 capture file¹: - Record: CP01 TCR6 - Position: 80 - Field: Mastercard Electronic Commerce Indicators	ccAuthService	String (1)
voidReply_amount	Amount that was voided.	voidReply	String (15)
voidReply_currency	Currency used for the order. For the possible values, see the ISO Standard Currency Codes .	voidReply	String (5)
¹ The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Table 56 Reply Fields (Continued)

Field	Description	Returned By	Data Type & Length
voidReply_reasonCode	Numeric value corresponding to the result of the void request. See Reason Codes .	voidReply	Integer (5)
voidReply_requestDateTime	Date and time when the service was requested.	voidReply	String (20)
voidReply_reversalSubmitted	Flag indicating whether a full authorization reversal was successfully submitted. Possible values: ■ <code>true</code>: The authorization reversal was successfully submitted. ■ <code>false</code>: The authorization reversal was not successfully submitted. You must send a credit request for a refund. This field is supported only for FDC Nashville Global.	voidReply	String (5)
1 The TC 33 Capture file contains information about the purchases and refunds that a merchant submits to CyberSource. CyberSource through VisaNet creates the TC 33 Capture file at the end of the day and sends it to the merchant's acquirer, who uses this information to facilitate end-of-day clearing processing with payment networks.			

Examples

Name-Value Pair Examples

Basic Credit Card Examples

Example 4 Credit Card Authorization Request

```
ccAuthService_run=true
merchantID=Napa Valley Vacations
merchantReferenceCode=482046C3A7E94F5
billTo_firstName=John
billTo_lastName=Doe
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_postalCode=94043
billTo_country=US
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
```

Example 5 Credit Card Authorization Reply

```
requestID=0305782650000167905080
decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5
purchaseTotals_currency=USD
ccAuthReply_reasonCode=100
ccAuthReply_amount=49.95
ccAuthReply_accountBalance=50.05
ccAuthReply_authorizationCode=123456
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_processorResponse=A
```

Example 6 Credit Card Capture Request

```
ccCaptureService_authRequestID=0305782650000167905080
merchantID=Napa Valley Vacations
merchantReferenceCode=482046C3A7E94F5BD1FE3C66C
ccCaptureService_run=true
item_0_unitPrice=49.95
purchaseTotals_currency=USD
```

Example 7 Credit Card Capture Reply

```
requestID=1019827520348290570293
merchantReferenceCode=482046C3A7E94F5BD1FE3C66C
decision=ACCEPT
reasonCode=100
ccCaptureReply_amount=49.95
purchaseTotals_currency=USD
ccCaptureReply_reasonCode=100
ccCaptureReply_reconciliationID=1094820975023470
```

Example 8 Credit Card Follow-on Credit Request

```
merchantID=Napa Valley Vacations
merchantReferenceCode=482046C3A7E94F5BD1FE3C66C
purchaseTotals_grandTotalAmount=1694.00
purchaseTotals_currency=EUR
ccCreditService_run=true
ccCreditService_captureRequestID=1019827520348290570293
```

Example 9 Credit Card Follow-on Credit Reply

```

merchantReferenceCode=482046C3A7E94F5BD1FE3C66C
requestID=1019827520348290570293
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=EUR
ccCreditReply_reasonCode=100
ccCreditReply_amount=1694.00
ccCreditReply_reconciliationID=C3A7E94F5BD1FE3C64820466C

```

Asia, Middle East, and Africa Gateway Examples**Example 10 Credit Card Authorization Request with Payer Authentication Data**

```

shipTo_firstName=Jane
shipTo_lastName=Smith
shipTo_street1=1234 ABCD Street
shipTo_city=Mountain View
shipTo_state=CA
shipTo_country=US
shipTo_postalCode=94043
billTo_firstName=John
billTo_lastName=Doe
billTo_street1=1295 Charleston Road
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
billTo_ipAddress=10.7.7.7
billTo_email=jdoe@example.com
billTo_phoneNumber=650-965-6000
merchantReferenceCode=0123456789
purchaseTotals_currency=USD
card_accountNumber=4111111111111111
card_expirationMonth=12
card_expirationYear=2020
card_cardType=001
ccAuthService_commerceIndicator=vbv
ccAuthService_xid=WhPlErd9WE2pb12345HlewUIQwQ
ccAuthService_veresEnrolled=Y
ccAuthService_paresStatus=Y
ccAuthService_cavv=PpmBUYXt2uyt12345mAb8XgnOk
ccAuthService_run=true
item_0_unitPrice=12.34
item_1_unitPrice=56.78

```

Example 11 Credit Card Authorization Reply

```

ccAuthReply_avsCode=2
ccAuthReply_amount=69.12
ccAuthReply_reasonCode=100
ccAuthReply_reconciliationID=19119123440
ccAuthReply_processorResponse=0
ccAuthReply_authorizationCode=ABC12345
requestID=1921371701234567904567
reasonCode=100
decision=ACCEPT
merchantReferenceCode=0123456789
purchaseTotals_currency=USD

```

Cielo Examples**Example 12 Automatic Capture Request with Elo**

```

merchantID=Foster City Flowers
merchantReferenceCode=Transaction-Cielo-NTA-3
billTo_firstName=Júlia
billTo_lastName=Fernández
billTo_buildingNumber=1024
billTo_street1=R. Augustã
billTo_street2=Bloco 01
billTo_city=São Paulo
billTo_district=Bela Vista
billTo_state=SP
billTo_postalCode=01310-000
billTo_country=BR
billTo_phoneNumber=999-999-9999
billTo_email=jfernandez@example.com
purchaseTotals_currency=usd
purchaseTotals_grandTotalAmount=104.00
card_accountNumber=1234567812345678
card_expirationMonth=03
card_expirationYear=2031
card_cardType=054
ccAuthService_run=true
ccAuthService_authType=AUTOCAPTURE
ccCaptureService_run=true

```

Example 13 Automatic Capture Reply with Elo

```

merchantReferenceCode=Transaction-Cielo-NTA-3
requestID=4231489930765000001540
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=usd
ccAuthReply_reasonCode=100
ccAuthReply_amount=104.00
ccAuthReply_authorizationCode=123456
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=CC
ccAuthReply_processorResponse=00
ccAuthReply_reconciliationID=Auth12345678
ccAuthReply_paymentNetworkTransactionID=333138
ccAuthReply_processorTransactionID=00142308609746028231
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=104.00
ccCaptureReply_reconciliationID=Auth12345678

```

Example 14 Debit Card Request with Maestro (International)

```

merchantID=Foster City Flowers
merchantReferenceCode=Transaction-Cielo-NTA-4
billTo_firstName=Júlia
billTo_lastName=Fernández
billTo_buildingNumber=1024
billTo_street1=R. Augusta
billTo_street2=Bloco 01
billTo_city=São Paulo
billTo_district=Bela Vista
billTo_state=SP
billTo_postalCode=01310-000
billTo_country=BR
billTo_phoneNumber=999-999-9999
billTo_email=jfernandez@example.com
purchaseTotals_currency=brl
purchaseTotals_grandTotalAmount=106.00
card_accountNumber=123456781234567812
card_expirationMonth=03
card_expirationYear=2031
card_cvIndicator=1
card_cvNumber=123
card_cardType=042
ucaf_authenticationData=WhPlErd9WE2pblyFjFHlewUIQwQ=
ucaf_collectionIndicator=2
ccAuthService_run=true
ccAuthService_commerceIndicator=spa
ccAuthService_xid=lEmYpm61EduaVZjPG1/HsgkAAQc=
ccAuthService_overridePaymentMethod=DB
ccCaptureService_run=true

```

Example 15 Debit Card Reply with Maestro (International)

```

merchantReferenceCode=Transaction-Cielo-NTA-4
requestID=4231489990775000001540
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=brl
ccAuthReply_reasonCode=100
ccAuthReply_amount=106.00
ccAuthReply_authorizationCode=123456
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=CC
ccAuthReply_processorResponse=00
ccAuthReply_reconciliationID=Auth12345678
ccAuthReply_paymentNetworkTransactionID=333138
ccAuthReply_processorTransactionID=00142308609746028231
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=106.00
ccCaptureReply_reconciliationID=Auth12345678

```

Example 16 Installment Request with Visa

```

merchantID=Foster City Flowers
merchantReferenceCode=Transaction-Cielo-NTA-1
billTo_firstName=Júlia
billTo_lastName=Fernández
billTo_buildingNumber=1024
billTo_street1=R. Augusta
billTo_street2=Bloco 01
billTo_city=São Paulo
billTo_district=Bela Vista
billTo_state=SP
billTo_postalCode=01310-000
billTo_country=BR
billTo_phoneNumber=999-999-9999
billTo_email=jfernandez@example.com
item_0_unitPrice=51025.00
item_0_quantity=1
purchaseTotals_currency=brl
installment_totalCount=4
installment_planType=1
card_accountNumber=4111111111111111
card_expirationMonth=12
card_expirationYear=2018
card_cardType=001
ccAuthService_run=true

```

Example 17 Installment Reply with Visa

```

merchantReferenceCode=Transaction-Cielo-NTA-1
requestID=4231493140785000001540
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=brl
ccAuthReply_reasonCode=100
ccAuthReply_amount=51025.00
ccAuthReply_authorizationCode=123456
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=CC
ccAuthReply_processorResponse=00
ccAuthReply_reconciliationID=Auth12345678
ccAuthReply_paymentNetworkTransactionID=333138
ccAuthReply_processorTransactionID=00142308609746028231

```

CyberSource Latin American Processing Examples**Note**

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. These examples are for the specific processing connection called *CyberSource Latin American Processing*. They are not for any other Latin American processors that CyberSource supports.

Example 18 Credit Card Authorization Request for Redecard in Brazil with AVS

```

ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=1234567890
billTo_firstName=Adriana
billTo_lastName=Tavares da Silva
billTo_street1=Rua da Quitanda 187
billTo_buildingNumber=187
billTo_city=Rio de Janeiro
billTo_postalCode=20091-005
billTo_country=BR
billTo_phoneNumber=+552121114700
billTo_email=asilva@example.com
billTo_personalID=987654321
item_0_quantity=1
item_0_unitPrice=49.95
purchaseTotals_currency=BRL
card_cardType=052
card_accountNumber=5432543254325432
card_expirationMonth=12
card_expirationYear=2015

```

Example 19 Credit Card Authorization Reply

```

decision=ACCEPT
reasonCode=100
requestID=12345678901234567890
merchantReferenceCode=1234567
purchaseTotals_currency=BRL
ccAuthReply_reasonCode=100
ccAuthReply_personalIDCode=Y
ccAuthReply_amount=49.95
ccAuthReply_authorizationCode=123456
ccAuthReply_reconciliationID=1911912456
ccAuthReply_avsCode=V

```

Incremental Authorization Examples**Example 20 Incremental Authorization Request**

```

ccIncrementalAuthService_run=true
purchaseTotals_grandTotalAmount=101.00
purchaseTotals_currency=USD
merchantID=Napa Valley Vacations
merchantReferenceCode=33557799
ccIncrementalAuthService_authRequestID=3434254100000181552556
subsequentAuthReason=2

```

Example 21 Incremental Authorization Reply

```

requestID=3524892270000181552556
decision=ACCEPT
reasonCode=100
merchantReferenceCode=33557799
purchaseTotals_currency=usd
additionalData=00612012345678910000000008160003359
ccIncrementalAuthReply_amount=101.00
ccIncrementalAuthReply_authorizationCode=831000
ccIncrementalAuthReply_processorResponse=A
ccIncrementalAuthReply_paymentNetworkTransactionID=016153570198200
ccIncrementalAuthReply_reasonCode=100
ccIncrementalAuthReply_reconciliationID=3524892270000181552556
ccIncrementalAuthReply_cardCategory=J1
receiptNumber=841215

```

Partial Authorization Examples

Fully Approved Request

The following two examples consist of an authorization request that is fully approved, and the subsequent authorization reply, which includes balance information:

- Original request amount: 1500.00 USD
- Approved amount: 1500.00 USD
- Balance amount: 23.62 USD positive

Example 22 Fully Approved Authorization Request

```
ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=AB1234.1-1
billTo_firstName=John
billTo_lastName=Smith
billTo_street1=201 S. Division St.
billTo_street2=Suite 500
billTo_city=Ann Arbor
billTo_state=MI
billTo_country=US
billTo_postalCode=48104-2201
billTo_email=okgo@example.com
billTo_phoneNumber=123-456-7890
card_accountNumber=4111111111111111
card_cardType=001
card_cvNumber=xxx
card_expirationMonth=12
card_expirationYear=2015
purchaseTotals_currency=USD
purchaseTotals_grandTotalAmount=1500.00
```

Example 23 Fully Approved Authorization Reply

```

merchantReferenceCode=AB1234.1-1
requestID=2688497722340000852964
decision=ACCEPT
reasonCode=100
ccAuthReply_reasonCode=100
ccAuthReply_amount=1500.00
ccAuthReply_avsCode=A
ccAuthReply_avsCodeRaw=A
ccAuthReply_authorizationCode=831000
ccAuthReply_processorResponse=000
ccAuthReply_accountBalance=23.62
ccAuthReply_accountBalanceCurrency=USD
ccAuthReply_accountBalanceSign=positive
ccAuthReply_cardCategory=J1
ccAuthReply_cvCode=3
ccAuthReply_merchantAdviceCode=00
purchaseTotals_currency=USD

```

Partially Approved Request

The following two examples consist of an authorization request that is partially approved and the subsequent authorization reply:

- Original request amount: 1401.00 USD
- Approved amount: 500.00 USD

Example 24 Partially Approved Authorization Request

```

ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=AB1234.1-1
billTo_firstName=John
billTo_lastName=Smith
billTo_street1=201 S. Division St.
billTo_street2=Suite 500
billTo_city=Ann Arbor
billTo_state=MI
billTo_country=US
billTo_postalCode=48104-2201
billTo_email=okgo@example.com
billTo_phoneNumber=123-456-7890
card_accountNumber=4111111111111111
card_cardType=001
card_cvNumber=xxx
card_expirationMonth=12
card_expirationYear=2015
purchaseTotals_currency=USD
purchaseTotals_grandTotalAmount=1401.00

```

Example 25 Partially Approved Authorization Reply

```

merchantReferenceCode=AB1234.1-1
requestID=2688497722340000852964
decision=REJECT
reasonCode=110
ccAuthReply_reasonCode=110
ccAuthReply_amount=500.00
ccAuthReply_avsCode=A
ccAuthReply_avsCodeRaw=A
ccAuthReply_authorizationCode=831000
ccAuthReply_processorResponse=010
ccAuthReply_requestAmount=1401.00
ccAuthReply_requestCurrency=USD
ccAuthReply_cardCategory=J1
ccAuthReply_cvCode=3
ccAuthReply_merchantAdviceCode=00
purchaseTotals_currency=USD

```

RuPay Examples

Example 26 Credit Card Authorization Request

```

merchantID=RuPayMerchant1
merchantReferenceCode=ABCDE12345
billTo_firstName=Jane
billTo_lastName=Smith
billTo_street1=123 Main Street
billTo_city=Any Town
billTo_state=CA
billTo_postalCode=99999-1234
billTo_country=IN
billTo_phoneNumber=999-999-9999
billTo_email=jsmith@example.com
billTo_ipAddress=123.4.567.89
purchaseTotals_currency=INR
purchaseTotals_grandTotalAmount=120.00
card_accountNumber=50888888888888
card_expirationMonth=12
card_expirationYear=2025
card_cvNumber=123
card_cardType=061
ccAuthService_run="true"
ccAuthService_cavv=XYZAV54321XYZAV54321XYZAV54321XYZAV54321
ccAuthService_commerceIndicator=rpy
ccAuthService_xid=ODI4MTEyMjU5ODAy

```

Example 27 Credit Card Authorization Reply

```
merchantReferenceCode=ABCDE12345
requestID=5391708182936018401540
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=INR
ccAuthReply_reasonCode=100
ccAuthReply_amount=120.00
ccAuthReply_authorizationCode=183217
ccAuthReply_processorResponse=ABCDE12345
ccAuthReply_reconciliationID=1094820975023470
```

Example 28 Check Status Request

```
merchantID=RuPayMerchant1
merchantReferenceCode=ABCDE12345
ccCheckStatusService_run="true"
ccCheckStatusService_authRequestID=5391708182936018401540
```

Example 29 Check Status Reply

```
merchantReferenceCode=ABCDE12345
requestID=3951708182936018401450
decision=ACCEPT
reasonCode=100
ccCheckStatusReply_reasonCode=100
ccCheckStatusReply_paymentStatus=AUTHORIZED
ccCheckStatusReply_authorizationCode=183217
```

Split-Shipment Examples

One Authorization and One Sale

Example 30 Credit Card Authorization Request

```

ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
billTo_firstName=John
billTo_lastName=Doe
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 31 Credit Card Authorization Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=0305782650000167905080
ccAuthReply_reasonCode=100
ccAuthReply_amount=49.95
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_authorizationCode=123456
ccAuthReply_processorResponse=A
purchaseTotals_currency=USD

```

Example 32 Sale Request

```

ccAuthService_run=true
ccCaptureService_run=true
linkToRequest=0305782650000167905080
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
billTo_firstName=John
billTo_lastName=Doe
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 33 Sale Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=1416783769994859
ccAuthReply_reasonCode=100
ccAuthReply_amount=49.95
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_authorizationCode=123456
ccAuthReply_processorResponse=A
purchaseTotals_currency=USD
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=49.95
ccCaptureReply_reconciliationID=02850840187309570

```

One Authorization and Two Captures

Example 34 Credit Card Authorization Request

```

ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
billTo_firstName=John
billTo_lastName=Doe
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
item_0_unitPrice=52.00
item_0_quantity=1
item_1_unitPrice=16.00
item_1_quantity=1
purchaseTotals_currency=USD

```

Example 35 Credit Card Authorization Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=0305782650000167905080
ccAuthReply_reasonCode=100
ccAuthReply_amount=68.00
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_authorizationCode=123456
ccAuthReply_processorResponse=A
purchaseTotals_currency=USD

```

Example 36 First Credit Card Capture Request

```

ccCaptureService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
ccCaptureService_authRequestID=0305782650000167905080
item_0_unitPrice=52.00
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 37 First Credit Card Capture Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=1019827520348290570293
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=52.00
ccCaptureReply_reconciliationID=02850840187309570
purchaseTotals_currency=USD

```

Example 38 Second Credit Card Capture Request

```

ccCaptureService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
ccCaptureService_authRequestID=0305782650000167905080
item_0_unitPrice=16.00
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 39 Second Credit Card Capture Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=49601835arb1569cj
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=16.00
ccCaptureReply_reconciliationID=s159vu2nh4ek9lq
purchaseTotals_currency=USD

```

Two Authorizations and One Capture

Example 40 First Credit Card Authorization Request

```
ccAuthService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
billTo_firstName=John
billTo_lastName=Doe
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD
```

Example 41 First Credit Card Authorization Reply

```
decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=0305782650000167905080
ccAuthReply_reasonCode=100
ccAuthReply_amount=49.95
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_authorizationCode=123456
ccAuthReply_processorResponse=A
purchaseTotals_currency=USD
```

Example 42 Second Credit Card Authorization Request

```

ccAuthService_run=true
linkToRequest=0305782650000167905080
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
billTo_firstName=John
billTo_lastName=Doe
billTo_phoneNumber=650-965-6000
billTo_email=jdoe@example.com
billTo_street1=1295 Charleston Rd.
billTo_city=Mountain View
billTo_state=CA
billTo_country=US
billTo_postalCode=94043
card_expirationMonth=12
card_expirationYear=2015
card_accountNumber=4111111111111111
card_cardType=001
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 43 Second Credit Card Authorization Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=1416783769994859
ccAuthReply_reasonCode=100
ccAuthReply_amount=49.95
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=YYY
ccAuthReply_authorizationCode=123456
ccAuthReply_processorResponse=A
purchaseTotals_currency=USD

```

Example 44 Credit Card Capture Request

```

ccCaptureService_run=true
merchantID=Foster City Flowers
merchantReferenceCode=482046C3A7E94F5BD1
ccCaptureService_authRequestID=1416783769994859
item_0_unitPrice=49.95
item_0_quantity=1
purchaseTotals_currency=USD

```

Example 45 Credit Card Capture Reply

```

decision=ACCEPT
reasonCode=100
merchantReferenceCode=482046C3A7E94F5BD1
requestID=1019827520348290570293
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=49.95
ccCaptureReply_reconciliationID=02850840187309570
purchaseTotals_currency=USD

```

Visa Checkout Examples**Example 46 Credit Card Authorization Request**

```

ccAuthService_run=true
merchantID=Foster_City_Flowers
merchantReferenceCode=123456
purchaseTotals_currency=USD
purchaseTotals_grandTotalAmount=25.00
paymentSolution=visacheckout
vc_orderID=335161017227386762

```

Example 47 Credit Card Authorization Reply

```

ccAuthReply_amount=25.00
ccAuthReply_avsCode=Y
ccAuthReply_authorizationCode=831000
ccAuthReply_processorResponse=00
ccAuthReply_avsCodeRaw=Y
ccAuthReply_reasonCode=100
purchaseTotals_currency=USD
decision=ACCEPT
reasonCode=100
merchantReferenceCode=123456
requestID=4068437426340172492292

```

Worldpay VAP Examples

Example 48 Credit Card Sale Request with Least-Cost Routing

```

merchantID=My_Store
merchantReferenceCode=Order_158
invoiceHeader_merchantDescriptor=Business_Name
invoiceHeader_merchantDescriptorContact=999-999-9999
invoiceHeader_merchantDescriptorAlternate=Business_Email@example.com
invoiceHeader_merchantDescriptorCity=Business_City
billTo_firstName=Jane
billTo_middleName=Anne
billTo_lastName=Smith
billTo_street1=1234 Main St.
billTo_street2=Suite 500
billTo_city=Foster City
billTo_state=CA
billTo_postalCode=94404
billTo_country=US
billTo_phoneNumber=650-999-9999
billTo_email=jsmith@example.com
billTo_ipAddress=111.111.111.111
billTo_customerID=Customer_1234
purchaseTotals_currency=USD
purchaseTotals_grandTotalAmount=5000
pos_entryMode=keyed
pos_cardPresent=y
pos_terminalCapability=2
card_accountNumber=4111111111111111
card_expirationMonth=12
card_expirationYear=2022
card_cvIndicator=1
card_cvNumber=000
ccAuthService_run=true
ccAuthService_commerceIndicator=retail
ccAuthService_leastCostRouting=y
ccCaptureService_run=true
reportGroup=Report_Group_1234

```

Example 49 Credit Card Sale Reply

```
merchantReferenceCode=Order_158
requestID=5459942086076547903095
decision=ACCEPT
reasonCode=100
purchaseTotals_currency=USD
ccAuthReply_reasonCode=100
ccAuthReply_amount=5000.00
ccAuthReply_authorizationCode=123456
ccAuthReply_avsCode=Y
ccAuthReply_avsCodeRaw=00
ccAuthReply_cvCode=M
ccAuthReply_cvCodeRaw=M
ccAuthReply_processorResponse=000
ccAuthReply_reconciliationID=5459942086076547903095
ccAuthReply_accountBalance=0.00
ccAuthReply_paymentNetworkTransactionID=1234567890123456
ccAuthReply_cardCategory=PREPAID
ccAuthReply_affluenceIndicator=MASS AFFLUENT
ccAuthReply_cardIssuerCountry=USA
ccAuthReply_processorTransactionID=1234567
ccCaptureReply_reasonCode=100
ccCaptureReply_amount=5000.00
ccCaptureReply_reconciliationID=5459942086076547903095
card_cardType=001
card_suffix=1111
card_virtual=true
card_prepaidReloadable=NO
card_prepaidType=GIFT
```

XML Examples

Basic Credit Card Examples

Example 50 Credit Card Authorization Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.23">
  <merchantID>Napa Valley Vacations</merchantID>
  <merchantReferenceCode>482046C3A7E94F5</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>
```

Example 51 Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.23">
  <c:merchantReferenceCode>482046C3A7E94F5</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
    <c:accountBalance>50.05</c:accountBalance>
  </c:ccAuthReply>
</c:replyMessage>

```

Example 52 Credit Card Capture Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.37">
  <merchantID>Napa Valley Vacations</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1FE3C66C</merchantReferenceCode>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <ccCaptureService run="true">
    <authRequestID>0305782650000167905080</authRequestID>
  </ccCaptureService>
</requestMessage>

```

Example 53 Credit Card Capture Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.37">
  <c:merchantReferenceCode>482046C3A7E94F5BD1FE3C66C</c:merchantReferenceCode>
  <c:requestID>1019827520348290570293</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:reconciliationID>1094820975023470</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>
```

Example 54 Credit Card Follow-on Credit Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.86">
  <merchantID>Napa Valley Vacations</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BDC66C</merchantReferenceCode>
  <purchaseTotals>
    <currency>EUR</currency>
    <grandTotalAmount>1694.00</grandTotalAmount>
  </purchaseTotals>
  <ccCreditService run="true">
    <captureRequestID>1019827520348290570293</captureRequestID>
  </ccCreditService>
</requestMessage>
```

Example 55 Credit Card Follow-on Credit Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.86">
  <c:merchantReferenceCode>482046C3A75BD1FE3C66C</c:merchantReferenceCode>
  <c:requestID>9057101982752034820293</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>EUR</c:currency>
  </c:purchaseTotals>
  <c:ccCreditReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>1694.00</c:amount>
    <c:reconciliationID>C3A7E94F5BD1FE3C64820466C</c:reconciliationID>
  </c:ccCreditReply>
</c:replyMessage>
```

Asia, Middle East, and Africa Gateway Examples

Example 56 Credit Card Authorization Request with Payer Authentication Data

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.32">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>0123456789</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Road</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
    <ipAddress>10.7.7.7</ipAddress>
  </billTo>
  <shipTo>
    <firstName>Jane</firstName>
    <lastName>Smith</lastName>
    <street1>1234 ABCD Street</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
  </shipTo>
  <item id="0">
    <unitPrice>12.34</unitPrice>
  </item>
  <item id="1">
    <unitPrice>56.78</unitPrice>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2020</expirationYear>
    <cvNumber>1234</cvNumber>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true">
    <cavv>PpmBUYXt2uytV6p12345KuImAb8XgnOk</cavv>
    <commerceIndicator>vbv</commerceIndicator>
    <xid>WhPlErd9WE1234562pb1yFjFHlewUIQwQ</xid>
    <veresEnrolled>Y</veresEnrolled>
    <paresStatus>Y</paresStatus>
  </ccAuthService>
</requestMessage>

```

Example 57 Credit Card Authorization Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.32">
  <c:merchantReferenceCode>0123456789</c:merchantReferenceCode>
  <c:requestID>1921312345620167904567</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>69.12</c:amount>
    <c:authorizationCode>ABC12345</c:authorizationCode>
    <c:avsCode>2</c:avsCode>
    <c:cvCode>2</c:cvCode>
    <c:cvCodeRaw>Q</c:cvCodeRaw>
    <c:processorResponse>0</c:processorResponse>
    <c:reconciliationID>19119123438</c:reconciliationID>
  </c:ccAuthReply>
</c:replyMessage>
```

Cielo Examples

Example 58 Automatic Capture Request with Elo

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.111">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>Transaction-Cielo-NTA-3</merchantReferenceCode>
  <billTo>
    <firstName>Júlia</firstName>
    <lastName>Fernández</lastName>
    <buildingNumber>1024</buildingNumber>
    <street1>R. Augustã</street1>
    <street2>Bloco 01</street2>
    <city>São Paulo</city>
    <district>Bela Vista</district>
    <state>SP</state>
    <postalCode>01310-000</postalCode>
    <country>BR</country>
    <phoneNumber>999-999-9999</phoneNumber>
    <email>jfernandez@example.com</email>
  </billTo>
  <purchaseTotals>
    <currency>usd</currency>
    <grandTotalAmount>104.00</grandTotalAmount>
  </purchaseTotals>
  <card>
    <accountNumber>1234567812345678</accountNumber>
    <expirationMonth>03</expirationMonth>
    <expirationYear>2031</expirationYear>
    <cardType>054</cardType>
  </card>
  <ccAuthService run="true">
    <authType>AUTOCAPTURE</authType>
  </ccAuthService>
  <ccCaptureService run="true"></ccCaptureService>
</requestMessage>
```

Example 59 Automatic Capture Reply with Elo

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.111">
  <c:merchantReferenceCode>Transaction-Cielo-NTA-3</c:merchantReferenceCode>
  <c:requestID>4231489930765000001540</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>usd</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>104.00</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>CC</c:avsCodeRaw>
    <c:processorResponse>00</c:processorResponse>
    <c:reconciliationID>Auth12345678</c:reconciliationID>
    <c:paymentNetworkTransactionID>333138</c:paymentNetworkTransactionID>
    <c:processorTransactionID>00142308609746028231</c:processorTransactionID>
  </c:ccAuthReply>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>104.00</c:amount>
    <c:reconciliationID>Auth12345678</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>

```

Example 60 Debit Card Request with Maestro (International)

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.111">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>Transaction-Cielo-NTA-4</merchantReferenceCode>
  <billTo>
    <firstName>Júlia</firstName>
    <lastName>Fernández</lastName>
    <buildingNumber>1024</buildingNumber>
    <street1>R. Augustã</street1>
    <street2>Bloco 01</street2>
    <city>São Paulo</city>
    <district>Bela Vista</district>
    <state>SP</state>
    <postalCode>01310-000</postalCode>
    <country>BR</country>
    <phoneNumber>999-999-9999</phoneNumber>
    <email>jfernandez@example.com</email>
  </billTo>
  <purchaseTotals>
    <currency>brl</currency>
    <grandTotalAmount>106.00</grandTotalAmount>
  </purchaseTotals>
  <card>
    <accountNumber>123456781234567812</accountNumber>
    <expirationMonth>03</expirationMonth>
    <expirationYear>2031</expirationYear>
    <cvIndicator>1</cvIndicator>
    <cvNumber>123</cvNumber>
    <cardType>042</cardType>
  </card>
  <ucaf>
    <authenticationData>WhPlErd9WE2pblyFjFHlewUIQwQ=</authenticationData>
    <collectionIndicator>2</collectionIndicator>
  </ucaf>
  <ccAuthService run="true">
    <commerceIndicator>spa</commerceIndicator>
    <xid>lEmYpm6lEduaVZjPG1/HsgkAAQc=</xid>
    <overridePaymentMethod>DB</overridePaymentMethod>
  </ccAuthService>
  <ccCaptureService run="true"></ccCaptureService>
</requestMessage>

```

Example 61 Debit Card Reply with Maestro (International)

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.111">
  <c:merchantReferenceCode>Transaction-Cielo-NTA-4</c:merchantReferenceCode>
  <c:requestID>4231489990775000001540</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>brl</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>106.00</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>CC</c:avsCodeRaw>
    <c:processorResponse>00</c:processorResponse>
    <c:reconciliationID>Auth12345678</c:reconciliationID>
    <c:paymentNetworkTransactionID>333138</c:paymentNetworkTransactionID>
    <c:processorTransactionID>00142308609746028231</c:processorTransactionID>
  </c:ccAuthReply>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>106.00</c:amount>
    <c:reconciliationID>Auth12345678</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>

```

Example 62 Installment Request with Visa

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.111">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>Transaction-Cielo-NTA-1</merchantReferenceCode>
  <billTo>
    <firstName>Júlia</firstName>
    <lastName>Fernández</lastName>
    <buildingNumber>1024</buildingNumber>
    <street1>R. Augustã</street1>
    <street2>Bloco 01</street2>
    <city>São Paulo</city>
    <district>Bela Vista</district>
    <state>SP</state>
    <postalCode>01310-000</postalCode>
    <country>BR</country>
    <phoneNumber>999-999-9999</phoneNumber>
    <email>jfernandez@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>51025.00</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>brl</currency>
  </purchaseTotals>
  <installment>
    <totalCount>4</totalCount>
    <planType>1</planType>
  </installment>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2018</expirationYear>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"></ccAuthService>
</requestMessage>

```

Example 63 Installment Reply with Visa

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.111">
  <c:merchantReferenceCode>Transaction-Cielo-NTA-1</c:merchantReferenceCode>
  <c:requestID>4231493140785000001540</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>brl</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>51025.00</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>CC</c:avsCodeRaw>
    <c:processorResponse>00</c:processorResponse>
    <c:reconciliationID>Auth12345678</c:reconciliationID>
    <c:paymentNetworkTransactionID>333138</c:paymentNetworkTransactionID>
    <c:processorTransactionID>00142308609746028231</c:processorTransactionID>
  </c:ccAuthReply>
</c:replyMessage>

```

CyberSource Latin American Processing Examples

**Note**

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. These examples are for the specific processing connection called *CyberSource Latin American Processing*. They are not for any other Latin American processors that CyberSource supports.

Example 64 Credit Card Authorization Request for Redecard in Brazil with AVS

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.41">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>1234567890</merchantReferenceCode>
  <billTo>
    <firstName>Adriana</firstName>
    <lastName>Tavares da Silva</lastName>
    <street1>Rua da Quitanda 187</street1>
    <city>Rio de Janeiro</city>
    <postalCode>20091-005</postalCode>
    <country>BR</country>
    <phoneNumber>+552121114700</phoneNumber>
    <email>asilva@example.com</email>
    <personalID>987654321</personalID>
    <buildingNumber>187</buildingNumber>
  </billTo>
  <item id="0"><unitPrice>49.95</unitPrice></item>
  <purchaseTotals><currency>BRL</currency></purchaseTotals>
  <card>
    <accountNumber>5432543254325432</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>052</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>

```

Example 65 Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.41">
  <c:merchantReferenceCode>1234567</c:merchantReferenceCode>
  <c:requestID>12345678901234567890</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals><c:currency>BRL</c:currency></c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>V</c:avsCode>
    <c:personalIDCode>Y</c:personalIDCode>
    <c:reconciliationID>19119123456</c:reconciliationID>
  </c:ccAuthReply>
</c:replyMessage>

```

Incremental Authorization Examples

Example 66 Incremental Authorization Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.23">
  <merchantID>Napa Valley Vacations</merchantID>
  <merchantReferenceCode>33557799</merchantReferenceCode>
  <purchaseTotals>
    <currency>USD</currency>
    <grandTotalAmount>101.00</grandTotalAmount>
  </purchaseTotals>
  <ccIncrementalAuthService run="true">
    <authRequestID>3434254100000181552556</authRequestID>
  </ccIncrementalAuthService>
  <subsequentAuthReason>2</subsequentAuthReason>
</requestMessage>
```

Example 67 Incremental Authorization Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.23">
  <c:merchantReferenceCode>482046C3A7E94F5</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccIncrementalAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>101.00</c:amount>
    <c:authorizationCode>831000</c:authorizationCode>
    <c:processorResponse>A</c:processorResponse>
    <c:reconciliationID>3524892270000181552556</c:reconciliationID>
    <c:paymentNetworkTransactionID>016153570198200</c:paymentNetworkTransactionID>
    <c:cardCategory>J1</c:cardCategory>
  </c:ccIncrementalAuthReply>
  <c:receiptNumber>841215</c:receiptNumber>
  <c:additionalData>00612012345678910000000008160003359</c:additionalData>
</c:replyMessage>
```

Partial Authorization Examples

Fully Approved Request

The following two examples consist of an authorization request that is fully approved, and the subsequent authorization reply, which includes balance information:

- Original request amount: 1500.00 USD
- Approved amount: 1500.00 USD
- Balance amount: 23.62 USD positive

Example 68 Fully Approved Authorization Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.52">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>AB1234.1-1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Smith</lastName>
    <street1>201 S. Division St.</street1>
    <street2>Suite 500</street2>
    <city>Ann Arbor</city>
    <state>MI</state>
    <postalCode>48104-2201</postalCode>
    <country>US</country>
    <phoneNumber>123-456-7890</phoneNumber>
    <email>okgo@example.com</email>
  </billTo>
  <purchaseTotals>
    <currency>USD</currency>
    <grandTotalAmount>1500.00</grandTotalAmount>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cvNumber>xxx</cvNumber>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>
```

Example 69 Fully Approved Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.52">
  <c:merchantReferenceCode>AB1234.1-1</c:merchantReferenceCode>
  <c:requestID>2688497722340000852964</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals><c:currency>USD</c:currency></c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>1500.00</c:amount>
    <c:authorizationCode>831000</c:authorizationCode>
    <c:avsCode>A</c:avsCode>
    <c:avsCodeRaw>A</c:avsCodeRaw>
    <c:cvCode>3</c:cvCode>
    <c:processorResponse>000</c:processorResponse>
    <c:merchantAdviceCode>00</c:merchantAdviceCode>
    <c:accountBalance>23.62</c:accountBalance>
    <c:cardCategory>J1</c:cardCategory>
    <c:accountBalanceCurrency>USD</c:accountBalanceCurrency>
    <c:accountBalanceSign>positive</c:accountBalanceSign>
  </c:ccAuthReply>
</c:replyMessage>

```

Partially Approved Request

The following two examples consist of an authorization request that is partially approved and the subsequent authorization reply:

- Original request amount: 1401.00 USD
- Approved amount: 500.00 USD

Example 70 Partially Approved Authorization Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.52">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>AB1234.1-1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Smith</lastName>
    <street1>201 S. Division St.</street1>
    <street2>Suite 500</street2>
    <city>Ann Arbor</city>
    <state>MI</state>
    <postalCode>48104-2201</postalCode>
    <country>US</country>
    <phoneNumber>123-456-7890</phoneNumber>
    <email>okgo@example.com</email>
  </billTo>
  <purchaseTotals>
    <currency>USD</currency>
    <grandTotalAmount>1401.00</grandTotalAmount>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cvNumber>xxx</cvNumber>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>
```

Example 71 Partially Approved Authorization Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.52">
  <c:merchantReferenceCode>AB1234.1-1</c:merchantReferenceCode>
  <c:requestID>2688497722340000852964</c:requestID>
  <c:decision>REJECT</c:decision>
  <c:reasonCode>110</c:reasonCode>
  <c:purchaseTotals><c:currency>USD</c:currency></c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>110</c:reasonCode>
    <c:amount>500.00</c:amount>
    <c:authorizationCode>831000</c:authorizationCode>
    <c:avsCode>A</c:avsCode>
    <c:avsCodeRaw>A</c:avsCodeRaw>
    <c:cvCode>3</c:cvCode>
    <c:processorResponse>010</c:processorResponse>
    <c:merchantAdviceCode>00</c:merchantAdviceCode>
    <c:cardCategory>J1</c:cardCategory>
    <c:requestAmount>1401.00</c:requestAmount>
    <c:requestCurrency>USD</c:requestCurrency>
  </c:ccAuthReply>
</c:replyMessage>
```

RuPay Examples

Example 72 Credit Card Authorization Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.151">
  <merchantID>RuPayMerchant1</merchantID>
  <merchantReferenceCode>ABCDE12345</merchantReferenceCode>
  <billTo>
    <firstName>Jane</firstName>
    <lastName>Smith</lastName>
    <street1>123 Main Street</street1>
    <city>Any Town</city>
    <state>CA</state>
    <postalCode>99999-1234</postalCode>
    <country>IN</country>
    <phoneNumber>999-999-9999</phoneNumber>
    <email>jsmith@example.com</email>
    <ipAddress>123.4.567.89</ipAddress>
  </billTo>
  <purchaseTotals>
    <currency>INR</currency>
    <grandTotalAmount>120.00</grandTotalAmount>
  </purchaseTotals>
  <card>
    <accountNumber>50888888888888</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2025</expirationYear>
    <cvNumber>123</cvNumber>
    <cardType>061</cardType>
  </card>
  <ccAuthService run="true">
    <cavv>XYZAV54321XYZAV54321XYZAV54321XYZAV54321</cavv>
    <commerceIndicator>rpy</commerceIndicator>
    <xid>ODI4MTEyMjU5ODAy</xid>
  </ccAuthService>
</requestMessage>

```

Example 73 Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.151">
  <c:merchantReferenceCode>ABCDE12345</c:merchantReferenceCode>
  <c:requestID>5391708182936018401540</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>INR</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>120.00</c:amount>
    <c:authorizationCode>183217</c:authorizationCode>
    <c:processorResponse>ABCDE12345</c:processorResponse>
    <c:reconciliationID>1094820975023470</c:reconciliationID>
  </c:ccAuthReply>
</c:replyMessage>

```

Example 74 Check Status Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.151">
  <merchantID>RuPayMerchant1</merchantID>
  <merchantReferenceCode>ABCDE12345</merchantReferenceCode>
  <ccCheckStatusService run="true">
    <authRequestID>5391708182936018401540</authRequestID>
  </ccCheckStatusService>
</requestMessage>

```

Example 75 Check Status Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.151">
  <c:merchantReferenceCode>ABCDE12345</c:merchantReferenceCode>
  <c:requestID>3951708182936018401450</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:ccCheckStatusReply>
    <c:reasonCode>100</c:reasonCode>
    <c:paymentStatus>AUTHORIZED</c:paymentStatus>
    <c:authorizationCode>183217</c:authorizationCode>
  </c:ccCheckStatusReply>
</c:replyMessage>

```

Split-Shipment Examples

One Authorization and One Sale

Example 76 Credit Card Authorization Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>
```

Example 77 Credit Card Authorization Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
  </c:ccAuthReply>
</c:replyMessage>
```

Example 78 Sale Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <linkToRequest>0305782650000167905080</linkToRequest>
  <ccAuthService run="true"/>
  <ccCaptureService run="true"/>
</requestMessage>

```

Example 79 Sale Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
  </c:ccAuthReply>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:reconciliationID>02850840187309570</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>

```

One Authorization and Two Captures

Example 80 Credit Card Authorization Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>52.00</unitPrice>
    <quantity>1</quantity>
  </item>
  <item id="1">
    <unitPrice>16.00</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>

```

Example 81 Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>68.00</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
  </c:ccAuthReply>
</c:replyMessage>

```

Example 82 First Credit Card Capture Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <item id="0">
    <unitPrice>52.00</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <ccCaptureService run="true">
    <authRequestID>0305782650000167905080</authRequestID>
  </ccCaptureService>
</requestMessage>

```

Example 83 First Credit Card Capture Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>1019827520348290570293</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>52.00</c:amount>
    <c:reconciliationID>02850840187309570</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>
```

Example 84 Second Credit Card Capture Request

```
<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <item id="0">
    <unitPrice>16.00</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <ccCaptureService run="true">
    <authRequestID>0305782650000167905080</authRequestID>
  </ccCaptureService>
</requestMessage>
```

Example 85 Second Credit Card Capture Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>1019827520348290570293</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>16.00</c:amount>
    <c:reconciliationID>sl59vu2nh4ek9lq</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>
```

Two Authorizations and One Capture

Example 86 First Credit Card Authorization Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <ccAuthService run="true"/>
</requestMessage>

```

Example 87 First Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>0305782650000167905080</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
  </c:ccAuthReply>
</c:replyMessage>

```

Example 88 Second Credit Card Authorization Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <billTo>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <street1>1295 Charleston Rd.</street1>
    <city>Mountain View</city>
    <state>CA</state>
    <postalCode>94043</postalCode>
    <country>US</country>
    <phoneNumber>650-965-6000</phoneNumber>
    <email>jdoe@example.com</email>
  </billTo>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2015</expirationYear>
    <cardType>001</cardType>
  </card>
  <linkToRequest>0305782650000167905080</linkToRequest>
  <ccAuthService run="true"/>
</requestMessage>

```

Example 89 Second Credit Card Authorization Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>1416783769994859</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>YYY</c:avsCodeRaw>
    <c:processorResponse>A</c:processorResponse>
  </c:ccAuthReply>
</c:replyMessage>

```

Example 90 Credit Card Capture Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.40">
  <merchantID>Foster City Flowers</merchantID>
  <merchantReferenceCode>482046C3A7E94F5BD1</merchantReferenceCode>
  <item id="0">
    <unitPrice>49.95</unitPrice>
    <quantity>1</quantity>
  </item>
  <purchaseTotals>
    <currency>USD</currency>
  </purchaseTotals>
  <ccCaptureService run="true">
    <authRequestID>1416783769994859</authRequestID>
  </ccCaptureService>
</requestMessage>

```

Example 91 Credit Card Capture Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.40">
  <c:merchantReferenceCode>482046C3A7E94F5BD1</c:merchantReferenceCode>
  <c:requestID>1019827520348290570293</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>49.95</c:amount>
    <c:reconciliationID>02850840187309570</c:reconciliationID>
  </c:ccCaptureReply>
</c:replyMessage>

```

Visa Checkout Examples

Example 92 Credit Card Authorization Request

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.105">
  <merchantID>Foster_City_Flowers</merchantID>
  <merchantReferenceCode>123456</merchantReferenceCode>
  <purchaseTotals>
    <currency>USD</currency>
    <grandTotalAmount>25.00</grandTotalAmount>
  </purchaseTotals>
  <ccAuthService_run="true"/>
  <paymentSolution>visacheckout</paymentSolution>
  <vc><orderID>335161017227386762</orderID></vc>
</requestMessage>

```

Example 93 Credit Card Authorization Reply

```
<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.105">
  <c:merchantReferenceCode>123456</c:merchantReferenceCode>
  <c:requestID>4068437426340172492292</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>25.00</c:amount>
    <c:authorizationCode>831000</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>Y</c:avsCodeRaw>
    <c:processorResponse>00</c:processorResponse>
  </c:ccAuthReply>
</c:replyMessage>
```

Worldpay VAP Examples

Example 94 Credit Card Sale Request with Least-Cost Routing

```

<requestMessage xmlns="urn:schemas-cybersource-com:transaction-data-1.151">
  <merchantID>My_Store</merchantID>
  <merchantReferenceCode>Order_158</merchantReferenceCode>
  <invoiceHeader>
    <merchantDescriptor>Business_Name</merchantDescriptor>
    <merchantDescriptorContact>999-999-9999</merchantDescriptorContact>
    <merchantDescriptorAlternate>Business_Email@example.com</merchantDescriptorAlternate>
    <merchantDescriptorCity>Business_City</merchantDescriptorCity>
  </invoiceHeader>
  <billTo>
    <firstName>Jane</firstName>
    <middleName>Anne</middleName>
    <lastName>Smith</lastName>
    <street1>1234 Main St.</street1>
    <street2>Suite 500</street2>
    <city>Foster City</city>
    <state>CA</state>
    <postalCode>94404</postalCode>
    <country>US</country>
    <phoneNumber>650-999-9999</phoneNumber>
    <email>jsmith@example.com</email>
    <ipAddress>111.111.111.111</ipAddress>
    <customerID>Customer_1234</customerID>
  </billTo>
  <purchaseTotals>
    <currency>USD</currency>
    <grandTotalAmount>5000</grandTotalAmount>
  </purchaseTotals>
  <pos>
    <entryMode>keyed</entryMode>
    <cardPresent>y</cardPresent>
    <terminalCapability>2</terminalCapability>
  </pos>
  <card>
    <accountNumber>4111111111111111</accountNumber>
    <expirationMonth>12</expirationMonth>
    <expirationYear>2022</expirationYear>
    <cvIndicator>1</cvIndicator>
    <cvNumber>000</cvNumber>
  </card>
  <ccAuthService run="true">
    <commerceIndicator>retail</commerceIndicator>
    <leastCostRouting>y</leastCostRouting>
  </ccAuthService>
  <ccCaptureService run="true"/>
  <reportGroup>Report_Group_1234</reportGroup>
</requestMessage>

```

Example 95 Credit Card Sale Reply

```

<c:replyMessage xmlns:c="urn:schemas-cybersource-com:transaction-data-1.151">
  <c:merchantReferenceCode>Order_158</c:merchantReferenceCode>
  <c:requestID>5459942086076547903095</c:requestID>
  <c:decision>ACCEPT</c:decision>
  <c:reasonCode>100</c:reasonCode>
  <c:purchaseTotals>
    <c:currency>USD</c:currency>
  </c:purchaseTotals>
  <c:ccAuthReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>5000.00</c:amount>
    <c:authorizationCode>123456</c:authorizationCode>
    <c:avsCode>Y</c:avsCode>
    <c:avsCodeRaw>00</c:avsCodeRaw>
    <c:cvCode>M</c:cvCode>
    <c:cvCodeRaw>M</c:cvCodeRaw>
    <c:processorResponse>000</c:processorResponse>
    <c:reconciliationID>5459942086076547903095</c:reconciliationID>
    <c:accountBalance>0.00</c:accountBalance>
    <c:paymentNetworkTransactionID>1234567890123456</c:paymentNetworkTransactionID>
    <c:cardCategory>PREPAID</c:cardCategory>
    <c:affluenceIndicator>MASS AFFLUENT</c:affluenceIndicator>
    <c:cardIssuerCountry>USA</c:cardIssuerCountry>
    <c:processorTransactionID>1234567</c:processorTransactionID>
  </c:ccAuthReply>
  <c:ccCaptureReply>
    <c:reasonCode>100</c:reasonCode>
    <c:amount>5000.00</c:amount>
    <c:reconciliationID>5459942086076547903095</c:reconciliationID>
  </c:ccCaptureReply>
  <c:card>
    <c:cardType>001</c:cardType>
    <c:suffix>1111</c:suffix>
    <c:virtual>true</c:virtual>
    <c:prepaidReloadable>NO</c:prepaidReloadable>
    <c:prepaidType>GIFT</c:prepaidType>
  </c:card>
</c:replyMessage>

```

Additional Amount Types

Additional amount types are used with additional amounts, which are described in ["Additional Amounts," page 116](#).

Table 57 Additional Amount Types for Goods and Services

Goods and Services	Code
Additional collection	102
Advance purchase for cabin upgrade	186
Advance purchase for excess baggage	184
Advance purchase for optional services	187
Advance purchase for preferred seat upgrade	185
Agency exchange order	119
Agency group ticket	120
Agency Miscellaneous charge order (MCO)	118
Agency passenger ticket	116
Agency tour order or voucher	117
Agent-assisted reservation or ticketing fee	168
Animal transportation charge	127
Bar	019
Bar or mini-bar	023
Barber or beauty salon	028
Beverage	017
Business center	036
Cabin update	166
Catalogue merchandise order	123
Catering charges	022
Checked sporting or special equipment	176
Club membership fee	143
Club membership fee or temporary trial	179
Convention fees	037
Coupon book	144

Table 57 Additional Amount Types for Goods and Services (Continued)

Goods and Services	Code
Credit for overcharge adjustment	136
Credit for unused transportation	130
Credit for class of service adjustment	131
Credit for denied boarding	132
Credit for exchange	135
Credit for lost ticket	134
Credit for miscellaneous refund	133
Credit for multiple unused tickets	137
Curbside baggage	173
Debit adjustment or duplicate refund	121
Dry ice fee	177
Duty-free sale	164
Excess baggage	103
Exchange adjustment	113
Exchange order	138
Firearms case	128
Food	016
Food and beverages	018
Frequent flyer fee or purchase	125
Frequent flyer fulfillment	148
Frequent flyer activation or reinstatement	180
Frequent flyer overnight delivery charge	147
Gift certificate	181
Gift shop	030
Group ticket	112
Health and fitness	029
In-flight adjustment	115
In-flight charges	145
In-flight duty-free purchase	141
In-flight merchandise order	122
In-flight phone charges	124
In-flight entertainment or internet access	190
In-flight medical equipment	174
Insurance purchased	052
Internet service	025

Table 57 Additional Amount Types for Goods and Services (Continued)

Goods and Services	Code
Kennel charge	126
Laundry or dry cleaning	027
Lodging	020
Lost ticket application	108
Lounge or club access or day pass	167
Mail or postage fee	178
Miscellaneous charge order (MCO) or prepaid ticket authorization	104
Miscellaneous taxes and fees	151
Movies or pay-per-view	026
Onboard/in-flight prepaid voucher	182
Optional services fee	183
Overweight bag fee	191
Packages	189
Passenger ticket	101
Pet fees	033
Phone	024
Preferred seat upgrade	165
Pro shop	031
Restaurant or room service	021
Reward program transaction	047
Self-service tickets	139
Senior citizen discount booklets	142
Sleep sets	192
Small package delivery	149
SPD or air freight	114
Special purchase fee	193
Special service ticket	105
Standby fee	172
Supported refund	106
Ticket by mail	110
Ticket change or cancel fee	169
Ticket or pass print fee	175
Tip or gratuity	058
Tour order voucher	109

Table 57 Additional Amount Types for Goods and Services (Continued)

Goods and Services	Code
Tour deposit	146
Tours	034
Travel agency fee	152
Trip insurance	170
Unaccompanied minor	171
Undercharge adjustment	111
Unsupported refund	107
Upgrade charge	129
Vendor refund	160
Vendor sale	150
WIFI	188

Table 58 Additional Amount Types for Charges and Fees

Charges and Fees	Code
Additional miles / kilometers / distance	062
Auto rental adjustment	060
Cancellation adjustment	065
Charges added after check-out or departure	041
Convenience charge	050
Delivery charge	051
Discount	053
Equipment rental	035
Express service charge	040
Freight / shipping / handling	055
Fuel charge	061
Late return	063
Meeting or conference charges	038
Miscellaneous charges and fees	042
No show charge	039
Order processing charge	049
Parking fee	032
Policy adjustment	066
Repairs	064
Surcharge	048

Table 58 Additional Amount Types for Charges and Fees (Continued)

Charges and Fees	Code
Tickets and violations	054
Travel miscellaneous charges	075
Travel transaction fees	074

Table 59 Additional Amount Types for Taxes

Taxes	Code
Airport tax	015
Circulation of merchandise and service tax (ICMS)	007
Consumption tax	002
Exempt—no GST charged	068
Goods and services tax (GST)	001
Harmonized sales tax (HST)	005
Income security and reform tax (ISR Income Tax)	011
Industrialized products federal tributary tax (IPI Federal Tributary)	008
Inland revenue income tax (IR Income Tax)	009
Insurance premium tax (IPT)	006
International students and scholars income tax (ISS Income Tax)	010
Miscellaneous tax	046
Occupancy tax	012
Provincial sales tax (PST)	003
Quebec sales tax (QST)	004
Room tax	013
Sales tax	056
Stamp duty	067
Surcharge tax	014
Ticket tax	043
Value added tax (VAT)	057

American Express SafeKey Response Codes

The American Express SafeKey response code is returned in **ccAuthReply_cavvResponseCode** in the reply message for an authorization request. See ["American Express SafeKey," page 184](#), for a description of American Express SafeKey.

Table 60 American Express SafeKey Response Codes

Response Code	Description
1	CAVV failed validation and authentication.
2	CAVV passed validation and authentication.
3	CAVV passed the validation attempt.
4	CAVV failed the validation attempt.
7	CAVV failed the validation attempt and the issuer is available.
8	CAVV passed the validation attempt and the issuer is available.
9	CAVV failed the validation attempt and the issuer is not available.
A	CAVV passed the validation attempt and the issuer is not available.
U	Issuer does not participate or 3D secure data was not used.
99	An unknown value was returned from the processor.

AVS Codes

The AVS code is returned in **ccAuthReply_avsCode** in the authorization reply message. See ["Address Verification System \(AVS\)," page 88](#), for a description of AVS.

AVS Codes for Cielo 3.0 and CyberSource Latin American Processing



Note

CyberSource Latin American Processing is the name of a specific processing connection that CyberSource supports. In the CyberSource API documentation, *CyberSource Latin American Processing* does not refer to the general topic of processing in Latin America. The information in this section is for the specific processing connection called *CyberSource Latin American Processing*. It is not for any other Latin American processors that CyberSource supports.

Table 61 AVS Codes for Cielo 3.0 and CyberSource Latin American Processing

Code	Description
D	Partial match: postal code and address match.
E	Not supported: AVS is not supported for this card type. or Invalid: the acquirer returned an unrecognized value for the AVS response.
F	Partial match: postal code matches, but CPF and address do not match. ¹
G	Not supported: AVS not supported or not verified.
I	No match: AVS information is not available.
K	Partial match: CPF matches, but postal code and address do not match. ¹
L	Partial match: postal code and CPF match, but address does not match. ¹
N	No match: postal code, CPF, and address do not match. ¹
¹ CPF (Cadastro de Pessoas Físicas) is required only for Redecard in Brazil.	

Table 61 AVS Codes for Cielo 3.0 and CyberSource Latin American Processing (Continued)

Code	Description
O	Partial match: CPF and address match, but postal code does not match. ¹
R	Not supported: your implementation does not support AVS. or System unavailable.
T	Partial match: address matches, but postal code and CPF do not match. ¹
V	Match: postal code, CPF, and address match. ¹
¹ CPF (Cadastro de Pessoas Físicas) is required only for Redecard in Brazil.	

AVS Codes for All Other Processors

Table 62 Types of AVS Codes

Type of Codes	Codes	Description
Codes for American Express Cards	F, H, K, L, O, T, V	For American Express cards only. For American Express cards, you can receive Visa and CyberSource AVS codes in addition to the American Express AVS codes. Note For CyberSource through VisaNet, the American Express AVS codes are converted to Visa AVS codes before they are returned to you. As a result, you will not receive American Express AVS codes for the American Express card type.

Table 62 Types of AVS Codes (Continued)

Type of Codes	Codes	Description
International Visa Codes	B, C, D, G, I, M, P	The international and domestic alphabetic AVS codes are the Visa standard AVS codes. CyberSource maps the standard AVS return codes for other types of payment cards, including American Express cards, to the Visa standard AVS codes. AVS is considered either domestic or international, depending on the location of the bank that issued the customer's payment card: ■ When the bank is in the U.S., the AVS is domestic. ■ When the bank is outside the U.S., the AVS is international. You should be prepared to handle domestic and international AVS result codes: ■ For international cards, you can receive domestic AVS codes in addition to the international AVS codes. ■ For domestic cards, you can receive international AVS codes in addition to the domestic AVS codes.
Domestic Visa Codes	A, E, N, R, S, U, W, X, Y, Z	
CyberSource Codes	1, 2, 3, 4	The numeric AVS codes are created by CyberSource and are not standard Visa codes. These AVS codes can be returned for any card type.

Table 63 AVS Codes

Code	Description
A	Partial match: street address matches, but 5-digit and 9-digit postal codes do not match.
B	Partial match: street address matches, but postal code is not verified. Returned only for Visa cards not issued in the U.S.
C	No match: street address and postal code do not match. Returned only for Visa cards not issued in the U.S.
D & M	Match: street address and postal code match. Returned only for Visa cards not issued in the U.S.
E	Invalid: AVS data is invalid or AVS is not allowed for this card type.
F	Partial match: card member's name does not match, but billing postal code matches.
G	Not supported: issuing bank outside the U.S. does not support AVS.
H	Partial match: card member's name does not match, but street address and postal code match. Returned only for the American Express card type.
I	No match: address not verified. Returned only for Visa cards not issued in the U.S.
K	Partial match: card member's name matches, but billing address and billing postal code do not match. Returned only for the American Express card type.

Table 63 AVS Codes (Continued)

Code	Description
L	Partial match: card member's name and billing postal code match, but billing address does not match. Returned only for the American Express card type.
M	See the entry for D & M.
N	No match: one of the following: ■ Street address and postal code do not match. ■ Card member's name, street address, and postal code do not match. Returned only for the American Express card type.
O	Partial match: card member's name and billing address match, but billing postal code does not match. Returned only for the American Express card type.
P	Partial match: postal code matches, but street address not verified. Returned only for Visa cards not issued in the U.S.
R	System unavailable.
S	Not supported: issuing bank in the U.S. does not support AVS.
T	Partial match: card member's name does not match, but street address matches. Returned only for the American Express card type.
U	System unavailable: address information unavailable for one of these reasons: ■ The U.S. bank does not support AVS outside the U.S. ■ The AVS in a U.S. bank is not functioning properly.
V	Match: card member's name, billing address, and billing postal code match. Returned only for the American Express card type.
W	Partial match: street address does not match, but 9-digit postal code matches.
X	Match: street address and 9-digit postal code match.
Y	Match: street address and 5-digit postal code match.
Z	Partial match: street address does not match, but 5-digit postal code matches.
1	Not supported: one of the following: ■ AVS is not supported for this processor or card type. ■ AVS is disabled for your CyberSource account. To enable AVS, contact CyberSource Customer Support.
2	Unrecognized: the processor returned an unrecognized value for the AVS response.
3	Match: address is confirmed. Returned only for PayPal Express Checkout.
4	No match: address is not confirmed. Returned only for PayPal Express Checkout.
5	No match: no AVS code was returned by the processor.

Business Application Identifiers

Send a business application identifier in the **invoiceHeader_businessApplicationID** field.

Table 64 Business Application Identifiers

Identifier	Description
AA	Account to account
BB	Business to business
BI	Bank-initiated money transfer
BP	Non-card bill payment
CC	Cash claim
CI	Cash in
CO	Cash out
CP	Card bill payment
FD	Funds disbursement (general)
GD	Government disbursement
GP	Gambling payout other than online gambling
LO	Loyalty and offers
MA	Mobile air time payment
MD	Merchant disbursement
MI	Merchant-initiated money transfer
MP	Face-to-face merchant payment
OG	Online gambling payout
PD	Payroll pension disbursement
PG	Payment to government
PP	Person to person
PS	Payment for goods and services (general)
TU	Top-up for enhanced prepaid loads
WT	Wallet transfer

Card Types

The following table lists the card type values to use in **ccAuthService** and **ccCreditService** requests. To see which cards can be handled by each processor, see ["Payment Processors," page 32](#).



Important

CyberSource strongly recommends that you send the card type even if it is optional for your processor and card type. Omitting the card type can cause the transaction to be processed with the wrong card type.

Table 65 Card Types for Authorizations and Credits

Value	Card Type
001	Visa For all processors except Ingenico ePayments and SIX, the Visa Electron card type is processed the same way that the Visa debit card is processed. Use card type value 001 for Visa Electron. Note Ingenico ePayments was previously called <i>Global Collect</i> .
002	Mastercard, Eurocard ¹ : European regional brand of Mastercard.
003	American Express
004	Discover
005	Diners Club: see "Discover Acquisitions and Alliances," page 23 .
006	Carte Blanche ¹
007	JCB ¹
014	EnRoute ¹
021	JAL ¹
024	Maestro (UK Domestic) ¹
1 For this card type, you must include the card_cardType field in your request for an authorization or a stand-alone credit. 2 For this card type on Cielo 3.0, you must include the card_cardType field in your request for an authorization or a stand-alone credit. This card type is not supported on Cielo 1.5. 3 For this card type on Getnet and Rede, you must include the card_cardType field in your request for an authorization or a stand-alone credit.	

Table 65 Card Types for Authorizations and Credits (Continued)

Value	Card Type
031	Delta ¹ : use this value only for Ingenico ePayments. For other processors, use 001 for all Visa card types. Note Ingenico ePayments was previously called <i>Global Collect</i> .
033	Visa Electron ¹ : Use this value only for Ingenico ePayments and SIX. For other processors, use 001 for all Visa card types. Note Ingenico ePayments was previously called <i>Global Collect</i> .
034	Dankort ¹
036	Cartes Bancaires ¹
037	Carta Si ¹
039	Encoded account number ¹
040	UATP ¹
042	Maestro (International) ¹
050	Hipercard ^{2,3}
051	Aura
054	Elo ³
061	RuPay
062	China UnionPay
1	For this card type, you must include the card_cardType field in your request for an authorization or a stand-alone credit.
2	For this card type on Cielo 3.0, you must include the card_cardType field in your request for an authorization or a stand-alone credit. This card type is not supported on Cielo 1.5.
3	For this card type on Getnet and Rede, you must include the card_cardType field in your request for an authorization or a stand-alone credit.

Chargeback Reason Codes

Chargeback Reason Codes for Visa

Table 66 Chargeback Reason Codes for Visa

Reason Code	Description
30	Services Not Provided or Merchandise Not Received
31	Error in Addition
41	Cancelled Recurring Transaction
50	Credit Posted as Purchase
53	Not as Described
56	Defective Merchandise
60	Requested Copy Illegible
61	Fraudulent Mail/Phone Order Transaction
71	Authorization Request Declined / Authorization Declined
72	No Authorization / Transaction Exceeds Floor Limit
74	Late Presentment
75	Cardholder Does Not Recognize the Transaction
79	Requested Transaction Information Not Received
82	Duplicate Processing
83	Nonpossession of Card
85	Credit Not Processed
86	Paid by Other Means
90	Nonreceipt of Merchandise

Chargeback Reason Codes for Mastercard

Table 67 Chargeback Reason Codes for Mastercard

Reason Code	Description
01	Requested Transaction Data Not Received
02	Requested Item Illegible
08	Requested / Required Authorization Not Obtained
12	Account Number Not on File
31	Transaction Amount Differs
34	Duplicate Processing
35	Card Not Valid or Expired
37	Fraudulent Mail/Phone Order Transaction
41	Cancelled Recurring Transaction
42	Late Presentment
47	Exceeds Floor Limit, Not Authorized, and Fraudulent Transactions
50	Credit Posted as a Debit
53	Cardholder Dispute Defective / Not as Described
54	Cardholder Dispute-Not Elsewhere (U.S. only)
55	Nonreceipt of Merchandise
59	Services Not Rendered
60	Credit Not Processed
63	Cardholder Does Not Recognize - Potential Fraud

Commerce Indicators

The commerce indicator is a request value that you send in the **ccAuthService_commerceIndicator** and **ccCreditService_commerceIndicator** fields.

Table 68 Commerce Indicators

Values	Description
aesk and aesg_attempted	See "American Express SafeKey," page 184.
install and install_internet	See "Installment Payments," page 145.
internet (default)	E-commerce order placed using a web site. On Ingenico ePayments, <code>internet</code> is supported only for Cartes Bancaires transactions. Note Ingenico ePayments was previously called <i>Global Collect</i> .
js and js_attempted	See "JCB J/Secure," page 185.
moto	Mail order or telephone order. Not supported on Cielo or UATP. On Ingenico ePayments, <code>moto</code> is supported only for Cartes Bancaires transactions. Note Ingenico ePayments was previously called <i>Global Collect</i> .
moto_cc	Mail order or telephone order from a call center. This value is available only on the Asia, Middle East, and Africa Gateway.
pb and pb_attempted	See "ProtectBuy," page 193.
recurring and recurring_internet	See "Recurring Payments," page 205. ■ <code>recurring</code>—U.S. transaction or non-U.S. mail order / telephone order (MOTO) transaction ■ <code>recurring_internet</code>—non-U.S. e-commerce (Internet) transaction
retail	See Card-Present Processing Using the Simple Order API.

Table 68 Commerce Indicators (Continued)

Values	Description
spa and spa_failure	See "Mastercard Identity Check," page 186.
vbv, vbv_attempted, and vbv_failure	See "Visa Secure," page 195.

CVN Codes

The CVN code is returned in **ccAuthReply_cvCode** in the authorization reply message. See ["Card Verification Numbers \(CVNs\)," page 97](#), for a description of CVN.

Table 69 CVN Codes

Code	Description
D	The transaction was determined to be suspicious by the issuing bank.
I	The CVN failed the processor's data validation check.
M	The CVN matched.
N	The CVN did not match.
P	The CVN was not processed by the processor for an unspecified reason.
S	The CVN is on the card but was not included in the request.
U	Card verification is not supported by the issuing bank.
X	Card verification is not supported by the payment card company.
1	Card verification is not supported for this processor or card type.
2	An unrecognized result code was returned by the processor for the card verification response.
3	No result code was returned by the processor.

CyberSource through VisaNet Acquirers

**Note**

The Visa Electron card type is processed the same way that the Visa debit card is processed. Use card type value 001 (Visa) for Visa Electron.

The following acquirers are supported for CyberSource through VisaNet:

- Absa Bank: Visa, Mastercard, JCB, Diners Club
- Agricultural Bank of China (ABC): Visa, Mastercard, American Express, JCB, Diners Club

**Note**

CyberSource through VisaNet cannot process domestic transactions in China. CyberSource through VisaNet can process only cross-border transactions. A *cross-border* transaction is a transaction for which the payment card is issued in another country and accepted by a merchant in China.

- Ahli United Bank in Bahrain: Visa, Mastercard, JCB, Diners Club
- Arab African International Bank (AAIB): Visa, Mastercard, JCB
- Asia Commercial Bank (ACB): Visa, Mastercard, JCB
- Auckland Savings Bank (ASB): Visa, Mastercard
- Australia and New Zealand Banking Group Limited (ANZ): Visa, Mastercard
- Axis Bank Ltd. of India: Visa, Mastercard, Diners Club
- Banco Nacional de México (Banamex): Visa, Mastercard, American Express, Discover, JCB, Diners Club
- Banco Safra: Visa, Mastercard, American Express
- Bangkok Bank Ltd.: Visa, Mastercard, JCB
- Bank Muscat of Oman: Visa, Mastercard, American Express, Diners Club

- Bank of Ayudhya (BAY): Visa, Mastercard, JCB
- Bank of China (BOC): Visa, Mastercard
- Bank of Communications: Visa, Mastercard

**Note**

CyberSource through VisaNet cannot process domestic transactions in China. CyberSource through VisaNet can process only cross-border transactions. A *cross-border* transaction is a transaction for which the payment card is issued in another country and accepted by a merchant in China.

- Bank Sinarmas (Omise Ltd.): Visa, Mastercard
- Banque Pour Le Commerce Exterieur Lao (BCEL): Visa, Mastercard, American Express, JCB
- Barclays Bank Botswana: Visa, Mastercard, American Express
- Barclays Bank Mauritius Limited: Visa, Mastercard, American Express
- Barclays Bank of Ghana Limited, Barclays Bank of Tanzania Limited, and Barclays Bank of Uganda Limited: Visa, Mastercard, American Express
- Barclays Bank of Kenya: Visa, Mastercard, American Express
- Barclays Bank of Zambia: Visa, Mastercard, American Express
- Barclays Bank Seychelles: Visa, Mastercard, American Express
- BC Card Co., Ltd.: Visa, Mastercard, American Express, JCB
- BLOM Bank: Visa, Mastercard
- Cathay United Bank (CUB): Visa, Mastercard, JCB
- Citibank Hongkong and Macau: Visa, Mastercard, Diners Club, JCB
- Citibank Malaysia: Visa, Mastercard
- Citibank Singapore Ltd.: Visa, Mastercard, JCB
- Commercial Bank of Qatar: Visa, Mastercard, American Express, JCB, Diners Club
- CrediMax (Bahrain): Visa, Mastercard, American Express, JCB, Diners Club
- CTBC Bank Ltd.: Visa, Mastercard, JCB
- First Data Merchant Solutions in Brunei: Visa, Mastercard, JCB

- First Data Merchant Solutions in Hong Kong: Visa, Mastercard, JCB
- First Data Merchant Solutions in Malaysia: Visa, Mastercard, JCB
- First Data Merchant Solutions in Singapore: Visa, Mastercard, JCB
- FirstRand Bank: Visa, Mastercard, American Express, Diners Club
- Global Payments Asia Pacific: Visa, Mastercard, JCB

**Note**

In India, the only supported card types are Visa and Mastercard. All three card types (Visa, Mastercard, JCB) are supported in all other countries that Global Payments Asia Pacific covers.

- Habib Bank Ltd. (HBL): Visa, Mastercard, American Express, JCB, Diners Club
- HDFC Bank Ltd. of India: Visa, Mastercard, Diners Club
- I&M Bank: Visa, Mastercard
- ICICI of India: Visa, Mastercard
- Korea Exchange Bank (KEB): Visa, Mastercard, JCB

**Note**

CyberSource through VisaNet cannot process domestic transactions in Korea. CyberSource through VisaNet can process only cross-border transactions. A *cross-border* transaction is a transaction for which the payment card is issued in another country and accepted by a merchant in Korea.

- Mashreq: Visa, Mastercard, American Express, JCB, Diners Club
- Maybank: Visa, Mastercard, American Express, JCB
- National Bank of Abu Dhabi (NBAD): Visa, Mastercard, JCB, Diners Club
- National Bank of Kuwait (NBK): Visa, Mastercard, Diners Club
- National Commercial Bank (NCB): Visa, Mastercard, mada as described in ["mada Debit Cards," page 24](#)
- Network International: Visa, Mastercard, American Express, JCB, Diners Club
- Overseas Chinese Banking Corp (OCBC): Visa, Mastercard
- Promerica in Honduras and Nicaragua: Visa, Mastercard
- PT Bank Negara Indonesia: Visa, Mastercard

- Qatar National Bank (QNB Group): Visa, Mastercard, American Express, JCB, Diners Club
- Sacombank: Visa, Mastercard, JCB
- Sumitomo Mitsui Card Co.: Visa
- Taishin Bank Ltd.: Visa, Mastercard, American Express, JCB
- United Overseas Bank (UOB) in Singapore and Vietnam: Visa, Mastercard, JCB
- United Overseas Bank (UOB) in Thailand: Visa, Mastercard
- Vantiv: Visa, Mastercard, American Express, Discover, JCB, Diners Club
- Vietcombank: Visa, Mastercard, American Express, JCB, Diners Club
- VietinBank: Visa, Mastercard, JCB, Diners Club
- Vietnam Technological and Commercial Joint-stock Bank: Visa, Mastercard, American Express, JCB, Diners Club
- Visa Guatemala: Visa
- VisaNet Uruguay: Visa
- Westpac: Visa, Mastercard
- Wing Hang Bank: Visa, Mastercard
- Wing Lung Bank: Visa, Mastercard

Expert Monitoring Solutions (EMS) Reason Codes

The following table describes the reason codes returned in positions 4 through 5 of the **ccAuthReply_emsTransactionRiskScore** field. See ["Mastercard Expert Monitoring Solutions \(EMS\)," page 168](#).

Table 70 EMS Reason Codes

Reason Code	Description
01	Suspicious cross border activity
02	Suspicious transaction
03	High number of transactions
04	High number of transactions at an unattended terminal
05	Suspicious recent history of transactions
06	Suspicious activity and high number of transactions
07	Suspicious cardholder not present activity
08	Suspicious activity and low number of transactions
09	Suspicious service station activity
10	Suspicious online activity
11	High amount transaction or high cumulated amount recently spent
12	Suspicious gambling activity
13	Suspicious phone or mail order activity
14	Suspicious grocery store activity
15	High risk country
16	High amount, high number of transactions, and cross border
17	Suspicious activity including previous declined transactions
18	Suspicious airline activity
19	Score forced to be 001 because the transaction being scored was a 04xx message
20	Not a financial transaction
21	Abnormal geographic activity
22	Abnormal, high frequency at the same MCC

Table 70 EMS Reason Codes (Continued)

Reason Code	Description
23	High amount recent ATM activity
24	Suspicious recent ATM activity or suspicious ATM activity following a recent abnormal activity
25	Suspicious telecom activity
26	High number of international ATM transactions
27	High cumulated withdrawal amount on international ATM
28	High velocity of domestic ATM transactions
29	High risk MCC

Electronic Verification Response Codes

See ["Electronic Verification \(EV\)," page 94](#), for a list of the fields in which the Electronic Verification response codes are returned. The following table describes the mapped response codes.

Table 71 Electronic Verification Mapped Response Codes

Response Code	Description
F	First name matches; last name does not match.
L	Last name matches; first name does not match.
M	First name and last name match.
N	No, the data does not match.
P	The processor did not return verification information.
R	The system is unavailable, so retry.
S	The verification service is not available.
U	Verification information is not available.
Y	Yes, the data matches.
1	Electronic verification did not generate a response.
2	The processor returned an unrecognized value.

Formats for Discretionary Data

This appendix provides examples of the formats for discretionary data for specific acquirers. In request messages, you can include discretionary data in the **issuer_additionalData** field. In reply messages, discretionary data can be sent to you in the same field.

**Note**

CyberSource recommends that you contact your acquirer for information about the formats to use.

Example for Visa Guatemala

This example is for issuer-funded installment payments. Additional formats exist; the issuers and acquirers work together to develop and reach consensus on the formats.

Example 96 Discretionary Data Format for Issuer-Funded Installment Payments with Visa Guatemala

VC10000000050000

Table 72 Discretionary Data Format for Issuer-Funded Installment Payments with Visa Guatemala

Position (Character or Digit #)	Number of Characters or Digits	Description
1-2	2	Prefix. Set this value to VC.
3-4	2	Total number of installments.
5-16	12	Total amount.

Example for VisaNet Uruguay

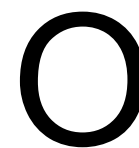
This example is for issuer-funded installment payments. Additional formats exist; the issuers and acquirers work together to develop and reach consensus on the formats.

Example 97 Discretionary Data Format for Issuer-Funded Installment Payments with VisaNet Uruguay

00612012345678910000000008160003359

Table 73 Discretionary Data Format for Issuer-Funded Installment Payments with VisaNet Uruguay

Position (Character or Digit #)	Number of Characters or Digits	Description
1-2	2	Plan type. Set this value to 00. Specifies that the transaction is an e-commerce transaction.
3	1	Grace period. Number of months that the issuer waits before charging customers.
4-5	2	Total number of installments. Possible values: 00 through 99.
6	1	POS entry mode. Set this value to 0. Specifies that the transaction is an e-commerce transaction.
7-15	9	Identity document number. Set this value to the number on the customer's identity document or leave it blank. Format: right justified with 0 (zero) padding on the left.
16	1	Financial inclusion law indicator. Possible values: ■ 1: Law 17934 ■ 2: Law 18099 ■ 3: Asignaciones familiares (AFAM) (family allowance program) ■ 4: Real state law ■ 5: Law 19210
17-28	12	Financial inclusion amount. This value is the amount the bank returns to the customer.
29-35	7	Merchant-generated invoice number.



Frequently Asked Questions

What kind of bank account do I need to accept payment cards?

You need a merchant bank account that is configured to process card-not-present or mail order/telephone order (MOTO) transactions. See ["Acquiring \(Merchant\) Banks," page 28](#).

What types of payment cards can my customers use?

CyberSource can accept payments made with numerous types of payment cards, including Visa, Mastercard, Discover, and American Express. In addition, CyberSource can accept most offline debit cards, which are also known as check cards, many private label cards, and Level II purchasing cards. Your payment processor can limit the types of cards that you can accept. See ["Payment Processors," page 32](#), or contact your CyberSource account representative.

Do I need to sign agreements with the payment card companies?

Some payment card companies, such as American Express and Discover, require you to sign agreements with them. For other card types, such as Visa and Mastercard, you can usually sign a single contract with your acquiring bank or payment processor. Your acquiring bank can help ensure that you sign all of the necessary agreements.

Can I use more than one payment processor or merchant account provider?

Yes. CyberSource can provide you with multiple CyberSource merchant IDs and configure each one to use a different payment processor or merchant account provider.

What happens when my customers commit fraud?

You could be liable for fraudulent transactions. When customers complain that you charged their accounts improperly, you might be required to return their money at your expense; this is known as a chargeback. If you receive a large number of chargebacks, or if a large number of your customers commit fraud, your acquiring bank might raise your fees or revoke your merchant bank account. Contact your CyberSource account representative for information about CyberSource products that can help prevent fraud.

When do authorizations expire?

Most authorizations expire within five to seven days, but the bank or company that issued the card decides how long an authorization lasts.

When an authorization expires, will I be able to charge my customer?

Yes. CyberSource is not notified when an authorization expires, so it is possible to capture an expired authorization. However, the capture might be downgraded, which would increase your fees for the transaction. Additionally, the payment card company can decide not to capture expired authorizations.

If you believe that an authorization expired, you can request a new authorization, then capture the new authorization. However, the new authorization could be denied if the customer's credit limit has been exceeded, if the card has expired, or if the card has been cancelled.

Can I reverse an authorization?

Yes. Some processors allow you to reverse an authorization, which releases the hold that the authorization placed on the customer's payment card funds. For the list of processors that allow you to reverse an authorization, see "[Reversing an Authorization](#)," page 52.

If your processor does not support authorization reversals and you need to reverse an authorization, contact the customer's issuing bank or wait for the authorization to expire.

Can I cancel a capture or credit?

Yes. For some processors, you can use the void service to cancel a capture or credit that you have previously requested. You must request the void before CyberSource submits the capture or credit request to your payment processor. See "[Voiding a Capture or Credit](#)," page 85.

How can I prevent my customers from clicking the "Buy" button more than once?

Use one or more of these options:

- After a customer clicks the "Buy" button, send the customer to a new web page
- After a customer clicks the "Buy" button, hide or disable the button

The Support Center provides sample JavaScript code to disable the "Buy" button after a customer clicks it. The code is available at:

http://www.cybersource.com/support_center/implementation/best_practices/view.xml?page_id=415

Can I change the company name and phone number that appears on my customers' payment card statements?

CyberSource permits you to change these values, which are called merchant descriptors, when you use a payment processor that supports this feature. After your processor configures the merchant descriptors for your account, you can choose which merchant descriptor to use every time you request a transaction. You must also contact CyberSource and your processor to specify default merchant descriptors for your account. See ["Merchant Descriptors," page 169](#).

When do my capture and credit transactions appear on my CyberSource reports?

Capture and credit transactions usually appear on your reports two calendar days after you request them. However, it might take longer for funds to be transferred.

When are funds transferred between my customer's bank account and my company's bank account?

Funds are usually transferred within two to three days after you request a capture or credit.

Ingenico ePayments Credit Card Reversals

**Note**

Ingenico ePayments was previously called *Global Collect*.

Credit card reversals and requests for information, which are also called retrieval requests, are business transactions initiated by your customers through their banks.

The information in this section is generally applicable to all card types and all operating regions although certain details can vary.

Requests for Information

Credit card reversals and requests for information involve communication:

- Between your customer and the acquiring bank
- Between you and Ingenico ePayments
- Between Ingenico ePayments and the acquiring bank

The process is:

- 1 The acquiring bank notifies Ingenico ePayments of your customer's request for information.
- 2 Ingenico ePayments searches for refunds already processed for the transaction identified by your customer.
- 3 Ingenico ePayments responds to the acquiring bank stating "already refunded." Ingenico ePayments does not take any further action because the information request has been satisfied. Requests for information are not documented within any report.

- 4 If Ingenico ePayments's research determines that a refund for the inquiry has not been initiated, Ingenico ePayments forwards the retrieval request to you. All requests received before midnight PT (Pacific Time) are forwarded to you by 0800 PT by email with a request for additional information. See "[Request for Information Example](#)," page 489.
- 5 A request for information is an impending chargeback. If Ingenico ePayments does not receive your answer by midnight PT before the fifth day, your customer's bank initiates a chargeback.

When you receive a request for information, you must respond promptly and with as much detail as possible:

- 1 Respond to your customer's request for information:
 - Address your email to dispute.management@ingenico.com.
 - There is no standard format to follow. However, you should provide as much information as you have. You should provide scanned copies of delivery receipts or official banking information with bank letterheads, bank logos, or other official bank insignia.
- 2 Ingenico ePayments forwards your response by email to the acquiring bank which then communicates with your customer's issuing bank.
- 3 If the information in the response is sufficient in the judgment of the issuing bank or customer in accordance with Mastercard/Visa/American Express rules, the chargeback is not executed. The dispute is dropped without further notification to the acquirer, Ingenico ePayments, or you.

Chargebacks

If one of the following situations occurs, then the issuing bank sends a chargeback (refund) to the customer's card and debits your account.:

- You do not send your response in a timely manner
- The information does not satisfy the reasons defined by the card type
- Your customer submits a valid claim for refund

If the information you provided in response to the request for information is not satisfactory or if your customer decides to charge the item back for any reason as defined by the specific card types, the issuing bank executes a chargeback. This adverse movement of funds is unavoidable, but can be reversed in some cases. See "[Representments](#)," page 487.

If Ingenico ePayments receives a chargeback by 0800 PT, the amount of the chargeback is deducted from your account the next business day and is reflected in:

- The Transaction Search in the Business Center
- The Payment Events Report for that processing day

The chargeback entry includes the reason code for the chargeback. The card types do not circulate lists of reason codes to merchants. However, notable merchant banks freely provide detailed explanations of chargeback reason codes on their web sites. This document provides:

- ["Chargeback Reason Codes for Visa," page 468](#)
- ["Chargeback Reason Codes for Mastercard," page 469](#)

Additionally, you can search the Internet for these phrases:

- Mastercard chargeback reason code
- Visa chargeback reason code

Whenever you receive a chargeback, your account is debited by the full or partial transaction amount associated with the chargeback. Chargebacks are deducted from the funding you would normally receive.

Representments

When you or Ingenico ePayments disputes the legitimacy of a chargeback, a *representation* case is initiated:

- 1 Ingenico ePayments automatically initiates a representment case if your customer initiates a chargeback for a transaction that has already been refunded by you.

As in all representment cases, there is no assurance that the issuing bank will reverse the chargeback even in the face of the evidence. However, the chances of success are excellent. Submitting a representment case does not automatically result in the debiting of your customer's account and the crediting of yours.

- 2 If you want to challenge a chargeback, in other words represent it, then you must do so very quickly. To optimize your chances for success, you must document your facts and submit them to Ingenico ePayments in five or fewer days after receiving notification of the chargeback.

Additionally, you can search the Internet for these phrases:

- fight chargebacks
- representment

- 3** If your representment case is approved by your customer's issuing bank, the bank notifies you by refunding your account for amount of the chargeback. Although it is inconvenient, the payment card companies and issuing banks do not provide any other method of notification.

The notification appears as a chargeback withdrawal that is noted in the Payment Events Report. This event generally takes place 11 to 15 business days after you submit the representment case information to Ingenico ePayments. A chargeback withdrawal credits the financial status and the subsequent funding event.

Request for Information Example

This example illustrates an email you might receive from Ingenico ePayments requesting information. In this example, the Xs represent values for the request.

Dear Sir/Madam,

With regards to the transactions below, we have been requested by the cardholders/ cardholders' banks to provide photocopies of the transaction receipts.

Please reply within 5 days from the date of this e-mail with:

- legible copies of the transaction receipts;
- a manually imprinted & signed voucher in the case of a hand keyed transaction;
- signed delivery information;
- any other relevant documentation to support these charges;
- or any information regarding a possible refund;
- together with a copy of this e-mail.

```
Ingenico ePayments Call-ID      : XXXXX
Bank Case ID                   : XXXXXXXXXX
Credit Card Number             : *****XXXX
External Order Number          : XXXXXXXXXXXX
Merchant Reference              :
Merchant Number                 : XXXXXXXXXXXX
Contract-ID                    : XXXX
```

Transaction history

Transaction	Curr	Amount	Date

Original order amount	USD	XX.XX	DD-MM-YYYY

Total	USD	XX.XX	
Amount currently in question	USD	XX.XX	

Visa and Mastercard International Rules and Regulations specify that Ingenico ePayments's bank must provide a copy of a sales voucher when requested by a cardholder or bank. Under these regulations, failure to provide a fully legible transaction receipt will result in the item being returned unpaid to you. In the event that this transaction was hand keyed into your terminal, you must also supply us with a copy of the manual imprinted voucher you took, to prove the presence of the card.

Remember to keep all original vouchers for 12 months as per your merchant agreement.

Kind regards,

Dispute Management
Ingenico ePayments
P.O. Box 2001
2130 GE Hoofddorp
The Netherlands
Fax: +31 23 554 8663
Email: dispute.management@globalcollect.com

Network Transaction Identifiers

The network transaction identifier is returned in **ccAuthReply_paymentNetworkTransactionID** in the authorization reply message.

CyberSource through VisaNet

For CyberSource through VisaNet, the following values are returned for each card type:

- American Express: American Express generates this value. It is included in all replies from the American Express Global Network (AEGN).
- Mastercard: This value is the qualification information for the Mastercard Interchange Compliance (MIC) program. It is used for all Mastercard responses coming from Banknet through Visa to certified acquirers. Format:
 - Bits 1-4: Banknet date
 - Bits 5-7: Mastercard product ID. See "[Mastercard Product IDs](#)," page 494.
 - Bits 8-13: Banknet reference number generated by Mastercard for each transaction
 - Bits 14-15: Spaces
- Visa and Other Card Types: The payment card company generates this value. It is unique for each original authorization and identifies a transaction throughout its life cycle.

GPN

For GPN, the following values are returned for each card type:

- American Express: The payment card company generates this value. CyberSource saves this value and sends it to the processor in all subsequent capture requests.
- Discover: The payment card company generates this value. CyberSource saves this value and sends it to the processor in all subsequent requests for full authorization reversals and captures.

- Mastercard: The payment card company generates this value. CyberSource saves it and sends it to the processor in all subsequent requests for full authorization reversals and captures. Format:
 - Bits 1-9: Banknet reference number generated by Mastercard for each transaction
 - Bits 10-13: Banknet date
 - Bits 14-15: Spaces
- Visa: The payment card company generates this value. CyberSource saves it and sends it to the processor in all subsequent requests for full authorization reversals and captures.
- Other Card Types: Not used.

Product Codes

The following table lists the values you can use for the product code in the **item_#_productCode** request field.

Table 74 Product Codes

Product Code	Definition
adult_content	Adult content.
coupon	Coupon applied to the entire order.
default	Default value for the product code. CyberSource uses <code>default</code> when a request message does not include a value for the product code.
electronic_good	Electronic product other than software.
electronic_software	Software distributed electronically rather than on disks or other media.
gift_certificate	Gift certificate.
handling_only	Fee that you charge your customer to cover your administrative selling costs.
service	Service that you perform for your customer.
shipping_and_handling	The shipping portion is the charge for shipping the product to your customer. The handling portion is the fee you charge your customer to cover your administrative selling costs.
shipping_only	Charge for transporting tangible personal property from your location to your customer. You must maintain documentation that clearly establishes the location where the title to the property passed from you to your customer.
subscription	Subscription to a web site or other content.

Product IDs

The Visa or Mastercard product ID is returned in **ccAuthReply_cardCategory** in the authorization reply message for all processors except CyberSource through VisaNet.

For CyberSource through VisaNet:

- The Visa product ID is returned in **ccAuthReply_cardCategory** in the authorization reply message.
- The Mastercard product ID is returned in **ccAuthReply_paymentNetworkTransactionID** in the authorization reply message.

For descriptions of the Visa product IDs, see the Product ID table on the Visa Request & Response Codes web page:

https://developer.visa.com/guides/request_response_codes

Mastercard Product IDs



Note

Mastercard can introduce new values for this field without advance notice. See the Mastercard technical documentation for additional information. CyberSource through VisaNet does not edit or validate field content.

Table 75 Mastercard Product IDs

Value	Description	Value	Description
CBL	Carte Blanche	MOC	Standard Maestro Social
DAG	Gold Debit Mastercard Salary	MPA	Prepaid Mastercard Payroll Card
DAP	Platinum Debit Mastercard Salary	MPB	Mastercard Preferred Business Card
DAS	Standard Debit Mastercard Salary	MPC	Mastercard Professional Card
DCC	Diners Club	MPD	Mastercard Flex Prepaid (Canada only)
DOS	Standard Debit Mastercard Social	MPF	Prepaid Mastercard Gift Card
JCB	Japanese Credit Bureau	MPG	Prepaid Mastercard Consumer Reloadable Card

Table 75 Mastercard Product IDs (Continued)

Value	Description	Value	Description
MAB	World Elite Mastercard for Business	MPJ	Prepaid Debit Mastercard Card Gold
MAC	Mastercard Corporate World Elite	MPK	Prepaid Mastercard Government Commercial Card
MAP	Mastercard Commercial Payments Account product	MPL	Platinum Mastercard Card
MAQ	Mastercard Prepaid Commercial Payments Account	MPM	Prepaid Mastercard Consumer Promotion Card
MAV	Mastercard Activation Verification	MPN	Prepaid Mastercard Insurance Card
MBB	Mastercard Prepaid Consumer	MPO	Prepaid Mastercard Other Card
MBC	Mastercard Prepaid Voucher	MPR	Prepaid Mastercard Travel Card
MBD	Deferred Debit Mastercard BusinessCard	MPT	Prepaid Mastercard Teen Card
MBE	Mastercard Electronic Business Card	MPV	Prepaid Mastercard Government Benefit Card
MBP	Mastercard Corporate Prepaid	MPW	Prepaid Mastercard Corporate Card
MBT	Mastercard Corporate Prepaid Travel	MPX	Prepaid Mastercard Flex Benefit Card
MCB	Mastercard Business Card/ Mastercard Corporate Card	MPY	Prepaid Mastercard Employee Incentive Card
MCC	Mastercard Card	MPZ	Prepaid Mastercard Emergency Assistance Card
MCE	Mastercard Electronic Card	MRB	Prepaid Mastercard Electronic BusinessCard
MCF	Mastercard Electronic Fleet Card	MRC	Prepaid Mastercard Electronic Card
MCG	Gold Mastercard Card	MRG	Prepaid Mastercard Card Outside U.S.
MCM	Mastercard Corporate Meeting Card	MRH	Mastercard Platinum Prepaid Travel Card
MCO	Mastercard Corporate	MRJ	Prepaid Mastercard Gold Card
MCP	Mastercard Corporate Purchasing Card	MRK	Prepaid Mastercard Electronic Commercial
MCS	Mastercard Standard Card	MRL	Prepaid Mastercard Electronic Commercial
MCW	World Mastercard Card	MRS	Prepaid Mastercard ISIC Student Card
MCX	Mastercard Card (international use)	MRW	Prepaid Mastercard BusinessCard Credit Outside U.S.
MDB	Debit Mastercard BusinessCard	MSI	Maestro point-of-sale debit program
MDG	Debit Gold Mastercard	MTP	Mastercard Platinum Prepaid Travel Card

Table 75 Mastercard Product IDs (Continued)

Value	Description	Value	Description
MDL	Business Debit Other Embossed	MUS	Prepaid Mastercard Unembossed U.S.
MDM	Middle Market Fleet Card	MWB	World Mastercard for Business
MDN	Middle Market Purchasing Card	MWE	Mastercard World Elite
MDO	Debit Mastercard Other	MWO	Mastercard Corporate World
MDP	Debit Mastercard Platinum	PRO	Proprietary Card
MDQ	Middle Market Corporate Card	PVL	Private label card
MDS	Debit Mastercard	SAG	Gold Mastercard Salary-Immediate Debit
MDT	Mastercard Business Debit	SAL	Standard Maestro Salary
MDW	Mastercard Black Debit/World Elite Debit Mastercard	SAP	Platinum Mastercard Salary-Immediate Debit
MEB	Mastercard Executive BusinessCard	SAS	Standard Mastercard Salary-Immediate Debit
MEC	Mastercard Electronic Commercial	SOS	Standard Mastercard Social-Immediate Debit
MEF	Mastercard Electronic Payment Account	SUR	Prepaid Mastercard Unembossed Outside U.S.
MEO	Mastercard Corporate Executive Card	TBE	Business-Immediate Debit
MET	Titanium Debit Mastercard	TCB	Mastercard Business Card-Immediate Debit
MGF	Mastercard Government Commercial Card	TCF	Mastercard Fleet Card-Immediate Debit
MHA	Mastercard Healthcare Prepaid Non-tax	TCO	Mastercard Corporate-Immediate Debit
MHB	Mastercard HSA Substantiated	TCP	Mastercard Purchasing Card-Immediate Debit
MHC	Mastercard Healthcare Credit Non-substantiated	TDN	Middle Market Mastercard Purchasing Card-Immediate Debit
MHH	Mastercard HSA Non-substantiated	TEB	Mastercard Executive BusinessCard-Immediate Debit
MIA	Mastercard Unembossed Prepaid Student Card	TEC	Mastercard Electronic Commercial-Immediate Debit
MIK	Mastercard Electronic Consumer Prepaid Non U.S. Student Card	TEO	Mastercard Corporate Executive Card-Immediate Debit
MIL	Mastercard Unembossed Prepaid Non U.S. Student Card	TLA	Mastercard Central Travel Solutions Air-Immediate Debit
MIP	Mastercard Debit Prepaid Student Card	TNF	Mastercard Public Sector Commercial Card-Immediate Debit

Table 75 Mastercard Product IDs (Continued)

Value	Description	Value	Description
MLA	Mastercard Central Travel Solutions Air	TPB	Mastercard Preferred Business Card-Immediate Debit
MLC	Mastercard Micro-Business Card	TPC	Mastercard Professional Card-Immediate Debit
MLD	Mastercard Distribution Card	WDR	World Debit Mastercard Rewards
MLL	Mastercard Central Travel Solutions Land	WMR	World Mastercard Rewards
MNF	Mastercard Public Sector Commercial Card		

Visa Secure Response Codes

The Visa Secure response code is returned in **ccAuthReply_cavvResponseCode** in the reply message for an authorization request. See ["Visa Secure," page 195](#), for a description of Visa Secure.

Table 76 Visa Secure Response Codes

Response Code	Description
0	CAVV not validated because erroneous data was submitted.
1	CAVV failed validation and authentication.
2	CAVV passed validation and authentication.
3	CAVV passed the validation attempt.
4	CAVV failed the validation attempt.
6	CAVV not validated because the issuer does not participate.
7	CAVV failed the validation attempt and the issuer is available.
8	CAVV passed the validation attempt and the issuer is available.
9	CAVV failed the validation attempt and the issuer is not available.
A	CAVV passed the validation attempt and the issuer is not available.
B	CAVV passed the validation with information only; no liability shift.
C	CAVV attempted but not validated; issuer did not return CAVV code.
D	CAVV not validated or authenticated; issuer did not return CAVV code.
I	Invalid security data.
U	Issuer does not participate or 3-D secure data was not used.
99	An unknown value was returned from the processor.

Values for the Wallet Type Field

The wallet type is sent in the **wallet_type** field in authorization requests and credit requests. Possible value are:

- 101: Masterpass remote payment. The customer created the wallet by manually interacting with a customer-controlled device such as a computer, tablet, or phone. This value is supported only for Masterpass transactions on Chase Paymentech Solutions and CyberSource through VisaNet.
- 102: Masterpass remote near field communication (NFC) payment. The customer created the wallet by tapping a PayPass card or customer-controlled device at a contactless card reader. This value is supported only for card-present Masterpass transactions on CyberSource through VisaNet.
- 103: Masterpass Apple Pay payment. The payment was made with a combination of Masterpass and Apple Pay. This value is supported only for Masterpass Apple Pay transactions on CyberSource through VisaNet. See [Apple Pay Using the Simple Order API](#).
- 216: Masterpass Google Pay payment. The payment was made with a combination of Masterpass and Google Pay. This value is supported only for Masterpass Google Pay transactions on CyberSource through VisaNet. See [Google Pay Using the Simple Order API](#).
- 217: Masterpass Samsung Pay payment. The payment was made with a combination of Masterpass and Samsung Pay. This value is supported only for Masterpass Samsung Pay transactions on CyberSource through VisaNet. See [Samsung Pay Using the Simple Order API](#).
- SDW: Staged digital wallet. An issuer or operator created the wallet. This value is supported only for Masterpass transactions on Chase Paymentech Solutions.
- VCIND: Visa Checkout payment. This value is supported only on CyberSource through VisaNet, FDC Compass, FDC Nashville Global, FDI Australia, and TSYS Acquiring Solutions. See [Getting Started with Visa Checkout](#).

For additional information about the **wallet_type** field, see [Appendix A, "API Fields," on page 238](#).

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